

```
{ 'xy-ric2': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': False,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
'file_name_prefix': 'xy_noise_off'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1
Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X
6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
'initial_cost_history': [ 4.118038614266066,
2.900590937440303,
2.874944433071395,
3.9296977116518517,
3.1841729388967344,
3.695928991667305,
3.180227821654077,
3.0475045550195774,
2.4669562192398296,
3.4124916083581294],
'noise_details': { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [0, 0, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': False,
'type': 'depolarizing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -8.433468022982382,
-8.464942606128835,
-8.43123094137328,
-8.435401809869298,
-7.9812014353834515,
-8.371080050195895,
-8.46198057547588,
```

```
-8.19415563172473,  
-8.438078664602543,  
-8.481886373274177],  
'optimized_parameters': [ [ 0.4631008843346199,  
0.8090636887146863,  
1.1101507421405439,  
1.3400184709676968,  
1.7452332568602784,  
1.9308700883292553,  
2.1752679291102544,  
2.1784212211010603,  
3.800207092295721,  
3.800207092295721,  
3.800207092295721,  
3.800207092295721,  
3.800207092295721,  
3.800207092295721,  
4.798448841518561,  
4.812026604031267,  
6.02543601067661,  
6.02543601067661,  
6.5265343452987,  
6.5265343452987,  
6.526534345298701,  
6.528381158760148,  
6.528381158760148,  
7.113023968622042,  
7.113023968622042,  
7.593654062389684,  
7.5949106716383605,  
7.780058793305066,  
9.000370275113886,  
9.005799799983558,  
-0.22105281285875572,  
0.07196296170715184,  
1.0527781347416576,  
-0.4594806493129553,  
1.0526039239081737,  
-0.01709270509934428,  
0.3199675106146978,  
-0.0972220164715271,  
0.9145097677998977,  
0.09808103992344623,  
0.5373974901276602,  
-0.5786059880820694,  
-0.02594763303680847,  
-0.22132394348491485,  
-0.7274785930689547,  
-0.41090279602128,  
-0.520884263329148,  
-1.0260835302846771,  
0.1692597388493397,  
1.3250313995601488,  
1.510357845739394,  
-0.9233604808876308,  
-0.2049157632594096,  
-0.9894281051832,  
0.25772120620526445,  
-0.04822142329545429,  
-0.3912638297585932,  
-0.48486712444934904,  
1.4836649713082644,  
1.566996182191827,  
0.08093959953023218,  
-2.3562926055425963,  
-0.21876618997457167,  
0.10911407834571141,  
-1.3776955808248483,  
-0.31816531211057325,  
0.7083857256813267,  
0.5593482503842677,  
0.7483381318097894,  
0.010042025170353972,  
0.05290425882351868,  
0.20616876613540233,
```

-0.8673924682871486,
-0.08017548043713153,
-0.3056222778255953,
2.3981163154468383,
0.37079365521682633,
0.3870558721039623,
0.19341028278759992,
-0.3154248344945196,
-0.12073260037561208,
-1.5858414285227351,
0.59830347262823,
0.30388710178795286,
-1.4558796844932331,
-0.3225618511944297,
0.09243390755822743,
-0.0006426173071943948,
0.7561662109631148,
-1.3902324585417218,
0.1677701056843372,
2.7311884232201735,
-1.1122880815211051,
0.0467986017971284,
-0.020089420529231013,
0.005481851366742631,
1.3205596999819047,
-0.1290403664286252,
0.3870444394307043,
-0.6549197738372419,
-0.17656835190090397,
-0.0008013451895215781,
-0.8769883624496301,
1.6929520910231297,
0.7288671099890552,
-0.6169673343691255,
1.667217439782634,
0.2554635335546482,
-0.029515171705053127,
-0.8290153205719569,
0.042733616172741226,
-0.5148227135977747,
1.2301836750084596,
-0.9175908300980525,
0.5229779878549122,
-1.5901934710811256,
-0.07316585886439231,
1.1195018365185199,
-0.31111306362203306,
0.7254709796877604,
1.1995417479543053,
0.6908225815252583,
-0.2919630814941808,
0.15955998483086176,
1.2871422030548714,
-1.2683991057896014,
-0.3768598572966417,
1.4863954796333156,
-0.4300298743216642,
0.19151460300148976,
0.9665838873771405,
-0.07244306377732504,
-0.04169546816073936,
-0.389029635704794,
1.5218961812832925,
-0.1772066326032777,
-0.70949107362336,
-0.015222466074935298,
0.3587128629768407,
1.5809494716454662,
0.12428164288740734,
1.5083554487395696,
-0.044379618902227067,
-0.03505579308192811,
-0.6786538117837789,
-0.007428157845337949,
1.5281494732580478,

-0.14124856089608578,
2.726718429443767,
-0.003106328084445967],
[0.5270475437101116,
0.7815425625485376,
1.6469610445191378,
1.6712876404917778,
2.122904188877425,
2.131106816606521,
3.7367798442214175,
3.7367798442219424,
4.750276145421292,
4.853973384996016,
4.865098072058574,
4.873980651141647,
4.987599328262889,
5.00246239884106,
5.520902021852277,
5.793284849001337,
5.8087832398624775,
7.432729128414403,
7.432953787975026,
8.144896858715603,
8.172580744083415,
8.788071761144659,
8.792288306719156,
8.828260688191317,
9.011586808740137,
9.011995976067807,
9.538729699920484,
9.576884709106812,
10.486750161390887,
10.48675016212107,
-0.15531374899360717,
-0.0849039080739262,
-0.8434555658556069,
0.9447826609611266,
-0.19482082661427938,
2.4999420394249707,
-2.7164831941210483,
-1.5408169364026967,
-0.8621482593978547,
0.31547458537730877,
1.4220435826823787,
1.348860553316339,
0.40827474166943256,
1.063960383917828,
-0.21906798993221652,
-0.33263312803097533,
0.4889869181158972,
-0.25524084746075687,
-1.3622146811064544,
0.996105149433398,
1.4161210728452394,
-0.6491827941870941,
0.7399329880002848,
-0.11994422281276146,
3.179573825543474,
-0.034943024466848674,
0.6075464711124173,
-0.3019825794696199,
0.029392952712390567,
0.14840732209332994,
0.08592470568620861,
0.016725634393736574,
-1.3697962205589145,
-0.06155375586805023,
0.16508080354919497,
-0.5975014914051775,
0.09635736997937351,
1.254101402191796,
-0.6193953337438176,
0.47602269510847217,
-1.7251540483535392,
0.5378274350019089,

-1.0159384367751108,
1.0968223266795405,
-0.79878134010976,
0.10468891520322528,
-0.2910173593847854,
-0.7078981306941186,
-0.3510837282628223,
0.5371002854277007,
-0.4911274293534503,
1.7273270693440552,
1.3500555714793954,
0.9052059235505011,
0.4771526028272018,
-0.1893747959421651,
0.13989050072973697,
-0.5930955187300607,
0.36780210947560876,
-0.5043370473487999,
-0.4851893292987325,
0.1987440888222681,
-0.5806445532873666,
2.3171646404124093,
0.06032126128170586,
-0.6389920437730593,
0.9985359186149056,
0.04870155999532171,
0.16447399991313041,
0.05315090520136875,
0.15727255736970833,
-1.3722292706433088,
-0.33233649668322635,
0.7758243461945761,
-0.20476613404890107,
0.06972046040604139,
0.3143879960336948,
-0.01452323353492752,
-1.2901366074359686,
-0.31033000747276407,
-0.16264485166390844,
-1.3914094473719103,
-0.5273128620655095,
0.02465459151953875,
0.10942766294132272,
1.0292094835171124,
-1.882689768994833,
-1.553216293090225,
1.3287846024814622,
0.20963913600213377,
0.7195732642990699,
0.0969787732183028,
-0.9744344415633356,
1.5822487402774768,
0.7077471503862973,
-0.766345302327988,
0.3831437091022152,
-0.12028853613988252,
-0.8209919023748664,
-0.21164805512089607,
0.5290684330656528,
-0.1040878870845114,
1.1956945887281514,
0.25703931501419025,
0.6344733295912399,
-0.02497584874729833,
-0.20937697561532842,
-1.4409411147319051,
0.9707320454927133,
1.4039378085761043,
0.9760826987262126,
-0.17314228405542512,
0.16734556762922406,
0.07881248116464283,
-0.2924467918405051,
1.4946430081327182,
0.7516806171134963,

0.799700486923224,
-0.8440600324500036,
-0.01069194371181784],
[0.3388610072211759,
0.3930923343746358,
0.4059472295013711,
0.4597995545870322,
0.4597995545870322,
0.4667083780713653,
0.9924683364283498,
1.054118787032341,
1.3839145947816414,
1.4469616998432806,
3.044048229936016,
3.044048229936016,
4.682975233106521,
4.682975233106521,
5.287925895396742,
5.287925895396742,
5.625677830869467,
5.625677830869467,
6.519005703020383,
6.519005703020383,
6.519005703020383,
6.854665591577039,
6.854665591577039,
8.092962128458302,
8.097830997780322,
8.102159968176109,
8.111954675742565,
8.727038821401255,
8.727038821401255,
8.784424594255395,
-0.8636093730812797,
-0.04742973062196874,
-0.37989856483295564,
0.3750798322441465,
-0.509087961968837,
0.5366875438436245,
0.1232704940065221,
0.5249134835872401,
0.13594138148710633,
0.10802292582448766,
-0.7156772546121367,
0.34756877087651405,
-0.5704812075854907,
0.289395233495819,
0.2674862930849357,
0.8724143866961164,
0.5770602988816299,
0.658777432128999,
-0.47288232304292416,
-0.029663566814993374,
-1.323180236232167,
-0.2890901656039231,
-0.015141732930961212,
1.1846850771343012,
-0.4506688471965015,
0.22241980016839272,
-0.015748484493019736,
1.1522499025192263,
0.2996907655672081,
-1.7370631162182948,
-0.35306232532663057,
0.17322471105923057,
-1.1855180134814352,
-0.31537295018088835,
-1.8120366955290261,
0.6663892425075478,
0.21874911670355193,
1.2729693967005706,
-0.14230256773273223,
0.2012427441067739,
-2.3269317425793643,
-0.03299848401271634,

-0.4074853136565566,
-0.30371905431664253,
-0.12037074206123735,
0.8456676988075262,
-0.21838328071319946,
0.019494764922420988,
0.16383099728733078,
0.056056403895700845,
-3.5632063709291613,
0.6579904510057321,
0.5207861046034454,
-0.017180990359968364,
0.2532807007765998,
-0.03315495616728509,
0.7564905992030696,
-0.11091568858030387,
-0.9281890577202553,
-0.6082181315906088,
-0.4480511658411004,
2.1953283079942025,
-0.30446713497464617,
-0.03359602431311142,
0.28972943746372876,
0.17817064395522006,
0.7965754083210965,
-0.9520780563921613,
0.10150445450145726,
1.3510884636720888,
-0.37849615173878054,
0.17729737647006832,
0.1884683806301832,
-0.03234547530918763,
-1.338763456177623,
-0.6711058962200585,
-0.012976043431998471,
0.027578392175021227,
0.10161286285089129,
-0.00939135247962935,
-1.302164623174096,
1.3715576399721376,
0.737767307868418,
1.5939269674119705,
1.4291758244230544,
0.5601859764306084,
0.3488764286390935,
-1.093762224545935,
-1.1976831420922067,
1.4557381665683378,
0.8924472305311927,
0.5300221300338881,
1.0785280788144638,
-0.6029509417428172,
-0.6790632925762424,
0.4138865114731201,
0.2917474006869537,
1.8050154062536312,
0.05177383423831921,
0.25467777497912664,
1.4671356779039024,
0.4422247342165525,
-1.5431607378377095,
1.8101814873938125,
1.6870467346899987,
0.7517920825884995,
1.3622912362338087,
0.13653050459472865,
1.5210183141257738,
-0.004247168330480351,
0.26137218980413907,
0.5451511562235306,
-0.6705596749039198,
1.2136085126593865,
0.5231529087420831,
-0.03279466120532818,
0.2548954759886831,

1.5797820110558691,
1.075840789720537,
1.5541818084195007],
[3.821020346740443e-17,
1.386496864345847,
1.386496864345847,
1.7951472089989222,
1.8559400139285378,
2.7980544315911504,
2.7980544315911504,
2.82954781812891,
2.82954781812891,
2.82954781812891,
2.956183634338348,
3.840561875918348,
3.840561875918348,
4.79501776182173,
4.79501776182173,
5.670460018404704,
5.670460018404704,
5.670460018404704,
6.865392250052202,
6.865392250052202,
7.506826147384778,
7.506826147384778,
8.737685173635683,
8.737685173635683,
8.737685173635683,
8.737685173635683,
9.601023092540274,
9.601023092540274,
9.853738132880634,
9.853738132880634,
0.5611658219809542,
1.427331325859137,
0.8881601644240952,
-0.8937497791603879,
0.5712044928850715,
1.3195951619089188,
0.5467792160605555,
-0.6816630744322637,
0.06707477836950569,
0.6798692668930644,
0.19576168860111362,
0.5094724850631845,
1.1424414477256812,
0.30471145260454696,
-0.08158911485577636,
0.9585661478313002,
1.0836238583568754,
-0.3718197048926503,
0.7841948506971846,
0.3023515457437066,
0.7662240328883452,
0.9970048707689876,
-0.8946064024093777,
0.27649390200762114,
-0.15342626519190924,
-0.4015720487025046,
1.460926694270634,
-0.055421168101734586,
0.6077050880702092,
-0.7932081335349191,
1.3188332541813446,
0.01929555634963922,
-1.1301845164589392,
-0.1718174564221745,
2.26154163970611,
-0.9882028269623031,
0.8038643847286138,
0.6849304102208338,
0.874798983022982,
-0.016486064274954207,
0.0264221098096002,
-0.3625000384278827,

1.4376331477886695,
0.8499992740594865,
0.10042171561200945,
0.14542648933357244,
1.3513090878714353,
-0.22248922869259902,
-0.4483175327546578,
1.2345758678942387,
-0.18983522552694923,
0.019591210301167533,
-1.3607248526627624,
0.018375533581887468,
-0.3730738776375715,
1.0123537416913542,
0.7412601688930508,
-1.6525919628732422,
-0.8341682190885938,
0.04519729634722039,
-0.06718028327556408,
-1.5649719285752928,
-0.37531851249345394,
-0.11822841604899964,
0.008154106549025474,
-1.4601449692077721,
-0.0030993446167428395,
1.462573870839384,
-0.11887757571245174,
-0.09361960188417112,
-1.2543408854038138,
-1.525280802664792,
-1.241413372948363,
0.008758808605552393,
-1.6676332942965022,
0.13826170532334445,
0.7537363962746654,
0.0833397476047736,
-0.12216949704941273,
0.02084807234445201,
-0.3421949354726438,
0.0017882611822728064,
1.5186851123669147,
-0.4603163345929323,
-0.09956972276089109,
0.7247597830417082,
0.5304935691839561,
0.003193152207841171,
0.5581346990750625,
0.010542051444639388,
0.7553167297004638,
0.5956891015800863,
-0.05566254332595716,
0.5362461335652635,
0.009895282042162012,
-0.3029901133286735,
0.09399206534914867,
0.8818812838713691,
-0.06795832284115338,
-1.580933930101608,
0.3014975711017802,
1.2092022074751856,
-0.019692531397995948,
0.9250981578351692,
0.4673383297638096,
0.03344339272872125,
0.13058472572385768,
0.036262684184263205,
0.458095375934523,
0.6492846129896516,
1.3039221898344189,
0.024764812872192767,
0.7236926243101844,
0.01444840086266692,
-1.0388939706148563,
-0.08359693344439546,
-0.4698004238001188,

-0.48055768545836347,
0.4662000815300196,
-0.041204969991243484],
[0.01798320141016013,
0.3393419127512466,
0.7047697703765823,
0.7064789163091142,
0.7191656785855808,
0.7198672068604162,
0.719867206860416,
1.0840512961729545,
1.0900003904895101,
1.1926153174924548,
1.867054342490295,
1.867054342490295,
2.368094916985588,
2.3683751276938714,
3.6383003552329964,
3.642527762446636,
4.505845683894636,
4.602446412902637,
4.972427659100928,
4.972427659100928,
5.1148164150678035,
5.8066136392845,
5.820681023327439,
6.727088267818048,
8.352653016300794,
8.773334122756978,
9.065902540064004,
9.065902540064004,
9.74880092287043,
11.30145464466351,
-0.13904770480884485,
0.23493722133539818,
0.4497803172772882,
-0.640584995353188,
0.9779304528121034,
0.0011745349607135795,
-0.7175493520168287,
0.373620380836484,
0.08111255661159392,
0.6653458367314794,
0.0037967443269442437,
0.6286122183467432,
-0.03825426016359485,
0.6896824961465278,
-0.23916946602734285,
0.4624850989736848,
0.1419493525208794,
0.7145176662978019,
-0.316909964913161,
0.6605458698076423,
-0.39309909050950365,
0.7623048158369822,
0.7469736169282462,
1.2400892032597353,
0.12266455668869777,
0.19578648318487166,
-0.784043232212839,
-0.10334183265831844,
0.32774480702026265,
-0.011491906972883237,
0.322988869457402,
-0.2027870428092001,
0.1281571522083961,
-0.1947961113336468,
1.7114513053475526,
0.005425684515457419,
0.8136825826883561,
-0.25903007981373005,
-0.5015428579892063,
-0.2600978115589891,
-0.15405473476720474,
-0.4207498663011112,

1.04622682040645,
-0.8642339256273757,
0.6248691295250803,
-0.3539240130838131,
0.011689879721248103,
-0.3068283940685192,
-0.6048660879777619,
-0.11704784670887185,
-1.2745062338511486,
-1.7890038533223112,
-0.7818671277256136,
-0.8877455887286905,
0.3164818198590212,
-0.27477307924208777,
1.8083251930535746,
-0.5031479272825263,
-0.9648993597903888,
-0.41682962131770707,
0.27040705073896654,
1.0892900815418987,
0.17779960873468054,
0.020337033950376556,
1.0551529464315492,
-0.10705558118147128,
-0.8851977322505518,
0.2232945485845247,
-0.5850403569085746,
-0.7690595025978197,
-0.4391673030332381,
-0.9309566421919284,
0.2973030450025941,
0.6780545449896137,
0.14045373030826291,
-0.5109313900859794,
-0.44070010469428217,
1.6940975579604587,
0.17062304524159472,
-0.402681393200546,
-0.7091187255059627,
1.3132357546586793,
0.2729391261599054,
1.0115940175088693,
1.0046673925246778,
0.4982257464819494,
0.04454679168285006,
0.4863023994043421,
-1.1433261311581624,
-1.820308437409531,
0.6035405700431521,
0.04003907881711185,
2.99381748600971,
-1.0437523456620463,
-0.22517480393070408,
0.10746937069246698,
-0.8788513722547997,
1.286531186523234,
-1.3157160458746164,
-0.6979748907629274,
-0.16573593981190854,
0.2529194783090185,
-0.2562517800846549,
-0.7966921479109774,
-0.4446940958312971,
-1.052310031662883,
-0.1506245930250076,
0.1615975776996014,
2.9220364155549214,
-1.7197843955783065,
-0.6592103621593355,
-0.2007498223803255,
-1.6188873615119244,
-0.15193115939751467,
-0.20866967970071015,
1.2910791163230286,
-0.4917152660712325,

-0.42348313202027765,
0.3518303936962707,
-0.10034486495238693],
[0.009039772647041902,
1.1400526086001903,
1.466905028683042,
1.6524670085313564,
1.987797816914966,
1.9900950346150876,
2.1413202332332344,
2.390935514217652,
2.402429399816812,
2.81913686416337,
2.81913686416337,
3.5656810200683298,
3.5656810200683298,
3.5656810200683298,
3.76603447051701,
3.76603447051701,
5.499236323276389,
5.4992363232763894,
5.4992363232763894,
5.532789537471719,
5.532789537471719,
7.371733777832876,
7.371733777832876,
7.680783486275433,
7.706623781446164,
7.706623781446164,
8.457742149729329,
8.457742149729329,
8.95222736351662,
8.95222736351662,
0.18928393791812126,
1.0354889043700823,
0.3703113207409527,
0.30988589523494625,
-0.34669340410550803,
0.09615284519636855,
0.1011898884094612,
0.6160115432573767,
0.6148540393435122,
0.07934943483664206,
0.8557624329627399,
-0.7672633290674925,
-0.14188111026081618,
0.1995320641320393,
0.4594888802421805,
-0.03593187081620776,
1.2926591415659605,
0.31433950962447343,
0.8053272546604554,
0.6634815559763872,
0.311660786273531,
0.0294653444200854,
0.3044394736040779,
0.7599259037372076,
0.22962735656034647,
0.22562971232722423,
0.29511485598499815,
0.02260075636691185,
-0.9497096253675098,
-0.5331371571917215,
0.2192512323868535,
1.104832430155296,
0.8636617903941131,
-0.6756684630428217,
0.4169844309263729,
1.1297527651723116,
-1.160118439748996,
-0.7971836651224,
1.9169656561991728,
-0.007260796290839314,
0.3614132074474742,
0.6243022778229537,

1.8335025630018775,
0.35139722497115833,
-0.7247870129359816,
-0.6744811297738377,
0.49997538785346773,
1.4139521212248116,
-3.0799818401546553,
-0.14768564076593044,
0.06251500852230953,
-0.8547134606957393,
-0.30107856776262504,
-0.7654972812734481,
0.28603020918627825,
0.2129303248959857,
-1.2077141264570097,
-0.8488549318812054,
0.6492574804166855,
0.7751290676155554,
0.40909519807061084,
-1.2750630737105486,
0.4187798718836698,
0.33498424523924863,
-0.19017028209895162,
-0.03232931911977397,
1.0621483440417678,
0.452017889610544,
1.0192767515800205,
0.7290563184502946,
-0.5286462788982386,
-0.05512372609304388,
1.3189151919853837,
0.5865235617306253,
-0.579843833868396,
0.6980158293666034,
0.027004860579276476,
1.2876675233834314,
-0.17185617277774212,
-0.3827017785428823,
-0.7286404996479297,
1.1682112525473984,
0.22765379123755217,
1.4870969768845252,
1.1642898801914527,
-0.03976972562460009,
-0.8163296820868416,
-0.15318057329999143,
0.0482918949035788,
-0.06014345828968787,
1.1478583262090496,
-0.005141116181289305,
-0.5820859555621626,
1.086519910329415,
-1.08558016317555,
1.531511042803357,
0.042734048269580674,
0.7452599788242152,
-1.3094923903018465,
-0.9012205175381317,
-0.0755166851159581,
1.5493661520767013,
-0.2278023712186688,
0.8627212805492942,
1.8523276191392382,
-0.10961709979291216,
0.6746606852631131,
0.22566845572735392,
0.5920154791154506,
1.2709342533151342,
0.26903045177842116,
0.11724485476711638,
0.621942277674204,
0.15491924869980966,
1.472217084758381,
0.8975149794611956,
0.3601026339731485,

1.3234513983254743,
0.0182465793142515,
-0.11914786829986494],
[-5.765348397008027e-18,
0.3428984748224433,
0.3655450682584724,
0.704784031551808,
1.5409081472811441,
1.549300366088437,
2.3787666619954635,
2.3787666619954635,
2.8808650819472406,
2.884987740396357,
3.1308808834009203,
3.1363836212549696,
4.869321728501938,
4.869321728501938,
6.1150174264690085,
6.2582278136815175,
6.271636236200362,
6.551203581243674,
6.59741701640448,
8.230564833747417,
8.294003346338926,
8.31431753208387,
8.369586036193128,
8.415319617883744,
8.70447046242739,
8.709120596281233,
8.918659758853744,
8.99410216914391,
9.002256766523654,
10.218690251738174,
-0.6798627919679444,
-1.1436724543472427,
-0.1278269606092795,
-0.9286552360068459,
0.6524046490818174,
0.06134572949387961,
-0.49761179900021657,
-1.3918471119443894,
0.5858287319319103,
0.18669127045720346,
-0.4586789737568629,
0.2400608953322058,
0.022222124883641912,
0.9282004764805514,
0.6952358394741303,
0.805684979006905,
0.6146729452742722,
0.6777524139082742,
-0.8117842884117104,
0.3426849285038029,
0.7100299235979358,
0.7658186509139059,
-0.05418633990950139,
0.1761917490887145,
0.5744825473612316,
0.31705937472296297,
-0.9188523464356001,
-0.7337446596965294,
0.2061368312243915,
-0.6645694031850036,
-0.009871104765786294,
-0.14388187083546433,
-1.2485707106311008,
2.849540809212292,
-0.5119215846255833,
-0.19381894062193808,
-0.17310717674563775,
0.8465397629359774,
-0.46757240698237135,
-0.05607605828560263,
-0.40986137011816176,
0.07649505002491204,

-2.1522907800348112,
-0.6653346582594515,
0.4084801543759574,
0.4028897820911683,
0.16908117843954765,
-0.07217026591146015,
-0.72804755296126,
-0.13548738477012795,
-2.590679566047418,
-0.40743076961744734,
1.1030856800887578,
0.08388380440455438,
0.25224805905642333,
0.10644300798076774,
1.0474125729404944,
-1.1459016177859003,
-1.6916287310468103,
-0.9886067106540645,
0.24182274960650893,
-0.25703819491337826,
-1.1942818044595898,
-0.7763678643520989,
-0.916718557142268,
0.20086626876005667,
-0.3711023890817823,
-0.006962349053602254,
2.687031223814704,
0.3970791704566904,
-0.6890591654410289,
1.0504536719091986,
-1.2458238157294268,
-1.197330311624414,
-0.12155955689525438,
-0.2969718406476498,
0.15406921803787998,
0.9984735505197111,
-0.2734456365103056,
0.0100494123830882,
0.16172873483083283,
-0.6780409734277772,
0.4539997995464368,
0.7558576390626953,
0.9556605206819043,
-0.10572075101403118,
1.5564628686733721,
0.05397552837650759,
0.7415987024402694,
-0.1648500682327342,
1.052699139325594,
0.6776572023171561,
0.15219630541397192,
0.8843933196559655,
1.3948099389813178,
-0.16710315713046545,
1.2199200746863228,
-0.3825678575328573,
-0.26766985045935626,
2.5777182352113908,
0.23156136798610805,
-0.3377548185037014,
1.0689111835439342,
-0.17216022970038714,
-0.29381807805976945,
-0.5689533077540762,
0.30596827309967156,
0.5260993417881752,
-0.6768480303494689,
-0.1092430795457648,
1.5442082488768039,
-0.10826504240549732,
-0.9555598065181787,
0.5334835330029429,
0.5569227134357198,
1.2665483894271157,
-1.9677332823608626,

1.5559931182978979,
0.7709453943956464,
1.5746506000831744],
[1.7625089228179507e-05,
1.298540182320117,
1.3141870420194457,
1.3305471045820527,
2.0686735179487985,
2.23810597518972,
2.267373737425337,
2.295441377067579,
2.3325388050051976,
2.352484360014207,
2.370891483796666,
3.3117124844820194,
3.363605896627787,
3.858981460194034,
3.8589814603490344,
3.859537867870235,
3.8595381713444072,
4.26458571773699,
4.264585717738593,
6.599549149586233,
6.655963438031886,
6.939612414177192,
6.939612414268938,
7.004302986969205,
7.010544991093791,
7.97314828104894,
7.999280802101656,
8.022628296732579,
8.027896471579398,
10.077254342713585,
-1.0733166032270363,
0.4264387306682,
-0.29490479134596703,
-0.45181113351566254,
-0.9496712390342118,
-0.24412788786515252,
-0.17005038539710493,
2.131747001055919,
-0.8801477965160032,
0.1985900676052623,
-0.17829180427317373,
0.1513896817808816,
-0.2996418627583153,
0.09152392337611555,
1.0155053498240294,
0.6713280773462803,
-0.5915672184524843,
-0.21780814909773427,
-0.23676595566314207,
-0.1539223229682568,
0.9733737893230769,
-0.8325690111887051,
-0.6684349693104433,
0.944923850675747,
-0.8270803600282979,
0.4869668407293288,
0.07342964610078141,
-0.8928903181234079,
-0.04262248204617692,
-0.7602498290394286,
-0.48858825463899247,
1.7340528561515622,
0.7340502315717261,
0.5163401187759543,
-0.5015858040228415,
0.2890335055344555,
-0.46033506221820203,
-0.8185907107929625,
0.4557079379705311,
1.6601520871164626,
0.07125306723268252,
-1.4035119795986863,

-0.6741575848895364,
1.84360490583597,
1.6836338044245691,
-0.33329370458317326,
0.19854317161902654,
2.03917897859608,
-0.3342251938073038,
-1.6015864477635742,
0.35422488479060155,
0.1487121545506934,
-0.2821114888507261,
0.21869934797564045,
0.7523961573071125,
1.3021785537923125,
0.26184859002187644,
1.2441238228144538,
0.039670539596101485,
-0.15687777352236798,
1.1386642271664005,
-0.13559116971858365,
0.39167499360215746,
2.4453261619465274,
-1.0890320328395309,
-0.4289422005081893,
1.425249318116838,
-0.0016733698174337221,
0.023121567661497415,
-0.28325087531692295,
0.940069260590291,
1.163437407349777,
-0.4256527399896147,
-1.4972731615803316,
-0.12624975701746496,
-0.10144651691714318,
1.3307378561466523,
0.041778278155808674,
-1.2746374184036902,
-1.2216848741140482,
1.4048769664929757,
0.8957112974734468,
1.0058068981677006,
-3.0965653057688316,
-0.3344448337959029,
-1.0948102803209163,
-0.038478236121346335,
0.12618687504486675,
-0.0850775611793361,
1.33550885346982,
-0.04334772196309596,
0.2406908654957122,
0.7327249608913058,
0.9966290850109706,
2.2852565843192423,
-0.030570948934034583,
-0.5566407461361556,
0.813246653395058,
0.5946355273642772,
0.07519254658407079,
0.27120989268694123,
-0.15415722381241237,
-1.1239288818585278,
1.893751281002709,
0.7530205722182245,
1.4454588921082565,
0.15037300343080462,
0.1793867334099716,
0.3587259643057371,
-0.006799349268608,
1.0907160355013774,
-0.30186717991666434,
0.8507589785411056,
0.009155709268365956,
-0.8165113788656561,
0.011816515180360782,
1.116430418883973,

1.5343789763879045,
-0.1682835166498154,
-1.5032827361947747],
[0.10956969985098665,
0.4360115975982133,
0.5932904704915389,
0.6050451872391112,
1.3688142364348175,
1.3737597367805747,
1.5226773273508583,
1.5589271816733294,
3.150320547140947,
3.1613053536766147,
4.7635234433865605,
4.823148382609575,
4.823188243925802,
7.171157860626202,
7.171157878485542,
7.171158529865571,
7.175073431930883,
7.183070822592498,
8.102400589897519,
8.10392263292654,
8.103940780725882,
9.021308924987686,
9.02133947231205,
9.082831297490506,
9.094868565409332,
9.236804949782064,
9.237029573386183,
9.26430291432718,
10.00744744746223,
10.032040374677464,
-0.5042382729215367,
1.0180015436638659,
0.10423420212375085,
-0.3355687578372037,
-0.3902319222280216,
-0.09867602213718178,
0.3860296418386605,
0.8522828003851446,
-0.7730596965193267,
0.08803782521712378,
1.175277695996097,
-0.3769225558693855,
-0.4553519820435133,
0.4758962487375975,
-1.0517577693418567,
0.3712029741888282,
0.27334544561774066,
0.03397030216872572,
0.3891396532807883,
0.12152393226873999,
1.5192516885791116,
-0.5124931636551364,
-0.5667003913709334,
-0.08425393582345661,
-0.8152956002657638,
-0.008087141654348068,
-0.20833380477584987,
-0.19186344764752156,
0.309463849209865,
-0.16555810903618162,
-0.28534485935787846,
0.025710068868521854,
-0.9172960495760906,
0.027100436879331523,
-0.9806057417125867,
0.46877133554227174,
0.22597640762440013,
-1.8700902599315918,
-0.6222133609404971,
0.023922176634194332,
-0.6777524378245746,
-0.3883818313368169,

-0.19921738892417568,
-1.3418496115619054,
-0.05365669225714106,
0.38835985529462186,
0.46741217514915123,
-0.1388983196762152,
1.0002905156718758,
1.6633296566254185,
-1.0097136763815113,
1.203957772036831,
0.34033273848251194,
1.1456357751380877,
-0.4637199651146574,
-0.13624703361320328,
0.6785440200843139,
0.21691345067474185,
-0.4166036808107477,
1.6003722740710606,
-0.294881370724797,
0.04198878790454511,
-0.689532047804753,
-1.3911901894570817,
-1.1667039833298627,
0.824824800006074,
0.8047687705752488,
0.5727196814509768,
0.2385364014798047,
-0.512164832149354,
0.17925141295137917,
-0.19704824673887428,
-1.6372757558452697,
0.5441675736491897,
-0.13936070677778437,
1.3935144064781402,
0.10763856884501717,
0.6679788244416961,
-0.33118654939932857,
-0.35796695002153883,
0.2632208027041665,
1.5831503864671255,
-0.25140146740224,
1.135774876905931,
0.5897962040478675,
-0.7664904031063712,
1.092649076983624,
-0.9862462517151122,
0.010216869101717933,
0.5705833807099128,
0.09839127014019662,
-1.034770130526523,
0.20582133073730807,
-0.44428721001967564,
0.5272444819968604,
-0.2627660867388947,
-0.9857421779520247,
-1.3646538516413942,
0.14779052759756187,
1.1207932022043874,
0.30427416833245535,
-1.055109005337564,
0.30548258474769546,
0.8153937738463674,
0.5082537250510629,
-0.2973639305797175,
0.0034742209991892575,
1.5168338311712208,
0.0971904670135119,
0.5305672799342332,
-0.08170159551874663,
0.4909203029282298,
-1.1771354207712392,
0.038725619801271274,
-0.6655680746562402,
1.312626970471223,
-1.0115568894576918,

0.16290676730453105,
0.3074286838975462,
0.013117345918778788],
[0.4866654913190249,
0.4866654913190249,
1.4314116389487666,
1.4314116389487666,
1.548608655381947,
1.5552664901547235,
1.662920497545568,
1.676155881591226,
1.6952532774325564,
1.6952532774325564,
2.46741851740979,
2.50102877426778,
3.636145431120801,
3.6361454311208004,
3.6361454311208004,
3.6361454311208004,
4.0161273218482085,
4.0161273218482085,
4.5411636896776235,
4.5411636896776235,
4.922545565757144,
5.624338470516907,
5.642847901355697,
5.904037934138086,
5.915382265322609,
6.860930229950661,
6.860930229950661,
9.422517992567101,
9.467623480195014,
9.467623480195014,
0.40091998571272736,
-0.3581846438287994,
-0.32033246941904975,
0.24409866553117068,
0.176560178011825,
0.4452285262890565,
-0.37150326481454765,
-1.0185408309777293,
1.6197684873006286,
2.0095194673452026,
-0.20342804182482532,
-0.4471079471088122,
0.37127155242298737,
0.4199607045713712,
0.11145558426636207,
-0.289968410859463,
0.9752004498173846,
-0.11972987215087512,
-0.8215210043660294,
-0.4144105549344152,
0.04966095047933562,
0.6091224412599766,
-0.2481186610805342,
-0.9036735819055022,
0.22922493224752044,
-0.04567884568114995,
-0.40529506738584414,
-0.44609185937563545,
1.1000542207972743,
0.8995834380092431,
-0.4391184290327733,
-1.623814742575614,
0.11121717167805684,
-0.15344289124808663,
0.7072346226342329,
0.013911394934765222,
1.1008239636292285,
1.1342299046285917,
-0.1605718532112884,
-0.8521154510350495,
-0.9726520318623231,
-0.11078467990736285,

1.2893067940177063,
-0.8656847782514395,
-0.4554157535410911,
0.46752563535079134,
1.0115326820592876,
-0.017465678177200683,
0.7555243091795097,
0.2225241128439446,
-0.6129237770264853,
0.36249038141661666,
-0.1470816778834455,
-0.23781939745964606,
0.7410640700988803,
-0.10219672891678527,
-0.9043863030697507,
-0.054446492384892985,
-0.23618927113386157,
-0.3971482428910102,
0.42980979959963467,
-0.4511512430809565,
-0.3322123782853245,
0.15031459204805087,
-1.4287283715133035,
-0.026859055017604255,
0.20826999854501216,
-0.016397767746330064,
-0.026711091547853946,
0.15899307013846248,
0.18349959987180348,
-0.019181103141429404,
-1.3777401647916119,
0.05578672583947999,
-0.17191397249005735,
0.08492618339903452,
0.2781511329827625,
1.6853098582374997,
-0.7964285169174421,
0.13628687772581302,
-0.26524580358677924,
1.3615146942088054,
-0.18163849035743365,
-1.5476669195398023,
0.2740670870741208,
0.19740142577560438,
0.18567840551061798,
0.43288784764736754,
-0.8069341399605817,
1.5139336080928714,
-0.7594440385529729,
-0.20443587392639553,
0.0916888700165231,
-0.03922249489745327,
0.4135801157921025,
-0.06731598775715468,
0.5038720973518589,
-0.7705715736697758,
-1.4775324565516457,
0.061950004909204714,
-0.9555167624183505,
3.1277455162888304,
-1.137887484425651,
-0.09852947784569874,
0.042597844143281625,
0.06134264537966588,
0.3384848933516047,
0.012002169875906457,
1.5732417764972468,
-0.3550969359952667,
0.37821230776206083,
1.4914211446267915,
-1.2027836619303884,
0.22421848814312553,
-0.2072711571643093,
0.05122655754886873,
-1.5578500765322636,

```
1.5918641150141506,
1.2635397829742223,
-1.2452354025631063]],
'run_time_sec': 43720.81102514267}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': { 'value': [ 3.750846163374531e-05,
0.22285241391868899,
0.26252411232084344,
0.26252411232084344,
0.26252411232084344,
1.459545906754556,
1.4595459067545562,
1.6094835194149661,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
2.5927782099427623,
3.3552499939702187,
3.3552499939702187,
5.033284106467877,
5.033284106467877,
5.056477210452712,
6.768913439946703,
6.768913439946703,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
```

1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,

```

-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.4606782650454075],
[12, 6, 21, 8, -3.2738854608106305]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,

```



```
        'type': 'depolarizing'},
        'odd_wires': 4}},
    'run_time_sec': 8.591975212097168}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
        'cn': [ 0.5,
            0.5,
            0.5,
            0.5,
            0.5,
            0.5],
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
    'init_param': { 'value': [ 0.00030922277578629,
        0.881375055029912,
        0.881375055029912,
        0.8813750550299119,
        0.9803845151714838,
        0.9803845151714838,
        1.201629973822729,
        1.2441077564052723,
        1.8660677974280824,
        1.8660677974280824,
        2.651002314516478,
        2.651002314516478,
        3.233637461790533,
        4.36384995486056,
        4.36384995486056,
        4.975714528653046,
        4.975714528653046,
        6.591840750700408,
        6.591840750700408,
        8.234009055219385,
        8.238223602720305,
        8.238223602720305,
        8.238223602720305,
        8.903406631369169,
        8.903406631369169,
        8.903406631369169,
        9.126449234760914,
        9.126449234760914,
        9.441918167657187,
        9.441918167657187,
        0.4384156956352651,
        -0.017407483354635628,
        -0.13579407977759564,
        -0.08181457210659056,
        -0.5635031413839604,
        0.570197125887921,
        -0.07389454022208618,
        0.18950484918514415,
        0.23442314823895152,
        0.5187197914643009,
        0.03760777637611914,
        1.2080522066616035,
        -0.006796933498365149,
        0.7597395812711474,
        0.3165595784185247,
        0.09234381717322694,
        -0.047116958459538144,
        0.02161370846364295,
        -0.3330735946096464,
        -0.310519881157395,
        -0.1321730433969341,
        -0.32004747583182785,
        0.38880240494533963,
        0.5587916266845782,
        -0.04068226055752174,
        -0.04929253976644778,
        0.7350044083650977,
        -0.3347542084226466,
        0.07994172439872818,
        0.177027499128215,
```

-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,

```

-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisy_time_evo_ric2_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.5111758480272135],
[12, 6, 21, 8, -3.5131084000754242]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,

```

```

        'type': 'depolarizing'},
        'odd_wires': 4}},
    'run_time_sec': 8.775294542312622}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
        'cn': [ 0.5,
            0.5,
            0.5,
            0.5,
            0.5,
            0.5],
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
    'init_param': { 'value': [ 0.0019047787007362745,
        0.7201496918445128,
        0.7201496918445128,
        0.7201496918445127,
        0.7226708680142584,
        0.8068041864232433,
        1.3847174673528184,
        1.6031895944534578,
        2.458865333289775,
        2.458865333289775,
        2.721120236828938,
        2.721120236828938,
        3.554687304897056,
        3.554687304897056,
        4.537252190232355,
        4.537252190232355,
        5.239770593633025,
        5.242363659304797,
        5.566731623636481,
        5.566731623636481,
        5.566731623636481,
        5.566731623636481,
        5.566731623636481,
        7.6484324534494865,
        7.6484324534494865,
        8.96757120343865,
        9.352637118466301,
        9.381932224950582,
        9.966925955531599,
        9.966925955531599,
        0.29840004251263574,
        0.3207087147269936,
        -0.22478909909228914,
        -1.2242611593120782,
        0.061748844824383295,
        1.9931455249507801,
        0.02449362327304914,
        -0.23402928130657288,
        0.3399922671860315,
        0.2355005268966802,
        0.7912945290539132,
        -0.9147564287830934,
        0.12280724330167514,
        0.42016875545313104,
        0.6183328009701774,
        0.9098148543333162,
        0.2439503202728901,
        0.4941740976390827,
        0.0721485641264495,
        0.20051214721509877,
        0.14273239312798908,
        -0.03846070968482946,
        0.11474641891065286,
        0.1552743298359438,
        -0.050455466832234855,
        0.24385918138398863,
        0.0646818550964626,
        1.0379575003266166,
        0.4438839418702803,
        0.6652939620943935,

```

0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,


```

        'odd_wires': 4]],
        'run_time_sec': 8.882729768753052}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.003375513600300809,
                              0.4884127443235871,
                              0.6719352108785641,
                              0.9073170135093351,
                              1.4946633690060278,
                              1.5076957064847494,
                              3.872917838142945,
                              3.872917838142945,
                              3.872917838142945,
                              3.872917838142945,
                              3.872917838142945,
                              5.047622474508209,
                              5.054336886650782,
                              5.067626550795242,
                              5.0733174878702085,
                              5.094629387163461,
                              5.749571576552799,
                              5.7731554023304374,
                              7.144946717661993,
                              7.144946717661993,
                              7.144946717661993,
                              7.144946717661993,
                              7.144946717661993,
                              7.602314289267704,
                              7.602314289267704,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              9.661071929582006,
                              0.2735947252101626,
                              2.471262065567802,
                              0.18954222370300397,
                              0.2633058348660961,
                              0.07494334922808346,
                              1.4980157261935538,
                              0.3538857424122425,
                              0.26341770410626436,
                              0.17174804107002956,
                              -0.22912718472194316,
                              0.11060714764109056,
                              -0.30804859927082595,
                              2.8406850614439736,
                              1.0823654569610508,
                              0.9668529413216363,
                              1.8727663516543385,
                              -0.0901659930896485,
                              0.05868560690604535,
                              0.24212775097742179,
                              1.0092515209898247,
                              -0.7556969386491817,
                              0.3279350524218167,
                              1.078117272505244,
                              0.10391815458190773,
                              -0.7434930272741371,
                              -1.6965388530236032,
                              -1.360748554100242,
                              0.34939560927302815,
                              0.03188634842980519,
                              -1.2447472675125983,
                              0.06092342743494014,

```

1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,


```

0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#7'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.397817460329861],
[12, 6, 21, 8, -3.2115716730808748]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],

```

False,

0},

False,

0},

```

'run_time_sec': 8.9400794506073}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
  'init_param': { 'value': [ -5.647934061474703e-18,
                              1.4787834906243755,
                              1.478918229526628,
                              1.478918229526628,
                              2.0077351764243123,
                              2.4302809101456218,
                              2.4302809101456218,
                              3.009709563144601,
                              3.009709563144601,
                              3.239471043198664,
                              3.2405283031129133,
                              3.2405283031129133,
                              3.2656598154222434,
                              3.5353392937639825,
                              6.198713539858257,
                              6.213604956994236,
                              6.73397854893658,
                              6.73397854893658,
                              6.743547791194661,
                              6.744609103740889,
                              6.744609103740889,
                              6.749543065628153,
                              6.749543065628153,
                              8.615256229309818,
                              8.635090640841518,
                              9.012323512360366,
                              9.012323512360366,
                              9.930483613907942,
                              9.930483613907942,
                              11.782061805135282,
                              0.6870051227789642,
                              0.5234095533264759,
                              0.11216801162302814,
                              0.044591843687496906,
                              -0.747159787599118,
                              1.5727279855217766,
                              -0.16958009141159439,
                              -0.7234855106318131,
                              0.45999589631996707,
                              1.3613822619347415,
                              0.5234464983029046,
                              0.18337466193510712,
                              1.0660408011194182,
                              1.2813123812864906,
                              0.34899885807903364,
                              1.2204971120849089,
                              0.37250493843195553,
                              0.3861568591798728,
                              -0.4881036169368887,
                              0.23720347697856276,
                              0.8136369801063428,
                              -0.8595161765585049,
                              0.6710512785075716,
                              0.5103741134651811,
                              -0.48491784736765847,
                              -1.939727949979543,
                              0.5221979500550953,
                              -0.39618913325979604,
                              -0.5258671175956018,
                              0.08753210163723862,
                              1.3619602585721242,
                              -1.2518915867901346,

```

0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,

```

0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#8'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
[12, 6, 21, 8, -3.1990719612158447]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 8.73015832901001}},

```

```

{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.3773346757087749,
                              0.5734233884194113,
                              0.5734233884194113,
                              1.2094818154974225,
                              1.2528672337957094,
                              1.3073966674132076,
                              4.193348178434801,
                              4.228670385547864,
                              4.228670385547864,
                              4.228670385547864,
                              4.228670385547864,
                              4.228670385547864,
                              4.274742575727675,
                              5.343872214974395,
                              5.354523589318104,
                              5.449133113124401,
                              5.449133113124401,
                              5.449133113124401,
                              5.449133113124401,
                              6.208341300319731,
                              6.208341300319731,
                              6.738723848778546,
                              6.7473872692498285,
                              8.515456463543387,
                              8.552181104315652,
                              8.552181104315652,
                              8.552181104315652,
                              9.363042150968939,
                              9.363042150968939,
                              11.478464454984158,
                              -0.2683676450988651,
                              -0.9370321768196137,
                              0.03790413469005285,
                              -0.3520989510779511,
                              0.5131804204688105,
                              0.767748989456867,
                              -1.1529933056041268,
                              1.0195402939041247,
                              -0.5189860890713169,
                              1.700377968800765,
                              0.48912124484584857,
                              1.506730848787325,
                              0.4485612726223177,
                              -1.6248073049289613,
                              0.10079320215400162,
                              1.7454590745233427,
                              1.474284546585253,
                              0.08837986403943651,
                              1.215839965531947,
                              -0.42666568442503866,
                              0.04298065245990749,
                              -1.495484471060033,
                              0.39055472334393304,
                              -0.06187227108331559,
                              1.0010296249342066,
                              -0.33745943355443286,
                              -0.19046242319241546,
                              -0.4884459619425607,
                              0.8930564792122229,
                              0.09947441181287657,
                              -1.4049039336860292,
                              1.0413688199593338,
                              0.027348986225077598,

```

1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,

```

0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#4'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.44845457619454],
                             [12, 6, 21, 8, -3.24468946816992]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 7, 0],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                'value':
0},
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [12, 3, 3, 8],
                                'identity_factors': [1, 1, 1, 0],
                                'noise_level': [12, 6, 21, 8],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                'value':
0},
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4}]],
            'run_time_sec': 11.057755708694458}},
{ 'config': { 'ansatz': { 'gateset': 1,

```

```
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                              'cn': [ 0.5,
                                       0.5,
                                       0.5,
                                       0.5,
                                       0.5,
                                       0.5],
                              'r': 0},
          'time': {'max': 10.0, 'min': 0.0},
          'type': 'xy-iss'}},
'init_param': { 'value': [ 0.08591881204796584,
                            1.5026023651616802,
                            1.9179097588555833,
                            1.9179097588555831,
                            1.9179097588555831,
                            1.9179097588555831,
                            2.8800636946459592,
                            2.9523818900173797,
                            2.9523818900173797,
                            2.954896407422686,
                            3.0487820562708485,
                            3.0487820562708485,
                            3.9462134206000328,
                            3.9462134206000328,
                            5.170248653928018,
                            5.170248653928018,
                            6.5066905594049205,
                            6.5066905594049205,
                            6.5066905594049205,
                            6.914119522580601,
                            6.914119522580601,
                            6.9842551957162815,
                            7.300516913785898,
                            8.782483894556266,
                            8.782483894556266,
                            8.999633503017922,
                            8.999633503017922,
                            8.999633503017922,
                            9.414122362394847,
                            9.414122362394847,
                            0.09854676674938925,
                            -0.05173289612974198,
                            0.30094803169630163,
                            -1.1984142138894447,
                            0.08610670053567002,
                            0.2851928692216035,
                            0.26177875477440216,
                            0.11764473243628538,
                            0.3495067876706816,
                            -0.9678438601279628,
                            -0.08762050677252062,
                            0.014826088936787857,
                            -0.30347729159143555,
                            -1.1730530788097624,
                            -0.12550631929596553,
                            2.258056275075482,
                            -0.5160508087356981,
                            0.004541093619226378,
                            -0.9345420797218197,
                            0.47154522212338745,
                            -0.9129446894819914,
                            0.2768683391341781,
                            0.29975752794333804,
                            -0.16177915588279423,
                            0.6021148977823381,
                            0.35405737535162146,
                            0.8149645603473178,
                            -0.7074306860917636,
                            -0.8281028342440522,
                            -0.20840122350984505,
                            -0.19593362207831977,
                            1.2208184674045053,
                            -0.10408257731449277,
                            0.9878855326441833,
```


0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,

```

-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406]],
'noise_profile': { 'noise_on_init_param': { 'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0.001, 0.001],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                  'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                  'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#3'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.410492799212101],
                             [12, 6, 21, 8, -3.2577133670016694]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 7, 0],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                'value':
0},
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [12, 3, 3, 8],
                                'identity_factors': [1, 1, 1, 0],
                                'noise_level': [12, 6, 21, 8],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                'value':
0},
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4}],
            'run_time_sec': 10.180011987686157}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,

```

```
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 0},
            'time': {'max': 10.0, 'min': 0.0},
            'type': 'xy-iss'}},
'init_param': { 'value': [ 0.33528943002425843,
                             0.8608925283648261,
                             2.8309935233222867,
                             2.8382233121900358,
                             2.847558790018716,
                             2.8477263794127468,
                             3.9337993695450666,
                             3.9337993695450666,
                             3.9337993695450666,
                             3.9337993695450666,
                             3.9337993695450666,
                             5.127462455084763,
                             5.128383696390232,
                             5.132051241075024,
                             5.132231848473024,
                             5.132297186784173,
                             5.132297186784173,
                             5.132297186784173,
                             5.132320898992942,
                             5.134903164275234,
                             5.734798022200061,
                             5.734798022200061,
                             5.734798022200061,
                             6.71748919093495,
                             6.71748919093495,
                             9.773430860122208,
                             9.791656954875775,
                             9.791980525916317,
                             9.804782964874805,
                             9.947289619134542,
                             0.6818618664375632,
                             0.42253658736184846,
                             0.19067339209915313,
                             -2.314563573529872,
                             -1.7784854373672185,
                             -0.10945476580518654,
                             0.08980428543050965,
                             0.6265702304745648,
                             -0.12016219313034968,
                             0.005450784512528824,
                             0.9621316606784164,
                             -0.01940541263687908,
                             -1.2890091107885369,
                             -0.09216272929775945,
                             -0.8615487717558121,
                             0.021600758375482327,
                             1.1068545366678577,
                             -0.06436428812394061,
                             -1.8186253546793636,
                             -0.2455724158610742,
                             0.7444478114283775,
                             1.0647343146888184,
                             -1.6388802958603008,
                             -0.05032235091912311,
                             0.7589104637111997,
                             0.0934759886788552,
                             0.7280377617711271,
                             -0.10012682423470079,
                             -2.7917715562202794,
                             0.5843828201862561,
                             1.2003794612588048,
                             -0.2782652692858759,
                             1.253725179404932,
                             -0.831140674623039,
                             -1.3898002860152843,
```

1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,

```

0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#2'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.333005581610797],
[12, 6, 21, 8, -3.165101785859742]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 10.927413940429688}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],

```

```
        'cn': [ 0.5,
                0.5,
                0.5,
                0.5,
                0.5,
                0.5],
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 1.5680106786222674e-17,
                            1.0238650999177359,
                            1.0238650999177359,
                            1.0238650999177359,
                            1.7041550191762513,
                            2.070931805530806,
                            2.586234954306046,
                            2.6265616878606783,
                            2.7566136785306155,
                            2.7689271757054255,
                            4.271831894498143,
                            4.271831894498143,
                            4.271831894498143,
                            5.004110362344359,
                            5.004110362344359,
                            5.529689350278954,
                            5.529689350278954,
                            5.529689350278954,
                            5.5493166460873375,
                            7.855838962179162,
                            7.855838962179162,
                            8.508023697610218,
                            8.540796404638465,
                            8.940056981382925,
                            8.940056981382925,
                            8.940056981382925,
                            8.940056981382925,
                            9.524293982069336,
                            9.524293982069336,
                            9.815045396471412,
                            0.717685470636315,
                            0.29125971197830336,
                            0.013619559122678915,
                            -0.46174394022911175,
                            -0.7802589318548062,
                            1.8785332138943256,
                            -0.07718385116900997,
                            -0.4181315830102034,
                            0.3859499067718677,
                            0.7218523694763775,
                            0.16350787496293626,
                            -0.06274337543165032,
                            1.048639096314506,
                            0.9454500907927006,
                            -0.18328208251477485,
                            0.5336296017995852,
                            0.7047960543548563,
                            -0.4717130073686535,
                            -0.14327049451726778,
                            0.5646728024946054,
                            0.1377201649736573,
                            1.0845662084088512,
                            0.7117671017527167,
                            1.8562916393289055,
                            -1.0958742208007046,
                            0.314756410152813,
                            1.0055290931424188,
                            -1.2653176839886926,
                            -0.6435463058390287,
                            -0.13689785210195535,
                            0.5185042269644529,
                            -0.07994120596020922,
                            -1.3394312085309903,
                            -0.01354724643952521,
                            -0.8508707890759795,
                            -0.7258020036409927,
```

-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,

```

-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455}},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#1'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.515742960091852],
[12, 6, 21, 8, -3.36492111003702]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 10.475180864334106}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,

```



```
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ -6.331146189926128e-17,
0.7069927066774304,
0.7069927066774304,
0.7069927066774305,
0.7069927066774304,
1.0044153364322925,
1.0092418692996383,
2.470333794214341,
2.478143749866421,
3.5776564981759003,
3.5776564981759003,
3.5776564981759003,
3.61745967653339,
3.6981780389472076,
3.755720232685741,
3.9712286023902594,
3.9712286023902594,
4.519037432417192,
4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,
6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
```

3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,

```

-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.311563719128683],
[12, 6, 21, 8, -3.2712898811023576]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 9.587943077087402}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},

```

False,
0},

False,
0},

4.271831894498143,
5.004110362344359,
5.004110362344359,
5.529689350278954,
5.529689350278954,
5.529689350278954,
5.5493166460873375,
7.855838962179162,
7.855838962179162,
8.508023697610218,
8.540796404638465,
8.940056981382925,
8.940056981382925,
8.940056981382925,
8.940056981382925,
9.524293982069336,
9.524293982069336,
9.815045396471412,
0.717685470636315,
0.29125971197830336,
0.013619559122678915,
-0.46174394022911175,
-0.7802589318548062,
1.8785332138943256,
-0.07718385116900997,
-0.4181315830102034,
0.3859499067718677,
0.7218523694763775,
0.16350787496293626,
-0.06274337543165032,
1.048639096314506,
0.9454500907927006,
-0.18328208251477485,
0.5336296017995852,
0.7047960543548563,
-0.4717130073686535,
-0.14327049451726778,
0.5646728024946054,
0.1377201649736573,
1.0845662084088512,
0.7117671017527167,
1.8562916393289055,
-1.0958742208007046,
0.314756410152813,
1.0055290931424188,
-1.2653176839886926,
-0.6435463058390287,
-0.13689785210195535,
0.5185042269644529,
-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,

0.755528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455],
[0.33528943002425843,
0.8608925283648261,
2.8309935233222867,
2.8382233121900358,
2.847558790018716,
2.8477263794127468,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
5.127462455084763,

5.128383696390232,
5.132051241075024,
5.132231848473024,
5.132297186784173,
5.132297186784173,
5.132297186784173,
5.132320898992942,
5.134903164275234,
5.734798022200061,
5.734798022200061,
5.734798022200061,
6.71748919093495,
6.71748919093495,
9.773430860122208,
9.791656954875775,
9.791980525916317,
9.804782964874805,
9.947289619134542,
0.6818618664375632,
0.42253658736184846,
0.19067339209915313,
-2.314563573529872,
-1.7784854373672185,
-0.10945476580518654,
0.08980428543050965,
0.6265702304745648,
-0.12016219313034968,
0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,
-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,

0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358],
[0.08591881204796584,
1.5026023651616802,
1.9179097588555833,
1.9179097588555831,
1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,

3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,

-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406],
[0.3773346757087749,
0.5734233884194113,
0.5734233884194113,
1.2094818154974225,
1.2528672337957094,
1.3073966674132076,
4.193348178434801,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,

4.274742575727675,
5.343872214974395,
5.354523589318104,
5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,

1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244],
[-6.331146189926128e-17,
0.7069927066774304,
0.7069927066774304,
0.7069927066774305,
0.7069927066774304,
1.0044153364322925,
1.0092418692996383,
2.470333794214341,
2.478143749866421,
3.5776564981759003,
3.5776564981759003,
3.5776564981759003,

3.61745967653339,
3.6981780389472076,
3.755720232685741,
3.9712286023902594,
3.9712286023902594,
4.519037432417192,
4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,
6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,

-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176],
[3.750846163374531e-05,
0.22285241391868899,
0.26252411232084344,
0.26252411232084344,
0.26252411232084344,
1.459545906754556,
1.4595459067545562,
1.6094835194149661,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
2.5927782099427623,

3.3552499939702187,
3.3552499939702187,
5.033284106467877,
5.033284106467877,
5.056477210452712,
6.768913439946703,
6.768913439946703,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,

-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138],
[0.003375513600300809,
0.4884127443235871,
0.6719352108785641,
0.9073170135093351,
1.4946633690060278,
1.5076957064847494,
3.872917838142945,
3.872917838142945,
3.872917838142945,
3.872917838142945,
5.047622474508209,
5.054336886650782,

5.067626550795242,
5.0733174878702085,
5.094629387163461,
5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,

1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989],
[-5.647934061474703e-18,
1.4787834906243755,
1.478918229526628,
1.478918229526628,
2.0077351764243123,
2.4302809101456218,
2.4302809101456218,
3.009709563144601,
3.009709563144601,
3.239471043198664,
3.2405283031129133,
3.2405283031129133,

3.2656598154222434,
3.5353392937639825,
6.198713539858257,
6.213604956994236,
6.73397854893658,
6.73397854893658,
6.743547791194661,
6.744609103740889,
6.744609103740889,
6.749543065628153,
6.749543065628153,
8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,

0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935],
[0.0019047787007362745,
0.7201496918445128,
0.7201496918445128,
0.7201496918445127,
0.7226708680142584,
0.8068041864232433,
1.3847174673528184,
1.6031895944534578,
2.458865333289775,
2.458865333289775,
2.721120236828938,
2.721120236828938,

3.554687304897056,
3.554687304897056,
4.537252190232355,
4.537252190232355,
5.239770593633025,
5.242363659304797,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
7.6484324534494865,
7.6484324534494865,
8.96757120343865,
9.352637118466301,
9.381932224950582,
9.966925955531599,
9.966925955531599,
0.29840004251263574,
0.3207087147269936,
-0.22478909909228914,
-1.2242611593120782,
0.061748844824383295,
1.9931455249507801,
0.02449362327304914,
-0.23402928130657288,
0.3399922671860315,
0.2355005268966802,
0.7912945290539132,
-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,

-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815],
[0.00030922277578629,
0.881375055029912,
0.881375055029912,
0.8813750550299119,
0.9803845151714838,
0.9803845151714838,
1.201629973822729,
1.2441077564052723,
1.8660677974280824,
1.8660677974280824,
2.651002314516478,
2.651002314516478,

3.233637461790533,
4.36384995486056,
4.36384995486056,
4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,
8.903406631369169,
8.903406631369169,
8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.13579407977759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,

```

-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'run_time_sec': 68892.59480500221}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
                                'layer': 30,
                                'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}}},
'init_param': { 'value': [ -6.331146189926128e-17,
0.7069927066774304,
0.7069927066774304,
0.7069927066774305,

```


0.7069927066774304,
1.0044153364322925,
1.0092418692996383,
2.470333794214341,
2.478143749866421,
3.5776564981759003,
3.5776564981759003,
3.5776564981759003,
3.61745967653339,
3.6981780389472076,
3.755720232685741,
3.9712286023902594,
3.9712286023902594,
4.519037432417192,
4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,
6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,

```
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
```

```

'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.311563719128683],
                           [12, 6, 21, 8, -3.2712898811023576]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.458427429199219e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -7.474021363079925,
                            'others': { 'beta_coefficients': [ 1.3823529411764706,
                                                                -0.38235294117647056],
                                          'cost_richardson_zne': 2.057093425605536,
                                          'sorted_expectation_vals': [ -6.311563719128683,
                                                                        -3.2712898811023576],
                                          'sorted_noise': [13, 47]},
                            'sampled data': [ [4, 2, 7, 0, -6.311563719128683],
                                                [12, 6, 21, 8, -3.2712898811023576]],
                            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ -5.647934061474703e-18,
                              1.4787834906243755,
                              1.478918229526628,
                              1.478918229526628,
                              2.0077351764243123,
                              2.4302809101456218,
                              2.4302809101456218,
                              3.009709563144601,
                              3.009709563144601,
                              3.239471043198664,
                              3.2405283031129133,
                              3.2405283031129133,
                              3.2656598154222434,
                              3.5353392937639825,
                              6.198713539858257,
                              6.213604956994236,
                              6.73397854893658,
                              6.73397854893658,
                              6.743547791194661,
                              6.744609103740889,
                              6.744609103740889,
                              6.749543065628153,
                              6.749543065628153,
                              8.615256229309818,
                              8.635090640841518,
                              9.012323512360366,
                              9.012323512360366,
                              9.930483613907942,
                              9.930483613907942,
                              11.782061805135282,
                              0.6870051227789642,
                              0.5234095533264759,
                              0.11216801162302814,
                              0.044591843687496906,
                              -0.747159787599118,
                              1.5727279855217766,

```

-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,

```

-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#8'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
                          [12, 6, 21, 8, -3.1990719612158447]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.361701965332031e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.665262291630623,
                           'others': { 'beta_coeffieicients': [ 1.3823529411764706,
                                                                -0.38235294117647056],
                                         'cost_richardson_zne': 2.057093425605536,
                                         'sorted_expectation_vals': [ -6.429933051303131,
                                                                      -3.1990719612158447],
                                         'sorted_noise': [13, 47]},
                           'sampled data': [ [4, 2, 7, 0, -6.429933051303131],
                                              [12, 6, 21, 8, -3.1990719612158447]],
                           'sampling': 'default'}}},

```

```
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
  'init_param': { 'value': [ 3.750846163374531e-05,
                              0.22285241391868899,
                              0.26252411232084344,
                              0.26252411232084344,
                              0.26252411232084344,
                              1.459545906754556,
                              1.4595459067545562,
                              1.6094835194149661,
                              2.5927782099427628,
                              2.5927782099427628,
                              2.5927782099427628,
                              2.5927782099427623,
                              3.3552499939702187,
                              3.3552499939702187,
                              5.033284106467877,
                              5.033284106467877,
                              5.056477210452712,
                              6.768913439946703,
                              6.768913439946703,
                              7.2782492670747585,
                              7.2782492670747585,
                              7.2782492670747585,
                              7.2782492670747585,
                              7.2782492670747585,
                              7.2782492670747585,
                              8.934198246659292,
                              8.937369801451712,
                              9.544406408519471,
                              9.544406408519471,
                              10.23170722265548,
                              -0.4501443853573418,
                              0.06519513561037263,
                              -0.13386490845883003,
                              -0.01542333988020039,
                              0.3988203537526196,
                              0.9163371790808124,
                              0.08159154039220383,
                              -0.11665429084087826,
                              -0.21130662780886203,
                              1.515236372470638,
                              0.4436142129357345,
                              -0.28780404896369716,
                              0.3681425383234808,
                              0.648176841676201,
                              0.3046875502986337,
                              0.05897820457762899,
                              0.09316951897603123,
                              1.5277657654344678,
                              -0.655141628591867,
                              0.073084468489131,
                              -0.11423944001370272,
                              0.3986898081963471,
                              1.0974832156349317,
                              1.8462557861113,
                              0.5998463368708868,
                              -0.0014214294560794885,
                              0.4026665434776887,
                              -0.03860309342262787,
                              0.3728818533964108,
                              1.7322816292529104,
                              0.42156018976657544,
                              0.9872712539907578,
                              0.5749190583224287,
                              0.1654949387796794,
                              1.0577776390475342,
                              -0.5892377479966974,
                              0.4324416138472369,
                              0.7909276553155474,
```

1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,

```

0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.4606782650454075],
[12, 6, 21, 8, -3.2738854608106305]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.337860107421875e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.679157866664586,
'others': { 'beta_coefficients': [ 1.3823529411764706,
-0.38235294117647056],
'cost_richardson_zne': 2.057093425605536,
'sorted_expectation_vals': [ -6.4606782650454075,
-3.2738854608106305],
'sorted_noise': [13, 47]},
'sampled data': [ [4, 2, 7, 0, -6.4606782650454075],
[12, 6, 21, 8, -3.2738854608106305]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.00030922277578629,
0.881375055029912,
0.881375055029912,
0.8813750550299119,
0.9803845151714838,
0.9803845151714838,
1.201629973822729,
1.2441077564052723,
1.8660677974280824,
1.8660677974280824,
2.651002314516478,
2.651002314516478,
3.233637461790533,
4.36384995486056,
4.36384995486056,
4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,
8.903406631369169,
8.903406631369169,

```


8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.13579407977759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,

```

-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.5111758480272135],
                          [12, 6, 21, 8, -3.5131084000754242]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.814697265625e-05,

```

```

'zne_values': { 'degree': 0,
                'extrapolated_value': -7.657495754597014,
                'others': { 'beta_coefficients': [ 1.3823529411764706,
                                                    -0.38235294117647056],
                            'cost_richardson_zne': 2.057093425605536,
                            'sorted_expectation_vals': [ -6.5111758480272135,
                                                         -3.5131084000754242],
                            'sorted_noise': [13, 47]},
                'sampled data': [ [4, 2, 7, 0, -6.5111758480272135],
                                   [12, 6, 21, 8, -3.5131084000754242]],
                'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.33528943002425843,
                              0.8608925283648261,
                              2.8309935233222867,
                              2.8382233121900358,
                              2.847558790018716,
                              2.8477263794127468,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              5.127462455084763,
                              5.128383696390232,
                              5.132051241075024,
                              5.132231848473024,
                              5.132297186784173,
                              5.132297186784173,
                              5.132297186784173,
                              5.132320898992942,
                              5.134903164275234,
                              5.734798022200061,
                              5.734798022200061,
                              5.734798022200061,
                              6.71748919093495,
                              6.71748919093495,
                              9.773430860122208,
                              9.791656954875775,
                              9.791980525916317,
                              9.804782964874805,
                              9.947289619134542,
                              0.6818618664375632,
                              0.42253658736184846,
                              0.19067339209915313,
                              -2.314563573529872,
                              -1.7784854373672185,
                              -0.10945476580518654,
                              0.08980428543050965,
                              0.6265702304745648,
                              -0.12016219313034968,
                              0.005450784512528824,
                              0.9621316606784164,
                              -0.01940541263687908,
                              -1.2890091107885369,
                              -0.09216272929775945,
                              -0.8615487717558121,
                              0.021600758375482327,
                              1.1068545366678577,
                              -0.06436428812394061,
                              -1.8186253546793636,
                              -0.2455724158610742,
                              0.7444478114283775,
                              1.0647343146888184,
                              -1.6388802958603008,
                              -0.05032235091912311,
                              0.7589104637111997,
                              0.0934759886788552,
                              0.7280377617711271,

```

-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,

4.537252190232355,
4.537252190232355,
5.239770593633025,
5.242363659304797,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
7.6484324534494865,
7.6484324534494865,
8.96757120343865,
9.352637118466301,
9.381932224950582,
9.966925955531599,
9.966925955531599,
0.29840004251263574,
0.3207087147269936,
-0.22478909909228914,
-1.2242611593120782,
0.061748844824383295,
1.9931455249507801,
0.02449362327304914,
-0.23402928130657288,
0.3399922671860315,
0.2355005268966802,
0.7912945290539132,
-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,

```

-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.753559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'zne',
'state': 'dmatrix',

```

```

'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.536182288327376],
                          [12, 6, 21, 8, -3.450081350248876]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.62396240234375e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -7.7161620587691555,
                          'others': { 'beta_coefficients': [ 1.3823529411764706,
                                                            -0.38235294117647056],
                          'cost_richardson_zne': 2.057093425605536,
                          'sorted_expectation_vals': [ -6.536182288327376,
                                                         -3.450081350248876],
                          'sorted_noise': [13, 47]},
            'sampled data': [ [4, 2, 7, 0, -6.536182288327376],
                              [12, 6, 21, 8, -3.450081350248876]],
            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                       'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                       'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.08591881204796584,
                              1.5026023651616802,
                              1.9179097588555833,
                              1.9179097588555831,
                              1.9179097588555831,
                              1.9179097588555831,
                              2.8800636946459592,
                              2.9523818900173797,
                              2.9523818900173797,
                              2.954896407422686,
                              3.0487820562708485,
                              3.0487820562708485,
                              3.9462134206000328,
                              3.9462134206000328,
                              5.170248653928018,
                              5.170248653928018,
                              6.5066905594049205,
                              6.5066905594049205,
                              6.5066905594049205,
                              6.914119522580601,
                              6.914119522580601,
                              6.9842551957162815,
                              7.300516913785898,
                              8.782483894556266,
                              8.782483894556266,
                              8.999633503017922,
                              8.999633503017922,
                              8.999633503017922,
                              9.414122362394847,
                              9.414122362394847,
                              0.09854676674938925,
                              -0.05173289612974198,
                              0.30094803169630163,
                              -1.1984142138894447,
                              0.08610670053567002,
                              0.2851928692216035,
                              0.26177875477440216,
                              0.11764473243628538,
                              0.3495067876706816,
                              -0.9678438601279628,
                              -0.08762050677252062,
                              0.014826088936787857,
                              -0.30347729159143555,
                              -1.1730530788097624,
                              -0.12550631929596553,
                              2.258056275075482,

```


-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,

```

0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#3'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.410492799212101],
                          [12, 6, 21, 8, -3.2577133670016694]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.62396240234375e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.615967287998442,
                           'others': { 'beta_coefficients': [ 1.3823529411764706,
                                                             -0.38235294117647056],
                                         'cost_richardson_zne': 2.057093425605536,
                                         'sorted_expectation_vals': [ -6.410492799212101,
                                                                      -3.2577133670016694],
                                         'sorted_noise': [13, 47]},
                           'sampled data': [ [4, 2, 7, 0, -6.410492799212101],
                                              [12, 6, 21, 8, -3.2577133670016694]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ 1.5680106786222674e-17,
                             1.0238650999177359,
                             1.0238650999177359,

```

1.0238650999177359,
1.7041550191762513,
2.070931805530806,
2.586234954306046,
2.6265616878606783,
2.7566136785306155,
2.7689271757054255,
4.271831894498143,
4.271831894498143,
4.271831894498143,
5.004110362344359,
5.004110362344359,
5.529689350278954,
5.529689350278954,
5.529689350278954,
5.5493166460873375,
7.855838962179162,
7.855838962179162,
8.508023697610218,
8.540796404638465,
8.940056981382925,
8.940056981382925,
8.940056981382925,
8.940056981382925,
9.524293982069336,
9.524293982069336,
9.815045396471412,
0.717685470636315,
0.29125971197830336,
0.013619559122678915,
-0.46174394022911175,
-0.7802589318548062,
1.8785332138943256,
-0.07718385116900997,
-0.4181315830102034,
0.3859499067718677,
0.7218523694763775,
0.16350787496293626,
-0.06274337543165032,
1.048639096314506,
0.9454500907927006,
-0.18328208251477485,
0.5336296017995852,
0.7047960543548563,
-0.4717130073686535,
-0.14327049451726778,
0.5646728024946054,
0.1377201649736573,
1.0845662084088512,
0.7117671017527167,
1.8562916393289055,
-1.0958742208007046,
0.314756410152813,
1.0055290931424188,
-1.2653176839886926,
-0.6435463058390287,
-0.13689785210195535,
0.5185042269644529,
-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,

```
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.755528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
```

```

        'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#1'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.515742960091852],
                          [12, 6, 21, 8, -3.36492111003702]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 0.0001556873321533203,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -7.720468961583405,
                          'others': { 'beta_coefficients': [ 1.3823529411764706,
                                                            -0.38235294117647056],
                                      'cost_richardson_zne': 2.057093425605536,
                                      'sorted_expectation_vals': [ -6.515742960091852,
                                                                  -3.36492111003702],
                                      'sorted_noise': [13, 47]},
                          'sampled data': [ [4, 2, 7, 0, -6.515742960091852],
                                             [12, 6, 21, 8, -3.36492111003702]],
                          'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                      'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                      'r': 0},
                                  'time': {'max': 10.0, 'min': 0.0},
                                  'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.3773346757087749,
                             0.5734233884194113,
                             0.5734233884194113,
                             1.2094818154974225,
                             1.2528672337957094,
                             1.3073966674132076,
                             4.193348178434801,
                             4.228670385547864,
                             4.228670385547864,
                             4.228670385547864,
                             4.228670385547864,
                             4.228670385547864,
                             4.228670385547864,
                             4.274742575727675,
                             5.343872214974395,
                             5.354523589318104,
                             5.449133113124401,
                             5.449133113124401,
                             5.449133113124401,
                             5.449133113124401,
                             6.208341300319731,
                             6.208341300319731,
                             6.738723848778546,
                             6.7473872692498285,
                             8.515456463543387,
                             8.552181104315652,
                             8.552181104315652,
                             8.552181104315652,
                             9.363042150968939,
                             9.363042150968939,
                             11.478464454984158,
                             -0.2683676450988651,
                             -0.9370321768196137,
                             0.03790413469005285,
                             -0.3520989510779511,
                             0.5131804204688105,

```

0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,

[illegible]

```
        'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.003375513600300809,
                              0.4884127443235871,
                              0.6719352108785641,
                              0.9073170135093351,
                              1.4946633690060278,
                              1.5076957064847494,
                              3.872917838142945,
                              3.872917838142945,
                              3.872917838142945,
                              3.872917838142945,
                              3.872917838142945,
                              5.047622474508209,
                              5.054336886650782,
                              5.067626550795242,
                              5.0733174878702085,
                              5.094629387163461,
                              5.749571576552799,
                              5.7731554023304374,
                              7.144946717661993,
                              7.144946717661993,
                              7.144946717661993,
                              7.144946717661993,
                              7.602314289267704,
                              7.602314289267704,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              9.661071929582006,
                              0.2735947252101626,
                              2.471262065567802,
                              0.18954222370300397,
                              0.2633058348660961,
                              0.07494334922808346,
                              1.4980157261935538,
                              0.3538857424122425,
                              0.26341770410626436,
                              0.17174804107002956,
                              -0.22912718472194316,
                              0.11060714764109056,
                              -0.30804859927082595,
                              2.8406850614439736,
                              1.0823654569610508,
                              0.9668529413216363,
                              1.8727663516543385,
                              -0.0901659930896485,
                              0.05868560690604535,
                              0.24212775097742179,
                              1.0092515209898247,
                              -0.7556969386491817,
                              0.3279350524218167,
                              1.078117272505244,
                              0.10391815458190773,
                              -0.7434930272741371,
                              -1.6965388530236032,
                              -1.360748554100242,
                              0.34939560927302815,
                              0.03188634842980519,
                              -1.2447472675125983,
                              0.06092342743494014,
                              1.017082923902293,
                              -1.007905562333318,
                              -0.8576057837691866,
                              0.5321038189070756,
                              1.300652060018237,
                              -1.8122591145188616,
```


-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,

```

0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989}],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric2_sample#7'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.397817460329861],
[12, 6, 21, 8, -3.2115716730808748]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 0.00014829635620117188,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.6160879083956505,
'others': { 'beta_coefficients': [ 1.3823529411764706,
-0.38235294117647056],
'cost_richardson_zne': 2.057093425605536,
'sorted_expectation_vals': [ -6.397817460329861,
-3.2115716730808748],
'sorted_noise': [13, 47]},
'sampled data': [ [4, 2, 7, 0, -6.397817460329861],
[12, 6, 21, 8, -3.2115716730808748]],
'sampling': 'default'}}}],
'xy-ric3': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': False,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
'file_name_prefix': 'xy_noise_off'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},

```

```

'zne': { 'data_points': None,
        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1 '
                                'Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                    '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                    '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                    '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                    '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                    '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},

'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 4.118038614266066,
                                     2.900590937440303,
                                     2.874944433071395,
                                     3.9296977116518517,
                                     3.1841729388967344,
                                     3.695928991667305,
                                     3.180227821654077,
                                     3.0475045550195774,
                                     2.4669562192398296,
                                     3.4124916083581294],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [0, 0, 0, 0],
                              'noise_profile': { 'noise_on_init_param': { 'status':
                                                                              'value': 0}},
                              'noise_prob': [ 0.001,
                                              0.001,
                                              0.001,
                                              0.001],
                              'status': False,
                              'type': 'depolarizing'},
            'odd_wires': 4},
'optimized_minimum_cost': [ -8.433468022982382,
                           -8.464942606128835,
                           -8.43123094137328,
                           -8.435401809869298,
                           -7.9812014353834515,
                           -8.371080050195895,
                           -8.46198057547588,
                           -8.19415563172473,
                           -8.438078664602543,
                           -8.481886373274177],
'optimized_parameters': [ [ 0.4631008843346199,
                           0.8090636887146863,
                           1.1101507421405439,
                           1.3400184709676968,
                           1.7452332568602784,
                           1.9308700883292553,
                           2.1752679291102544,
                           2.1784212211010603,
                           3.800207092295721,
                           3.800207092295721,
                           3.800207092295721,
                           3.800207092295721,
                           3.800207092295721,
                           3.800207092295721,
                           3.800207092295721,
                           4.798448841518561,
                           4.812026604031267,
                           6.02543601067661,
                           6.02543601067661,
                           6.5265343452987,
                           6.5265343452987,
                           6.526534345298701,
                           6.528381158760148,
                           6.528381158760148,
                           7.113023968622042,
                           7.113023968622042,
                           7.593654062389684,
                           7.5949106716383605,
                           7.780058793305066,

```

False,

9.000370275113886,
9.005799799983558,
-0.22105281285875572,
0.07196296170715184,
1.0527781347416576,
-0.4594806493129553,
1.0526039239081737,
-0.01709270509934428,
0.3199675106146978,
-0.0972220164715271,
0.9145097677998977,
0.09808103992344623,
0.5373974901276602,
-0.5786059880820694,
-0.02594763303680847,
-0.22132394348491485,
-0.7274785930689547,
-0.41090279602128,
-0.520884263329148,
-1.0260835302846771,
0.1692597388493397,
1.3250313995601488,
1.510357845739394,
-0.9233604808876308,
-0.2049157632594096,
-0.9894281051832,
0.25772120620526445,
-0.04822142329545429,
-0.3912638297585932,
-0.48486712444934904,
1.4836649713082644,
1.566996182191827,
0.08093959953023218,
-2.3562926055425963,
-0.21876618997457167,
0.10911407834571141,
-1.3776955808248483,
-0.31816531211057325,
0.7083857256813267,
0.5593482503842677,
0.7483381318097894,
0.010042025170353972,
0.05290425882351868,
0.20616876613540233,
-0.8673924682871486,
-0.08017548043713153,
-0.3056222778255953,
2.3981163154468383,
0.37079365521682633,
0.3870558721039623,
0.19341028278759992,
-0.3154248344945196,
-0.12073260037561208,
-1.5858414285227351,
0.59830347262823,
0.30388710178795286,
-1.4558796844932331,
-0.3225618511944297,
0.09243390755822743,
-0.0006426173071943948,
0.7561662109631148,
-1.3902324585417218,
0.1677701056843372,
2.7311884232201735,
-1.1122880815211051,
0.0467986017971284,
-0.020089420529231013,
0.005481851366742631,
1.3205596999819047,
-0.1290403664286252,
0.3870444394307043,
-0.6549197738372419,
-0.17656835190090397,
-0.0008013451895215781,
-0.8769883624496301,

1.6929520910231297,
0.7288671099890552,
-0.6169673343691255,
1.667217439782634,
0.2554635335546482,
-0.029515171705053127,
-0.8290153205719569,
0.042733616172741226,
-0.5148227135977747,
1.2301836750084596,
-0.9175908300980525,
0.5229779878549122,
-1.5901934710811256,
-0.07316585886439231,
1.1195018365185199,
-0.31111306362203306,
0.7254709796877604,
1.1995417479543053,
0.6908225815252583,
-0.2919630814941808,
0.15955998483086176,
1.2871422030548714,
-1.2683991057896014,
-0.3768598572966417,
1.4863954796333156,
-0.4300298743216642,
0.19151460300148976,
0.9665838873771405,
-0.07244306377732504,
-0.04169546816073936,
-0.389029635704794,
1.5218961812832925,
-0.1772066326032777,
-0.70949107362336,
-0.015222466074935298,
0.3587128629768407,
1.5809494716454662,
0.12428164288740734,
1.5083554487395696,
-0.044379618902227067,
-0.03505579308192811,
-0.6786538117837789,
-0.007428157845337949,
1.5281494732580478,
-0.14124856089608578,
2.726718429443767,
-0.003106328084445967],
[0.5270475437101116,
0.7815425625485376,
1.6469610445191378,
1.6712876404917778,
2.122904188877425,
2.131106816606521,
3.7367798442214175,
3.7367798442219424,
4.750276145421292,
4.853973384996016,
4.865098072058574,
4.873980651141647,
4.987599328262889,
5.00246239884106,
5.520902021852277,
5.793284849001337,
5.8087832398624775,
7.432729128414403,
7.432953787975026,
8.144896858715603,
8.172580744083415,
8.788071761144659,
8.792288306719156,
8.828260688191317,
9.011586808740137,
9.011995976067807,
9.538729699920484,
9.576884709106812,

10.486750161390887,
10.48675016212107,
-0.15531374899360717,
-0.0849039080739262,
-0.8434555658556069,
0.9447826609611266,
-0.19482082661427938,
2.4999420394249707,
-2.7164831941210483,
-1.5408169364026967,
-0.8621482593978547,
0.31547458537730877,
1.4220435826823787,
1.348860553316339,
0.40827474166943256,
1.063960383917828,
-0.21906798993221652,
-0.33263312803097533,
0.4889869181158972,
-0.25524084746075687,
-1.3622146811064544,
0.996105149433398,
1.4161210728452394,
-0.6491827941870941,
0.7399329880002848,
-0.11994422281276146,
3.179573825543474,
-0.034943024466848674,
0.6075464711124173,
-0.3019825794696199,
0.029392952712390567,
0.14840732209332994,
0.08592470568620861,
0.016725634393736574,
-1.3697962205589145,
-0.06155375586805023,
0.16508080354919497,
-0.5975014914051775,
0.09635736997937351,
1.254101402191796,
-0.6193953337438176,
0.47602269510847217,
-1.7251540483535392,
0.5378274350019089,
-1.0159384367751108,
1.0968223266795405,
-0.79878134010976,
0.10468891520322528,
-0.2910173593847854,
-0.7078981306941186,
-0.3510837282628223,
0.5371002854277007,
-0.4911274293534503,
1.7273270693440552,
1.3500555714793954,
0.9052059235505011,
0.4771526028272018,
-0.1893747959421651,
0.13989050072973697,
-0.5930955187300607,
0.36780210947560876,
-0.5043370473487999,
-0.4851893292987325,
0.1987440888222681,
-0.5806445532873666,
2.3171646404124093,
0.06032126128170586,
-0.6389920437730593,
0.9985359186149056,
0.04870155999532171,
0.16447399991313041,
0.05315090520136875,
0.15727255736970833,
-1.3722292706433088,
-0.33233649668322635,

0.7758243461945761,
-0.20476613404890107,
0.06972046040604139,
0.3143879960336948,
-0.01452323353492752,
-1.2901366074359686,
-0.31033000747276407,
-0.16264485166390844,
-1.3914094473719103,
-0.5273128620655095,
0.02465459151953875,
0.10942766294132272,
1.0292094835171124,
-1.882689768994833,
-1.553216293090225,
1.3287846024814622,
0.20963913600213377,
0.7195732642990699,
0.0969787732183028,
-0.9744344415633356,
1.5822487402774768,
0.7077471503862973,
-0.766345302327988,
0.3831437091022152,
-0.12028853613988252,
-0.8209919023748664,
-0.21164805512089607,
0.5290684330656528,
-0.1040878870845114,
1.1956945887281514,
0.25703931501419025,
0.6344733295912399,
-0.02497584874729833,
-0.20937697561532842,
-1.4409411147319051,
0.9707320454927133,
1.4039378085761043,
0.9760826987262126,
-0.17314228405542512,
0.16734556762922406,
0.07881248116464283,
-0.2924467918405051,
1.4946430081327182,
0.7516806171134963,
0.799700486923224,
-0.8440600324500036,
-0.01069194371181784],
[0.3388610072211759,
0.3930923343746358,
0.4059472295013711,
0.4597995545870322,
0.4597995545870322,
0.4667083780713653,
0.9924683364283498,
1.054118787032341,
1.3839145947816414,
1.4469616998432806,
3.044048229936016,
3.044048229936016,
4.682975233106521,
4.682975233106521,
5.287925895396742,
5.287925895396742,
5.625677830869467,
5.625677830869467,
6.519005703020383,
6.519005703020383,
6.519005703020383,
6.854665591577039,
6.854665591577039,
8.092962128458302,
8.097830997780322,
8.102159968176109,
8.111954675742565,
8.727038821401255,

8.727038821401255,
8.784424594255395,
-0.8636093730812797,
-0.04742973062196874,
-0.37989856483295564,
0.3750798322441465,
-0.509087961968837,
0.5366875438436245,
0.1232704940065221,
0.5249134835872401,
0.13594138148710633,
0.10802292582448766,
-0.7156772546121367,
0.34756877087651405,
-0.5704812075854907,
0.289395233495819,
0.2674862930849357,
0.8724143866961164,
0.5770602988816299,
0.658777432128999,
-0.47288232304292416,
-0.029663566814993374,
-1.323180236232167,
-0.2890901656039231,
-0.015141732930961212,
1.1846850771343012,
-0.4506688471965015,
0.22241980016839272,
-0.015748484493019736,
1.1522499025192263,
0.2996907655672081,
-1.7370631162182948,
-0.35306232532663057,
0.17322471105923057,
-1.1855180134814352,
-0.31537295018088835,
-1.8120366955290261,
0.6663892425075478,
0.21874911670355193,
1.2729693967005706,
-0.14230256773273223,
0.2012427441067739,
-2.3269317425793643,
-0.03299848401271634,
-0.4074853136565566,
-0.30371905431664253,
-0.12037074206123735,
0.8456676988075262,
-0.21838328071319946,
0.019494764922420988,
0.16383099728733078,
0.056056403895700845,
-3.5632063709291613,
0.6579904510057321,
0.5207861046034454,
-0.017180990359968364,
0.2532807007765998,
-0.03315495616728509,
0.7564905992030696,
-0.11091568858030387,
-0.9281890577202553,
-0.6082181315906088,
-0.4480511658411004,
2.1953283079942025,
-0.30446713497464617,
-0.03359602431311142,
0.28972943746372876,
0.17817064395522006,
0.7965754083210965,
-0.9520780563921613,
0.10150445450145726,
1.3510884636720888,
-0.37849615173878054,
0.17729737647006832,
0.1884683806301832,

-0.03234547530918763,
-1.338763456177623,
-0.6711058962200585,
-0.012976043431998471,
0.027578392175021227,
0.10161286285089129,
-0.00939135247962935,
-1.302164623174096,
1.3715576399721376,
0.737767307868418,
1.5939269674119705,
1.4291758244230544,
0.5601859764306084,
0.3488764286390935,
-1.093762224545935,
-1.1976831420922067,
1.4557381665683378,
0.8924472305311927,
0.5300221300338881,
1.0785280788144638,
-0.6029509417428172,
-0.6790632925762424,
0.4138865114731201,
0.2917474006869537,
1.8050154062536312,
0.05177383423831921,
0.25467777497912664,
1.4671356779039024,
0.4422247342165525,
-1.5431607378377095,
1.8101814873938125,
1.6870467346899987,
0.7517920825884995,
1.3622912362338087,
0.13653050459472865,
1.5210183141257738,
-0.004247168330480351,
0.26137218980413907,
0.5451511562235306,
-0.6705596749039198,
1.2136085126593865,
0.5231529087420831,
-0.03279466120532818,
0.2548954759886831,
1.5797820110558691,
1.075840789720537,
1.5541818084195007],
[3.821020346740443e-17,
1.386496864345847,
1.386496864345847,
1.7951472089989222,
1.8559400139285378,
2.7980544315911504,
2.7980544315911504,
2.82954781812891,
2.82954781812891,
2.82954781812891,
2.956183634338348,
3.840561875918348,
3.840561875918348,
4.79501776182173,
4.79501776182173,
5.670460018404704,
5.670460018404704,
5.670460018404704,
6.865392250052202,
6.865392250052202,
7.506826147384778,
7.506826147384778,
8.737685173635683,
8.737685173635683,
8.737685173635683,
8.737685173635683,
9.601023092540274,
9.601023092540274,

9.853738132880634,
9.853738132880634,
0.5611658219809542,
1.427331325859137,
0.8881601644240952,
-0.8937497791603879,
0.5712044928850715,
1.3195951619089188,
0.5467792160605555,
-0.6816630744322637,
0.06707477836950569,
0.6798692668930644,
0.19576168860111362,
0.5094724850631845,
1.1424414477256812,
0.30471145260454696,
-0.08158911485577636,
0.9585661478313002,
1.0836238583568754,
-0.3718197048926503,
0.7841948506971846,
0.3023515457437066,
0.7662240328883452,
0.9970048707689876,
-0.8946064024093777,
0.27649390200762114,
-0.15342626519190924,
-0.4015720487025046,
1.460926694270634,
-0.055421168101734586,
0.6077050880702092,
-0.7932081335349191,
1.3188332541813446,
0.01929555634963922,
-1.1301845164589392,
-0.1718174564221745,
2.26154163970611,
-0.9882028269623031,
0.8038643847286138,
0.6849304102208338,
0.874798983022982,
-0.016486064274954207,
0.0264221098096002,
-0.3625000384278827,
1.4376331477886695,
0.8499992740594865,
0.10042171561200945,
0.14542648933357244,
1.3513090878714353,
-0.22248922869259902,
-0.4483175327546578,
1.2345758678942387,
-0.18983522552694923,
0.019591210301167533,
-1.3607248526627624,
0.018375533581887468,
-0.3730738776375715,
1.0123537416913542,
0.7412601688930508,
-1.6525919628732422,
-0.8341682190885938,
0.04519729634722039,
-0.06718028327556408,
-1.5649719285752928,
-0.37531851249345394,
-0.11822841604899964,
0.008154106549025474,
-1.4601449692077721,
-0.0030993446167428395,
1.462573870839384,
-0.11887757571245174,
-0.09361960188417112,
-1.2543408854038138,
-1.525280802664792,
-1.241413372948363,

0.008758808605552393,
-1.6676332942965022,
0.13826170532334445,
0.7537363962746654,
0.0833397476047736,
-0.12216949704941273,
0.02084807234445201,
-0.3421949354726438,
0.0017882611822728064,
1.5186851123669147,
-0.4603163345929323,
-0.09956972276089109,
0.7247597830417082,
0.5304935691839561,
0.003193152207841171,
0.5581346990750625,
0.010542051444639388,
0.7553167297004638,
0.5956891015800863,
-0.05566254332595716,
0.5362461335652635,
0.009895282042162012,
-0.3029901133286735,
0.09399206534914867,
0.8818812838713691,
-0.06795832284115338,
-1.580933930101608,
0.3014975711017802,
1.2092022074751856,
-0.019692531397995948,
0.9250981578351692,
0.4673383297638096,
0.03344339272872125,
0.13058472572385768,
0.036262684184263205,
0.458095375934523,
0.6492846129896516,
1.3039221898344189,
0.024764812872192767,
0.7236926243101844,
0.01444840086266692,
-1.0388939706148563,
-0.08359693344439546,
-0.4698004238001188,
-0.48055768545836347,
0.4662000815300196,
-0.041204969991243484],
[0.01798320141016013,
0.3393419127512466,
0.7047697703765823,
0.7064789163091142,
0.7191656785855808,
0.7198672068604162,
0.719867206860416,
1.0840512961729545,
1.0900003904895101,
1.1926153174924548,
1.867054342490295,
1.867054342490295,
2.368094916985588,
2.3683751276938714,
3.6383003552329964,
3.642527762446636,
4.505845683894636,
4.602446412902637,
4.972427659100928,
4.972427659100928,
5.1148164150678035,
5.8066136392845,
5.820681023327439,
6.727088267818048,
8.352653016300794,
8.773334122756978,
9.065902540064004,
9.065902540064004,

9.74880092287043,
11.30145464466351,
-0.13904770480884485,
0.23493722133539818,
0.4497803172772882,
-0.640584995353188,
0.9779304528121034,
0.0011745349607135795,
-0.7175493520168287,
0.373620380836484,
0.08111255661159392,
0.6653458367314794,
0.0037967443269442437,
0.6286122183467432,
-0.03825426016359485,
0.6896824961465278,
-0.23916946602734285,
0.4624850989736848,
0.1419493525208794,
0.7145176662978019,
-0.316909964913161,
0.6605458698076423,
-0.39309909050950365,
0.7623048158369822,
0.7469736169282462,
1.2400892032597353,
0.12266455668869777,
0.19578648318487166,
-0.784043232212839,
-0.10334183265831844,
0.32774480702026265,
-0.011491906972883237,
0.322988869457402,
-0.2027870428092001,
0.1281571522083961,
-0.1947961113336468,
1.7114513053475526,
0.005425684515457419,
0.8136825826883561,
-0.25903007981373005,
-0.5015428579892063,
-0.2600978115589891,
-0.15405473476720474,
-0.4207498663011112,
1.04622682040645,
-0.8642339256273757,
0.6248691295250803,
-0.3539240130838131,
0.011689879721248103,
-0.3068283940685192,
-0.6048660879777619,
-0.11704784670887185,
-1.2745062338511486,
-1.7890038533223112,
-0.7818671277256136,
-0.8877455887286905,
0.3164818198590212,
-0.27477307924208777,
1.8083251930535746,
-0.5031479272825263,
-0.9648993597903888,
-0.41682962131770707,
0.27040705073896654,
1.0892900815418987,
0.17779960873468054,
0.020337033950376556,
1.0551529464315492,
-0.10705558118147128,
-0.8851977322505518,
0.2232945485845247,
-0.5850403569085746,
-0.7690595025978197,
-0.4391673030332381,
-0.9309566421919284,
0.2973030450025941,

0.6780545449896137,
0.14045373030826291,
-0.5109313900859794,
-0.44070010469428217,
1.6940975579604587,
0.17062304524159472,
-0.402681393200546,
-0.7091187255059627,
1.3132357546586793,
0.2729391261599054,
1.0115940175088693,
1.0046673925246778,
0.4982257464819494,
0.04454679168285006,
0.4863023994043421,
-1.1433261311581624,
-1.820308437409531,
0.6035405700431521,
0.04003907881711185,
2.99381748600971,
-1.0437523456620463,
-0.22517480393070408,
0.10746937069246698,
-0.8788513722547997,
1.286531186523234,
-1.3157160458746164,
-0.6979748907629274,
-0.16573593981190854,
0.2529194783090185,
-0.2562517800846549,
-0.7966921479109774,
-0.4446940958312971,
-1.052310031662883,
-0.1506245930250076,
0.1615975776996014,
2.9220364155549214,
-1.7197843955783065,
-0.6592103621593355,
-0.2007498223803255,
-1.6188873615119244,
-0.15193115939751467,
-0.20866967970071015,
1.2910791163230286,
-0.4917152660712325,
-0.42348313202027765,
0.3518303936962707,
-0.10034486495238693],
[0.009039772647041902,
1.1400526086001903,
1.466905028683042,
1.6524670085313564,
1.987797816914966,
1.9900950346150876,
2.1413202332332344,
2.390935514217652,
2.402429399816812,
2.81913686416337,
2.81913686416337,
3.5656810200683298,
3.5656810200683298,
3.5656810200683298,
3.76603447051701,
3.76603447051701,
5.499236323276389,
5.4992363232763894,
5.4992363232763894,
5.532789537471719,
5.532789537471719,
7.371733777832876,
7.371733777832876,
7.680783486275433,
7.706623781446164,
7.706623781446164,
8.457742149729329,
8.457742149729329,

8.95222736351662,
8.95222736351662,
0.18928393791812126,
1.0354889043700823,
0.3703113207409527,
0.30988589523494625,
-0.34669340410550803,
0.09615284519636855,
0.1011898884094612,
0.6160115432573767,
0.6148540393435122,
0.07934943483664206,
0.8557624329627399,
-0.7672633290674925,
-0.14188111026081618,
0.1995320641320393,
0.4594888802421805,
-0.03593187081620776,
1.2926591415659605,
0.31433950962447343,
0.8053272546604554,
0.6634815559763872,
0.311660786273531,
0.0294653444200854,
0.3044394736040779,
0.7599259037372076,
0.22962735656034647,
0.22562971232722423,
0.29511485598499815,
0.02260075636691185,
-0.9497096253675098,
-0.5331371571917215,
0.2192512323868535,
1.104832430155296,
0.8636617903941131,
-0.6756684630428217,
0.4169844309263729,
1.1297527651723116,
-1.160118439748996,
-0.7971836651224,
1.9169656561991728,
-0.007260796290839314,
0.3614132074474742,
0.6243022778229537,
1.8335025630018775,
0.35139722497115833,
-0.7247870129359816,
-0.6744811297738377,
0.49997538785346773,
1.4139521212248116,
-3.0799818401546553,
-0.14768564076593044,
0.06251500852230953,
-0.8547134606957393,
-0.30107856776262504,
-0.7654972812734481,
0.28603020918627825,
0.2129303248959857,
-1.2077141264570097,
-0.8488549318812054,
0.6492574804166855,
0.7751290676155554,
0.40909519807061084,
-1.2750630737105486,
0.4187798718836698,
0.33498424523924863,
-0.19017028209895162,
-0.03232931911977397,
1.0621483440417678,
0.452017889610544,
1.0192767515800205,
0.7290563184502946,
-0.5286462788982386,
-0.05512372609304388,
1.3189151919853837,

0.5865235617306253,
-0.579843833868396,
0.6980158293666034,
0.027004860579276476,
1.2876675233834314,
-0.17185617277774212,
-0.3827017785428823,
-0.7286404996479297,
1.1682112525473984,
0.22765379123755217,
1.4870969768845252,
1.1642898801914527,
-0.03976972562460009,
-0.8163296820868416,
-0.15318057329999143,
0.0482918949035788,
-0.06014345828968787,
1.1478583262090496,
-0.005141116181289305,
-0.5820859555621626,
1.086519910329415,
-1.08558016317555,
1.531511042803357,
0.042734048269580674,
0.7452599788242152,
-1.3094923903018465,
-0.9012205175381317,
-0.0755166851159581,
1.5493661520767013,
-0.2278023712186688,
0.8627212805492942,
1.8523276191392382,
-0.10961709979291216,
0.6746606852631131,
0.22566845572735392,
0.5920154791154506,
1.2709342533151342,
0.26903045177842116,
0.11724485476711638,
0.621942277674204,
0.15491924869980966,
1.472217084758381,
0.8975149794611956,
0.3601026339731485,
1.3234513983254743,
0.0182465793142515,
-0.11914786829986494],
[-5.765348397008027e-18,
0.3428984748224433,
0.3655450682584724,
0.704784031551808,
1.5409081472811441,
1.549300366088437,
2.3787666619954635,
2.3787666619954635,
2.8808650819472406,
2.884987740396357,
3.1308808834009203,
3.1363836212549696,
4.869321728501938,
4.869321728501938,
6.1150174264690085,
6.2582278136815175,
6.271636236200362,
6.551203581243674,
6.59741701640448,
8.230564833747417,
8.294003346338926,
8.31431753208387,
8.369586036193128,
8.415319617883744,
8.70447046242739,
8.709120596281233,
8.918659758853744,
8.99410216914391,

9.002256766523654,
10.218690251738174,
-0.6798627919679444,
-1.1436724543472427,
-0.1278269606092795,
-0.9286552360068459,
0.6524046490818174,
0.06134572949387961,
-0.49761179900021657,
-1.3918471119443894,
0.5858287319319103,
0.18669127045720346,
-0.4586789737568629,
0.2400608953322058,
0.022222124883641912,
0.9282004764805514,
0.6952358394741303,
0.805684979006905,
0.6146729452742722,
0.6777524139082742,
-0.8117842884117104,
0.3426849285038029,
0.7100299235979358,
0.7658186509139059,
-0.05418633990950139,
0.1761917490887145,
0.5744825473612316,
0.31705937472296297,
-0.9188523464356001,
-0.7337446596965294,
0.2061368312243915,
-0.6645694031850036,
-0.009871104765786294,
-0.14388187083546433,
-1.2485707106311008,
2.849540809212292,
-0.5119215846255833,
-0.19381894062193808,
-0.17310717674563775,
0.8465397629359774,
-0.46757240698237135,
-0.05607605828560263,
-0.40986137011816176,
0.07649505002491204,
-2.1522907800348112,
-0.6653346582594515,
0.4084801543759574,
0.4028897820911683,
0.16908117843954765,
-0.07217026591146015,
-0.72804755296126,
-0.13548738477012795,
-2.590679566047418,
-0.40743076961744734,
1.1030856800887578,
0.08388380440455438,
0.25224805905642333,
0.10644300798076774,
1.0474125729404944,
-1.1459016177859003,
-1.6916287310468103,
-0.9886067106540645,
0.24182274960650893,
-0.25703819491337826,
-1.1942818044595898,
-0.7763678643520989,
-0.916718557142268,
0.20086626876005667,
-0.3711023890817823,
-0.006962349053602254,
2.687031223814704,
0.3970791704566904,
-0.6890591654410289,
1.0504536719091986,
-1.2458238157294268,

-1.197330311624414,
-0.12155955689525438,
-0.2969718406476498,
0.15406921803787998,
0.9984735505197111,
-0.2734456365103056,
0.0100494123830882,
0.16172873483083283,
-0.6780409734277772,
0.4539997995464368,
0.7558576390626953,
0.9556605206819043,
-0.10572075101403118,
1.5564628686733721,
0.05397552837650759,
0.7415987024402694,
-0.1648500682327342,
1.052699139325594,
0.6776572023171561,
0.15219630541397192,
0.8843933196559655,
1.3948099389813178,
-0.16710315713046545,
1.2199200746863228,
-0.3825678575328573,
-0.26766985045935626,
2.5777182352113908,
0.23156136798610805,
-0.3377548185037014,
1.0689111835439342,
-0.17216022970038714,
-0.29381807805976945,
-0.5689533077540762,
0.30596827309967156,
0.5260993417881752,
-0.6768480303494689,
-0.1092430795457648,
1.5442082488768039,
-0.10826504240549732,
-0.9555598065181787,
0.5334835330029429,
0.5569227134357198,
1.2665483894271157,
-1.9677332823608626,
1.5559931182978979,
0.7709453943956464,
1.5746506000831744],
[1.7625089228179507e-05,
1.298540182320117,
1.3141870420194457,
1.3305471045820527,
2.0686735179487985,
2.23810597518972,
2.267373737425337,
2.295441377067579,
2.3325388050051976,
2.352484360014207,
2.370891483796666,
3.3117124844820194,
3.363605896627787,
3.858981460194034,
3.8589814603490344,
3.859537867870235,
3.8595381713444072,
4.26458571773699,
4.264585717738593,
6.599549149586233,
6.655963438031886,
6.939612414177192,
6.939612414268938,
7.004302986969205,
7.010544991093791,
7.97314828104894,
7.999280802101656,
8.022628296732579,

8.027896471579398,
10.077254342713585,
-1.0733166032270363,
0.4264387306682,
-0.29490479134596703,
-0.45181113351566254,
-0.9496712390342118,
-0.24412788786515252,
-0.17005038539710493,
2.131747001055919,
-0.8801477965160032,
0.1985900676052623,
-0.17829180427317373,
0.1513896817808816,
-0.2996418627583153,
0.09152392337611555,
1.0155053498240294,
0.6713280773462803,
-0.5915672184524843,
-0.21780814909773427,
-0.23676595566314207,
-0.1539223229682568,
0.9733737893230769,
-0.8325690111887051,
-0.6684349693104433,
0.944923850675747,
-0.8270803600282979,
0.4869668407293288,
0.07342964610078141,
-0.8928903181234079,
-0.04262248204617692,
-0.7602498290394286,
-0.48858825463899247,
1.7340528561515622,
0.7340502315717261,
0.5163401187759543,
-0.5015858040228415,
0.2890335055344555,
-0.46033506221820203,
-0.8185907107929625,
0.4557079379705311,
1.6601520871164626,
0.07125306723268252,
-1.4035119795986863,
-0.6741575848895364,
1.84360490583597,
1.6836338044245691,
-0.33329370458317326,
0.19854317161902654,
2.03917897859608,
-0.3342251938073038,
-1.6015864477635742,
0.35422488479060155,
0.1487121545506934,
-0.2821114888507261,
0.21869934797564045,
0.7523961573071125,
1.3021785537923125,
0.26184859002187644,
1.2441238228144538,
0.039670539596101485,
-0.15687777352236798,
1.1386642271664005,
-0.13559116971858365,
0.39167499360215746,
2.4453261619465274,
-1.0890320328395309,
-0.4289422005081893,
1.425249318116838,
-0.0016733698174337221,
0.023121567661497415,
-0.28325087531692295,
0.940069260590291,
1.163437407349777,
-0.4256527399896147,

-1.4972731615803316,
-0.12624975701746496,
-0.10144651691714318,
1.3307378561466523,
0.041778278155808674,
-1.2746374184036902,
-1.2216848741140482,
1.4048769664929757,
0.8957112974734468,
1.0058068981677006,
-3.0965653057688316,
-0.3344448337959029,
-1.0948102803209163,
-0.038478236121346335,
0.12618687504486675,
-0.0850775611793361,
1.33550885346982,
-0.04334772196309596,
0.2406908654957122,
0.7327249608913058,
0.9966290850109706,
2.2852565843192423,
-0.030570948934034583,
-0.5566407461361556,
0.813246653395058,
0.5946355273642772,
0.07519254658407079,
0.27120989268694123,
-0.15415722381241237,
-1.1239288818585278,
1.893751281002709,
0.7530205722182245,
1.4454588921082565,
0.15037300343080462,
0.1793867334099716,
0.3587259643057371,
-0.006799349268608,
1.0907160355013774,
-0.30186717991666434,
0.8507589785411056,
0.009155709268365956,
-0.8165113788656561,
0.011816515180360782,
1.116430418883973,
1.5343789763879045,
-0.1682835166498154,
-1.5032827361947747],
[0.10956969985098665,
0.4360115975982133,
0.5932904704915389,
0.6050451872391112,
1.3688142364348175,
1.3737597367805747,
1.5226773273508583,
1.5589271816733294,
3.1503205471140947,
3.1613053536766147,
4.7635234433865605,
4.823148382609575,
4.823188243925802,
7.171157860626202,
7.171157878485542,
7.171158529865571,
7.175073431930883,
7.183070822592498,
8.102400589897519,
8.10392263292654,
8.103940780725882,
9.021308924987686,
9.02133947231205,
9.082831297490506,
9.094868565409332,
9.236804949782064,
9.237029573386183,
9.26430291432718,

10.00744744746223,
10.032040374677464,
-0.5042382729215367,
1.0180015436638659,
0.10423420212375085,
-0.3355687578372037,
-0.3902319222280216,
-0.09867602213718178,
0.3860296418386605,
0.8522828003851446,
-0.7730596965193267,
0.08803782521712378,
1.175277695996097,
-0.3769225558693855,
-0.4553519820435133,
0.4758962487375975,
-1.0517577693418567,
0.3712029741888282,
0.27334544561774066,
0.03397030216872572,
0.3891396532807883,
0.12152393226873999,
1.5192516885791116,
-0.5124931636551364,
-0.5667003913709334,
-0.08425393582345661,
-0.8152956002657638,
-0.008087141654348068,
-0.20833380477584987,
-0.19186344764752156,
0.309463849209865,
-0.16555810903618162,
-0.28534485935787846,
0.025710068868521854,
-0.9172960495760906,
0.027100436879331523,
-0.9806057417125867,
0.46877133554227174,
0.22597640762440013,
-1.8700902599315918,
-0.6222133609404971,
0.023922176634194332,
-0.6777524378245746,
-0.3883818313368169,
-0.19921738892417568,
-1.3418496115619054,
-0.05365669225714106,
0.38835985529462186,
0.46741217514915123,
-0.1388983196762152,
1.0002905156718758,
1.6633296566254185,
-1.0097136763815113,
1.203957772036831,
0.34033273848251194,
1.1456357751380877,
-0.4637199651146574,
-0.13624703361320328,
0.6785440200843139,
0.21691345067474185,
-0.4166036808107477,
1.6003722740710606,
-0.294881370724797,
0.04198878790454511,
-0.689532047804753,
-1.3911901894570817,
-1.1667039833298627,
0.824824800006074,
0.8047687705752488,
0.5727196814509768,
0.2385364014798047,
-0.512164832149354,
0.17925141295137917,
-0.19704824673887428,
-1.6372757558452697,

0.5441675736491897,
-0.13936070677778437,
1.3935144064781402,
0.10763856884501717,
0.6679788244416961,
-0.33118654939932857,
-0.35796695002153883,
0.2632208027041665,
1.5831503864671255,
-0.25140146740224,
1.135774876905931,
0.5897962040478675,
-0.7664904031063712,
1.092649076983624,
-0.9862462517151122,
0.010216869101717933,
0.5705833807099128,
0.09839127014019662,
-1.034770130526523,
0.20582133073730807,
-0.44428721001967564,
0.5272444819968604,
-0.2627660867388947,
-0.9857421779520247,
-1.3646538516413942,
0.14779052759756187,
1.1207932022043874,
0.30427416833245535,
-1.055109005337564,
0.30548258474769546,
0.8153937738463674,
0.5082537250510629,
-0.2973639305797175,
0.0034742209991892575,
1.5168338311712208,
0.0971904670135119,
0.5305672799342332,
-0.08170159551874663,
0.4909203029282298,
-1.1771354207712392,
0.038725619801271274,
-0.6655680746562402,
1.312626970471223,
-1.0115568894576918,
0.16290676730453105,
0.3074286838975462,
0.013117345918778788],
[0.4866654913190249,
0.4866654913190249,
1.4314116389487666,
1.4314116389487666,
1.548608655381947,
1.5552664901547235,
1.662920497545568,
1.676155881591226,
1.6952532774325564,
1.6952532774325564,
2.46741851740979,
2.50102877426778,
3.636145431120801,
3.6361454311208004,
3.6361454311208004,
3.6361454311208004,
4.0161273218482085,
4.0161273218482085,
4.5411636896776235,
4.5411636896776235,
4.922545565757144,
5.624338470516907,
5.642847901355697,
5.904037934138086,
5.915382265322609,
6.860930229950661,
6.860930229950661,
9.422517992567101,

9.467623480195014,
9.467623480195014,
0.40091998571272736,
-0.3581846438287994,
-0.32033246941904975,
0.24409866553117068,
0.176560178011825,
0.4452285262890565,
-0.37150326481454765,
-1.0185408309777293,
1.6197684873006286,
2.0095194673452026,
-0.20342804182482532,
-0.4471079471088122,
0.37127155242298737,
0.4199607045713712,
0.11145558426636207,
-0.289968410859463,
0.9752004498173846,
-0.11972987215087512,
-0.8215210043660294,
-0.4144105549344152,
0.04966095047933562,
0.6091224412599766,
-0.2481186610805342,
-0.9036735819055022,
0.22922493224752044,
-0.04567884568114995,
-0.40529506738584414,
-0.44609185937563545,
1.1000542207972743,
0.8995834380092431,
-0.4391184290327733,
-1.623814742575614,
0.11121717167805684,
-0.15344289124808663,
0.7072346226342329,
0.013911394934765222,
1.1008239636292285,
1.1342299046285917,
-0.1605718532112884,
-0.8521154510350495,
-0.9726520318623231,
-0.11078467990736285,
1.2893067940177063,
-0.8656847782514395,
-0.4554157535410911,
0.46752563535079134,
1.0115326820592876,
-0.017465678177200683,
0.7555243091795097,
0.2225241128439446,
-0.6129237770264853,
0.36249038141661666,
-0.1470816778834455,
-0.23781939745964606,
0.7410640700988803,
-0.10219672891678527,
-0.9043863030697507,
-0.054446492384892985,
-0.23618927113386157,
-0.3971482428910102,
0.42980979959963467,
-0.4511512430809565,
-0.3322123782853245,
0.15031459204805087,
-1.4287283715133035,
-0.026859055017604255,
0.20826999854501216,
-0.016397767746330064,
-0.026711091547853946,
0.15899307013846248,
0.18349959987180348,
-0.019181103141429404,
-1.3777401647916119,

```

0.05578672583947999,
-0.17191397249005735,
0.08492618339903452,
0.2781511329827625,
1.6853098582374997,
-0.7964285169174421,
0.13628687772581302,
-0.26524580358677924,
1.3615146942088054,
-0.18163849035743365,
-1.5476669195398023,
0.2740670870741208,
0.19740142577560438,
0.18567840551061798,
0.43288784764736754,
-0.8069341399605817,
1.5139336080928714,
-0.7594440385529729,
-0.20443587392639553,
0.0916888700165231,
-0.03922249489745327,
0.4135801157921025,
-0.06731598775715468,
0.5038720973518589,
-0.7705715736697758,
-1.4775324565516457,
0.061950004909204714,
-0.9555167624183505,
3.1277455162888304,
-1.137887484425651,
-0.09852947784569874,
0.042597844143281625,
0.06134264537966588,
0.3384848933516047,
0.012002169875906457,
1.5732417764972468,
-0.3550969359952667,
0.37821230776206083,
1.4914211446267915,
-1.2027836619303884,
0.22421848814312553,
-0.2072711571643093,
0.05122655754886873,
-1.5578500765322636,
1.5918641150141506,
1.2635397829742223,
-1.2452354025631063]],
'run_time_sec': 43720.81102514267}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 3.750846163374531e-05,
0.22285241391868899,
0.26252411232084344,
0.26252411232084344,
0.26252411232084344,
1.459545906754556,
1.4595459067545562,
1.6094835194149661,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
3.3552499939702187,
3.3552499939702187,
5.033284106467877,

```

5.033284106467877,
5.056477210452712,
6.768913439946703,
6.768913439946703,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,


```
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0]]},
'run': 'redundant',
```



```
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.33528943002425843,
0.8608925283648261,
2.8309935233222867,
2.8382233121900358,
2.847558790018716,
2.8477263794127468,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
5.127462455084763,
5.128383696390232,
5.132051241075024,
5.132231848473024,
5.132297186784173,
5.132297186784173,
5.132297186784173,
5.132320898992942,
5.134903164275234,
5.734798022200061,
5.734798022200061,
5.734798022200061,
6.71748919093495,
6.71748919093495,
9.773430860122208,
9.791656954875775,
9.791980525916317,
9.804782964874805,
9.947289619134542,
0.6818618664375632,
0.42253658736184846,
0.19067339209915313,
-2.314563573529872,
-1.7784854373672185,
-0.10945476580518654,
0.08980428543050965,
0.6265702304745648,
-0.12016219313034968,
0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,
-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
```

-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,


```

                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4]],
                                'run_time_sec': 17.832751750946045}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}}},
  'init_param': { 'value': [ 1.5680106786222674e-17,
                              1.0238650999177359,
                              1.0238650999177359,
                              1.0238650999177359,
                              1.7041550191762513,
                              2.070931805530806,
                              2.586234954306046,
                              2.6265616878606783,
                              2.7566136785306155,
                              2.7689271757054255,
                              4.271831894498143,
                              4.271831894498143,
                              4.271831894498143,
                              5.004110362344359,
                              5.004110362344359,
                              5.529689350278954,
                              5.529689350278954,
                              5.529689350278954,
                              5.5493166460873375,
                              7.855838962179162,
                              7.855838962179162,
                              8.508023697610218,
                              8.540796404638465,
                              8.940056981382925,
                              8.940056981382925,
                              8.940056981382925,
                              8.940056981382925,
                              9.524293982069336,
                              9.524293982069336,
                              9.815045396471412,
                              0.717685470636315,
                              0.29125971197830336,
                              0.013619559122678915,
                              -0.46174394022911175,
                              -0.7802589318548062,
                              1.8785332138943256,
                              -0.07718385116900997,
                              -0.4181315830102034,
                              0.3859499067718677,
                              0.7218523694763775,
                              0.16350787496293626,
                              -0.06274337543165032,
                              1.048639096314506,
                              0.9454500907927006,
                              -0.18328208251477485,
                              0.5336296017995852,
                              0.7047960543548563,
                              -0.4717130073686535,
                              -0.14327049451726778,
                              0.5646728024946054,
                              0.1377201649736573,
                              1.0845662084088512,
                              0.7117671017527167,
                              1.8562916393289055,
                              -1.0958742208007046,
                              0.314756410152813,
                              1.0055290931424188,
                              -1.2653176839886926,

```

-0.6435463058390287,
-0.13689785210195535,
0.5185042269644529,
-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,


```

0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
'identity_factors': [2, 2, 2, 0],
'noise_level': [20, 10, 35, 16],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'value':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 16.81493878364563}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.0019047787007362745,
0.7201496918445128,
0.7201496918445128,
0.7201496918445127,
0.7226708680142584,
0.8068041864232433,
1.3847174673528184,
1.6031895944534578,
2.458865333289775,
2.458865333289775,
2.721120236828938,
2.721120236828938,
3.554687304897056,
3.554687304897056,
4.537252190232355,
4.537252190232355,
5.239770593633025,
5.242363659304797,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
7.6484324534494865,
7.6484324534494865,
8.96757120343865,
9.352637118466301,
9.381932224950582,
9.966925955531599,
9.966925955531599,
0.29840004251263574,
0.3207087147269936,
-0.22478909909228914,
-1.2242611593120782,
0.061748844824383295,
1.9931455249507801,
0.02449362327304914,
-0.23402928130657288,
0.3399922671860315,
0.2355005268966802,
0.7912945290539132,
-0.9147564287830934,

```

0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,

```

-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.536182288327376],
[12, 6, 21, 8, -3.450081350248876],
[20, 10, 35, 16, -1.8650932993173606]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':

```

False,

'value':

```

0},

'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
  'identity_factors': [1, 1, 1, 0],
  'noise_level': [12, 6, 21, 8],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                              'value':
                                              'noise_prob': [ 0.001,
                                                              0.001,
                                                              0.001,
                                                              0.001],
                                                              'status': True,
                                                              'type': 'depolarizing'},
                                                              'odd_wires': 4},
  { 'gates_num': [20, 5, 5, 16],
    'identity_factors': [2, 2, 2, 0],
    'noise_level': [20, 10, 35, 16],
    'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                                'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                                'status': True,
                                                                'type': 'depolarizing'},
                                                                'odd_wires': 4}],
    'run_time_sec': 18.563719034194946}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ -6.331146189926128e-17,
                             0.7069927066774304,
                             0.7069927066774304,
                             0.7069927066774305,
                             0.7069927066774304,
                             1.0044153364322925,
                             1.0092418692996383,
                             2.470333794214341,
                             2.478143749866421,
                             3.5776564981759003,
                             3.5776564981759003,
                             3.5776564981759003,
                             3.61745967653339,
                             3.6981780389472076,
                             3.755720232685741,
                             3.9712286023902594,
                             3.9712286023902594,
                             4.519037432417192,
                             4.519037432417193,
                             4.519037432417193,
                             5.280953883299406,
                             5.280953883299406,
                             6.620356980277175,
                             6.620356980277175,
                             6.620356980277175,
                             6.620356980277176,

```

8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,

```

1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '

```


5.047622474508209,
5.054336886650782,
5.067626550795242,
5.0733174878702085,
5.094629387163461,
5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,


```

-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},

```



```
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.08591881204796584,
1.5026023651616802,
1.9179097588555833,
1.9179097588555831,
1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
```

-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,


```

0},
'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4]],
'run_time_sec': 17.06001567840576}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
'init_param': { 'value': [ 0.00030922277578629,
                            0.881375055029912,
                            0.881375055029912,
                            0.8813750550299119,
                            0.9803845151714838,
                            0.9803845151714838,
                            1.201629973822729,
                            1.2441077564052723,
                            1.8660677974280824,
                            1.8660677974280824,
                            2.651002314516478,
                            2.651002314516478,
                            3.233637461790533,
                            4.36384995486056,
                            4.36384995486056,
                            4.975714528653046,
                            4.975714528653046,
                            6.591840750700408,
                            6.591840750700408,
                            8.234009055219385,
                            8.238223602720305,
                            8.238223602720305,
                            8.238223602720305,
                            8.903406631369169,
                            8.903406631369169,
                            8.903406631369169,
                            9.126449234760914,
                            9.126449234760914,
                            9.441918167657187,
                            9.441918167657187,
                            0.4384156956352651,
                            -0.017407483354635628,
                            -0.13579407977759564,
                            -0.08181457210659056,
                            -0.5635031413839604,
                            0.570197125887921,
                            -0.07389454022208618,
                            0.18950484918514415,
                            0.23442314823895152,
                            0.5187197914643009,
                            0.03760777637611914,
                            1.2080522066616035,
                            -0.006796933498365149,
                            0.7597395812711474,
                            0.3165595784185247,
                            0.09234381717322694,
                            -0.047116958459538144,
                            0.02161370846364295,
                            -0.3330735946096464,
                            -0.310519881157395,
                            -0.1321730433969341,
                            -0.32004747583182785,
                            0.38880240494533963,

```

0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,

```

0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisy_time_evo_ric3_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.5111758480272135],
[12, 6, 21, 8, -3.513108400075424],
[20, 10, 35, 16, -1.9256894730075684]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],

```



```

False,
0},
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
'identity_factors': [2, 2, 2, 0],
'noise_level': [20, 10, 35, 16],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 18.43172597885132}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.3773346757087749,
0.5734233884194113,
0.5734233884194113,
1.2094818154974225,
1.2528672337957094,
1.3073966674132076,
4.193348178434801,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.274742575727675,
5.343872214974395,
5.354523589318104,
5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,

```

-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,

```

1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.44845457619454],
                             [12, 6, 21, 8, -3.24468946816992],
                             [20, 10, 35, 16, -1.686607616211824]],

```

```

'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 7, 0],
                    'noise_profile': { 'noise_on_init_param': { 'status':
False,                                                                'value':
0},                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                    'status': True,
                    'type': 'depolarizing'},
                    'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
  'identity_factors': [1, 1, 1, 0],
  'noise_level': [12, 6, 21, 8],
  'noise_profile': { 'noise_on_init_param': { 'status':
False,                                                                'value':
0},                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                    'status': True,
                    'type': 'depolarizing'},
                    'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
  'identity_factors': [2, 2, 2, 0],
  'noise_level': [20, 10, 35, 16],
  'noise_profile': { 'noise_on_init_param': { 'status':
False,                                                                'value':
0},                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                    'status': True,
                    'type': 'depolarizing'},
                    'odd_wires': 4}],
'run_time_sec': 17.2414448261261}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
False,                                                                'cn': [ 0.5,
                                                0.5,
                                                0.5,
                                                0.5,
                                                0.5,
                                                0.5],
                                                'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
'init_param': { 'value': [ -5.647934061474703e-18,
1.4787834906243755,
1.478918229526628,
1.478918229526628,
2.0077351764243123,
2.4302809101456218,
2.4302809101456218,
3.009709563144601,
3.009709563144601,
3.239471043198664,
3.2405283031129133,
3.2405283031129133,
3.2656598154222434,
3.5353392937639825,
6.198713539858257,
6.213604956994236,
6.73397854893658,
6.73397854893658,
6.743547791194661,
6.744609103740889,

```

6.744609103740889,
6.749543065628153,
6.749543065628153,
8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,

```

1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 0],
                                      [2, 2, 2, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},

```

```

        'zne': { 'data_points': None,
                  'degree': 0,
                  'method': 'richardson',
                  'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                      '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                      '5 X 6 + 1 Z 5 + 1 Z 6',
                  'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                              '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                              '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                              '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                              '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                              '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
    'output': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
                                   [12, 6, 21, 8, -3.199071961215845],
                                   [20, 10, 35, 16, -1.635137691104089]],
                  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                         'identity_factors': [0, 0, 0, 0],
                                         'noise_level': [4, 2, 7, 0],
                                         'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.001,
                                                                                       0.001],
                                                                                       'status': True,
                                                                                       'type': 'depolarizing'},
                                         'odd_wires': 4},
                                         { 'gates_num': [12, 3, 3, 8],
                                         'identity_factors': [1, 1, 1, 0],
                                         'noise_level': [12, 6, 21, 8],
                                         'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.001,
                                                                                       0.001],
                                                                                       'status': True,
                                                                                       'type': 'depolarizing'},
                                         'odd_wires': 4},
                                         { 'gates_num': [20, 5, 5, 16],
                                         'identity_factors': [2, 2, 2, 0],
                                         'noise_level': [20, 10, 35, 16],
                                         'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.001,
                                                                                       0.001],
                                                                                       'status': True,
                                                                                       'type': 'depolarizing'},
                                         'odd_wires': 4}],
                                         'run_time_sec': 18.288968086242676}}],
    'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
                                         'layer': 30,
                                         'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                         'r': 0},
                                         'time': {'max': 10.0, 'min': 0.0},
                                         'type': 'xy-iss'}},
                                         'init_param': {'value': 'random'},
                                         'noise_profile': { 'noise_on_init_param': { 'status': False, 'value': 0},
                                         'noise_prob': [0.001, 0.001, 0.001, 0.001],
                                         'status': True,
                                         'type': 'depolarizing'},
                                         'nqubits': 7,
                                         'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                         'r': 1},

```


8.940056981382925,
8.940056981382925,
8.940056981382925,
8.940056981382925,
9.524293982069336,
9.524293982069336,
9.815045396471412,
0.717685470636315,
0.29125971197830336,
0.013619559122678915,
-0.46174394022911175,
-0.7802589318548062,
1.8785332138943256,
-0.07718385116900997,
-0.4181315830102034,
0.3859499067718677,
0.7218523694763775,
0.16350787496293626,
-0.06274337543165032,
1.048639096314506,
0.9454500907927006,
-0.18328208251477485,
0.5336296017995852,
0.7047960543548563,
-0.4717130073686535,
-0.14327049451726778,
0.5646728024946054,
0.1377201649736573,
1.0845662084088512,
0.7117671017527167,
1.8562916393289055,
-1.0958742208007046,
0.314756410152813,
1.0055290931424188,
-1.2653176839886926,
-0.6435463058390287,
-0.13689785210195535,
0.5185042269644529,
-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,

-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455],
[0.33528943002425843,
0.8608925283648261,
2.8309935233222867,
2.8382233121900358,
2.847558790018716,
2.8477263794127468,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
5.127462455084763,
5.128383696390232,
5.132051241075024,
5.132231848473024,
5.132297186784173,
5.132297186784173,
5.132297186784173,
5.132320898992942,
5.134903164275234,
5.734798022200061,
5.734798022200061,
5.734798022200061,

6.71748919093495,
6.71748919093495,
9.773430860122208,
9.791656954875775,
9.791980525916317,
9.804782964874805,
9.947289619134542,
0.6818618664375632,
0.42253658736184846,
0.19067339209915313,
-2.314563573529872,
-1.7784854373672185,
-0.10945476580518654,
0.08980428543050965,
0.6265702304745648,
-0.12016219313034968,
0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,
-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,

-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358],
[0.08591881204796584,
1.5026023651616802,
1.9179097588555833,
1.9179097588555831,
1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,

8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,

1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406],
[0.3773346757087749,
0.5734233884194113,
0.5734233884194113,
1.2094818154974225,
1.2528672337957094,
1.3073966674132076,
4.193348178434801,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.274742575727675,
5.343872214974395,
5.354523589318104,
5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,

8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,

-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244],
[-6.331146189926128e-17,
0.7069927066774304,
0.7069927066774304,
0.7069927066774305,
0.7069927066774304,
1.0044153364322925,
1.0092418692996383,
2.470333794214341,
2.478143749866421,
3.5776564981759003,
3.5776564981759003,
3.5776564981759003,
3.61745967653339,
3.6981780389472076,
3.755720232685741,
3.9712286023902594,
3.9712286023902594,
4.519037432417192,
4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,

6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,

-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176],
[3.750846163374531e-05,
0.22285241391868899,
0.26252411232084344,
0.26252411232084344,
0.26252411232084344,
1.459545906754556,
1.4595459067545562,
1.6094835194149661,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
2.5927782099427623,
3.3552499939702187,
3.3552499939702187,
5.033284106467877,
5.033284106467877,
5.056477210452712,
6.768913439946703,
6.768913439946703,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,

7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,

0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138],
[0.003375513600300809,
0.4884127443235871,
0.6719352108785641,
0.9073170135093351,
1.4946633690060278,
1.5076957064847494,
3.872917838142945,
3.872917838142945,
3.872917838142945,
3.872917838142945,
5.047622474508209,
5.054336886650782,
5.067626550795242,
5.0733174878702085,
5.094629387163461,
5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,

8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,

0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989],
[-5.647934061474703e-18,
1.4787834906243755,
1.478918229526628,
1.478918229526628,
2.0077351764243123,
2.4302809101456218,
2.4302809101456218,
3.009709563144601,
3.009709563144601,
3.239471043198664,
3.2405283031129133,
3.2405283031129133,
3.2656598154222434,
3.5353392937639825,
6.198713539858257,
6.213604956994236,
6.73397854893658,
6.73397854893658,
6.743547791194661,
6.744609103740889,
6.744609103740889,
6.749543065628153,
6.749543065628153,

8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,

-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935],
[0.0019047787007362745,
0.7201496918445128,
0.7201496918445128,
0.7201496918445127,
0.7226708680142584,
0.8068041864232433,
1.3847174673528184,
1.6031895944534578,
2.458865333289775,
2.458865333289775,
2.721120236828938,
2.721120236828938,
3.554687304897056,
3.554687304897056,
4.537252190232355,
4.537252190232355,
5.239770593633025,
5.242363659304797,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,

7.6484324534494865,
7.6484324534494865,
8.96757120343865,
9.352637118466301,
9.381932224950582,
9.966925955531599,
9.966925955531599,
0.29840004251263574,
0.3207087147269936,
-0.22478909909228914,
-1.2242611593120782,
0.061748844824383295,
1.9931455249507801,
0.02449362327304914,
-0.23402928130657288,
0.3399922671860315,
0.2355005268966802,
0.7912945290539132,
-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,

-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.753559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815],
[0.00030922277578629,
0.881375055029912,
0.881375055029912,
0.8813750550299119,
0.9803845151714838,
0.9803845151714838,
1.201629973822729,
1.2441077564052723,
1.8660677974280824,
1.8660677974280824,
2.651002314516478,
2.651002314516478,
3.233637461790533,
4.36384995486056,
4.36384995486056,
4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,

8.903406631369169,
8.903406631369169,
8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.1357940797759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,

```

-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'run_time_sec': 68892.59480500221}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.00030922277578629,
0.881375055029912,
0.881375055029912,
0.8813750550299119,
0.9803845151714838,
0.9803845151714838,
1.201629973822729,
1.2441077564052723,
1.8660677974280824,
1.8660677974280824,
2.651002314516478,
2.651002314516478,
3.233637461790533,
4.36384995486056,
4.36384995486056,

```

4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,
8.903406631369169,
8.903406631369169,
8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.13579407977759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,

```

0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0], [2, 2, 2, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,

```

```

optimization': { 'algorithm': 'SLSQP',
                  'constraint': True,
                  'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.5111758480272135],
                           [12, 6, 21, 8, -3.513108400075424],
                           [20, 10, 35, 16, -1.9256894730075684]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.981590270996094e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -8.03029257391366,
                            'others': { 'beta_coefficients': [ 1.6466262975778547,
                                                                -0.9108996539792387,
                                                                0.26427335640138405],
                                      'cost_richardson_zne': 3.6109567503981035,
                                      'sorted_expectation_vals': [ -6.5111758480272135,
                                                                    -3.513108400075424,
                                                                    -1.9256894730075684],
                                      'sorted_noise': [13, 47, 81]},
                            'sampled data': [ [4, 2, 7, 0, -6.5111758480272135],
                                                [12, 6, 21, 8, -3.513108400075424],
                                                [20, 10, 35, 16, -1.9256894730075684]],
                            'sampling': 'default'}}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                            'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                            'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.0019047787007362745,
                              0.7201496918445128,
                              0.7201496918445128,
                              0.7201496918445127,
                              0.7226708680142584,
                              0.8068041864232433,
                              1.3847174673528184,
                              1.6031895944534578,
                              2.458865333289775,
                              2.458865333289775,
                              2.721120236828938,
                              2.721120236828938,
                              3.554687304897056,
                              3.554687304897056,
                              4.537252190232355,
                              4.537252190232355,
                              5.239770593633025,
                              5.242363659304797,
                              5.566731623636481,
                              5.566731623636481,
                              5.566731623636481,
                              5.566731623636481,
                              5.566731623636481,
                              7.6484324534494865,
                              7.6484324534494865,
                              8.96757120343865,
                              9.352637118466301,
                              9.381932224950582,
                              9.96692595531599,
                              9.96692595531599,
                              0.29840004251263574,
                              0.3207087147269936,
                              -0.22478909909228914,
                              -1.2242611593120782,
                              0.061748844824383295,
                              1.9931455249507801,
                              0.02449362327304914,
                              -0.23402928130657288,
                              0.3399922671860315,
                              0.2355005268966802,
                              0.7912945290539132,
                              -0.9147564287830934,
                              0.12280724330167514,

```

0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,


```

0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0], [2, 2, 2, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.536182288327376],
[12, 6, 21, 8, -3.450081350248876],
[20, 10, 35, 16, -1.8650932993173606]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 0.00010824203491210938,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.112866199792863,
'others': { 'beta_coeffieicients': [ 1.6466262975778547,
-0.9108996539792387,
0.26427335640138405],
'cost_richardson_zne': 3.6109567503981035,
'sorted_expectation_vals': [ -6.536182288327376,
-3.450081350248876,
-1.8650932993173606],
'sorted_noise': [13, 47, 81]},
'sampled data': [ [4, 2, 7, 0, -6.536182288327376],
[12, 6, 21, 8, -3.450081350248876],
[20, 10, 35, 16, -1.8650932993173606]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],

```

```
        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 0.3773346757087749,
    0.5734233884194113,
    0.5734233884194113,
    1.2094818154974225,
    1.2528672337957094,
    1.3073966674132076,
    4.193348178434801,
    4.228670385547864,
    4.228670385547864,
    4.228670385547864,
    4.228670385547864,
    4.228670385547864,
    4.274742575727675,
    5.343872214974395,
    5.354523589318104,
    5.449133113124401,
    5.449133113124401,
    5.449133113124401,
    5.449133113124401,
    6.208341300319731,
    6.208341300319731,
    6.738723848778546,
    6.7473872692498285,
    8.515456463543387,
    8.552181104315652,
    8.552181104315652,
    8.552181104315652,
    9.363042150968939,
    9.363042150968939,
    11.478464454984158,
    -0.2683676450988651,
    -0.9370321768196137,
    0.03790413469005285,
    -0.3520989510779511,
    0.5131804204688105,
    0.767748989456867,
    -1.1529933056041268,
    1.0195402939041247,
    -0.5189860890713169,
    1.700377968800765,
    0.48912124484584857,
    1.506730848787325,
    0.4485612726223177,
    -1.6248073049289613,
    0.10079320215400162,
    1.7454590745233427,
    1.474284546585253,
    0.08837986403943651,
    1.215839965531947,
    -0.42666568442503866,
    0.04298065245990749,
    -1.495484471060033,
    0.39055472334393304,
    -0.06187227108331559,
    1.0010296249342066,
    -0.33745943355443286,
    -0.19046242319241546,
    -0.4884459619425607,
    0.8930564792122229,
    0.09947441181287657,
    -1.4049039336860292,
    1.0413688199593338,
    0.027348986225077598,
    1.5457890290583323,
    0.8022774572900748,
    -1.0377116853398,
    0.5652694406258009,
    0.9990396087940023,
    -0.3409174955760714,
    -0.8590373420111477,
    0.3662952964918326,
```

0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,

```

-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#4'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0], [2, 2, 2, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.44845457619454],
[12, 6, 21, 8, -3.24468946816992],
[20, 10, 35, 16, -1.686607616211824]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 7.605552673339844e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.108333825740566,
'others': { 'beta_coefficients': [ 1.6466262975778547,
-0.9108996539792387,
0.26427335640138405],
'cost_richardson_zne': 3.6109567503981035,
'sorted_expectation_vals': [ -6.44845457619454,
-3.24468946816992,
-1.686607616211824],
'sorted_noise': [13, 47, 81]},
'sampled data': [ [4, 2, 7, 0, -6.44845457619454],
[12, 6, 21, 8, -3.24468946816992],
[20, 10, 35, 16, -1.686607616211824]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.33528943002425843,
0.8608925283648261,
2.8309935233222867,
2.8382233121900358,
2.847558790018716,
2.8477263794127468,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
5.127462455084763,
5.128383696390232,
5.132051241075024,
5.132231848473024,
5.132297186784173,
5.132297186784173,
5.132297186784173,
5.132297186784173,
5.132320898992942,
5.134903164275234,
5.734798022200061,
5.734798022200061,
5.734798022200061,
6.71748919093495,

```

6.71748919093495,
9.773430860122208,
9.791656954875775,
9.791980525916317,
9.804782964874805,
9.947289619134542,
0.6818618664375632,
0.42253658736184846,
0.19067339209915313,
-2.314563573529872,
-1.7784854373672185,
-0.10945476580518654,
0.08980428543050965,
0.6265702304745648,
-0.12016219313034968,
0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,
-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,

```

0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#2'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0], [2, 2, 2, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.333005581610797],
                          [12, 6, 21, 8, -3.1651017858597426],
                          [20, 10, 35, 16, -1.6439583689682524]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}}},

```

```

'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.00543212890625e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -7.979457807790349,
                            'others': { 'beta_coeffieicients': [ 1.6466262975778547,
                                                                -0.9108996539792387,
                                                                0.26427335640138405],
                            'cost_richardson_zne': 3.6109567503981035,
                            'sorted_expectation_vals': [ -6.333005581610797,
                                                         -3.1651017858597426,
                                                         -1.6439583689682524],
                            'sorted_noise': [13, 47, 81]},
                            'sampled data': [ [4, 2, 7, 0, -6.333005581610797],
                                                [12, 6, 21, 8, -3.1651017858597426],
                                                [20, 10, 35, 16, -1.6439583689682524]],
                            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.08591881204796584,
                              1.5026023651616802,
                              1.9179097588555833,
                              1.9179097588555831,
                              1.9179097588555831,
                              1.9179097588555831,
                              2.8800636946459592,
                              2.9523818900173797,
                              2.9523818900173797,
                              2.954896407422686,
                              3.0487820562708485,
                              3.0487820562708485,
                              3.9462134206000328,
                              3.9462134206000328,
                              5.170248653928018,
                              5.170248653928018,
                              6.5066905594049205,
                              6.5066905594049205,
                              6.5066905594049205,
                              6.914119522580601,
                              6.914119522580601,
                              6.9842551957162815,
                              7.300516913785898,
                              8.782483894556266,
                              8.782483894556266,
                              8.999633503017922,
                              8.999633503017922,
                              8.999633503017922,
                              9.414122362394847,
                              9.414122362394847,
                              0.09854676674938925,
                              -0.05173289612974198,
                              0.30094803169630163,
                              -1.1984142138894447,
                              0.08610670053567002,
                              0.2851928692216035,
                              0.26177875477440216,
                              0.11764473243628538,
                              0.3495067876706816,
                              -0.9678438601279628,
                              -0.08762050677252062,
                              0.014826088936787857,
                              -0.30347729159143555,
                              -1.1730530788097624,
                              -0.12550631929596553,
                              2.258056275075482,
                              -0.5160508087356981,
                              0.004541093619226378,
                              -0.9345420797218197,
                              0.47154522212338745,
                              -0.9129446894819914,
                              0.2768683391341781,

```

0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,


```

0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#3'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0], [2, 2, 2, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.4104927992121],
                          [12, 6, 21, 8, -3.25771336700167],
                          [20, 10, 35, 16, -1.674813432318219]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.363059997558594e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.030844611955615,
                           'others': { 'beta_coeffieicients': [ 1.6466262975778547,
                                                                -0.9108996539792387,
                                                                0.26427335640138405],
                                       'cost_richardson_zne': 3.6109567503981035,
                                       'sorted_expectation_vals': [ -6.4104927992121,
                                                                    -3.25771336700167,
                                                                    -1.674813432318219],
                                       'sorted_noise': [13, 47, 81]},
                           'sampled data': [ [4, 2, 7, 0, -6.4104927992121],
                                              [12, 6, 21, 8, -3.25771336700167],
                                              [20, 10, 35, 16, -1.674813432318219]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                     'time': {'max': 10.0, 'min': 0.0},
                                     'type': 'xy-iss'}}},
  'init_param': { 'value': [ -5.647934061474703e-18,
                              1.4787834906243755,
                              1.478918229526628,
                              1.478918229526628,
                              2.0077351764243123,

```

2.4302809101456218,
2.4302809101456218,
3.009709563144601,
3.009709563144601,
3.239471043198664,
3.2405283031129133,
3.2405283031129133,
3.2656598154222434,
3.5353392937639825,
6.198713539858257,
6.213604956994236,
6.73397854893658,
6.73397854893658,
6.743547791194661,
6.744609103740889,
6.744609103740889,
6.749543065628153,
6.749543065628153,
8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,

```
-0.628541665091616,  
-1.1357659634071433,  
-0.5734897531715744,  
-1.6096579963890296,  
-0.8342055948126037,  
-1.5447846874108708,  
-1.1270394146554403,  
0.06392732561319403,  
0.7711512775812011,  
-1.8472478314151175,  
-0.46160141905881696,  
1.8963279721150403,  
-2.313663424814036,  
0.060952702837845166,  
-1.4971949982077863,  
1.3040610818975555,  
0.10605679464403613,  
-1.138800572315487,  
-0.03947636943653289,  
0.07839455865010819,  
-0.6470277815644087,  
0.946101637008616,  
-0.32144060478582315,  
-0.6371693991779968,  
0.30580409265523,  
-0.2617129381706641,  
0.4094287336545931,  
0.49837362733711055,  
-0.9058714721495094,  
0.6137893604415533,  
0.8546844695081749,  
-0.8205470722456557,  
-0.598852289145785,  
0.46823811794119646,  
1.2205247300232143,  
0.031629319060699435,  
1.1369587010491427,  
0.12298062775535874,  
-0.016960252964031076,  
1.1127338626212067,  
0.37972118532018134,  
-0.1475616440844684,  
0.32749925367563326,  
0.030050642250761914,  
0.17169542743574548,  
-0.5825729547476518,  
0.7052714548722974,  
0.9884296874822006,  
-0.500384308215816,  
0.01722431287494931,  
0.40290712722731054,  
-0.029200452046427447,  
1.2704708243944813,  
2.3709704014259345,  
-2.0267551524865275,  
0.7347873374577798,  
-0.34767223966060473,  
0.009778943736386289,  
0.221000841842164,  
-0.010995780695540691,  
-0.481889271583797,  
0.7008500159073542,  
2.3944065221124076,  
-0.08320445199579081,  
2.02470278988152,  
0.02142760118000664,  
1.6794429086669174,  
1.6046551807968945,  
0.2831685070910348,  
1.5805135598594935]],  
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},  
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],  
                    'status': True,  
                    'type': 'depolarizing'},  
'nqubits': 7,
```

```

'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#8'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0], [2, 2, 2, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
                          [12, 6, 21, 8, -3.199071961215845],
                          [20, 10, 35, 16, -1.635137691104089]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.38690185546875e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.10578663722104,
                           'others': { 'beta_coefficients': [ 1.6466262975778547,
                                                             -0.9108996539792387,
                                                             0.26427335640138405],
                                       'cost_richardson_zne': 3.6109567503981035,
                                       'sorted_expectation_vals': [ -6.429933051303131,
                                                                    -3.199071961215845,
                                                                    -1.635137691104089],
                                       'sorted_noise': [13, 47, 81]},
                           'sampled data': [ [4, 2, 7, 0, -6.429933051303131],
                                              [12, 6, 21, 8, -3.199071961215845],
                                              [20, 10, 35, 16, -1.635137691104089]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ 3.750846163374531e-05,
                             0.22285241391868899,
                             0.26252411232084344,
                             0.26252411232084344,
                             0.26252411232084344,
                             1.459545906754556,
                             1.459545906754556,
                             1.6094835194149661,
                             2.5927782099427628,
                             2.5927782099427628,
                             2.5927782099427628,
                             2.5927782099427628,
                             2.5927782099427623,
                             3.3552499939702187,
                             3.3552499939702187,
                             5.033284106467877,
                             5.033284106467877,
                             5.056477210452712,
                             6.768913439946703,
                             6.768913439946703,
                             7.2782492670747585,
                             7.2782492670747585,
                             7.2782492670747585,
                             7.2782492670747585,
                             7.2782492670747585,
                             7.2782492670747585,
                             8.934198246659292,
                             8.937369801451712,
                             9.544406408519471,
                             9.544406408519471,
                             10.23170722265548,
                             -0.4501443853573418,
                             0.06519513561037263,
                             -0.13386490845883003,

```

-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,

```

-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0], [2, 2, 2, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.4606782650454075],
                          [12, 6, 21, 8, -3.273885460810631],
                          [20, 10, 35, 16, -1.6974907774474755]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.124641418457031e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.104743183209813,
                           'others': { 'beta_coeffieicients': [ 1.6466262975778547,
                                                                -0.9108996539792387,
                                                                0.26427335640138405],
                                       'cost_richardson_zne': 3.6109567503981035,
                                       'sorted_expectation_vals': [ -6.4606782650454075,

```

```

-3.273885460810631,
-1.6974907774474755],
'sorted_noise': [13, 47, 81]],
'sampled data': [ [4, 2, 7, 0, -6.4606782650454075],
                  [12, 6, 21, 8, -3.273885460810631],
                  [20, 10, 35, 16, -1.6974907774474755]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
  'init_param': { 'value': [ 1.5680106786222674e-17,
                             1.0238650999177359,
                             1.0238650999177359,
                             1.0238650999177359,
                             1.7041550191762513,
                             2.070931805530806,
                             2.586234954306046,
                             2.6265616878606783,
                             2.7566136785306155,
                             2.7689271757054255,
                             4.271831894498143,
                             4.271831894498143,
                             4.271831894498143,
                             5.004110362344359,
                             5.004110362344359,
                             5.529689350278954,
                             5.529689350278954,
                             5.529689350278954,
                             5.529689350278954,
                             5.5493166460873375,
                             7.855838962179162,
                             7.855838962179162,
                             8.508023697610218,
                             8.540796404638465,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             9.524293982069336,
                             9.524293982069336,
                             9.815045396471412,
                             0.717685470636315,
                             0.29125971197830336,
                             0.013619559122678915,
                             -0.46174394022911175,
                             -0.7802589318548062,
                             1.8785332138943256,
                             -0.07718385116900997,
                             -0.4181315830102034,
                             0.3859499067718677,
                             0.7218523694763775,
                             0.16350787496293626,
                             -0.06274337543165032,
                             1.048639096314506,
                             0.9454500907927006,
                             -0.18328208251477485,
                             0.5336296017995852,
                             0.7047960543548563,
                             -0.4717130073686535,
                             -0.14327049451726778,
                             0.5646728024946054,
                             0.1377201649736573,
                             1.0845662084088512,
                             0.7117671017527167,
                             1.8562916393289055,
                             -1.0958742208007046,
                             0.314756410152813,
                             1.0055290931424188,
                             -1.2653176839886926,
                             -0.6435463058390287,
                             -0.13689785210195535,
                             0.5185042269644529,

```

-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,


```

0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#1'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0], [2, 2, 2, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.515742960091852],
[12, 6, 21, 8, -3.36492111003702],
[20, 10, 35, 16, -1.7545658776574033]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.1484832763671875e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.127573245060722,
'others': { 'beta_coeffieicients': [ 1.6466262975778547,
-0.9108996539792387,
0.26427335640138405],
'cost_richardson_zne': 3.6109567503981035,
'sorted_expectation_vals': [ -6.515742960091852,
-3.36492111003702,
-1.7545658776574033],
'sorted_noise': [13, 47, 81]},
'sampled data': [ [4, 2, 7, 0, -6.515742960091852],
[12, 6, 21, 8, -3.36492111003702],
[20, 10, 35, 16, -1.7545658776574033]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.003375513600300809,
0.4884127443235871,
0.6719352108785641,
0.9073170135093351,
1.4946633690060278,
1.5076957064847494,
3.872917838142945,
3.872917838142945,
3.872917838142945,
3.872917838142945,
5.047622474508209,
5.054336886650782,
5.067626550795242,
5.0733174878702085,

```

5.094629387163461,
5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,

```

0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989}],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric3_sample#7'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 0], [2, 2, 2, 0]]},
'run': 'zne',
'state': 'dmatrix',

```

```

'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.397817460329862],
                          [12, 6, 21, 8, -3.2115716730808748],
                          [20, 10, 35, 16, -1.641958817033255]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 8.606910705566406e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -8.043320919193246,
                          'others': { 'beta_coefficients': [ 1.6466262975778547,
                                                            -0.9108996539792387,
                                                            0.26427335640138405],
                                      'cost_richardson_zne': 3.6109567503981035,
                                      'sorted_expectation_vals': [ -6.397817460329862,
                                                                  -3.2115716730808748,
                                                                  -1.641958817033255],
                                      'sorted_noise': [13, 47, 81]},
                          'sampled data': [ [4, 2, 7, 0, -6.397817460329862],
                                              [12, 6, 21, 8, -3.2115716730808748],
                                              [20, 10, 35, 16, -1.641958817033255]],
                          'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                      'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                      'r': 0},
                        'time': { 'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}}},
  'init_param': { 'value': [ -6.331146189926128e-17,
                              0.7069927066774304,
                              0.7069927066774304,
                              0.7069927066774305,
                              0.7069927066774304,
                              1.0044153364322925,
                              1.0092418692996383,
                              2.470333794214341,
                              2.478143749866421,
                              3.5776564981759003,
                              3.5776564981759003,
                              3.5776564981759003,
                              3.61745967653339,
                              3.6981780389472076,
                              3.755720232685741,
                              3.9712286023902594,
                              3.9712286023902594,
                              4.519037432417192,
                              4.519037432417193,
                              4.519037432417193,
                              5.280953883299406,
                              5.280953883299406,
                              6.620356980277175,
                              6.620356980277175,
                              6.620356980277175,
                              6.620356980277176,
                              8.214004051881032,
                              8.214004051881032,
                              9.105951544537387,
                              10.829366164164977,
                              1.4490195266558639,
                              1.145329650589643,
                              -0.9374272684957279,
                              -1.2910248872233119,
                              0.9701069453007352,
                              2.0238893871034924,
                              -0.6731600678051081,
                              -0.7305037462514778,
                              0.6743483023845589,
                              1.611766624162738,
                              -0.162446688913207,
                              -0.2464021583836281,

```

1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.5612639525109002,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,

[illegible]

```

        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                      'cn': [ 0.5,
                                              0.5,
                                              0.5,
                                              0.5,
                                              0.5,
                                              0.5],
                                      'r': 0},
                  'time': {'max': 10.0, 'min': 0.0},
                  'type': 'xy-iss'}},
    'init_param': {'value': 'random'},
    'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                      'noise_prob': [0.001, 0.001, 0.001, 0.001],
                      'status': False,
                      'type': 'depolarizing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                      'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                      'r': 1},
                  'def': 'ising'},
    'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
                'file_name_prefix': 'xy_noise_off'},
    'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
    'run': 'vqe',
    'state': 'dmatrix',
    'vqe': { 'iteration': 10,
             'optimization': { 'algorithm': 'SLSQP',
                              'constraint': True,
                              'status': True}},
    'zne': { 'data_points': None,
            'degree': 1,
            'method': 'richardson',
            'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1 '
                                     'Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                     '6 + 1 Z 5 + 1 Z 6',
                'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                            '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                            '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                            '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                            '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                            '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
    'output': { 'exact_sol': -8.56677223350563,
                'initial_cost_history': [ 4.118038614266066,
                                         2.900590937440303,
                                         2.874944433071395,
                                         3.9296977116518517,
                                         3.1841729388967344,
                                         3.695928991667305,
                                         3.180227821654077,
                                         3.0475045550195774,
                                         2.4669562192398296,
                                         3.4124916083581294],
                'noise_details': { 'gates_num': [4, 1, 1, 0],
                                   'identity_factors': [0, 0, 0, 0],
                                   'noise_level': [0, 0, 0, 0],
                                   'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                   'value': 0},
                                                     'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                     'status': False,
                                                     'type': 'depolarizing'},
                                   'odd_wires': 4},
                'optimized_minimum_cost': [ -8.433468022982382,
                                           -8.464942606128835,
                                           -8.43123094137328,
                                           -8.435401809869298,
                                           -7.9812014353834515,
                                           -8.371080050195895,
                                           -8.46198057547588,
                                           -8.19415563172473,
                                           -8.438078664602543,

```

False,

```
-8.481886373274177],
'optimized_parameters': [ [ 0.4631008843346199,
0.8090636887146863,
1.1101507421405439,
1.3400184709676968,
1.7452332568602784,
1.9308700883292553,
2.1752679291102544,
2.1784212211010603,
3.800207092295721,
3.800207092295721,
3.800207092295721,
3.800207092295721,
3.800207092295721,
3.800207092295721,
4.798448841518561,
4.812026604031267,
6.02543601067661,
6.02543601067661,
6.5265343452987,
6.5265343452987,
6.526534345298701,
6.528381158760148,
6.528381158760148,
7.113023968622042,
7.113023968622042,
7.593654062389684,
7.5949106716383605,
7.780058793305066,
9.000370275113886,
9.005799799983558,
-0.22105281285875572,
0.07196296170715184,
1.0527781347416576,
-0.4594806493129553,
1.0526039239081737,
-0.01709270509934428,
0.3199675106146978,
-0.0972220164715271,
0.9145097677998977,
0.09808103992344623,
0.5373974901276602,
-0.5786059880820694,
-0.02594763303680847,
-0.22132394348491485,
-0.7274785930689547,
-0.41090279602128,
-0.520884263329148,
-1.0260835302846771,
0.1692597388493397,
1.3250313995601488,
1.510357845739394,
-0.9233604808876308,
-0.2049157632594096,
-0.9894281051832,
0.25772120620526445,
-0.04822142329545429,
-0.3912638297585932,
-0.48486712444934904,
1.4836649713082644,
1.566996182191827,
0.08093959953023218,
-2.3562926055425963,
-0.21876618997457167,
0.10911407834571141,
-1.3776955808248483,
-0.31816531211057325,
0.7083857256813267,
0.5593482503842677,
0.7483381318097894,
0.010042025170353972,
0.05290425882351868,
0.20616876613540233,
-0.8673924682871486,
-0.08017548043713153,
```


-0.3056222778255953,
2.3981163154468383,
0.37079365521682633,
0.3870558721039623,
0.19341028278759992,
-0.3154248344945196,
-0.12073260037561208,
-1.5858414285227351,
0.59830347262823,
0.30388710178795286,
-1.4558796844932331,
-0.3225618511944297,
0.09243390755822743,
-0.0006426173071943948,
0.7561662109631148,
-1.3902324585417218,
0.1677701056843372,
2.7311884232201735,
-1.1122880815211051,
0.0467986017971284,
-0.020089420529231013,
0.005481851366742631,
1.3205596999819047,
-0.1290403664286252,
0.3870444394307043,
-0.6549197738372419,
-0.17656835190090397,
-0.0008013451895215781,
-0.8769883624496301,
1.6929520910231297,
0.7288671099890552,
-0.6169673343691255,
1.667217439782634,
0.2554635335546482,
-0.029515171705053127,
-0.8290153205719569,
0.042733616172741226,
-0.5148227135977747,
1.2301836750084596,
-0.9175908300980525,
0.5229779878549122,
-1.5901934710811256,
-0.07316585886439231,
1.1195018365185199,
-0.31111306362203306,
0.7254709796877604,
1.1995417479543053,
0.6908225815252583,
-0.2919630814941808,
0.15955998483086176,
1.2871422030548714,
-1.2683991057896014,
-0.3768598572966417,
1.4863954796333156,
-0.4300298743216642,
0.19151460300148976,
0.9665838873771405,
-0.07244306377732504,
-0.04169546816073936,
-0.389029635704794,
1.5218961812832925,
-0.1772066326032777,
-0.70949107362336,
-0.015222466074935298,
0.3587128629768407,
1.5809494716454662,
0.12428164288740734,
1.5083554487395696,
-0.044379618902227067,
-0.03505579308192811,
-0.6786538117837789,
-0.007428157845337949,
1.5281494732580478,
-0.14124856089608578,
2.726718429443767,

-0.003106328084445967],
[0.5270475437101116,
0.7815425625485376,
1.6469610445191378,
1.6712876404917778,
2.122904188877425,
2.131106816606521,
3.7367798442214175,
3.7367798442219424,
4.750276145421292,
4.853973384996016,
4.865098072058574,
4.873980651141647,
4.987599328262889,
5.00246239884106,
5.520902021852277,
5.793284849001337,
5.8087832398624775,
7.432729128414403,
7.432953787975026,
8.144896858715603,
8.172580744083415,
8.788071761144659,
8.792288306719156,
8.828260688191317,
9.011586808740137,
9.011995976067807,
9.538729699920484,
9.576884709106812,
10.486750161390887,
10.48675016212107,
-0.15531374899360717,
-0.0849039080739262,
-0.8434555658556069,
0.9447826609611266,
-0.19482082661427938,
2.4999420394249707,
-2.7164831941210483,
-1.5408169364026967,
-0.8621482593978547,
0.31547458537730877,
1.4220435826823787,
1.348860553316339,
0.40827474166943256,
1.063960383917828,
-0.21906798993221652,
-0.33263312803097533,
0.4889869181158972,
-0.25524084746075687,
-1.3622146811064544,
0.996105149433398,
1.4161210728452394,
-0.6491827941870941,
0.7399329880002848,
-0.11994422281276146,
3.179573825543474,
-0.034943024466848674,
0.6075464711124173,
-0.3019825794696199,
0.029392952712390567,
0.14840732209332994,
0.08592470568620861,
0.016725634393736574,
-1.3697962205589145,
-0.06155375586805023,
0.16508080354919497,
-0.5975014914051775,
0.09635736997937351,
1.254101402191796,
-0.6193953337438176,
0.47602269510847217,
-1.7251540483535392,
0.5378274350019089,
-1.0159384367751108,
1.0968223266795405,

-0.79878134010976,
0.10468891520322528,
-0.2910173593847854,
-0.7078981306941186,
-0.3510837282628223,
0.5371002854277007,
-0.4911274293534503,
1.7273270693440552,
1.3500555714793954,
0.9052059235505011,
0.4771526028272018,
-0.1893747959421651,
0.13989050072973697,
-0.5930955187300607,
0.36780210947560876,
-0.5043370473487999,
-0.4851893292987325,
0.1987440888222681,
-0.5806445532873666,
2.3171646404124093,
0.06032126128170586,
-0.6389920437730593,
0.9985359186149056,
0.04870155999532171,
0.16447399991313041,
0.05315090520136875,
0.15727255736970833,
-1.3722292706433088,
-0.33233649668322635,
0.7758243461945761,
-0.20476613404890107,
0.06972046040604139,
0.3143879960336948,
-0.01452323353492752,
-1.2901366074359686,
-0.31033000747276407,
-0.16264485166390844,
-1.3914094473719103,
-0.5273128620655095,
0.02465459151953875,
0.10942766294132272,
1.0292094835171124,
-1.882689768994833,
-1.553216293090225,
1.3287846024814622,
0.20963913600213377,
0.7195732642990699,
0.0969787732183028,
-0.9744344415633356,
1.5822487402774768,
0.7077471503862973,
-0.766345302327988,
0.3831437091022152,
-0.12028853613988252,
-0.8209919023748664,
-0.21164805512089607,
0.5290684330656528,
-0.1040878870845114,
1.1956945887281514,
0.25703931501419025,
0.6344733295912399,
-0.02497584874729833,
-0.20937697561532842,
-1.4409411147319051,
0.9707320454927133,
1.4039378085761043,
0.9760826987262126,
-0.17314228405542512,
0.16734556762922406,
0.07881248116464283,
-0.2924467918405051,
1.4946430081327182,
0.7516806171134963,
0.799700486923224,
-0.8440600324500036,

-0.01069194371181784],
[0.3388610072211759,
0.3930923343746358,
0.4059472295013711,
0.4597995545870322,
0.4597995545870322,
0.4667083780713653,
0.9924683364283498,
1.054118787032341,
1.3839145947816414,
1.4469616998432806,
3.044048229936016,
3.044048229936016,
4.682975233106521,
4.682975233106521,
5.287925895396742,
5.287925895396742,
5.625677830869467,
5.625677830869467,
6.519005703020383,
6.519005703020383,
6.519005703020383,
6.854665591577039,
6.854665591577039,
8.092962128458302,
8.097830997780322,
8.102159968176109,
8.111954675742565,
8.727038821401255,
8.727038821401255,
8.784424594255395,
-0.8636093730812797,
-0.04742973062196874,
-0.37989856483295564,
0.3750798322441465,
-0.509087961968837,
0.5366875438436245,
0.1232704940065221,
0.5249134835872401,
0.13594138148710633,
0.10802292582448766,
-0.7156772546121367,
0.34756877087651405,
-0.5704812075854907,
0.289395233495819,
0.2674862930849357,
0.8724143866961164,
0.5770602988816299,
0.658777432128999,
-0.47288232304292416,
-0.029663566814993374,
-1.323180236232167,
-0.2890901656039231,
-0.015141732930961212,
1.1846850771343012,
-0.4506688471965015,
0.22241980016839272,
-0.015748484493019736,
1.1522499025192263,
0.2996907655672081,
-1.7370631162182948,
-0.35306232532663057,
0.17322471105923057,
-1.1855180134814352,
-0.31537295018088835,
-1.8120366955290261,
0.6663892425075478,
0.21874911670355193,
1.2729693967005706,
-0.14230256773273223,
0.2012427441067739,
-2.3269317425793643,
-0.03299848401271634,
-0.4074853136565566,
-0.30371905431664253,

-0.12037074206123735,
0.8456676988075262,
-0.21838328071319946,
0.019494764922420988,
0.16383099728733078,
0.056056403895700845,
-3.5632063709291613,
0.6579904510057321,
0.5207861046034454,
-0.017180990359968364,
0.2532807007765998,
-0.03315495616728509,
0.7564905992030696,
-0.11091568858030387,
-0.9281890577202553,
-0.6082181315906088,
-0.4480511658411004,
2.1953283079942025,
-0.30446713497464617,
-0.03359602431311142,
0.28972943746372876,
0.17817064395522006,
0.7965754083210965,
-0.9520780563921613,
0.10150445450145726,
1.3510884636720888,
-0.37849615173878054,
0.17729737647006832,
0.1884683806301832,
-0.03234547530918763,
-1.338763456177623,
-0.6711058962200585,
-0.012976043431998471,
0.027578392175021227,
0.10161286285089129,
-0.00939135247962935,
-1.302164623174096,
1.3715576399721376,
0.737767307868418,
1.5939269674119705,
1.4291758244230544,
0.5601859764306084,
0.3488764286390935,
-1.093762224545935,
-1.1976831420922067,
1.4557381665683378,
0.8924472305311927,
0.5300221300338881,
1.0785280788144638,
-0.6029509417428172,
-0.6790632925762424,
0.4138865114731201,
0.2917474006869537,
1.8050154062536312,
0.05177383423831921,
0.25467777497912664,
1.4671356779039024,
0.4422247342165525,
-1.5431607378377095,
1.8101814873938125,
1.6870467346899987,
0.7517920825884995,
1.3622912362338087,
0.13653050459472865,
1.5210183141257738,
-0.004247168330480351,
0.26137218980413907,
0.5451511562235306,
-0.6705596749039198,
1.2136085126593865,
0.5231529087420831,
-0.03279466120532818,
0.2548954759886831,
1.5797820110558691,
1.075840789720537,

1.5541818084195007],
[3.821020346740443e-17,
1.386496864345847,
1.386496864345847,
1.7951472089989222,
1.8559400139285378,
2.7980544315911504,
2.7980544315911504,
2.82954781812891,
2.82954781812891,
2.82954781812891,
2.82954781812891,
2.956183634338348,
3.840561875918348,
3.840561875918348,
4.79501776182173,
4.79501776182173,
5.670460018404704,
5.670460018404704,
5.670460018404704,
6.865392250052202,
6.865392250052202,
7.506826147384778,
7.506826147384778,
8.737685173635683,
8.737685173635683,
8.737685173635683,
8.737685173635683,
9.601023092540274,
9.601023092540274,
9.853738132880634,
9.853738132880634,
0.5611658219809542,
1.427331325859137,
0.8881601644240952,
-0.8937497791603879,
0.5712044928850715,
1.3195951619089188,
0.5467792160605555,
-0.6816630744322637,
0.06707477836950569,
0.6798692668930644,
0.19576168860111362,
0.5094724850631845,
1.1424414477256812,
0.30471145260454696,
-0.08158911485577636,
0.9585661478313002,
1.0836238583568754,
-0.3718197048926503,
0.7841948506971846,
0.3023515457437066,
0.7662240328883452,
0.9970048707689876,
-0.8946064024093777,
0.27649390200762114,
-0.15342626519190924,
-0.4015720487025046,
1.460926694270634,
-0.055421168101734586,
0.6077050880702092,
-0.7932081335349191,
1.3188332541813446,
0.01929555634963922,
-1.1301845164589392,
-0.1718174564221745,
2.26154163970611,
-0.9882028269623031,
0.8038643847286138,
0.6849304102208338,
0.874798983022982,
-0.016486064274954207,
0.0264221098096002,
-0.3625000384278827,
1.4376331477886695,
0.8499992740594865,

0.10042171561200945,
0.14542648933357244,
1.3513090878714353,
-0.22248922869259902,
-0.4483175327546578,
1.2345758678942387,
-0.18983522552694923,
0.019591210301167533,
-1.3607248526627624,
0.018375533581887468,
-0.3730738776375715,
1.0123537416913542,
0.7412601688930508,
-1.6525919628732422,
-0.8341682190885938,
0.04519729634722039,
-0.06718028327556408,
-1.5649719285752928,
-0.37531851249345394,
-0.11822841604899964,
0.008154106549025474,
-1.4601449692077721,
-0.0030993446167428395,
1.462573870839384,
-0.11887757571245174,
-0.09361960188417112,
-1.2543408854038138,
-1.525280802664792,
-1.241413372948363,
0.008758808605552393,
-1.6676332942965022,
0.13826170532334445,
0.7537363962746654,
0.0833397476047736,
-0.12216949704941273,
0.02084807234445201,
-0.3421949354726438,
0.0017882611822728064,
1.5186851123669147,
-0.4603163345929323,
-0.09956972276089109,
0.7247597830417082,
0.5304935691839561,
0.003193152207841171,
0.5581346990750625,
0.010542051444639388,
0.7553167297004638,
0.5956891015800863,
-0.05566254332595716,
0.5362461335652635,
0.009895282042162012,
-0.3029901133286735,
0.09399206534914867,
0.8818812838713691,
-0.06795832284115338,
-1.580933930101608,
0.3014975711017802,
1.2092022074751856,
-0.019692531397995948,
0.9250981578351692,
0.4673383297638096,
0.03344339272872125,
0.13058472572385768,
0.036262684184263205,
0.458095375934523,
0.6492846129896516,
1.3039221898344189,
0.024764812872192767,
0.7236926243101844,
0.01444840086266692,
-1.0388939706148563,
-0.08359693344439546,
-0.4698004238001188,
-0.48055768545836347,
0.4662000815300196,

-0.041204969991243484],
[0.01798320141016013,
0.3393419127512466,
0.7047697703765823,
0.7064789163091142,
0.7191656785855808,
0.7198672068604162,
0.719867206860416,
1.0840512961729545,
1.0900003904895101,
1.1926153174924548,
1.867054342490295,
1.867054342490295,
2.368094916985588,
2.3683751276938714,
3.6383003552329964,
3.642527762446636,
4.505845683894636,
4.602446412902637,
4.972427659100928,
4.972427659100928,
5.1148164150678035,
5.8066136392845,
5.820681023327439,
6.727088267818048,
8.352653016300794,
8.773334122756978,
9.065902540064004,
9.065902540064004,
9.74880092287043,
11.30145464466351,
-0.13904770480884485,
0.23493722133539818,
0.4497803172772882,
-0.640584995353188,
0.9779304528121034,
0.0011745349607135795,
-0.7175493520168287,
0.373620380836484,
0.08111255661159392,
0.6653458367314794,
0.0037967443269442437,
0.6286122183467432,
-0.03825426016359485,
0.6896824961465278,
-0.23916946602734285,
0.4624850989736848,
0.1419493525208794,
0.7145176662978019,
-0.316909964913161,
0.6605458698076423,
-0.39309909050950365,
0.7623048158369822,
0.7469736169282462,
1.2400892032597353,
0.12266455668869777,
0.19578648318487166,
-0.784043232212839,
-0.10334183265831844,
0.32774480702026265,
-0.011491906972883237,
0.322988869457402,
-0.2027870428092001,
0.1281571522083961,
-0.1947961113336468,
1.7114513053475526,
0.005425684515457419,
0.8136825826883561,
-0.25903007981373005,
-0.5015428579892063,
-0.2600978115589891,
-0.15405473476720474,
-0.4207498663011112,
1.04622682040645,
-0.8642339256273757,

0.6248691295250803,
-0.3539240130838131,
0.011689879721248103,
-0.3068283940685192,
-0.6048660879777619,
-0.11704784670887185,
-1.2745062338511486,
-1.7890038533223112,
-0.7818671277256136,
-0.8877455887286905,
0.3164818198590212,
-0.27477307924208777,
1.8083251930535746,
-0.5031479272825263,
-0.9648993597903888,
-0.41682962131770707,
0.27040705073896654,
1.0892900815418987,
0.17779960873468054,
0.020337033950376556,
1.0551529464315492,
-0.10705558118147128,
-0.8851977322505518,
0.2232945485845247,
-0.5850403569085746,
-0.7690595025978197,
-0.4391673030332381,
-0.9309566421919284,
0.2973030450025941,
0.6780545449896137,
0.14045373030826291,
-0.5109313900859794,
-0.44070010469428217,
1.6940975579604587,
0.17062304524159472,
-0.402681393200546,
-0.7091187255059627,
1.3132357546586793,
0.2729391261599054,
1.0115940175088693,
1.0046673925246778,
0.4982257464819494,
0.04454679168285006,
0.4863023994043421,
-1.1433261311581624,
-1.820308437409531,
0.6035405700431521,
0.04003907881711185,
2.99381748600971,
-1.0437523456620463,
-0.22517480393070408,
0.10746937069246698,
-0.8788513722547997,
1.286531186523234,
-1.3157160458746164,
-0.6979748907629274,
-0.16573593981190854,
0.2529194783090185,
-0.2562517800846549,
-0.7966921479109774,
-0.4446940958312971,
-1.052310031662883,
-0.1506245930250076,
0.1615975776996014,
2.9220364155549214,
-1.7197843955783065,
-0.6592103621593355,
-0.2007498223803255,
-1.6188873615119244,
-0.15193115939751467,
-0.20866967970071015,
1.2910791163230286,
-0.4917152660712325,
-0.42348313202027765,
0.3518303936962707,

-0.10034486495238693],
[0.009039772647041902,
1.1400526086001903,
1.466905028683042,
1.6524670085313564,
1.987797816914966,
1.9900950346150876,
2.1413202332332344,
2.390935514217652,
2.402429399816812,
2.81913686416337,
2.81913686416337,
3.5656810200683298,
3.5656810200683298,
3.5656810200683298,
3.76603447051701,
3.76603447051701,
5.499236323276389,
5.4992363232763894,
5.4992363232763894,
5.532789537471719,
5.532789537471719,
7.371733777832876,
7.371733777832876,
7.680783486275433,
7.706623781446164,
7.706623781446164,
8.457742149729329,
8.457742149729329,
8.95222736351662,
8.95222736351662,
0.18928393791812126,
1.0354889043700823,
0.3703113207409527,
0.30988589523494625,
-0.34669340410550803,
0.09615284519636855,
0.1011898884094612,
0.6160115432573767,
0.6148540393435122,
0.07934943483664206,
0.8557624329627399,
-0.7672633290674925,
-0.14188111026081618,
0.1995320641320393,
0.4594888802421805,
-0.03593187081620776,
1.2926591415659605,
0.31433950962447343,
0.8053272546604554,
0.6634815559763872,
0.311660786273531,
0.0294653444200854,
0.3044394736040779,
0.7599259037372076,
0.22962735656034647,
0.22562971232722423,
0.29511485598499815,
0.02260075636691185,
-0.9497096253675098,
-0.5331371571917215,
0.2192512323868535,
1.104832430155296,
0.8636617903941131,
-0.6756684630428217,
0.4169844309263729,
1.1297527651723116,
-1.160118439748996,
-0.7971836651224,
1.9169656561991728,
-0.007260796290839314,
0.3614132074474742,
0.6243022778229537,
1.8335025630018775,
0.35139722497115833,

-0.7247870129359816,
-0.6744811297738377,
0.49997538785346773,
1.4139521212248116,
-3.0799818401546553,
-0.14768564076593044,
0.06251500852230953,
-0.8547134606957393,
-0.30107856776262504,
-0.7654972812734481,
0.28603020918627825,
0.2129303248959857,
-1.2077141264570097,
-0.8488549318812054,
0.6492574804166855,
0.7751290676155554,
0.40909519807061084,
-1.2750630737105486,
0.4187798718836698,
0.33498424523924863,
-0.19017028209895162,
-0.03232931911977397,
1.0621483440417678,
0.452017889610544,
1.0192767515800205,
0.7290563184502946,
-0.5286462788982386,
-0.05512372609304388,
1.3189151919853837,
0.5865235617306253,
-0.579843833868396,
0.6980158293666034,
0.027004860579276476,
1.2876675233834314,
-0.1718561727774212,
-0.3827017785428823,
-0.7286404996479297,
1.1682112525473984,
0.22765379123755217,
1.4870969768845252,
1.1642898801914527,
-0.03976972562460009,
-0.8163296820868416,
-0.15318057329999143,
0.0482918949035788,
-0.06014345828968787,
1.1478583262090496,
-0.005141116181289305,
-0.5820859555621626,
1.086519910329415,
-1.08558016317555,
1.531511042803357,
0.042734048269580674,
0.7452599788242152,
-1.3094923903018465,
-0.9012205175381317,
-0.0755166851159581,
1.5493661520767013,
-0.2278023712186688,
0.8627212805492942,
1.8523276191392382,
-0.10961709979291216,
0.6746606852631131,
0.22566845572735392,
0.5920154791154506,
1.2709342533151342,
0.26903045177842116,
0.11724485476711638,
0.621942277674204,
0.15491924869980966,
1.472217084758381,
0.8975149794611956,
0.3601026339731485,
1.3234513983254743,
0.0182465793142515,

-0.11914786829986494],
[-5.765348397008027e-18,
0.3428984748224433,
0.3655450682584724,
0.704784031551808,
1.5409081472811441,
1.549300366088437,
2.3787666619954635,
2.3787666619954635,
2.8808650819472406,
2.884987740396357,
3.1308808834009203,
3.1363836212549696,
4.869321728501938,
4.869321728501938,
6.1150174264690085,
6.2582278136815175,
6.271636236200362,
6.551203581243674,
6.59741701640448,
8.230564833747417,
8.294003346338926,
8.31431753208387,
8.369586036193128,
8.415319617883744,
8.70447046242739,
8.709120596281233,
8.918659758853744,
8.99410216914391,
9.002256766523654,
10.218690251738174,
-0.6798627919679444,
-1.1436724543472427,
-0.1278269606092795,
-0.9286552360068459,
0.6524046490818174,
0.06134572949387961,
-0.49761179900021657,
-1.3918471119443894,
0.5858287319319103,
0.18669127045720346,
-0.4586789737568629,
0.2400608953322058,
0.022222124883641912,
0.9282004764805514,
0.6952358394741303,
0.805684979006905,
0.6146729452742722,
0.6777524139082742,
-0.8117842884117104,
0.3426849285038029,
0.7100299235979358,
0.7658186509139059,
-0.05418633990950139,
0.1761917490887145,
0.5744825473612316,
0.31705937472296297,
-0.9188523464356001,
-0.7337446596965294,
0.2061368312243915,
-0.6645694031850036,
-0.009871104765786294,
-0.14388187083546433,
-1.2485707106311008,
2.849540809212292,
-0.5119215846255833,
-0.19381894062193808,
-0.17310717674563775,
0.8465397629359774,
-0.46757240698237135,
-0.05607605828560263,
-0.40986137011816176,
0.07649505002491204,
-2.1522907800348112,
-0.6653346582594515,

0.4084801543759574,
0.4028897820911683,
0.16908117843954765,
-0.07217026591146015,
-0.72804755296126,
-0.13548738477012795,
-2.590679566047418,
-0.40743076961744734,
1.1030856800887578,
0.08388380440455438,
0.25224805905642333,
0.10644300798076774,
1.0474125729404944,
-1.1459016177859003,
-1.6916287310468103,
-0.9886067106540645,
0.24182274960650893,
-0.25703819491337826,
-1.1942818044595898,
-0.7763678643520989,
-0.916718557142268,
0.20086626876005667,
-0.3711023890817823,
-0.006962349053602254,
2.687031223814704,
0.3970791704566904,
-0.6890591654410289,
1.0504536719091986,
-1.2458238157294268,
-1.197330311624414,
-0.12155955689525438,
-0.2969718406476498,
0.15406921803787998,
0.9984735505197111,
-0.2734456365103056,
0.0100494123830882,
0.16172873483083283,
-0.6780409734277772,
0.4539997995464368,
0.7558576390626953,
0.9556605206819043,
-0.10572075101403118,
1.5564628686733721,
0.05397552837650759,
0.7415987024402694,
-0.1648500682327342,
1.052699139325594,
0.6776572023171561,
0.15219630541397192,
0.8843933196559655,
1.3948099389813178,
-0.16710315713046545,
1.2199200746863228,
-0.3825678575328573,
-0.26766985045935626,
2.5777182352113908,
0.23156136798610805,
-0.3377548185037014,
1.0689111835439342,
-0.17216022970038714,
-0.29381807805976945,
-0.5689533077540762,
0.30596827309967156,
0.5260993417881752,
-0.6768480303494689,
-0.1092430795457648,
1.5442082488768039,
-0.10826504240549732,
-0.9555598065181787,
0.5334835330029429,
0.5569227134357198,
1.2665483894271157,
-1.9677332823608626,
1.5559931182978979,
0.7709453943956464,

1.5746506000831744],
[1.7625089228179507e-05,
1.298540182320117,
1.3141870420194457,
1.3305471045820527,
2.0686735179487985,
2.23810597518972,
2.267373737425337,
2.295441377067579,
2.3325388050051976,
2.352484360014207,
2.370891483796666,
3.3117124844820194,
3.363605896627787,
3.858981460194034,
3.8589814603490344,
3.859537867870235,
3.8595381713444072,
4.26458571773699,
4.264585717738593,
6.599549149586233,
6.655963438031886,
6.939612414177192,
6.939612414268938,
7.004302986969205,
7.010544991093791,
7.97314828104894,
7.999280802101656,
8.022628296732579,
8.027896471579398,
10.077254342713585,
-1.0733166032270363,
0.4264387306682,
-0.29490479134596703,
-0.45181113351566254,
-0.9496712390342118,
-0.24412788786515252,
-0.17005038539710493,
2.131747001055919,
-0.8801477965160032,
0.1985900676052623,
-0.17829180427317373,
0.1513896817808816,
-0.2996418627583153,
0.09152392337611555,
1.0155053498240294,
0.6713280773462803,
-0.5915672184524843,
-0.21780814909773427,
-0.23676595566314207,
-0.1539223229682568,
0.9733737893230769,
-0.8325690111887051,
-0.6684349693104433,
0.944923850675747,
-0.8270803600282979,
0.4869668407293288,
0.07342964610078141,
-0.8928903181234079,
-0.04262248204617692,
-0.7602498290394286,
-0.48858825463899247,
1.7340528561515622,
0.7340502315717261,
0.5163401187759543,
-0.5015858040228415,
0.2890335055344555,
-0.46033506221820203,
-0.8185907107929625,
0.4557079379705311,
1.6601520871164626,
0.07125306723268252,
-1.4035119795986863,
-0.6741575848895364,
1.84360490583597,

1.6836338044245691,
-0.33329370458317326,
0.19854317161902654,
2.03917897859608,
-0.3342251938073038,
-1.6015864477635742,
0.35422488479060155,
0.1487121545506934,
-0.2821114888507261,
0.21869934797564045,
0.7523961573071125,
1.3021785537923125,
0.26184859002187644,
1.2441238228144538,
0.039670539596101485,
-0.15687777352236798,
1.1386642271664005,
-0.13559116971858365,
0.39167499360215746,
2.4453261619465274,
-1.0890320328395309,
-0.4289422005081893,
1.425249318116838,
-0.0016733698174337221,
0.023121567661497415,
-0.28325087531692295,
0.940069260590291,
1.163437407349777,
-0.4256527399896147,
-1.4972731615803316,
-0.12624975701746496,
-0.10144651691714318,
1.3307378561466523,
0.041778278155808674,
-1.2746374184036902,
-1.2216848741140482,
1.4048769664929757,
0.8957112974734468,
1.0058068981677006,
-3.0965653057688316,
-0.3344448337959029,
-1.0948102803209163,
-0.038478236121346335,
0.12618687504486675,
-0.0850775611793361,
1.33550885346982,
-0.04334772196309596,
0.2406908654957122,
0.7327249608913058,
0.9966290850109706,
2.2852565843192423,
-0.030570948934034583,
-0.5566407461361556,
0.813246653395058,
0.5946355273642772,
0.07519254658407079,
0.27120989268694123,
-0.15415722381241237,
-1.1239288818585278,
1.893751281002709,
0.7530205722182245,
1.4454588921082565,
0.15037300343080462,
0.1793867334099716,
0.3587259643057371,
-0.006799349268608,
1.0907160355013774,
-0.30186717991666434,
0.8507589785411056,
0.009155709268365956,
-0.8165113788656561,
0.011816515180360782,
1.116430418883973,
1.5343789763879045,
-0.1682835166498154,

-1.5032827361947747],
[0.10956969985098665,
0.4360115975982133,
0.5932904704915389,
0.6050451872391112,
1.3688142364348175,
1.3737597367805747,
1.5226773273508583,
1.5589271816733294,
3.150320547140947,
3.1613053536766147,
4.7635234433865605,
4.823148382609575,
4.823188243925802,
7.171157860626202,
7.171157878485542,
7.171158529865571,
7.175073431930883,
7.183070822592498,
8.102400589897519,
8.10392263292654,
8.103940780725882,
9.021308924987686,
9.02133947231205,
9.082831297490506,
9.094868565409332,
9.236804949782064,
9.237029573386183,
9.26430291432718,
10.00744744746223,
10.032040374677464,
-0.5042382729215367,
1.0180015436638659,
0.10423420212375085,
-0.3355687578372037,
-0.3902319222280216,
-0.09867602213718178,
0.3860296418386605,
0.8522828003851446,
-0.7730596965193267,
0.08803782521712378,
1.175277695996097,
-0.3769225558693855,
-0.4553519820435133,
0.4758962487375975,
-1.0517577693418567,
0.3712029741888282,
0.27334544561774066,
0.03397030216872572,
0.3891396532807883,
0.12152393226873999,
1.5192516885791116,
-0.5124931636551364,
-0.5667003913709334,
-0.08425393582345661,
-0.8152956002657638,
-0.008087141654348068,
-0.20833380477584987,
-0.19186344764752156,
0.309463849209865,
-0.16555810903618162,
-0.28534485935787846,
0.025710068868521854,
-0.9172960495760906,
0.027100436879331523,
-0.9806057417125867,
0.46877133554227174,
0.22597640762440013,
-1.8700902599315918,
-0.6222133609404971,
0.023922176634194332,
-0.6777524378245746,
-0.3883818313368169,
-0.19921738892417568,
-1.3418496115619054,

-0.05365669225714106,
0.38835985529462186,
0.46741217514915123,
-0.1388983196762152,
1.0002905156718758,
1.6633296566254185,
-1.0097136763815113,
1.203957772036831,
0.34033273848251194,
1.1456357751380877,
-0.4637199651146574,
-0.13624703361320328,
0.6785440200843139,
0.21691345067474185,
-0.4166036808107477,
1.6003722740710606,
-0.294881370724797,
0.04198878790454511,
-0.689532047804753,
-1.3911901894570817,
-1.1667039833298627,
0.824824800006074,
0.8047687705752488,
0.5727196814509768,
0.2385364014798047,
-0.512164832149354,
0.17925141295137917,
-0.19704824673887428,
-1.6372757558452697,
0.5441675736491897,
-0.13936070677778437,
1.3935144064781402,
0.10763856884501717,
0.6679788244416961,
-0.33118654939932857,
-0.35796695002153883,
0.2632208027041665,
1.5831503864671255,
-0.25140146740224,
1.135774876905931,
0.5897962040478675,
-0.7664904031063712,
1.092649076983624,
-0.9862462517151122,
0.010216869101717933,
0.5705833807099128,
0.09839127014019662,
-1.034770130526523,
0.20582133073730807,
-0.44428721001967564,
0.5272444819968604,
-0.2627660867388947,
-0.9857421779520247,
-1.3646538516413942,
0.14779052759756187,
1.1207932022043874,
0.30427416833245535,
-1.055109005337564,
0.30548258474769546,
0.8153937738463674,
0.5082537250510629,
-0.2973639305797175,
0.0034742209991892575,
1.5168338311712208,
0.0971904670135119,
0.5305672799342332,
-0.08170159551874663,
0.4909203029282298,
-1.1771354207712392,
0.038725619801271274,
-0.6655680746562402,
1.312626970471223,
-1.0115568894576918,
0.16290676730453105,
0.3074286838975462,

0.013117345918778788],
[0.4866654913190249,
0.4866654913190249,
1.4314116389487666,
1.4314116389487666,
1.548608655381947,
1.5552664901547235,
1.662920497545568,
1.676155881591226,
1.6952532774325564,
1.6952532774325564,
2.46741851740979,
2.50102877426778,
3.636145431120801,
3.6361454311208004,
3.6361454311208004,
3.6361454311208004,
4.0161273218482085,
4.0161273218482085,
4.5411636896776235,
4.5411636896776235,
4.922545565757144,
5.624338470516907,
5.642847901355697,
5.904037934138086,
5.915382265322609,
6.860930229950661,
6.860930229950661,
9.422517992567101,
9.467623480195014,
9.467623480195014,
0.40091998571272736,
-0.3581846438287994,
-0.32033246941904975,
0.24409866553117068,
0.176560178011825,
0.4452285262890565,
-0.37150326481454765,
-1.0185408309777293,
1.6197684873006286,
2.0095194673452026,
-0.20342804182482532,
-0.4471079471088122,
0.37127155242298737,
0.4199607045713712,
0.11145558426636207,
-0.289968410859463,
0.9752004498173846,
-0.11972987215087512,
-0.8215210043660294,
-0.4144105549344152,
0.04966095047933562,
0.6091224412599766,
-0.2481186610805342,
-0.9036735819055022,
0.22922493224752044,
-0.04567884568114995,
-0.40529506738584414,
-0.44609185937563545,
1.1000542207972743,
0.8995834380092431,
-0.4391184290327733,
-1.623814742575614,
0.11121717167805684,
-0.15344289124808663,
0.7072346226342329,
0.013911394934765222,
1.1008239636292285,
1.1342299046285917,
-0.1605718532112884,
-0.8521154510350495,
-0.9726520318623231,
-0.11078467990736285,
1.2893067940177063,
-0.8656847782514395,

-0.4554157535410911,
0.46752563535079134,
1.0115326820592876,
-0.017465678177200683,
0.7555243091795097,
0.2225241128439446,
-0.6129237770264853,
0.36249038141661666,
-0.1470816778834455,
-0.23781939745964606,
0.7410640700988803,
-0.10219672891678527,
-0.9043863030697507,
-0.054446492384892985,
-0.23618927113386157,
-0.3971482428910102,
0.42980979959963467,
-0.4511512430809565,
-0.3322123782853245,
0.15031459204805087,
-1.4287283715133035,
-0.026859055017604255,
0.20826999854501216,
-0.016397767746330064,
-0.026711091547853946,
0.15899307013846248,
0.18349959987180348,
-0.019181103141429404,
-1.3777401647916119,
0.05578672583947999,
-0.17191397249005735,
0.08492618339903452,
0.2781511329827625,
1.6853098582374997,
-0.7964285169174421,
0.13628687772581302,
-0.26524580358677924,
1.3615146942088054,
-0.18163849035743365,
-1.5476669195398023,
0.2740670870741208,
0.19740142577560438,
0.18567840551061798,
0.43288784764736754,
-0.8069341399605817,
1.5139336080928714,
-0.7594440385529729,
-0.20443587392639553,
0.0916888700165231,
-0.03922249489745327,
0.4135801157921025,
-0.06731598775715468,
0.5038720973518589,
-0.7705715736697758,
-1.4775324565516457,
0.061950004909204714,
-0.9555167624183505,
3.1277455162888304,
-1.137887484425651,
-0.09852947784569874,
0.042597844143281625,
0.06134264537966588,
0.3384848933516047,
0.012002169875906457,
1.5732417764972468,
-0.3550969359952667,
0.37821230776206083,
1.4914211446267915,
-1.2027836619303884,
0.22421848814312553,
-0.2072711571643093,
0.05122655754886873,
-1.5578500765322636,
1.5918641150141506,
1.2635397829742223,

```

-1.2452354025631063]],
'run_time_sec': 43720.81102514267}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': { 'value': [ 1.5680106786222674e-17,
1.0238650999177359,
1.0238650999177359,
1.0238650999177359,
1.7041550191762513,
2.070931805530806,
2.586234954306046,
2.6265616878606783,
2.7566136785306155,
2.7689271757054255,
4.271831894498143,
4.271831894498143,
4.271831894498143,
5.004110362344359,
5.004110362344359,
5.529689350278954,
5.529689350278954,
5.529689350278954,
5.5493166460873375,
7.855838962179162,
7.855838962179162,
8.508023697610218,
8.540796404638465,
8.940056981382925,
8.940056981382925,
8.940056981382925,
8.940056981382925,
9.524293982069336,
9.524293982069336,
9.815045396471412,
0.717685470636315,
0.29125971197830336,
0.013619559122678915,
-0.46174394022911175,
-0.7802589318548062,
1.8785332138943256,
-0.07718385116900997,
-0.4181315830102034,
0.3859499067718677,
0.7218523694763775,
0.16350787496293626,
-0.06274337543165032,
1.048639096314506,
0.9454500907927006,
-0.18328208251477485,
0.5336296017995852,
0.7047960543548563,
-0.4717130073686535,
-0.14327049451726778,
0.5646728024946054,
0.1377201649736573,
1.0845662084088512,
0.7117671017527167,
1.8562916393289055,
-1.0958742208007046,
0.314756410152813,
1.0055290931424188,
-1.2653176839886926,
-0.6435463058390287,
-0.13689785210195535,
0.5185042269644529,

```

-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,

```

0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.515742960091851],
                             [12, 6, 21, 8, -3.36492111003702],
                             [20, 10, 35, 16, -1.7545658776574033],
                             [28, 14, 49, 24, -0.9139478342631158]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 7, 0],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
'identity_factors': [1, 1, 1, 0],
'noise_level': [20, 10, 35, 16],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
'identity_factors': [1, 1, 1, 0],
'noise_level': [28, 14, 49, 24],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}]]}

```

```

                                0.001,
                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
                                { 'gates_num': [20, 5, 5, 16],
                                'identity_factors': [2, 2, 2, 0],
                                'noise_level': [20, 10, 35, 16],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                'value':
0},
                                'noise_prob': [ 0.001,
                                0.001,
                                0.001,
                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
                                { 'gates_num': [28, 7, 7, 24],
                                'identity_factors': [3, 3, 3, 0],
                                'noise_level': [28, 14, 49, 24],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                'value':
0},
                                'noise_prob': [ 0.001,
                                0.001,
                                0.001,
                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4}},
                                'run_time_sec': 28.50358009338379}},
                                { 'config': { 'ansatz': { 'gateset': 1,
                                'layer': 30,
                                'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                'cn': [ 0.5,
                                0.5,
                                0.5,
                                0.5,
                                0.5,
                                0.5],
                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}},
                                'init_param': { 'value': [ -6.331146189926128e-17,
                                0.7069927066774304,
                                0.7069927066774304,
                                0.7069927066774305,
                                0.7069927066774304,
                                1.0044153364322925,
                                1.0092418692996383,
                                2.470333794214341,
                                2.478143749866421,
                                3.5776564981759003,
                                3.5776564981759003,
                                3.5776564981759003,
                                3.61745967653339,
                                3.6981780389472076,
                                3.755720232685741,
                                3.9712286023902594,
                                3.9712286023902594,
                                4.519037432417192,
                                4.519037432417193,
                                4.519037432417193,
                                5.280953883299406,
                                5.280953883299406,
                                6.620356980277175,
                                6.620356980277175,
                                6.620356980277175,
                                6.620356980277176,
                                8.214004051881032,
                                8.214004051881032,
                                9.105951544537387,

```

10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,


```

1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 X 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '

```



```
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 0.3773346757087749,
    0.5734233884194113,
    0.5734233884194113,
    1.2094818154974225,
    1.2528672337957094,
    1.3073966674132076,
    4.193348178434801,
    4.228670385547864,
    4.228670385547864,
    4.228670385547864,
    4.228670385547864,
    4.228670385547864,
    4.274742575727675,
    5.343872214974395,
    5.354523589318104,
    5.449133113124401,
    5.449133113124401,
    5.449133113124401,
    5.449133113124401,
    6.208341300319731,
    6.208341300319731,
    6.738723848778546,
    6.7473872692498285,
    8.515456463543387,
    8.552181104315652,
    8.552181104315652,
    8.552181104315652,
    9.363042150968939,
    9.363042150968939,
    11.478464454984158,
    -0.2683676450988651,
    -0.9370321768196137,
    0.03790413469005285,
    -0.3520989510779511,
    0.5131804204688105,
    0.767748989456867,
    -1.1529933056041268,
    1.0195402939041247,
    -0.5189860890713169,
    1.700377968800765,
    0.48912124484584857,
    1.506730848787325,
    0.4485612726223177,
    -1.6248073049289613,
    0.10079320215400162,
    1.7454590745233427,
    1.474284546585253,
    0.08837986403943651,
    1.215839965531947,
    -0.42666568442503866,
    0.04298065245990749,
    -1.495484471060033,
    0.39055472334393304,
    -0.06187227108331559,
    1.0010296249342066,
    -0.33745943355443286,
    -0.19046242319241546,
    -0.4884459619425607,
    0.8930564792122229,
    0.09947441181287657,
    -1.4049039336860292,
    1.0413688199593338,
    0.027348986225077598,
    1.5457890290583323,
    0.8022774572900748,
    -1.0377116853398,
    0.5652694406258009,
    0.9990396087940023,
    -0.3409174955760714,
    -0.8590373420111477,
    0.3662952964918326,
    0.6492392290096071,
```

-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,

0},

False,

0},

```
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
  'identity_factors': [3, 3, 3, 0],
  'noise_level': [28, 14, 49, 24],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                                'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                                'status': True,
                                                                'type': 'depolarizing'},
                                                                'odd_wires': 4}},
  'run_time_sec': 31.118547201156616}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.0019047787007362745,
                              0.7201496918445128,
                              0.7201496918445128,
                              0.7201496918445127,
                              0.7226708680142584,
                              0.8068041864232433,
                              1.3847174673528184,
                              1.6031895944534578,
                              2.458865333289775,
                              2.458865333289775,
                              2.721120236828938,
                              2.721120236828938,
                              3.554687304897056,
                              3.554687304897056,
                              4.537252190232355,
                              4.537252190232355,
                              5.239770593633025,
                              5.242363659304797,
                              5.566731623636481,
                              5.566731623636481,
                              5.566731623636481,
                              5.566731623636481,
                              5.566731623636481,
                              7.6484324534494865,
                              7.6484324534494865,
                              8.96757120343865,
                              9.352637118466301,
                              9.381932224950582,
                              9.966925955531599,
                              9.966925955531599,
                              0.29840004251263574,
                              0.3207087147269936,
                              -0.22478909909228914,
                              -1.2242611593120782,
                              0.061748844824383295,
                              1.9931455249507801,
                              0.02449362327304914,
                              -0.23402928130657288,
                              0.3399922671860315,
                              0.2355005268966802,
```

0.7912945290539132,
-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,

```

1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.536182288327376],
                             [12, 6, 21, 8, -3.450081350248876],
                             [20, 10, 35, 16, -1.8650932993173606],
                             [28, 14, 49, 24, -1.0247351729167147]]},
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],

```



```

False,
0},
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
'identity_factors': [2, 2, 2, 0],
'noise_level': [20, 10, 35, 16],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
'identity_factors': [3, 3, 3, 0],
'noise_level': [28, 14, 49, 24],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 32.254499673843384}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.00030922277578629,
0.881375055029912,
0.881375055029912,
0.8813750550299119,
0.9803845151714838,
0.9803845151714838,
1.201629973822729,
1.2441077564052723,

```

1.8660677974280824,
1.8660677974280824,
2.651002314516478,
2.651002314516478,
3.233637461790533,
4.36384995486056,
4.36384995486056,
4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,
8.903406631369169,
8.903406631369169,
8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.13579407977759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,

```

-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},

```

```

    'def': 'ising'},
    'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
                'file_name_prefix':
'AUTOMATE_xy_noisy_time_evo_ric4_sample#10'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                         [1, 1, 1, 0],
                                         [2, 2, 2, 0],
                                         [3, 3, 3, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
              'optimization': { 'algorithm': 'SLSQP',
                                'constraint': True,
                                'status': False}},
    'zne': { 'data_points': None,
              'degree': 0,
              'method': 'richardson',
              'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                     '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                     '5 X 6 + 1 Z 5 + 1 Z 6',
                 'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                             '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                             '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                             '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                             '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                             '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
    'output': { 'data_points': [ [4, 2, 7, 0, -6.511175848027214],
                                 [12, 6, 21, 8, -3.5131084000754242],
                                 [20, 10, 35, 16, -1.9256894730075684],
                                 [28, 14, 49, 24, -1.064810006156379]],
                'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                     'identity_factors': [0, 0, 0, 0],
                                     'noise_level': [4, 2, 7, 0],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                     'value':
0},
                                     'noise_prob': [ 0.001,
                                                     0.001,
                                                     0.001,
                                                     0.001],
                                     'status': True,
                                     'type': 'depolarizing'},
                                     'odd_wires': 4},
                                   { 'gates_num': [12, 3, 3, 8],
                                     'identity_factors': [1, 1, 1, 0],
                                     'noise_level': [12, 6, 21, 8],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                     'value':
0},
                                     'noise_prob': [ 0.001,
                                                     0.001,
                                                     0.001,
                                                     0.001],
                                     'status': True,
                                     'type': 'depolarizing'},
                                     'odd_wires': 4},
                                   { 'gates_num': [20, 5, 5, 16],
                                     'identity_factors': [2, 2, 2, 0],
                                     'noise_level': [20, 10, 35, 16],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                     'value':
0},
                                     'noise_prob': [ 0.001,
                                                     0.001,
                                                     0.001,
                                                     0.001],
                                     'status': True,
                                     'type': 'depolarizing'},
                                     'odd_wires': 4},
                                   { 'gates_num': [28, 7, 7, 24],
                                     'identity_factors': [3, 3, 3, 0],
                                     'noise_level': [28, 14, 49, 24],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                     'value':
0},
                                     'noise_prob': [ 0.001,
                                                     0.001,
                                                     0.001,
                                                     0.001],
                                     'status': True,
                                     'type': 'depolarizing'},
                                     'odd_wires': 4}
                                ]
    }
}

```

```
'noise_level': [28, 14, 49, 24],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
'status': True,
'type': 'depolarizing'},
    'odd_wires': 4}},
'run_time_sec': 36.53622078895569}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.08591881204796584,
                              1.5026023651616802,
                              1.9179097588555833,
                              1.9179097588555831,
                              1.9179097588555831,
                              1.9179097588555831,
                              2.8800636946459592,
                              2.9523818900173797,
                              2.9523818900173797,
                              2.954896407422686,
                              3.0487820562708485,
                              3.0487820562708485,
                              3.9462134206000328,
                              3.9462134206000328,
                              5.170248653928018,
                              5.170248653928018,
                              6.5066905594049205,
                              6.5066905594049205,
                              6.5066905594049205,
                              6.914119522580601,
                              6.914119522580601,
                              6.9842551957162815,
                              7.300516913785898,
                              8.782483894556266,
                              8.782483894556266,
                              8.999633503017922,
                              8.999633503017922,
                              8.999633503017922,
                              9.414122362394847,
                              9.414122362394847,
                              0.09854676674938925,
                              -0.05173289612974198,
                              0.30094803169630163,
                              -1.1984142138894447,
                              0.08610670053567002,
                              0.2851928692216035,
                              0.26177875477440216,
                              0.11764473243628538,
                              0.3495067876706816,
                              -0.9678438601279628,
                              -0.08762050677252062,
                              0.014826088936787857,
                              -0.30347729159143555,
                              -1.1730530788097624,
                              -0.12550631929596553,
                              2.258056275075482,
                              -0.5160508087356981,
                              0.004541093619226378,
                              -0.9345420797218197,
                              0.47154522212338745,
```

-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,

```

-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.410492799212099],
[12, 6, 21, 8, -3.2577133670016694],
[20, 10, 35, 16, -1.674813432318219],
[28, 14, 49, 24, -0.8591076530612253]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,

```

```
False,
0},
    'odd_wires': 4},
    { 'gates_num': [12, 3, 3, 8],
      'identity_factors': [1, 1, 1, 0],
      'noise_level': [12, 6, 21, 8],
      'noise_profile': { 'noise_on_init_param': { 'status':
        'value':
          'noise_prob': [ 0.001,
                        0.001,
                        0.001,
                        0.001],
          'status': True,
          'type': 'depolarizing'}},
      'odd_wires': 4},
      { 'gates_num': [20, 5, 5, 16],
        'identity_factors': [2, 2, 2, 0],
        'noise_level': [20, 10, 35, 16],
        'noise_profile': { 'noise_on_init_param': { 'status':
          'value':
            'noise_prob': [ 0.001,
                          0.001,
                          0.001,
                          0.001],
            'status': True,
            'type': 'depolarizing'}},
          'odd_wires': 4},
          { 'gates_num': [28, 7, 7, 24],
            'identity_factors': [3, 3, 3, 0],
            'noise_level': [28, 14, 49, 24],
            'noise_profile': { 'noise_on_init_param': { 'status':
              'value':
                'noise_prob': [ 0.001,
                              0.001,
                              0.001,
                              0.001],
                'status': True,
                'type': 'depolarizing'}},
                'odd_wires': 4]],
                'run_time_sec': 31.50432777404785}},
                { 'config': { 'ansatz': { 'gateset': 1,
                  'layer': 30,
                  'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                    'cn': [ 0.5,
                      0.5,
                      0.5,
                      0.5,
                      0.5,
                      0.5],
                    'r': 0}},
                    'time': {'max': 10.0, 'min': 0.0},
                    'type': 'xy-iss'}},
                    'init_param': { 'value': [ -5.647934061474703e-18,
                      1.4787834906243755,
                      1.478918229526628,
                      1.478918229526628,
                      2.0077351764243123,
                      2.4302809101456218,
                      2.4302809101456218,
                      3.009709563144601,
                      3.009709563144601,
                      3.239471043198664,
                      3.2405283031129133,
                      3.2405283031129133,
                      3.2656598154222434,
                      3.5353392937639825,
                      6.198713539858257,
                      6.213604956994236,
                      6.73397854893658,
                      6.73397854893658,
```


6.743547791194661,
6.744609103740889,
6.744609103740889,
6.749543065628153,
6.749543065628153,
8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,

```

0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,

```

```

        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
    'zne': { 'data_points': None,
            'degree': 0,
            'method': 'richardson',
            'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                     '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                     '5 X 6 + 1 Z 5 + 1 Z 6',
                'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                            '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                            '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                            '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                            '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                            '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
    'output': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
                                  [12, 6, 21, 8, -3.199071961215845],
                                  [20, 10, 35, 16, -1.6351376911040885],
                                  [28, 14, 49, 24, -0.8526977600673815]],
                'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                     'identity_factors': [0, 0, 0, 0],
                                     'noise_level': [4, 2, 7, 0],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.001,
                                                                                       0.001],
                                                                                       'status': True,
                                                                                       'type': 'depolarizing'},
                                     'odd_wires': 4},
                                     { 'gates_num': [12, 3, 3, 8],
                                     'identity_factors': [1, 1, 1, 0],
                                     'noise_level': [12, 6, 21, 8],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.001,
                                                                                       0.001],
                                                                                       'status': True,
                                                                                       'type': 'depolarizing'},
                                     'odd_wires': 4},
                                     { 'gates_num': [20, 5, 5, 16],
                                     'identity_factors': [2, 2, 2, 0],
                                     'noise_level': [20, 10, 35, 16],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.001,
                                                                                       0.001],
                                                                                       'status': True,
                                                                                       'type': 'depolarizing'},
                                     'odd_wires': 4},
                                     { 'gates_num': [28, 7, 7, 24],
                                     'identity_factors': [3, 3, 3, 0],
                                     'noise_level': [28, 14, 49, 24],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.001,
                                                                                       0.001],
                                                                                       'status': True,
                                                                                       'type': 'depolarizing'},
                                     'odd_wires': 4},
                                     { 'gates_num': [4, 1, 1, 0],
                                     'identity_factors': [0, 0, 0, 0],
                                     'noise_level': [4, 2, 7, 0],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.001,
                                                                                       0.001],
                                                                                       'status': True,
                                                                                       'type': 'depolarizing'},
                                     'odd_wires': 4},
                                     { 'gates_num': [12, 3, 3, 8],
                                     'identity_factors': [1, 1, 1, 0],
                                     'noise_level': [12, 6, 21, 8],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.001,
                                                                                       0.001],
                                                                                       'status': True,
                                                                                       'type': 'depolarizing'},
                                     'odd_wires': 4},
                                     { 'gates_num': [20, 5, 5, 16],
                                     'identity_factors': [2, 2, 2, 0],
                                     'noise_level': [20, 10, 35, 16],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.001,
                                                                                       0.001],
                                                                                       'status': True,
                                                                                       'type': 'depolarizing'},
                                     'odd_wires': 4},
                                     { 'gates_num': [28, 7, 7, 24],
                                     'identity_factors': [3, 3, 3, 0],
                                     'noise_level': [28, 14, 49, 24],
                                     'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.001,
                                                                                       0.001],
                                                                                       'status': True,
                                                                                       'type': 'depolarizing'},
                                     'odd_wires': 4}
                                   ]
    }
}

```

```

        'odd_wires': 4]],
        'run_time_sec': 31.051584482192993}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.33528943002425843,
                           0.8608925283648261,
                           2.8309935233222867,
                           2.8382233121900358,
                           2.847558790018716,
                           2.8477263794127468,
                           3.9337993695450666,
                           3.9337993695450666,
                           3.9337993695450666,
                           3.9337993695450666,
                           3.9337993695450666,
                           5.127462455084763,
                           5.128383696390232,
                           5.132051241075024,
                           5.132231848473024,
                           5.132297186784173,
                           5.132297186784173,
                           5.132297186784173,
                           5.132320898992942,
                           5.134903164275234,
                           5.734798022200061,
                           5.734798022200061,
                           5.734798022200061,
                           6.71748919093495,
                           6.71748919093495,
                           9.773430860122208,
                           9.791656954875775,
                           9.791980525916317,
                           9.804782964874805,
                           9.947289619134542,
                           0.6818618664375632,
                           0.42253658736184846,
                           0.19067339209915313,
                           -2.314563573529872,
                           -1.7784854373672185,
                           -0.10945476580518654,
                           0.08980428543050965,
                           0.6265702304745648,
                           -0.12016219313034968,
                           0.005450784512528824,
                           0.9621316606784164,
                           -0.01940541263687908,
                           -1.2890091107885369,
                           -0.09216272929775945,
                           -0.8615487717558121,
                           0.021600758375482327,
                           1.1068545366678577,
                           -0.06436428812394061,
                           -1.8186253546793636,
                           -0.2455724158610742,
                           0.7444478114283775,
                           1.0647343146888184,
                           -1.6388802958603008,
                           -0.05032235091912311,
                           0.7589104637111997,
                           0.0934759886788552,
                           0.7280377617711271,
                           -0.10012682423470079,
                           -2.7917715562202794,
                           0.5843828201862561,
                           1.2003794612588048,

```

-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,

```

1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.333005581610795],
                             [12, 6, 21, 8, -3.1651017858597426],
                             [20, 10, 35, 16, -1.6439583689682524],
                             [28, 14, 49, 24, -0.8813072530521873]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                     'identity_factors': [0, 0, 0, 0],
                     'noise_level': [4, 2, 7, 0],
                     'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
'identity_factors': [1, 1, 1, 0],
'noise_level': [20, 10, 35, 16],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
'identity_factors': [1, 1, 1, 0],
'noise_level': [28, 14, 49, 24],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}]]}

```

```

                                0.001,
                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
                                { 'gates_num': [20, 5, 5, 16],
                                'identity_factors': [2, 2, 2, 0],
                                'noise_level': [20, 10, 35, 16],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                'value':
0},
                                'noise_prob': [ 0.001,
                                0.001,
                                0.001,
                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
                                { 'gates_num': [28, 7, 7, 24],
                                'identity_factors': [3, 3, 3, 0],
                                'noise_level': [28, 14, 49, 24],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                'value':
0},
                                'noise_prob': [ 0.001,
                                0.001,
                                0.001,
                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4}},
                                'run_time_sec': 33.18718099594116}},
                                { 'config': { 'ansatz': { 'gateset': 1,
                                'layer': 30,
                                'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                'cn': [ 0.5,
                                0.5,
                                0.5,
                                0.5,
                                0.5,
                                0.5],
                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}},
                                'init_param': { 'value': [ 3.750846163374531e-05,
                                0.22285241391868899,
                                0.26252411232084344,
                                0.26252411232084344,
                                0.26252411232084344,
                                1.459545906754556,
                                1.4595459067545562,
                                1.6094835194149661,
                                2.5927782099427628,
                                2.5927782099427628,
                                2.5927782099427628,
                                2.5927782099427623,
                                3.3552499939702187,
                                3.3552499939702187,
                                5.033284106467877,
                                5.033284106467877,
                                5.056477210452712,
                                6.768913439946703,
                                6.768913439946703,
                                7.2782492670747585,
                                7.2782492670747585,
                                7.2782492670747585,
                                7.2782492670747585,
                                7.2782492670747585,
                                7.2782492670747585,
                                8.934198246659292,
                                8.937369801451712,
                                9.544406408519471,
                                9.544406408519471,

```

10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,


```

-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 X 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '

```



```

        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 0.003375513600300809,
    0.4884127443235871,
    0.6719352108785641,
    0.9073170135093351,
    1.4946633690060278,
    1.5076957064847494,
    3.872917838142945,
    3.872917838142945,
    3.872917838142945,
    3.872917838142945,
    5.047622474508209,
    5.054336886650782,
    5.067626550795242,
    5.0733174878702085,
    5.094629387163461,
    5.749571576552799,
    5.7731554023304374,
    7.144946717661993,
    7.144946717661993,
    7.144946717661993,
    7.144946717661993,
    7.602314289267704,
    7.602314289267704,
    8.018775147517832,
    8.018775147517832,
    8.018775147517832,
    8.018775147517832,
    8.018775147517832,
    8.018775147517832,
    9.661071929582006,
    0.2735947252101626,
    2.471262065567802,
    0.18954222370300397,
    0.2633058348660961,
    0.07494334922808346,
    1.4980157261935538,
    0.3538857424122425,
    0.26341770410626436,
    0.17174804107002956,
    -0.22912718472194316,
    0.11060714764109056,
    -0.30804859927082595,
    2.8406850614439736,
    1.0823654569610508,
    0.9668529413216363,
    1.8727663516543385,
    -0.0901659930896485,
    0.05868560690604535,
    0.24212775097742179,
    1.0092515209898247,
    -0.7556969386491817,
    0.3279350524218167,
    1.078117272505244,
    0.10391815458190773,
    -0.7434930272741371,
    -1.6965388530236032,
    -1.360748554100242,
    0.34939560927302815,
    0.03188634842980519,
    -1.2447472675125983,
    0.06092342743494014,
    1.017082923902293,
    -1.007905562333318,
    -0.8576057837691866,
    0.5321038189070756,
    1.300652060018237,
    -1.8122591145188616,
    -1.247002191580596,
    1.2529274198051785,
    0.3750417535596218,
    0.2898066740735119,
    1.589675559301449,
```

2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,

```

-1.6023915303224998,
0.6733061844063541,
1.5755992624240989]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.397817460329862],
[12, 6, 21, 8, -3.2115716730808748],
[20, 10, 35, 16, -1.641958817033255],
[28, 14, 49, 24, -0.8464644349040011]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
'identity_factors': [2, 2, 2, 0],
'noise_level': [20, 10, 35, 16],
'noise_profile': { 'noise_on_init_param': { 'status':
'value':

```

False,

0},

False,

0},

False,

```

0},

'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
  'identity_factors': [3, 3, 3, 0],
  'noise_level': [28, 14, 49, 24],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                                'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                                'status': True,
                                                                'type': 'depolarizing'},
                                                                'odd_wires': 4}},
  'run_time_sec': 32.358261823654175}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
                                   'layer': 30,
                                   'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                                'r': 0},
                                   'time': { 'max': 10.0, 'min': 0.0},
                                   'type': 'xy-iss'}},
  'init_param': { 'value': 'random'},
  'noise_profile': { 'noise_on_init_param': { 'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
  'nqubits': 7,
  'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                    'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                    'r': 1},
  'def': 'ising'},
  'output': { 'draw': { 'fig_dpi': 100, 'status': True, 'type': 'png'},
              'file_name_prefix': 'xy_noisey_time_evo'},
  'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
  'run': 'vqe',
  'state': 'dmatrix',
  'vqe': { 'iteration': 10,
           'optimization': { 'algorithm': 'SLSQP',
                             'constraint': True,
                             'status': True}},
  'zne': { 'data_points': None,
           'degree': 1,
           'method': 'richardson',
           'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1 Z 2 '
                                ' + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z '
                                '5 + 1 Z 6',
  'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + 0.5 X 1 '
                                              'X 2 + 0.5 Z 1 Z 2 + 0.5 X 2 X 3 + '
                                              '0.5 Z 2 Z 3 + 0.5 X 3 X 4 + 0.5 Z 3 '
                                              'Z 4 + 0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                              '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
  'initial_cost_history': [ 3.5476707600867368,
                           2.3562053879970173,
                           1.7833099820235232,
                           2.4654982864165467,
                           3.6613143494662093,
                           3.6462371631182062,
                           3.2935768662401737,
                           3.7145398712422657,
                           2.5227446974102397,
                           2.4756634278387035],
  'noise_details': { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 7, 0],

```

```
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                  'noise_prob': [0.001, 0.001, 0.001, 0.001],
                  'status': True,
                  'type': 'depolarizing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -6.515742960091897,
                             -6.3330055816107915,
                             -6.410492799212151,
                             -6.4484545761945915,
                             -6.311563719128753,
                             -6.46067826504545,
                             -6.397817460329919,
                             -6.429933051303193,
                             -6.536182288327426,
                             -6.511175848027232],
'optimized_parameters': [ [ 1.5680106786222674e-17,
                             1.0238650999177359,
                             1.0238650999177359,
                             1.0238650999177359,
                             1.7041550191762513,
                             2.070931805530806,
                             2.586234954306046,
                             2.6265616878606783,
                             2.7566136785306155,
                             2.7689271757054255,
                             4.271831894498143,
                             4.271831894498143,
                             4.271831894498143,
                             5.004110362344359,
                             5.004110362344359,
                             5.529689350278954,
                             5.529689350278954,
                             5.529689350278954,
                             5.5493166460873375,
                             7.855838962179162,
                             7.855838962179162,
                             8.508023697610218,
                             8.540796404638465,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             9.524293982069336,
                             9.524293982069336,
                             9.815045396471412,
                             0.717685470636315,
                             0.29125971197830336,
                             0.013619559122678915,
                             -0.46174394022911175,
                             -0.7802589318548062,
                             1.8785332138943256,
                             -0.07718385116900997,
                             -0.4181315830102034,
                             0.3859499067718677,
                             0.7218523694763775,
                             0.16350787496293626,
                             -0.06274337543165032,
                             1.048639096314506,
                             0.9454500907927006,
                             -0.18328208251477485,
                             0.5336296017995852,
                             0.7047960543548563,
                             -0.4717130073686535,
                             -0.14327049451726778,
                             0.5646728024946054,
                             0.1377201649736573,
                             1.0845662084088512,
                             0.7117671017527167,
                             1.8562916393289055,
                             -1.0958742208007046,
                             0.314756410152813,
                             1.0055290931424188,
                             -1.2653176839886926,
                             -0.6435463058390287,
```

-0.13689785210195535,
0.5185042269644529,
-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,

0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455],
[0.33528943002425843,
0.8608925283648261,
2.8309935233222867,
2.8382233121900358,
2.847558790018716,
2.8477263794127468,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
5.127462455084763,
5.128383696390232,
5.132051241075024,
5.132231848473024,
5.132297186784173,
5.132297186784173,
5.132297186784173,
5.132320898992942,
5.134903164275234,
5.734798022200061,
5.734798022200061,
5.734798022200061,
6.71748919093495,
6.71748919093495,
9.773430860122208,
9.791656954875775,
9.791980525916317,
9.804782964874805,
9.947289619134542,
0.6818618664375632,
0.42253658736184846,
0.19067339209915313,
-2.314563573529872,
-1.7784854373672185,
-0.10945476580518654,
0.08980428543050965,
0.6265702304745648,
-0.12016219313034968,
0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,
-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,

0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,

0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358],
[0.08591881204796584,
1.5026023651616802,
1.9179097588555833,
1.9179097588555831,
1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,

-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,

0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406],
[0.3773346757087749,
0.5734233884194113,
0.5734233884194113,
1.2094818154974225,
1.2528672337957094,
1.3073966674132076,
4.193348178434801,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.274742575727675,
5.343872214974395,
5.354523589318104,
5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,

0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,

-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244],
[-6.331146189926128e-17,
0.7069927066774304,
0.7069927066774304,
0.7069927066774305,
0.7069927066774304,
1.0044153364322925,
1.0092418692996383,
2.470333794214341,
2.478143749866421,
3.5776564981759003,
3.5776564981759003,
3.5776564981759003,
3.61745967653339,
3.6981780389472076,
3.755720232685741,
3.9712286023902594,
3.9712286023902594,
4.519037432417192,
4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,
6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,

-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,

-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176],
[3.750846163374531e-05,
0.22285241391868899,
0.26252411232084344,
0.26252411232084344,
0.26252411232084344,
1.459545906754556,
1.4595459067545562,
1.6094835194149661,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
2.5927782099427623,
3.3552499939702187,
3.3552499939702187,
5.033284106467877,
5.033284106467877,
5.056477210452712,
6.768913439946703,
6.768913439946703,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,

1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,

-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138],
[0.003375513600300809,
0.4884127443235871,
0.6719352108785641,
0.9073170135093351,
1.4946633690060278,
1.5076957064847494,
3.872917838142945,
3.872917838142945,
3.872917838142945,
3.872917838142945,
5.047622474508209,
5.054336886650782,
5.067626550795242,
5.0733174878702085,
5.094629387163461,
5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,

-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,

0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989],
[-5.647934061474703e-18,
1.4787834906243755,
1.478918229526628,
1.478918229526628,
2.0077351764243123,
2.4302809101456218,
2.4302809101456218,
3.009709563144601,
3.009709563144601,
3.239471043198664,
3.2405283031129133,
3.2405283031129133,
3.2656598154222434,
3.5353392937639825,
6.198713539858257,
6.213604956994236,
6.73397854893658,
6.73397854893658,
6.743547791194661,
6.744609103740889,
6.744609103740889,
6.749543065628153,
6.749543065628153,
8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,

0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,

-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935],
[0.0019047787007362745,
0.7201496918445128,
0.7201496918445128,
0.7201496918445127,
0.7226708680142584,
0.8068041864232433,
1.3847174673528184,
1.6031895944534578,
2.458865333289775,
2.458865333289775,
2.721120236828938,
2.721120236828938,
3.554687304897056,
3.554687304897056,
4.537252190232355,
4.537252190232355,
5.239770593633025,
5.242363659304797,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
7.6484324534494865,
7.6484324534494865,
8.96757120343865,
9.352637118466301,
9.381932224950582,
9.966925955531599,
9.966925955531599,
0.29840004251263574,
0.3207087147269936,
-0.22478909909228914,
-1.2242611593120782,
0.061748844824383295,
1.9931455249507801,
0.02449362327304914,
-0.23402928130657288,
0.3399922671860315,
0.2355005268966802,
0.7912945290539132,
-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,

0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,

-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.753559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815],
[0.00030922277578629,
0.881375055029912,
0.881375055029912,
0.8813750550299119,
0.9803845151714838,
0.9803845151714838,
1.201629973822729,
1.2441077564052723,
1.8660677974280824,
1.8660677974280824,
2.651002314516478,
2.651002314516478,
3.233637461790533,
4.36384995486056,
4.36384995486056,
4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,
8.903406631369169,
8.903406631369169,
8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.1357940797759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,

0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,

```

-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'run_time_sec': 68892.59480500221}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.3773346757087749,
0.5734233884194113,
0.5734233884194113,
1.2094818154974225,
1.2528672337957094,
1.3073966674132076,
4.193348178434801,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.274742575727675,
5.343872214974395,
5.354523589318104,
5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,

```

-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,

```

-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.44845457619454],
                          [12, 6, 21, 8, -3.2446894681699203],
                          [20, 10, 35, 16, -1.6866076162118238],
                          [28, 14, 49, 24, -0.8984323471828195]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.100799560546875e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.292127940547555,
                           'others': { 'beta_coefficients': [ 1.856490433543659,
                                                             -1.5404920618766529,
                                                             0.8938657642987985,
                                                             -0.20986413596580492],
                                       'cost_richardson_zne': 6.662711482694253,
                                       'sorted_expectation_vals': [ -6.44845457619454,
                                                                    -3.2446894681699203,
                                                                    -1.6866076162118238,
                                                                    -0.8984323471828195],
                                       'sorted_noise': [13, 47, 81, 115]},
                           'sampled data': [ [4, 2, 7, 0, -6.44845457619454],
                                              [12, 6, 21, 8, -3.2446894681699203],
                                              [20, 10, 35, 16, -1.6866076162118238],
                                              [28, 14, 49, 24, -0.8984323471828195]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],

```

```
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 3.750846163374531e-05,
    0.22285241391868899,
    0.26252411232084344,
    0.26252411232084344,
    0.26252411232084344,
    1.459545906754556,
    1.4595459067545562,
    1.6094835194149661,
    2.5927782099427628,
    2.5927782099427628,
    2.5927782099427628,
    2.5927782099427623,
    3.3552499939702187,
    3.3552499939702187,
    5.033284106467877,
    5.033284106467877,
    5.056477210452712,
    6.768913439946703,
    6.768913439946703,
    7.2782492670747585,
    7.2782492670747585,
    7.2782492670747585,
    7.2782492670747585,
    7.2782492670747585,
    7.2782492670747585,
    8.934198246659292,
    8.937369801451712,
    9.544406408519471,
    9.544406408519471,
    10.23170722265548,
    -0.4501443853573418,
    0.06519513561037263,
    -0.13386490845883003,
    -0.01542333988020039,
    0.3988203537526196,
    0.9163371790808124,
    0.08159154039220383,
    -0.11665429084087826,
    -0.21130662780886203,
    1.515236372470638,
    0.4436142129357345,
    -0.28780404896369716,
    0.3681425383234808,
    0.648176841676201,
    0.3046875502986337,
    0.05897820457762899,
    0.09316951897603123,
    1.5277657654344678,
    -0.655141628591867,
    0.073084468489131,
    -0.11423944001370272,
    0.3986898081963471,
    1.0974832156349317,
    1.8462557861113,
    0.5998463368708868,
    -0.0014214294560794885,
    0.4026665434776887,
    -0.03860309342262787,
    0.3728818533964108,
    1.7322816292529104,
    0.42156018976657544,
    0.9872712539907578,
    0.5749190583224287,
    0.1654949387796794,
    1.0577776390475342,
    -0.5892377479966974,
    0.4324416138472369,
    0.7909276553155474,
    1.0132171075838867,
    0.008626434202972095,
    -1.4776636473003866,
    -0.0743310361925915,
```

1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,

```

1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0.001, 0.001],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                  'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                  'r': 1},
               'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.4606782650454075],
                          [12, 6, 21, 8, -3.2738854608106305],
                          [20, 10, 35, 16, -1.6974907774474755],
                          [28, 14, 49, 24, -0.8939947022135477]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.291534423828125e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.280504294823274,
                           'others': { 'beta_coefficients': [ 1.856490433543659,
                                                             -1.5404920618766529,
                                                             0.8938657642987985,
                                                             -0.20986413596580492],
                                       'cost_richardson_zne': 6.662711482694253,
                                       'sorted_expectation_vals': [ -6.4606782650454075,
                                                                    -3.2738854608106305,
                                                                    -1.6974907774474755,
                                                                    -0.8939947022135477],
                                       'sorted_noise': [13, 47, 81, 115]},
                           'sampled data': [ [4, 2, 7, 0, -6.4606782650454075],
                                              [12, 6, 21, 8, -3.2738854608106305],
                                              [20, 10, 35, 16, -1.6974907774474755],
                                              [28, 14, 49, 24, -0.8939947022135477]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                    'time': {'max': 10.0, 'min': 0.0},
                                    'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.08591881204796584,
                             1.5026023651616802,
                             1.9179097588555833,
                             1.9179097588555831,
                             1.9179097588555831,
                             1.9179097588555831,
                             2.8800636946459592,
                             2.9523818900173797,
                             2.9523818900173797,
                             2.954896407422686,
                             3.0487820562708485,
                             3.0487820562708485,
                             3.9462134206000328,
                             3.9462134206000328,
                             5.170248653928018,
                             5.170248653928018,
                             6.5066905594049205,
                             6.5066905594049205,

```


6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,

```

-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 0],
                                      [2, 2, 2, 0],
                                      [3, 3, 3, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,

```

[illegible]

0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,
-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,

```
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.333005581610795],
[12, 6, 21, 8, -3.1651017858597426],
[20, 10, 35, 16, -1.6439583689682524],
[28, 14, 49, 24, -0.8813072530521873]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.9577484130859375e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.165873420461505,
'others': { 'beta_coefficients': [ 1.856490433543659,
-1.5404920618766529,
0.8938657642987985,
-0.20986413596580492],
'cost_richardson_zne': 6.662711482694253,
'sorted_expectation_vals': [ -6.333005581610795,
-3.1651017858597426,
```

```

-1.6439583689682524,
-0.8813072530521873],
'sorted_noise': [13, 47, 81, 115]],
'sampled data': [ [4, 2, 7, 0, -6.333005581610795],
[12, 6, 21, 8, -3.1651017858597426],
[20, 10, 35, 16, -1.6439583689682524],
[28, 14, 49, 24, -0.8813072530521873]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ -6.331146189926128e-17,
0.7069927066774304,
0.7069927066774304,
0.7069927066774305,
0.7069927066774304,
1.0044153364322925,
1.0092418692996383,
2.470333794214341,
2.478143749866421,
3.5776564981759003,
3.5776564981759003,
3.5776564981759003,
3.61745967653339,
3.6981780389472076,
3.755720232685741,
3.9712286023902594,
3.9712286023902594,
4.519037432417192,
4.519037432417193,
4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,
6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,

```

1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,

```

-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 0],
                                      [2, 2, 2, 0],
                                      [3, 3, 3, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.3115637191286815],
                          [12, 6, 21, 8, -3.2712898811023576],
                          [20, 10, 35, 16, -1.7533390824025292],
                          [28, 14, 49, 24, -0.9662277022700592]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 6.437301635742188e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.042434808411594,
                           'others': { 'beta_coefficients': [ 1.856490433543659,
                                                             -1.5404920618766529,
                                                             0.8938657642987985,
                                                             -0.20986413596580492],
                                       'cost_richardson_zne': 6.662711482694253,
                                       'sorted_expectation_vals': [ -6.3115637191286815,
                                                                    -3.2712898811023576,
                                                                    -1.7533390824025292,
                                                                    -0.9662277022700592],
                                       'sorted_noise': [13, 47, 81, 115]},
                           'sampled data': [ [4, 2, 7, 0, -6.3115637191286815],
                                              [12, 6, 21, 8, -3.2712898811023576],
                                              [20, 10, 35, 16, -1.7533390824025292],
                                              [28, 14, 49, 24, -0.9662277022700592]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.0019047787007362745,
                             0.7201496918445128,
                             0.7201496918445128,
                             0.7201496918445127,
                             0.7226708680142584,
                             0.8068041864232433,

```


1.3847174673528184,
1.6031895944534578,
2.458865333289775,
2.458865333289775,
2.721120236828938,
2.721120236828938,
3.554687304897056,
3.554687304897056,
4.537252190232355,
4.537252190232355,
5.239770593633025,
5.242363659304797,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
7.6484324534494865,
7.6484324534494865,
8.96757120343865,
9.352637118466301,
9.381932224950582,
9.966925955531599,
9.966925955531599,
0.29840004251263574,
0.3207087147269936,
-0.22478909909228914,
-1.2242611593120782,
0.061748844824383295,
1.9931455249507801,
0.02449362327304914,
-0.23402928130657288,
0.3399922671860315,
0.2355005268966802,
0.7912945290539132,
-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,

```

-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],

```

```

        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 1},
        'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0]]},

'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.536182288327376],
                          [12, 6, 21, 8, -3.450081350248876],
                          [20, 10, 35, 16, -1.8650932993173606],
                          [28, 14, 49, 24, -1.0247351729167147]],

        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.054473876953125e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -8.271624843115143,
                          'others': { 'beta_coefficients': [ 1.856490433543659,
                                                            -1.5404920618766529,
                                                            0.8938657642987985,
                                                            -0.20986413596580492],
                          'cost_richardson_zne': 6.662711482694253,
                          'sorted_expectation_vals': [ -6.536182288327376,
                                                        -3.450081350248876,
                                                        -1.8650932993173606,
                                                        -1.0247351729167147],
                          'sorted_noise': [13, 47, 81, 115]},
                          'sampled data': [ [4, 2, 7, 0, -6.536182288327376],
                                              [12, 6, 21, 8, -3.450081350248876],
                                              [20, 10, 35, 16, -1.8650932993173606],
                                              [28, 14, 49, 24, -1.0247351729167147]],
                          'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}}},
  'init_param': { 'value': [ -5.647934061474703e-18,
                              1.4787834906243755,
                              1.478918229526628,
                              1.478918229526628,
                              2.0077351764243123,
                              2.4302809101456218,
                              2.4302809101456218,
                              3.009709563144601,
                              3.009709563144601,
                              3.239471043198664,
                              3.2405283031129133,
                              3.2405283031129133,
                              3.2656598154222434,
                              3.5353392937639825,
                              6.198713539858257,
                              6.213604956994236,
                              6.73397854893658,
                              6.73397854893658,
                              6.743547791194661,
                              6.744609103740889,
                              6.744609103740889,
                              6.749543065628153,
                              6.749543065628153,
                              8.615256229309818,
                              8.635090640841518,
                              9.012323512360366,
                              9.012323512360366,

```

9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,

```

-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 0],
                                      [2, 2, 2, 0],
                                      [3, 3, 3, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
                          [12, 6, 21, 8, -3.199071961215845],
                          [20, 10, 35, 16, -1.6351376911040885],
                          [28, 14, 49, 24, -0.8526977600673815]],
        'degree': 0,
        'method': 'richardson',

```

```

'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.482269287109375e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.291607159781275,
'others': { 'beta_coefficients': [ 1.856490433543659,
-1.5404920618766529,
0.8938657642987985,
-0.20986413596580492],
'cost_richardson_zne': 6.662711482694253,
'sorted_expectation_vals': [ -6.429933051303131,
-3.199071961215845,
-1.6351376911040885,
-0.8526977600673815],
'sorted_noise': [13, 47, 81, 115]},
'sampled data': [ [4, 2, 7, 0, -6.429933051303131],
[12, 6, 21, 8, -3.199071961215845],
[20, 10, 35, 16, -1.6351376911040885],
[28, 14, 49, 24, -0.8526977600673815]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.003375513600300809,
0.4884127443235871,
0.6719352108785641,
0.9073170135093351,
1.4946633690060278,
1.5076957064847494,
3.872917838142945,
3.872917838142945,
3.872917838142945,
3.872917838142945,
5.047622474508209,
5.054336886650782,
5.067626550795242,
5.0733174878702085,
5.094629387163461,
5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,

```

0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,

```

0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989}],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 0],
                                      [2, 2, 2, 0],
                                      [3, 3, 3, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.397817460329862],
                          [12, 6, 21, 8, -3.2115716730808748],
                          [20, 10, 35, 16, -1.641958817033255],
                          [28, 14, 49, 24, -0.8464644349040011]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.4345855712890625e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.22013448780964,
                           'others': { 'beta_coefficients': [ 1.856490433543659,
                                                             -1.5404920618766529,
                                                             0.8938657642987985,
                                                             -0.20986413596580492],
                                     'cost_richardson_zne': 6.662711482694253,
                                     'sorted_expectation_vals': [ -6.397817460329862,
                                                                -3.2115716730808748,
                                                                -1.641958817033255,
                                                                -0.8464644349040011],
                                     'sorted_noise': [13, 47, 81, 115]},
                           'sampled data': [ [4, 2, 7, 0, -6.397817460329862],
                                              [12, 6, 21, 8, -3.2115716730808748],
                                              [20, 10, 35, 16, -1.641958817033255],
                                              [28, 14, 49, 24, -0.8464644349040011]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,

```



```
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                              'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                              'r': 0},
           'time': {'max': 10.0, 'min': 0.0},
           'type': 'xy-iss'}},
'init_param': { 'value': [ 1.5680106786222674e-17,
                             1.0238650999177359,
                             1.0238650999177359,
                             1.0238650999177359,
                             1.7041550191762513,
                             2.070931805530806,
                             2.586234954306046,
                             2.6265616878606783,
                             2.7566136785306155,
                             2.7689271757054255,
                             4.271831894498143,
                             4.271831894498143,
                             4.271831894498143,
                             5.004110362344359,
                             5.004110362344359,
                             5.529689350278954,
                             5.529689350278954,
                             5.529689350278954,
                             5.5493166460873375,
                             7.855838962179162,
                             7.855838962179162,
                             8.508023697610218,
                             8.540796404638465,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             9.524293982069336,
                             9.524293982069336,
                             9.815045396471412,
                             0.717685470636315,
                             0.29125971197830336,
                             0.013619559122678915,
                             -0.46174394022911175,
                             -0.7802589318548062,
                             1.8785332138943256,
                             -0.07718385116900997,
                             -0.4181315830102034,
                             0.3859499067718677,
                             0.7218523694763775,
                             0.16350787496293626,
                             -0.06274337543165032,
                             1.048639096314506,
                             0.9454500907927006,
                             -0.18328208251477485,
                             0.5336296017995852,
                             0.7047960543548563,
                             -0.4717130073686535,
                             -0.14327049451726778,
                             0.5646728024946054,
                             0.1377201649736573,
                             1.0845662084088512,
                             0.7117671017527167,
                             1.8562916393289055,
                             -1.0958742208007046,
                             0.314756410152813,
                             1.0055290931424188,
                             -1.2653176839886926,
                             -0.6435463058390287,
                             -0.13689785210195535,
                             0.5185042269644529,
                             -0.07994120596020922,
                             -1.3394312085309903,
                             -0.01354724643952521,
                             -0.8508707890759795,
                             -0.7258020036409927,
                             -0.1513648764845159,
                             -0.5702686618598916,
                             -0.20804467461705725,
```

-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,

```

0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0.001, 0.001],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.515742960091851],
                          [12, 6, 21, 8, -3.36492111003702],
                          [20, 10, 35, 16, -1.7545658776574033],
                          [28, 14, 49, 24, -0.9139478342631158]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.933906555175781e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -8.289321710676138,
                          'others': { 'beta_coefficients': [ 1.856490433543659,
                                                            -1.5404920618766529,
                                                            0.8938657642987985,
                                                            -0.20986413596580492],
                          'cost_richardson_zne': 6.662711482694253,
                          'sorted_expectation_vals': [ -6.515742960091851,
                                                        -3.36492111003702,
                                                        -1.7545658776574033,
                                                        -0.9139478342631158],
                          'sorted_noise': [13, 47, 81, 115]},
                          'sampled data': [ [4, 2, 7, 0, -6.515742960091851],
                                             [12, 6, 21, 8, -3.36492111003702],
                                             [20, 10, 35, 16, -1.7545658776574033],
                                             [28, 14, 49, 24, -0.9139478342631158]],
                          'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                       'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                       'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.00030922277578629,
                             0.881375055029912,
                             0.881375055029912,
                             0.8813750550299119,
                             0.9803845151714838,
                             0.9803845151714838,
                             1.201629973822729,
                             1.2441077564052723,
                             1.8660677974280824,
                             1.8660677974280824,
                             2.651002314516478,
                             2.651002314516478,
                             3.233637461790533,
                             4.36384995486056,
                             4.36384995486056,

```

4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,
8.903406631369169,
8.903406631369169,
8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.13579407977759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,

```
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric4_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0]]},
```

[illegible]

```

'0.5 X 5 X 6 + 0.5 Z 5 Z 6']]],
'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 4.118038614266066,
                                     2.900590937440303,
                                     2.874944433071395,
                                     3.9296977116518517,
                                     3.1841729388967344,
                                     3.695928991667305,
                                     3.180227821654077,
                                     3.0475045550195774,
                                     2.4669562192398296,
                                     3.4124916083581294],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                               'identity_factors': [0, 0, 0, 0],
                               'noise_level': [0, 0, 0, 0],
                               'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                   'value': 0},
                                                  'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                  'status': False,
                                                  'type': 'depolarizing'},
                               'odd_wires': 4},
            'optimized_minimum_cost': [ -8.433468022982382,
                                       -8.464942606128835,
                                       -8.43123094137328,
                                       -8.435401809869298,
                                       -7.9812014353834515,
                                       -8.371080050195895,
                                       -8.46198057547588,
                                       -8.19415563172473,
                                       -8.438078664602543,
                                       -8.481886373274177],
            'optimized_parameters': [ [ 0.4631008843346199,
                                       0.8090636887146863,
                                       1.1101507421405439,
                                       1.3400184709676968,
                                       1.7452332568602784,
                                       1.9308700883292553,
                                       2.1752679291102544,
                                       2.1784212211010603,
                                       3.800207092295721,
                                       3.800207092295721,
                                       3.800207092295721,
                                       3.800207092295721,
                                       3.800207092295721,
                                       3.800207092295721,
                                       3.800207092295721,
                                       4.798448841518561,
                                       4.812026604031267,
                                       6.02543601067661,
                                       6.02543601067661,
                                       6.5265343452987,
                                       6.5265343452987,
                                       6.526534345298701,
                                       6.528381158760148,
                                       6.528381158760148,
                                       7.113023968622042,
                                       7.113023968622042,
                                       7.593654062389684,
                                       7.5949106716383605,
                                       7.780058793305066,
                                       9.000370275113886,
                                       9.005799799983558,
                                       -0.22105281285875572,
                                       0.07196296170715184,
                                       1.0527781347416576,
                                       -0.4594806493129553,
                                       1.0526039239081737,
                                       -0.01709270509934428,
                                       0.3199675106146978,
                                       -0.0972220164715271,
                                       0.9145097677998977,
                                       0.09808103992344623,

```

False,

0.5373974901276602,
-0.5786059880820694,
-0.02594763303680847,
-0.22132394348491485,
-0.7274785930689547,
-0.41090279602128,
-0.520884263329148,
-1.0260835302846771,
0.1692597388493397,
1.3250313995601488,
1.510357845739394,
-0.9233604808876308,
-0.2049157632594096,
-0.9894281051832,
0.25772120620526445,
-0.04822142329545429,
-0.3912638297585932,
-0.48486712444934904,
1.4836649713082644,
1.566996182191827,
0.08093959953023218,
-2.3562926055425963,
-0.21876618997457167,
0.10911407834571141,
-1.3776955808248483,
-0.31816531211057325,
0.7083857256813267,
0.5593482503842677,
0.7483381318097894,
0.010042025170353972,
0.05290425882351868,
0.20616876613540233,
-0.8673924682871486,
-0.08017548043713153,
-0.3056222778255953,
2.3981163154468383,
0.37079365521682633,
0.3870558721039623,
0.19341028278759992,
-0.3154248344945196,
-0.12073260037561208,
-1.5858414285227351,
0.59830347262823,
0.30388710178795286,
-1.4558796844932331,
-0.3225618511944297,
0.09243390755822743,
-0.0006426173071943948,
0.7561662109631148,
-1.3902324585417218,
0.1677701056843372,
2.7311884232201735,
-1.1122880815211051,
0.0467986017971284,
-0.020089420529231013,
0.005481851366742631,
1.3205596999819047,
-0.1290403664286252,
0.3870444394307043,
-0.6549197738372419,
-0.17656835190090397,
-0.0008013451895215781,
-0.8769883624496301,
1.6929520910231297,
0.7288671099890552,
-0.6169673343691255,
1.667217439782634,
0.2554635335546482,
-0.029515171705053127,
-0.8290153205719569,
0.042733616172741226,
-0.5148227135977747,
1.2301836750084596,
-0.9175908300980525,
0.5229779878549122,

-1.5901934710811256,
-0.07316585886439231,
1.1195018365185199,
-0.31111306362203306,
0.7254709796877604,
1.1995417479543053,
0.6908225815252583,
-0.2919630814941808,
0.15955998483086176,
1.2871422030548714,
-1.2683991057896014,
-0.3768598572966417,
1.4863954796333156,
-0.4300298743216642,
0.19151460300148976,
0.9665838873771405,
-0.07244306377732504,
-0.04169546816073936,
-0.389029635704794,
1.5218961812832925,
-0.1772066326032777,
-0.70949107362336,
-0.015222466074935298,
0.3587128629768407,
1.5809494716454662,
0.12428164288740734,
1.5083554487395696,
-0.044379618902227067,
-0.03505579308192811,
-0.6786538117837789,
-0.007428157845337949,
1.5281494732580478,
-0.14124856089608578,
2.726718429443767,
-0.003106328084445967],
[0.5270475437101116,
0.7815425625485376,
1.6469610445191378,
1.6712876404917778,
2.122904188877425,
2.131106816606521,
3.7367798442214175,
3.7367798442219424,
4.750276145421292,
4.853973384996016,
4.865098072058574,
4.873980651141647,
4.987599328262889,
5.00246239884106,
5.520902021852277,
5.793284849001337,
5.8087832398624775,
7.432729128414403,
7.432953787975026,
8.144896858715603,
8.172580744083415,
8.788071761144659,
8.792288306719156,
8.828260688191317,
9.011586808740137,
9.011995976067807,
9.538729699920484,
9.576884709106812,
10.486750161390887,
10.48675016212107,
-0.15531374899360717,
-0.0849039080739262,
-0.8434555658556069,
0.9447826609611266,
-0.19482082661427938,
2.4999420394249707,
-2.7164831941210483,
-1.5408169364026967,
-0.8621482593978547,
0.31547458537730877,

1.4220435826823787,
1.348860553316339,
0.40827474166943256,
1.063960383917828,
-0.21906798993221652,
-0.33263312803097533,
0.4889869181158972,
-0.25524084746075687,
-1.3622146811064544,
0.996105149433398,
1.4161210728452394,
-0.6491827941870941,
0.7399329880002848,
-0.11994422281276146,
3.179573825543474,
-0.034943024466848674,
0.6075464711124173,
-0.3019825794696199,
0.029392952712390567,
0.14840732209332994,
0.08592470568620861,
0.016725634393736574,
-1.3697962205589145,
-0.06155375586805023,
0.16508080354919497,
-0.5975014914051775,
0.09635736997937351,
1.254101402191796,
-0.6193953337438176,
0.47602269510847217,
-1.7251540483535392,
0.5378274350019089,
-1.0159384367751108,
1.0968223266795405,
-0.79878134010976,
0.10468891520322528,
-0.2910173593847854,
-0.7078981306941186,
-0.3510837282628223,
0.5371002854277007,
-0.4911274293534503,
1.7273270693440552,
1.3500555714793954,
0.9052059235505011,
0.4771526028272018,
-0.1893747959421651,
0.13989050072973697,
-0.5930955187300607,
0.36780210947560876,
-0.5043370473487999,
-0.4851893292987325,
0.1987440888222681,
-0.5806445532873666,
2.3171646404124093,
0.06032126128170586,
-0.6389920437730593,
0.9985359186149056,
0.04870155999532171,
0.16447399991313041,
0.05315090520136875,
0.15727255736970833,
-1.3722292706433088,
-0.33233649668322635,
0.7758243461945761,
-0.20476613404890107,
0.06972046040604139,
0.3143879960336948,
-0.01452323353492752,
-1.2901366074359686,
-0.31033000747276407,
-0.16264485166390844,
-1.3914094473719103,
-0.5273128620655095,
0.02465459151953875,
0.10942766294132272,

1.0292094835171124,
-1.882689768994833,
-1.553216293090225,
1.3287846024814622,
0.20963913600213377,
0.7195732642990699,
0.0969787732183028,
-0.9744344415633356,
1.5822487402774768,
0.7077471503862973,
-0.766345302327988,
0.3831437091022152,
-0.12028853613988252,
-0.8209919023748664,
-0.21164805512089607,
0.5290684330656528,
-0.1040878870845114,
1.1956945887281514,
0.25703931501419025,
0.6344733295912399,
-0.02497584874729833,
-0.20937697561532842,
-1.4409411147319051,
0.9707320454927133,
1.4039378085761043,
0.9760826987262126,
-0.17314228405542512,
0.16734556762922406,
0.07881248116464283,
-0.2924467918405051,
1.4946430081327182,
0.7516806171134963,
0.799700486923224,
-0.8440600324500036,
-0.01069194371181784],
[0.3388610072211759,
0.3930923343746358,
0.4059472295013711,
0.4597995545870322,
0.4597995545870322,
0.4667083780713653,
0.9924683364283498,
1.054118787032341,
1.3839145947816414,
1.4469616998432806,
3.044048229936016,
3.044048229936016,
4.682975233106521,
4.682975233106521,
5.287925895396742,
5.287925895396742,
5.625677830869467,
5.625677830869467,
6.519005703020383,
6.519005703020383,
6.519005703020383,
6.854665591577039,
6.854665591577039,
8.092962128458302,
8.097830997780322,
8.102159968176109,
8.111954675742565,
8.727038821401255,
8.727038821401255,
8.784424594255395,
-0.8636093730812797,
-0.04742973062196874,
-0.37989856483295564,
0.3750798322441465,
-0.509087961968837,
0.5366875438436245,
0.1232704940065221,
0.5249134835872401,
0.13594138148710633,
0.10802292582448766,

-0.7156772546121367,
0.34756877087651405,
-0.5704812075854907,
0.289395233495819,
0.2674862930849357,
0.8724143866961164,
0.5770602988816299,
0.658777432128999,
-0.47288232304292416,
-0.029663566814993374,
-1.323180236232167,
-0.2890901656039231,
-0.015141732930961212,
1.1846850771343012,
-0.4506688471965015,
0.22241980016839272,
-0.015748484493019736,
1.1522499025192263,
0.2996907655672081,
-1.7370631162182948,
-0.35306232532663057,
0.17322471105923057,
-1.1855180134814352,
-0.31537295018088835,
-1.8120366955290261,
0.6663892425075478,
0.21874911670355193,
1.2729693967005706,
-0.14230256773273223,
0.2012427441067739,
-2.3269317425793643,
-0.03299848401271634,
-0.4074853136565566,
-0.30371905431664253,
-0.12037074206123735,
0.8456676988075262,
-0.21838328071319946,
0.019494764922420988,
0.16383099728733078,
0.056056403895700845,
-3.5632063709291613,
0.6579904510057321,
0.5207861046034454,
-0.017180990359968364,
0.2532807007765998,
-0.03315495616728509,
0.7564905992030696,
-0.11091568858030387,
-0.9281890577202553,
-0.6082181315906088,
-0.4480511658411004,
2.1953283079942025,
-0.30446713497464617,
-0.03359602431311142,
0.28972943746372876,
0.17817064395522006,
0.7965754083210965,
-0.9520780563921613,
0.10150445450145726,
1.3510884636720888,
-0.37849615173878054,
0.17729737647006832,
0.1884683806301832,
-0.03234547530918763,
-1.338763456177623,
-0.6711058962200585,
-0.012976043431998471,
0.027578392175021227,
0.10161286285089129,
-0.00939135247962935,
-1.302164623174096,
1.3715576399721376,
0.737767307868418,
1.5939269674119705,
1.4291758244230544,

0.5601859764306084,
0.3488764286390935,
-1.093762224545935,
-1.1976831420922067,
1.4557381665683378,
0.8924472305311927,
0.5300221300338881,
1.0785280788144638,
-0.6029509417428172,
-0.6790632925762424,
0.4138865114731201,
0.2917474006869537,
1.8050154062536312,
0.05177383423831921,
0.25467777497912664,
1.4671356779039024,
0.4422247342165525,
-1.5431607378377095,
1.8101814873938125,
1.6870467346899987,
0.7517920825884995,
1.3622912362338087,
0.13653050459472865,
1.5210183141257738,
-0.004247168330480351,
0.26137218980413907,
0.5451511562235306,
-0.6705596749039198,
1.2136085126593865,
0.5231529087420831,
-0.03279466120532818,
0.2548954759886831,
1.5797820110558691,
1.075840789720537,
1.5541818084195007],
[3.821020346740443e-17,
1.386496864345847,
1.386496864345847,
1.7951472089989222,
1.8559400139285378,
2.7980544315911504,
2.7980544315911504,
2.82954781812891,
2.82954781812891,
2.82954781812891,
2.956183634338348,
3.840561875918348,
3.840561875918348,
4.79501776182173,
4.79501776182173,
5.670460018404704,
5.670460018404704,
5.670460018404704,
6.865392250052202,
6.865392250052202,
7.506826147384778,
7.506826147384778,
8.737685173635683,
8.737685173635683,
8.737685173635683,
8.737685173635683,
9.601023092540274,
9.601023092540274,
9.853738132880634,
9.853738132880634,
0.5611658219809542,
1.427331325859137,
0.8881601644240952,
-0.8937497791603879,
0.5712044928850715,
1.3195951619089188,
0.5467792160605555,
-0.6816630744322637,
0.06707477836950569,
0.6798692668930644,

0.19576168860111362,
0.5094724850631845,
1.1424414477256812,
0.30471145260454696,
-0.08158911485577636,
0.9585661478313002,
1.0836238583568754,
-0.3718197048926503,
0.7841948506971846,
0.3023515457437066,
0.7662240328883452,
0.9970048707689876,
-0.8946064024093777,
0.27649390200762114,
-0.15342626519190924,
-0.4015720487025046,
1.460926694270634,
-0.055421168101734586,
0.6077050880702092,
-0.7932081335349191,
1.3188332541813446,
0.01929555634963922,
-1.1301845164589392,
-0.1718174564221745,
2.26154163970611,
-0.9882028269623031,
0.8038643847286138,
0.6849304102208338,
0.874798983022982,
-0.016486064274954207,
0.0264221098096002,
-0.3625000384278827,
1.4376331477886695,
0.8499992740594865,
0.10042171561200945,
0.14542648933357244,
1.3513090878714353,
-0.22248922869259902,
-0.4483175327546578,
1.2345758678942387,
-0.18983522552694923,
0.019591210301167533,
-1.3607248526627624,
0.018375533581887468,
-0.3730738776375715,
1.0123537416913542,
0.7412601688930508,
-1.6525919628732422,
-0.8341682190885938,
0.04519729634722039,
-0.06718028327556408,
-1.5649719285752928,
-0.37531851249345394,
-0.11822841604899964,
0.008154106549025474,
-1.4601449692077721,
-0.0030993446167428395,
1.462573870839384,
-0.11887757571245174,
-0.09361960188417112,
-1.2543408854038138,
-1.525280802664792,
-1.241413372948363,
0.008758808605552393,
-1.6676332942965022,
0.13826170532334445,
0.7537363962746654,
0.0833397476047736,
-0.12216949704941273,
0.02084807234445201,
-0.3421949354726438,
0.0017882611822728064,
1.5186851123669147,
-0.4603163345929323,
-0.09956972276089109,

0.7247597830417082,
0.5304935691839561,
0.003193152207841171,
0.5581346990750625,
0.010542051444639388,
0.7553167297004638,
0.5956891015800863,
-0.05566254332595716,
0.5362461335652635,
0.009895282042162012,
-0.3029901133286735,
0.09399206534914867,
0.8818812838713691,
-0.06795832284115338,
-1.580933930101608,
0.3014975711017802,
1.2092022074751856,
-0.019692531397995948,
0.9250981578351692,
0.4673383297638096,
0.03344339272872125,
0.13058472572385768,
0.036262684184263205,
0.458095375934523,
0.6492846129896516,
1.3039221898344189,
0.024764812872192767,
0.7236926243101844,
0.01444840086266692,
-1.0388939706148563,
-0.08359693344439546,
-0.4698004238001188,
-0.48055768545836347,
0.4662000815300196,
-0.041204969991243484],
[0.01798320141016013,
0.3393419127512466,
0.7047697703765823,
0.7064789163091142,
0.7191656785855808,
0.7198672068604162,
0.719867206860416,
1.0840512961729545,
1.0900003904895101,
1.1926153174924548,
1.867054342490295,
1.867054342490295,
2.368094916985588,
2.3683751276938714,
3.6383003552329964,
3.642527762446636,
4.505845683894636,
4.602446412902637,
4.972427659100928,
4.972427659100928,
5.1148164150678035,
5.8066136392845,
5.820681023327439,
6.727088267818048,
8.352653016300794,
8.773334122756978,
9.065902540064004,
9.065902540064004,
9.74880092287043,
11.30145464466351,
-0.13904770480884485,
0.23493722133539818,
0.4497803172772882,
-0.640584995353188,
0.9779304528121034,
0.0011745349607135795,
-0.7175493520168287,
0.373620380836484,
0.08111255661159392,
0.6653458367314794,

0.0037967443269442437,
0.6286122183467432,
-0.03825426016359485,
0.6896824961465278,
-0.23916946602734285,
0.4624850989736848,
0.1419493525208794,
0.7145176662978019,
-0.316909964913161,
0.6605458698076423,
-0.39309909050950365,
0.7623048158369822,
0.7469736169282462,
1.2400892032597353,
0.12266455668869777,
0.19578648318487166,
-0.784043232212839,
-0.10334183265831844,
0.32774480702026265,
-0.011491906972883237,
0.322988869457402,
-0.2027870428092001,
0.1281571522083961,
-0.1947961113336468,
1.7114513053475526,
0.005425684515457419,
0.8136825826883561,
-0.25903007981373005,
-0.5015428579892063,
-0.2600978115589891,
-0.15405473476720474,
-0.4207498663011112,
1.04622682040645,
-0.8642339256273757,
0.6248691295250803,
-0.3539240130838131,
0.011689879721248103,
-0.3068283940685192,
-0.6048660879777619,
-0.11704784670887185,
-1.2745062338511486,
-1.7890038533223112,
-0.7818671277256136,
-0.8877455887286905,
0.3164818198590212,
-0.27477307924208777,
1.8083251930535746,
-0.5031479272825263,
-0.9648993597903888,
-0.41682962131770707,
0.27040705073896654,
1.0892900815418987,
0.17779960873468054,
0.020337033950376556,
1.0551529464315492,
-0.10705558118147128,
-0.8851977322505518,
0.2232945485845247,
-0.5850403569085746,
-0.7690595025978197,
-0.4391673030332381,
-0.9309566421919284,
0.2973030450025941,
0.6780545449896137,
0.14045373030826291,
-0.5109313900859794,
-0.44070010469428217,
1.6940975579604587,
0.17062304524159472,
-0.402681393200546,
-0.7091187255059627,
1.3132357546586793,
0.2729391261599054,
1.0115940175088693,
1.0046673925246778,

0.4982257464819494,
0.04454679168285006,
0.4863023994043421,
-1.1433261311581624,
-1.820308437409531,
0.6035405700431521,
0.04003907881711185,
2.99381748600971,
-1.0437523456620463,
-0.22517480393070408,
0.10746937069246698,
-0.8788513722547997,
1.286531186523234,
-1.3157160458746164,
-0.6979748907629274,
-0.16573593981190854,
0.2529194783090185,
-0.2562517800846549,
-0.7966921479109774,
-0.4446940958312971,
-1.052310031662883,
-0.1506245930250076,
0.1615975776996014,
2.9220364155549214,
-1.7197843955783065,
-0.6592103621593355,
-0.2007498223803255,
-1.6188873615119244,
-0.15193115939751467,
-0.20866967970071015,
1.2910791163230286,
-0.4917152660712325,
-0.42348313202027765,
0.3518303936962707,
-0.10034486495238693],
[0.009039772647041902,
1.1400526086001903,
1.466905028683042,
1.6524670085313564,
1.987797816914966,
1.9900950346150876,
2.1413202332332344,
2.390935514217652,
2.402429399816812,
2.81913686416337,
2.81913686416337,
3.5656810200683298,
3.5656810200683298,
3.5656810200683298,
3.76603447051701,
3.76603447051701,
5.499236323276389,
5.4992363232763894,
5.4992363232763894,
5.532789537471719,
5.532789537471719,
7.371733777832876,
7.371733777832876,
7.680783486275433,
7.706623781446164,
7.706623781446164,
8.457742149729329,
8.457742149729329,
8.95222736351662,
8.95222736351662,
0.18928393791812126,
1.0354889043700823,
0.3703113207409527,
0.30988589523494625,
-0.34669340410550803,
0.09615284519636855,
0.1011898884094612,
0.6160115432573767,
0.6148540393435122,
0.07934943483664206,

0.8557624329627399,
-0.7672633290674925,
-0.14188111026081618,
0.1995320641320393,
0.4594888802421805,
-0.03593187081620776,
1.2926591415659605,
0.31433950962447343,
0.8053272546604554,
0.6634815559763872,
0.311660786273531,
0.0294653444200854,
0.3044394736040779,
0.7599259037372076,
0.22962735656034647,
0.22562971232722423,
0.29511485598499815,
0.02260075636691185,
-0.9497096253675098,
-0.5331371571917215,
0.2192512323868535,
1.104832430155296,
0.8636617903941131,
-0.6756684630428217,
0.4169844309263729,
1.1297527651723116,
-1.160118439748996,
-0.7971836651224,
1.9169656561991728,
-0.007260796290839314,
0.3614132074474742,
0.6243022778229537,
1.8335025630018775,
0.35139722497115833,
-0.7247870129359816,
-0.6744811297738377,
0.49997538785346773,
1.4139521212248116,
-3.0799818401546553,
-0.14768564076593044,
0.06251500852230953,
-0.8547134606957393,
-0.30107856776262504,
-0.7654972812734481,
0.28603020918627825,
0.2129303248959857,
-1.2077141264570097,
-0.8488549318812054,
0.6492574804166855,
0.7751290676155554,
0.40909519807061084,
-1.2750630737105486,
0.4187798718836698,
0.33498424523924863,
-0.19017028209895162,
-0.03232931911977397,
1.0621483440417678,
0.452017889610544,
1.0192767515800205,
0.7290563184502946,
-0.5286462788982386,
-0.05512372609304388,
1.3189151919853837,
0.5865235617306253,
-0.579843833868396,
0.6980158293666034,
0.027004860579276476,
1.2876675233834314,
-0.1718561727774212,
-0.3827017785428823,
-0.7286404996479297,
1.1682112525473984,
0.22765379123755217,
1.4870969768845252,
1.1642898801914527,

-0.03976972562460009,
-0.8163296820868416,
-0.15318057329999143,
0.0482918949035788,
-0.06014345828968787,
1.1478583262090496,
-0.005141116181289305,
-0.5820859555621626,
1.086519910329415,
-1.08558016317555,
1.531511042803357,
0.042734048269580674,
0.7452599788242152,
-1.3094923903018465,
-0.9012205175381317,
-0.0755166851159581,
1.5493661520767013,
-0.2278023712186688,
0.8627212805492942,
1.8523276191392382,
-0.10961709979291216,
0.6746606852631131,
0.22566845572735392,
0.5920154791154506,
1.2709342533151342,
0.26903045177842116,
0.11724485476711638,
0.621942277674204,
0.15491924869980966,
1.472217084758381,
0.8975149794611956,
0.3601026339731485,
1.3234513983254743,
0.0182465793142515,
-0.11914786829986494],
[-5.765348397008027e-18,
0.3428984748224433,
0.3655450682584724,
0.704784031551808,
1.5409081472811441,
1.549300366088437,
2.3787666619954635,
2.3787666619954635,
2.8808650819472406,
2.884987740396357,
3.1308808834009203,
3.1363836212549696,
4.869321728501938,
4.869321728501938,
6.1150174264690085,
6.2582278136815175,
6.271636236200362,
6.551203581243674,
6.59741701640448,
8.230564833747417,
8.294003346338926,
8.31431753208387,
8.369586036193128,
8.415319617883744,
8.70447046242739,
8.709120596281233,
8.918659758853744,
8.99410216914391,
9.002256766523654,
10.218690251738174,
-0.6798627919679444,
-1.1436724543472427,
-0.1278269606092795,
-0.9286552360068459,
0.6524046490818174,
0.06134572949387961,
-0.49761179900021657,
-1.3918471119443894,
0.5858287319319103,
0.18669127045720346,

-0.4586789737568629,
0.2400608953322058,
0.022222124883641912,
0.9282004764805514,
0.6952358394741303,
0.805684979006905,
0.6146729452742722,
0.6777524139082742,
-0.8117842884117104,
0.3426849285038029,
0.7100299235979358,
0.7658186509139059,
-0.05418633990950139,
0.1761917490887145,
0.5744825473612316,
0.31705937472296297,
-0.9188523464356001,
-0.7337446596965294,
0.2061368312243915,
-0.6645694031850036,
-0.009871104765786294,
-0.14388187083546433,
-1.2485707106311008,
2.849540809212292,
-0.5119215846255833,
-0.19381894062193808,
-0.17310717674563775,
0.8465397629359774,
-0.46757240698237135,
-0.05607605828560263,
-0.40986137011816176,
0.07649505002491204,
-2.1522907800348112,
-0.6653346582594515,
0.4084801543759574,
0.4028897820911683,
0.16908117843954765,
-0.07217026591146015,
-0.72804755296126,
-0.13548738477012795,
-2.590679566047418,
-0.40743076961744734,
1.1030856800887578,
0.08388380440455438,
0.25224805905642333,
0.10644300798076774,
1.0474125729404944,
-1.1459016177859003,
-1.6916287310468103,
-0.9886067106540645,
0.24182274960650893,
-0.25703819491337826,
-1.1942818044595898,
-0.7763678643520989,
-0.916718557142268,
0.20086626876005667,
-0.3711023890817823,
-0.006962349053602254,
2.687031223814704,
0.3970791704566904,
-0.6890591654410289,
1.0504536719091986,
-1.2458238157294268,
-1.197330311624414,
-0.12155955689525438,
-0.2969718406476498,
0.15406921803787998,
0.9984735505197111,
-0.2734456365103056,
0.0100494123830882,
0.16172873483083283,
-0.6780409734277772,
0.4539997995464368,
0.7558576390626953,
0.9556605206819043,

-0.10572075101403118,
1.5564628686733721,
0.05397552837650759,
0.7415987024402694,
-0.1648500682327342,
1.052699139325594,
0.6776572023171561,
0.15219630541397192,
0.8843933196559655,
1.3948099389813178,
-0.16710315713046545,
1.2199200746863228,
-0.3825678575328573,
-0.26766985045935626,
2.5777182352113908,
0.23156136798610805,
-0.3377548185037014,
1.0689111835439342,
-0.17216022970038714,
-0.29381807805976945,
-0.5689533077540762,
0.30596827309967156,
0.5260993417881752,
-0.6768480303494689,
-0.1092430795457648,
1.5442082488768039,
-0.10826504240549732,
-0.9555598065181787,
0.5334835330029429,
0.5569227134357198,
1.2665483894271157,
-1.9677332823608626,
1.5559931182978979,
0.7709453943956464,
1.5746506000831744],
[1.7625089228179507e-05,
1.298540182320117,
1.3141870420194457,
1.3305471045820527,
2.0686735179487985,
2.23810597518972,
2.267373737425337,
2.295441377067579,
2.3325388050051976,
2.352484360014207,
2.370891483796666,
3.3117124844820194,
3.363605896627787,
3.858981460194034,
3.8589814603490344,
3.859537867870235,
3.8595381713444072,
4.26458571773699,
4.264585717738593,
6.599549149586233,
6.655963438031886,
6.939612414177192,
6.939612414268938,
7.004302986969205,
7.010544991093791,
7.97314828104894,
7.999280802101656,
8.022628296732579,
8.027896471579398,
10.077254342713585,
-1.0733166032270363,
0.4264387306682,
-0.29490479134596703,
-0.45181113351566254,
-0.9496712390342118,
-0.24412788786515252,
-0.17005038539710493,
2.131747001055919,
-0.8801477965160032,
0.1985900676052623,

-0.17829180427317373,
0.1513896817808816,
-0.2996418627583153,
0.09152392337611555,
1.0155053498240294,
0.6713280773462803,
-0.5915672184524843,
-0.21780814909773427,
-0.23676595566314207,
-0.1539223229682568,
0.9733737893230769,
-0.8325690111887051,
-0.6684349693104433,
0.944923850675747,
-0.8270803600282979,
0.4869668407293288,
0.07342964610078141,
-0.8928903181234079,
-0.04262248204617692,
-0.7602498290394286,
-0.48858825463899247,
1.7340528561515622,
0.7340502315717261,
0.5163401187759543,
-0.5015858040228415,
0.2890335055344555,
-0.46033506221820203,
-0.8185907107929625,
0.4557079379705311,
1.6601520871164626,
0.07125306723268252,
-1.4035119795986863,
-0.6741575848895364,
1.84360490583597,
1.6836338044245691,
-0.33329370458317326,
0.19854317161902654,
2.03917897859608,
-0.3342251938073038,
-1.6015864477635742,
0.35422488479060155,
0.1487121545506934,
-0.2821114888507261,
0.21869934797564045,
0.7523961573071125,
1.3021785537923125,
0.26184859002187644,
1.2441238228144538,
0.039670539596101485,
-0.15687777352236798,
1.1386642271664005,
-0.13559116971858365,
0.39167499360215746,
2.4453261619465274,
-1.0890320328395309,
-0.4289422005081893,
1.425249318116838,
-0.0016733698174337221,
0.023121567661497415,
-0.28325087531692295,
0.940069260590291,
1.163437407349777,
-0.4256527399896147,
-1.4972731615803316,
-0.12624975701746496,
-0.10144651691714318,
1.3307378561466523,
0.041778278155808674,
-1.2746374184036902,
-1.2216848741140482,
1.4048769664929757,
0.8957112974734468,
1.0058068981677006,
-3.0965653057688316,
-0.3344448337959029,

-1.0948102803209163,
-0.038478236121346335,
0.12618687504486675,
-0.0850775611793361,
1.33550885346982,
-0.04334772196309596,
0.2406908654957122,
0.7327249608913058,
0.9966290850109706,
2.2852565843192423,
-0.030570948934034583,
-0.5566407461361556,
0.813246653395058,
0.5946355273642772,
0.07519254658407079,
0.27120989268694123,
-0.15415722381241237,
-1.1239288818585278,
1.893751281002709,
0.7530205722182245,
1.4454588921082565,
0.15037300343080462,
0.1793867334099716,
0.3587259643057371,
-0.006799349268608,
1.0907160355013774,
-0.30186717991666434,
0.8507589785411056,
0.009155709268365956,
-0.8165113788656561,
0.011816515180360782,
1.116430418883973,
1.5343789763879045,
-0.1682835166498154,
-1.5032827361947747],
[0.10956969985098665,
0.4360115975982133,
0.5932904704915389,
0.6050451872391112,
1.3688142364348175,
1.3737597367805747,
1.5226773273508583,
1.5589271816733294,
3.150320547140947,
3.1613053536766147,
4.7635234433865605,
4.823148382609575,
4.823188243925802,
7.171157860626202,
7.171157878485542,
7.171158529865571,
7.175073431930883,
7.183070822592498,
8.102400589897519,
8.10392263292654,
8.103940780725882,
9.021308924987686,
9.02133947231205,
9.082831297490506,
9.094868565409332,
9.236804949782064,
9.237029573386183,
9.26430291432718,
10.00744744746223,
10.032040374677464,
-0.5042382729215367,
1.0180015436638659,
0.10423420212375085,
-0.3355687578372037,
-0.3902319222280216,
-0.09867602213718178,
0.3860296418386605,
0.8522828003851446,
-0.7730596965193267,
0.08803782521712378,

1.175277695996097,
-0.3769225558693855,
-0.4553519820435133,
0.4758962487375975,
-1.0517577693418567,
0.3712029741888282,
0.27334544561774066,
0.03397030216872572,
0.3891396532807883,
0.12152393226873999,
1.5192516885791116,
-0.5124931636551364,
-0.5667003913709334,
-0.08425393582345661,
-0.8152956002657638,
-0.008087141654348068,
-0.20833380477584987,
-0.19186344764752156,
0.309463849209865,
-0.16555810903618162,
-0.28534485935787846,
0.025710068868521854,
-0.9172960495760906,
0.027100436879331523,
-0.9806057417125867,
0.46877133554227174,
0.22597640762440013,
-1.8700902599315918,
-0.6222133609404971,
0.023922176634194332,
-0.6777524378245746,
-0.3883818313368169,
-0.19921738892417568,
-1.3418496115619054,
-0.05365669225714106,
0.38835985529462186,
0.46741217514915123,
-0.1388983196762152,
1.0002905156718758,
1.6633296566254185,
-1.0097136763815113,
1.203957772036831,
0.34033273848251194,
1.1456357751380877,
-0.4637199651146574,
-0.13624703361320328,
0.6785440200843139,
0.21691345067474185,
-0.4166036808107477,
1.6003722740710606,
-0.294881370724797,
0.04198878790454511,
-0.689532047804753,
-1.3911901894570817,
-1.1667039833298627,
0.824824800006074,
0.8047687705752488,
0.5727196814509768,
0.2385364014798047,
-0.512164832149354,
0.17925141295137917,
-0.19704824673887428,
-1.6372757558452697,
0.5441675736491897,
-0.13936070677778437,
1.3935144064781402,
0.10763856884501717,
0.6679788244416961,
-0.33118654939932857,
-0.35796695002153883,
0.2632208027041665,
1.5831503864671255,
-0.25140146740224,
1.135774876905931,
0.5897962040478675,

-0.7664904031063712,
1.092649076983624,
-0.9862462517151122,
0.010216869101717933,
0.5705833807099128,
0.09839127014019662,
-1.034770130526523,
0.20582133073730807,
-0.44428721001967564,
0.5272444819968604,
-0.2627660867388947,
-0.9857421779520247,
-1.3646538516413942,
0.14779052759756187,
1.1207932022043874,
0.30427416833245535,
-1.055109005337564,
0.30548258474769546,
0.8153937738463674,
0.5082537250510629,
-0.2973639305797175,
0.0034742209991892575,
1.5168338311712208,
0.0971904670135119,
0.5305672799342332,
-0.08170159551874663,
0.4909203029282298,
-1.1771354207712392,
0.038725619801271274,
-0.6655680746562402,
1.312626970471223,
-1.0115568894576918,
0.16290676730453105,
0.3074286838975462,
0.013117345918778788],
[0.4866654913190249,
0.4866654913190249,
1.4314116389487666,
1.4314116389487666,
1.548608655381947,
1.5552664901547235,
1.662920497545568,
1.676155881591226,
1.6952532774325564,
1.6952532774325564,
2.46741851740979,
2.50102877426778,
3.636145431120801,
3.6361454311208004,
3.6361454311208004,
3.6361454311208004,
4.0161273218482085,
4.0161273218482085,
4.5411636896776235,
4.5411636896776235,
4.922545565757144,
5.624338470516907,
5.642847901355697,
5.904037934138086,
5.915382265322609,
6.860930229950661,
6.860930229950661,
9.422517992567101,
9.467623480195014,
9.467623480195014,
0.40091998571272736,
-0.3581846438287994,
-0.32033246941904975,
0.24409866553117068,
0.176560178011825,
0.4452285262890565,
-0.37150326481454765,
-1.0185408309777293,
1.6197684873006286,
2.0095194673452026,

-0.20342804182482532,
-0.4471079471088122,
0.37127155242298737,
0.4199607045713712,
0.11145558426636207,
-0.289968410859463,
0.9752004498173846,
-0.11972987215087512,
-0.8215210043660294,
-0.4144105549344152,
0.04966095047933562,
0.6091224412599766,
-0.2481186610805342,
-0.9036735819055022,
0.22922493224752044,
-0.04567884568114995,
-0.40529506738584414,
-0.44609185937563545,
1.1000542207972743,
0.8995834380092431,
-0.4391184290327733,
-1.623814742575614,
0.11121717167805684,
-0.15344289124808663,
0.7072346226342329,
0.013911394934765222,
1.1008239636292285,
1.1342299046285917,
-0.1605718532112884,
-0.8521154510350495,
-0.9726520318623231,
-0.11078467990736285,
1.2893067940177063,
-0.8656847782514395,
-0.4554157535410911,
0.46752563535079134,
1.0115326820592876,
-0.017465678177200683,
0.7555243091795097,
0.2225241128439446,
-0.6129237770264853,
0.36249038141661666,
-0.1470816778834455,
-0.23781939745964606,
0.7410640700988803,
-0.10219672891678527,
-0.9043863030697507,
-0.054446492384892985,
-0.23618927113386157,
-0.3971482428910102,
0.42980979959963467,
-0.4511512430809565,
-0.3322123782853245,
0.15031459204805087,
-1.4287283715133035,
-0.026859055017604255,
0.20826999854501216,
-0.016397767746330064,
-0.026711091547853946,
0.15899307013846248,
0.18349959987180348,
-0.019181103141429404,
-1.3777401647916119,
0.05578672583947999,
-0.17191397249005735,
0.08492618339903452,
0.2781511329827625,
1.6853098582374997,
-0.7964285169174421,
0.13628687772581302,
-0.26524580358677924,
1.3615146942088054,
-0.18163849035743365,
-1.5476669195398023,
0.2740670870741208,

```
0.19740142577560438,  
0.18567840551061798,  
0.43288784764736754,  
-0.8069341399605817,  
1.5139336080928714,  
-0.7594440385529729,  
-0.20443587392639553,  
0.0916888700165231,  
-0.03922249489745327,  
0.4135801157921025,  
-0.06731598775715468,  
0.5038720973518589,  
-0.7705715736697758,  
-1.4775324565516457,  
0.061950004909204714,  
-0.9555167624183505,  
3.1277455162888304,  
-1.137887484425651,  
-0.09852947784569874,  
0.042597844143281625,  
0.06134264537966588,  
0.3384848933516047,  
0.012002169875906457,  
1.5732417764972468,  
-0.3550969359952667,  
0.37821230776206083,  
1.4914211446267915,  
-1.2027836619303884,  
0.22421848814312553,  
-0.2072711571643093,  
0.05122655754886873,  
-1.5578500765322636,  
1.5918641150141506,  
1.2635397829742223,  
-1.2452354025631063]],  
    'run_time_sec': 43720.81102514267}}],  
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,  
    'layer': 30,  
    'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],  
        'cn': [ 0.5,  
            0.5,  
            0.5,  
            0.5,  
            0.5,  
            0.5],  
        'r': 0},  
    'time': {'max': 10.0, 'min': 0.0},  
    'type': 'xy-iss'}},  
    'init_param': { 'value': [ 1.5680106786222674e-17,  
        1.0238650999177359,  
        1.0238650999177359,  
        1.0238650999177359,  
        1.7041550191762513,  
        2.070931805530806,  
        2.586234954306046,  
        2.6265616878606783,  
        2.7566136785306155,  
        2.7689271757054255,  
        4.271831894498143,  
        4.271831894498143,  
        4.271831894498143,  
        5.004110362344359,  
        5.004110362344359,  
        5.529689350278954,  
        5.529689350278954,  
        5.529689350278954,  
        5.5493166460873375,  
        7.855838962179162,  
        7.855838962179162,  
        8.508023697610218,  
        8.540796404638465,  
        8.940056981382925,  
        8.940056981382925,  
        8.940056981382925,  
        8.940056981382925,
```

9.524293982069336,
9.524293982069336,
9.815045396471412,
0.717685470636315,
0.29125971197830336,
0.013619559122678915,
-0.46174394022911175,
-0.7802589318548062,
1.8785332138943256,
-0.07718385116900997,
-0.4181315830102034,
0.3859499067718677,
0.7218523694763775,
0.16350787496293626,
-0.06274337543165032,
1.048639096314506,
0.9454500907927006,
-0.18328208251477485,
0.5336296017995852,
0.7047960543548563,
-0.4717130073686535,
-0.14327049451726778,
0.5646728024946054,
0.1377201649736573,
1.0845662084088512,
0.7117671017527167,
1.8562916393289055,
-1.0958742208007046,
0.314756410152813,
1.0055290931424188,
-1.2653176839886926,
-0.6435463058390287,
-0.13689785210195535,
0.5185042269644529,
-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,

```

-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455}},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '

```

```
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                             '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                             '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                             '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                             '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                             '0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.515742960091852],
                              [12, 6, 21, 8, -3.36492111003702],
                              [20, 10, 35, 16, -1.7545658776574036],
                              [28, 14, 49, 24, -0.9139478342631155],
                              [36, 18, 63, 32, -0.46803177808311025]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 7, 0],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                              False,
                                                                              'value':
                                                                              0},
                                                  'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                  'status': True,
                                                  'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [12, 3, 3, 8],
                                'identity_factors': [1, 1, 1, 0],
                                'noise_level': [12, 6, 21, 8],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                              False,
                                                                              'value':
                                                                              0},
                                                  'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                  'status': True,
                                                  'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [20, 5, 5, 16],
                                'identity_factors': [2, 2, 2, 0],
                                'noise_level': [20, 10, 35, 16],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                              False,
                                                                              'value':
                                                                              0},
                                                  'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                  'status': True,
                                                  'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [28, 7, 7, 24],
                                'identity_factors': [3, 3, 3, 0],
                                'noise_level': [28, 14, 49, 24],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                              False,
                                                                              'value':
                                                                              0},
                                                  'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                  'status': True,
                                                  'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [36, 9, 9, 32],
                                'identity_factors': [4, 4, 4, 0],
                                'noise_level': [36, 18, 63, 32],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                              False,
                                                                              'value':
                                                                              0}
                                                    }
                              ]
          }
```

0},

```
                                'noise_prob': [ 0.001,
                                                  0.001,
                                                  0.001,
                                                  0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4}},
    'run_time_sec': 60.80421566963196}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.0019047787007362745,
                              0.7201496918445128,
                              0.7201496918445128,
                              0.7201496918445127,
                              0.7226708680142584,
                              0.8068041864232433,
                              1.3847174673528184,
                              1.6031895944534578,
                              2.458865333289775,
                              2.458865333289775,
                              2.721120236828938,
                              2.721120236828938,
                              3.554687304897056,
                              3.554687304897056,
                              4.537252190232355,
                              4.537252190232355,
                              5.239770593633025,
                              5.242363659304797,
                              5.566731623636481,
                              5.566731623636481,
                              5.566731623636481,
                              5.566731623636481,
                              5.566731623636481,
                              7.6484324534494865,
                              7.6484324534494865,
                              8.96757120343865,
                              9.352637118466301,
                              9.381932224950582,
                              9.966925955531599,
                              9.966925955531599,
                              0.29840004251263574,
                              0.3207087147269936,
                              -0.22478909909228914,
                              -1.2242611593120782,
                              0.061748844824383295,
                              1.9931455249507801,
                              0.02449362327304914,
                              -0.23402928130657288,
                              0.3399922671860315,
                              0.2355005268966802,
                              0.7912945290539132,
                              -0.9147564287830934,
                              0.12280724330167514,
                              0.42016875545313104,
                              0.6183328009701774,
                              0.9098148543333162,
                              0.2439503202728901,
                              0.4941740976390827,
                              0.0721485641264495,
                              0.20051214721509877,
                              0.14273239312798908,
                              -0.03846070968482946,
                              0.11474641891065286,
                              0.1552743298359438,
```

-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,


```

-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.536182288327376],
[12, 6, 21, 8, -3.4500813502488756],
[20, 10, 35, 16, -1.8650932993173603],
[28, 14, 49, 24, -1.0247351729167147],
[36, 18, 63, 32, -0.5670015302617211]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
'value':

```

```

False,
0},
{
  'gates_num': [12, 3, 3, 8],
  'identity_factors': [1, 1, 1, 0],
  'noise_level': [12, 6, 21, 8],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
  { 'gates_num': [20, 5, 5, 16],
    'identity_factors': [2, 2, 2, 0],
    'noise_level': [20, 10, 35, 16],
    'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
    { 'gates_num': [28, 7, 7, 24],
      'identity_factors': [3, 3, 3, 0],
      'noise_level': [28, 14, 49, 24],
      'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
      { 'gates_num': [36, 9, 9, 32],
        'identity_factors': [4, 4, 4, 0],
        'noise_level': [36, 18, 63, 32],
        'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4}},
    'run_time_sec': 49.731422901153564}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.33528943002425843,
                              0.8608925283648261,
                              2.8309935233222867,
                              2.8382233121900358,
                              2.847558790018716,
                              2.8477263794127468,

```

3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
5.127462455084763,
5.128383696390232,
5.132051241075024,
5.132231848473024,
5.132297186784173,
5.132297186784173,
5.132297186784173,
5.132320898992942,
5.134903164275234,
5.734798022200061,
5.734798022200061,
5.734798022200061,
6.71748919093495,
6.71748919093495,
9.773430860122208,
9.791656954875775,
9.791980525916317,
9.804782964874805,
9.947289619134542,
0.6818618664375632,
0.42253658736184846,
0.19067339209915313,
-2.314563573529872,
-1.7784854373672185,
-0.10945476580518654,
0.08980428543050965,
0.6265702304745648,
-0.12016219313034968,
0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,
-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,

```
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
```



```

False,
0},

'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
  'identity_factors': [3, 3, 3, 0],
  'noise_level': [28, 14, 49, 24],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                                'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                'status': True,
                                                'type': 'depolarizing'},
  'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
  'identity_factors': [4, 4, 4, 0],
  'noise_level': [36, 18, 63, 32],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                                'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                'status': True,
                                                'type': 'depolarizing'},
  'odd_wires': 4}},
  'run_time_sec': 47.62909483909607}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.00030922277578629,
                              0.881375055029912,
                              0.881375055029912,
                              0.8813750550299119,
                              0.9803845151714838,
                              0.9803845151714838,
                              1.201629973822729,
                              1.2441077564052723,
                              1.8660677974280824,
                              1.8660677974280824,
                              2.651002314516478,
                              2.651002314516478,
                              3.233637461790533,
                              4.36384995486056,
                              4.36384995486056,
                              4.975714528653046,
                              4.975714528653046,
                              6.591840750700408,
                              6.591840750700408,
                              8.234009055219385,
                              8.238223602720305,
                              8.238223602720305,
                              8.238223602720305,
                              8.903406631369169,
                              8.903406631369169,
                              8.903406631369169,
                              9.126449234760914,
                              9.126449234760914,
                              9.441918167657187,
                              9.441918167657187,
                              0.4384156956352651,
                              -0.017407483354635628,
                              -0.1357940797759564,
```

-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,

[illegible]


```
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '  
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '  
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']}]},  
    'output': { 'data_points': [ [4, 2, 7, 0, -6.5111758480272135],  
                                  [12, 6, 21, 8, -3.513108400075424],  
                                  [20, 10, 35, 16, -1.9256894730075684],  
                                  [28, 14, 49, 24, -1.0648100061563792],  
                                  [36, 18, 63, 32, -0.5886022016151555]],  
      'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
                           'identity_factors': [0, 0, 0, 0],  
                           'noise_level': [4, 2, 7, 0],  
                           'noise_profile': { 'noise_on_init_param': { 'status':  
                                                                           'value':  
                                                                               'noise_prob': [ 0.001,  
                                                             0.001,  
                                                             0.001,  
                                                             0.001],  
                                                           'status': True,  
                                                           'type': 'depolarizing'}},  
                           'odd_wires': 4},  
                          { 'gates_num': [12, 3, 3, 8],  
                           'identity_factors': [1, 1, 1, 0],  
                           'noise_level': [12, 6, 21, 8],  
                           'noise_profile': { 'noise_on_init_param': { 'status':  
                                                                           'value':  
                                                                               'noise_prob': [ 0.001,  
                                                             0.001,  
                                                             0.001,  
                                                             0.001],  
                                                           'status': True,  
                                                           'type': 'depolarizing'}},  
                           'odd_wires': 4},  
                          { 'gates_num': [20, 5, 5, 16],  
                           'identity_factors': [2, 2, 2, 0],  
                           'noise_level': [20, 10, 35, 16],  
                           'noise_profile': { 'noise_on_init_param': { 'status':  
                                                                           'value':  
                                                                               'noise_prob': [ 0.001,  
                                                             0.001,  
                                                             0.001,  
                                                             0.001],  
                                                           'status': True,  
                                                           'type': 'depolarizing'}},  
                           'odd_wires': 4},  
                          { 'gates_num': [28, 7, 7, 24],  
                           'identity_factors': [3, 3, 3, 0],  
                           'noise_level': [28, 14, 49, 24],  
                           'noise_profile': { 'noise_on_init_param': { 'status':  
                                                                           'value':  
                                                                               'noise_prob': [ 0.001,  
                                                             0.001,  
                                                             0.001,  
                                                             0.001],  
                                                           'status': True,  
                                                           'type': 'depolarizing'}},  
                           'odd_wires': 4},  
                          { 'gates_num': [36, 9, 9, 32],  
                           'identity_factors': [4, 4, 4, 0],  
                           'noise_level': [36, 18, 63, 32],  
                           'noise_profile': { 'noise_on_init_param': { 'status':  
                                                                           'value':  
                                                                               'noise_prob': [ 0.001,  
                                                             0.001,  
                                                             0.001,  
                                                             0.001],  
                                                           'status': True,  
                                                           'type': 'depolarizing'}}}  
False,  
  
0},  
  
False,  
  
0},  
  
False,  
  
0},  
  
False,  
  
0},  
  
False,  
  
0},
```

```

        'status': True,
        'type': 'depolarizing'},
        'odd_wires': 4]],
    'run_time_sec': 48.79940581321716}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
        'cn': [ 0.5,
            0.5,
            0.5,
            0.5,
            0.5,
            0.5],
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 3.750846163374531e-05,
    0.22285241391868899,
    0.26252411232084344,
    0.26252411232084344,
    0.26252411232084344,
    1.459545906754556,
    1.4595459067545562,
    1.6094835194149661,
    2.5927782099427628,
    2.5927782099427628,
    2.5927782099427628,
    2.5927782099427623,
    3.3552499939702187,
    3.3552499939702187,
    5.033284106467877,
    5.033284106467877,
    5.056477210452712,
    6.768913439946703,
    6.768913439946703,
    7.2782492670747585,
    7.2782492670747585,
    7.2782492670747585,
    7.2782492670747585,
    7.2782492670747585,
    7.2782492670747585,
    8.934198246659292,
    8.937369801451712,
    9.544406408519471,
    9.544406408519471,
    10.23170722265548,
    -0.4501443853573418,
    0.06519513561037263,
    -0.13386490845883003,
    -0.01542333988020039,
    0.3988203537526196,
    0.9163371790808124,
    0.08159154039220383,
    -0.11665429084087826,
    -0.21130662780886203,
    1.515236372470638,
    0.4436142129357345,
    -0.28780404896369716,
    0.3681425383234808,
    0.648176841676201,
    0.3046875502986337,
    0.05897820457762899,
    0.09316951897603123,
    1.5277657654344678,
    -0.655141628591867,
    0.073084468489131,
    -0.11423944001370272,
    0.3986898081963471,
    1.0974832156349317,
    1.8462557861113,
    0.5998463368708868,
    -0.0014214294560794885,
    0.4026665434776887,
    -0.03860309342262787,
    0.3728818533964108,

```

1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,

```

-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.4606782650454075],
[12, 6, 21, 8, -3.2738854608106305],
[20, 10, 35, 16, -1.6974907774474755],
[28, 14, 49, 24, -0.8939947022135477],
[36, 18, 63, 32, -0.4736436630774058]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
False,

```

```

                                'value':
0},
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
  'identity_factors': [2, 2, 2, 0],
  'noise_level': [20, 10, 35, 16],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
  'identity_factors': [3, 3, 3, 0],
  'noise_level': [28, 14, 49, 24],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
  'identity_factors': [4, 4, 4, 0],
  'noise_level': [36, 18, 63, 32],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4}],
  'run_time_sec': 49.173208713531494}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.08591881204796584,
                             1.5026023651616802,
                             1.9179097588555833,
                             1.9179097588555831,
                             1.9179097588555831,
                             1.9179097588555831,
                             2.8800636946459592,
                             2.9523818900173797,
                             2.9523818900173797,
                             2.954896407422686,
                             3.0487820562708485,

```

3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,

```

-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#3'},

```

```

'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 0],
                                      [2, 2, 2, 0],
                                      [3, 3, 3, 0],
                                      [4, 4, 4, 0]]},

'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},

'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},

'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
         'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                    '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                    '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                    '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                    '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                    '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},

'output': { 'data_points': [ [4, 2, 7, 0, -6.4104927992121],
                              [12, 6, 21, 8, -3.25771336700167],
                              [20, 10, 35, 16, -1.674813432318219],
                              [28, 14, 49, 24, -0.8591076530612253],
                              [36, 18, 63, 32, -0.43058689568319575]],

'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 7, 0],
                    'noise_profile': { 'noise_on_init_param': { 'status':

False,
                                'value':

'noise_prob': [ 0.001,
                 0.001,
                 0.001,
                 0.001],
                'status': True,
                'type': 'depolarizing'},
                    'odd_wires': 4},
                  { 'gates_num': [12, 3, 3, 8],
                    'identity_factors': [1, 1, 1, 0],
                    'noise_level': [12, 6, 21, 8],
                    'noise_profile': { 'noise_on_init_param': { 'status':

False,
                                'value':

'noise_prob': [ 0.001,
                 0.001,
                 0.001,
                 0.001],
                'status': True,
                'type': 'depolarizing'},
                    'odd_wires': 4},
                  { 'gates_num': [20, 5, 5, 16],
                    'identity_factors': [2, 2, 2, 0],
                    'noise_level': [20, 10, 35, 16],
                    'noise_profile': { 'noise_on_init_param': { 'status':

False,
                                'value':

'noise_prob': [ 0.001,
                 0.001,
                 0.001,
                 0.001],
                'status': True,
                'type': 'depolarizing'},
                    'odd_wires': 4},
                  { 'gates_num': [28, 7, 7, 24],
                    'identity_factors': [3, 3, 3, 0],
                    'noise_level': [28, 14, 49, 24],
                    'noise_profile': { 'noise_on_init_param': { 'status':

```



```

False,
0},
                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
  'identity_factors': [4, 4, 4, 0],
  'noise_level': [36, 18, 63, 32],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                                'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                                'status': True,
                                                                'type': 'depolarizing'},
                                                                'odd_wires': 4}},
  'run_time_sec': 48.794405460357666}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
  'init_param': { 'value': [ -5.647934061474703e-18,
                             1.4787834906243755,
                             1.478918229526628,
                             1.478918229526628,
                             2.0077351764243123,
                             2.4302809101456218,
                             2.4302809101456218,
                             3.009709563144601,
                             3.009709563144601,
                             3.239471043198664,
                             3.2405283031129133,
                             3.2405283031129133,
                             3.2656598154222434,
                             3.5353392937639825,
                             6.198713539858257,
                             6.213604956994236,
                             6.73397854893658,
                             6.73397854893658,
                             6.743547791194661,
                             6.744609103740889,
                             6.744609103740889,
                             6.749543065628153,
                             6.749543065628153,
                             8.615256229309818,
                             8.635090640841518,
                             9.012323512360366,
                             9.012323512360366,
                             9.930483613907942,
                             9.930483613907942,
                             11.782061805135282,
                             0.6870051227789642,
                             0.5234095533264759,
                             0.11216801162302814,
                             0.044591843687496906,
                             -0.747159787599118,
                             1.5727279855217766,
                             -0.16958009141159439,
                             -0.7234855106318131,

```

0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,

```

0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
[12, 6, 21, 8, -3.1990719612158447],
[20, 10, 35, 16, -1.635137691104089],

```

```

[28, 14, 49, 24, -0.8526977600673817],
[36, 18, 63, 32, -0.44953303322849647]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 7, 0],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                    { 'gates_num': [12, 3, 3, 8],
                    'identity_factors': [1, 1, 1, 0],
                    'noise_level': [12, 6, 21, 8],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                    { 'gates_num': [20, 5, 5, 16],
                    'identity_factors': [2, 2, 2, 0],
                    'noise_level': [20, 10, 35, 16],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                    { 'gates_num': [28, 7, 7, 24],
                    'identity_factors': [3, 3, 3, 0],
                    'noise_level': [28, 14, 49, 24],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                    { 'gates_num': [36, 9, 9, 32],
                    'identity_factors': [4, 4, 4, 0],
                    'noise_level': [36, 18, 63, 32],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4}],
                    'run_time_sec': 49.53232932090759}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,

```

```
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                              'cn': [ 0.5,
                                       0.5,
                                       0.5,
                                       0.5,
                                       0.5,
                                       0.5],
                              'r': 0},
          'time': {'max': 10.0, 'min': 0.0},
          'type': 'xy-iss'}},
'init_param': { 'value': [ -6.331146189926128e-17,
                             0.7069927066774304,
                             0.7069927066774304,
                             0.7069927066774305,
                             0.7069927066774304,
                             1.0044153364322925,
                             1.0092418692996383,
                             2.470333794214341,
                             2.478143749866421,
                             3.5776564981759003,
                             3.5776564981759003,
                             3.5776564981759003,
                             3.61745967653339,
                             3.6981780389472076,
                             3.755720232685741,
                             3.9712286023902594,
                             3.9712286023902594,
                             4.519037432417192,
                             4.519037432417193,
                             4.519037432417193,
                             5.280953883299406,
                             5.280953883299406,
                             6.620356980277175,
                             6.620356980277175,
                             6.620356980277175,
                             6.620356980277176,
                             8.214004051881032,
                             8.214004051881032,
                             9.105951544537387,
                             10.829366164164977,
                             1.4490195266558639,
                             1.145329650589643,
                             -0.9374272684957279,
                             -1.2910248872233119,
                             0.9701069453007352,
                             2.0238893871034924,
                             -0.6731600678051081,
                             -0.7305037462514778,
                             0.6743483023845589,
                             1.611766624162738,
                             -0.162446688913207,
                             -0.2464021583836281,
                             1.3073653685111934,
                             1.560629053845934,
                             -0.09768282308379246,
                             -0.31393750682085847,
                             0.1625705045557987,
                             -0.1344649185634244,
                             -0.03516682581352061,
                             0.15934568227883428,
                             1.4484389719925783,
                             0.31346340320241645,
                             -1.5406853809043388,
                             0.3831701903776482,
                             0.07255447956326048,
                             -0.4619144712511035,
                             0.6220101497739178,
                             -0.08645510294766677,
                             -0.036774035023737585,
                             -0.02433291891195538,
                             1.2745777688613118,
                             0.806643386980706,
                             -0.1734465019831428,
                             -1.474099767063678,
                             -1.2534035864969235,
```

0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,


```

False,
0},
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4},
    { 'gates_num': [20, 5, 5, 16],
      'identity_factors': [2, 2, 2, 0],
      'noise_level': [20, 10, 35, 16],
      'noise_profile': { 'noise_on_init_param': { 'status':
                                                              'value':
                                                                  'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                'status': True,
                                                                'type': 'depolarizing'},
                        'odd_wires': 4},
      { 'gates_num': [28, 7, 7, 24],
        'identity_factors': [3, 3, 3, 0],
        'noise_level': [28, 14, 49, 24],
        'noise_profile': { 'noise_on_init_param': { 'status':
                                                            'value':
                                                                'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                'status': True,
                                                                'type': 'depolarizing'},
                          'odd_wires': 4},
        { 'gates_num': [36, 9, 9, 32],
          'identity_factors': [4, 4, 4, 0],
          'noise_level': [36, 18, 63, 32],
          'noise_profile': { 'noise_on_init_param': { 'status':
                                                            'value':
                                                                'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                'status': True,
                                                                'type': 'depolarizing'},
                            'odd_wires': 4}],
      'run_time_sec': 48.256184101104736}},
    { 'config': { 'ansatz': { 'gateset': 1,
                              'layer': 30,
                              'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                             'cn': [ 0.5,
                                                                 0.5,
                                                                 0.5,
                                                                 0.5,
                                                                 0.5],
                                                             'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}},
      'init_param': { 'value': [ 0.3773346757087749,
                                  0.5734233884194113,
                                  0.5734233884194113,
                                  1.2094818154974225,
                                  1.2528672337957094,
                                  1.3073966674132076,
                                  4.193348178434801,
                                  4.228670385547864,
                                  4.228670385547864,
                                  4.228670385547864,
                                  4.228670385547864,
                                  4.228670385547864,
                                  4.274742575727675,
                                  5.343872214974395,
                                  5.354523589318104,
                                  5.449133113124401,
                                  5.449133113124401,

```


5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,

[illegible]


```

                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
                                { 'gates_num': [36, 9, 9, 32],
                                'identity_factors': [4, 4, 4, 0],
                                'noise_level': [36, 18, 63, 32],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                'value':
                                'noise_prob': [ 0.001,
                                0.001,
                                0.001,
                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4}},
                                'run_time_sec': 47.56407403945923}},
                                { 'config': { 'ansatz': { 'gateset': 1,
                                'layer': 30,
                                'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                'cn': [ 0.5,
                                0.5,
                                0.5,
                                0.5,
                                0.5,
                                0.5],
                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}},
                                'init_param': { 'value': [ 0.003375513600300809,
                                0.4884127443235871,
                                0.6719352108785641,
                                0.9073170135093351,
                                1.4946633690060278,
                                1.5076957064847494,
                                3.872917838142945,
                                3.872917838142945,
                                3.872917838142945,
                                3.872917838142945,
                                5.047622474508209,
                                5.054336886650782,
                                5.067626550795242,
                                5.0733174878702085,
                                5.094629387163461,
                                5.749571576552799,
                                5.7731554023304374,
                                7.144946717661993,
                                7.144946717661993,
                                7.144946717661993,
                                7.144946717661993,
                                7.602314289267704,
                                7.602314289267704,
                                8.018775147517832,
                                8.018775147517832,
                                8.018775147517832,
                                8.018775147517832,
                                8.018775147517832,
                                8.018775147517832,
                                9.661071929582006,
                                0.2735947252101626,
                                2.471262065567802,
                                0.18954222370300397,
                                0.2633058348660961,
                                0.07494334922808346,
                                1.4980157261935538,
                                0.3538857424122425,
                                0.26341770410626436,
                                0.17174804107002956,
                                -0.22912718472194316,
                                0.11060714764109056,
                                -0.30804859927082595,
                                2.8406850614439736,
                                1.0823654569610508,

```

0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,

```

0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.397817460329861],
[12, 6, 21, 8, -3.2115716730808748],
[20, 10, 35, 16, -1.641958817033255],
[28, 14, 49, 24, -0.8464644349040011],
[36, 18, 63, 32, -0.4337371704190119]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':

```

```

'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
  'identity_factors': [1, 1, 1, 0],
  'noise_level': [12, 6, 21, 8],
  'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
  'identity_factors': [2, 2, 2, 0],
  'noise_level': [20, 10, 35, 16],
  'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
  'identity_factors': [3, 3, 3, 0],
  'noise_level': [28, 14, 49, 24],
  'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
  'identity_factors': [4, 4, 4, 0],
  'noise_level': [36, 18, 63, 32],
  'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 50.0461266040802}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': {'value': 'random'},

```


5.529689350278954,
5.529689350278954,
5.529689350278954,
5.5493166460873375,
7.855838962179162,
7.855838962179162,
8.508023697610218,
8.540796404638465,
8.940056981382925,
8.940056981382925,
8.940056981382925,
8.940056981382925,
9.524293982069336,
9.524293982069336,
9.815045396471412,
0.717685470636315,
0.29125971197830336,
0.013619559122678915,
-0.46174394022911175,
-0.7802589318548062,
1.8785332138943256,
-0.07718385116900997,
-0.4181315830102034,
0.3859499067718677,
0.7218523694763775,
0.16350787496293626,
-0.06274337543165032,
1.048639096314506,
0.9454500907927006,
-0.18328208251477485,
0.5336296017995852,
0.7047960543548563,
-0.4717130073686535,
-0.14327049451726778,
0.5646728024946054,
0.1377201649736573,
1.0845662084088512,
0.7117671017527167,
1.8562916393289055,
-1.0958742208007046,
0.314756410152813,
1.0055290931424188,
-1.2653176839886926,
-0.6435463058390287,
-0.13689785210195535,
0.5185042269644529,
-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,

0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455],
[0.33528943002425843,
0.8608925283648261,
2.8309935233222867,
2.8382233121900358,
2.847558790018716,
2.8477263794127468,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
5.127462455084763,
5.128383696390232,
5.132051241075024,
5.132231848473024,

5.132297186784173,
5.132297186784173,
5.132297186784173,
5.132320898992942,
5.134903164275234,
5.734798022200061,
5.734798022200061,
5.734798022200061,
6.71748919093495,
6.71748919093495,
9.773430860122208,
9.791656954875775,
9.791980525916317,
9.804782964874805,
9.947289619134542,
0.6818618664375632,
0.42253658736184846,
0.19067339209915313,
-2.314563573529872,
-1.7784854373672185,
-0.10945476580518654,
0.08980428543050965,
0.6265702304745648,
-0.12016219313034968,
0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,
-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,

0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358],
[0.08591881204796584,
1.5026023651616802,
1.9179097588555833,
1.9179097588555831,
1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,

5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,

-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406],
[0.3773346757087749,
0.5734233884194113,
0.5734233884194113,
1.2094818154974225,
1.2528672337957094,
1.3073966674132076,
4.193348178434801,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.274742575727675,
5.343872214974395,
5.354523589318104,

5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,

1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244],
[-6.331146189926128e-17,
0.7069927066774304,
0.7069927066774304,
0.7069927066774305,
0.7069927066774304,
1.0044153364322925,
1.0092418692996383,
2.470333794214341,
2.478143749866421,
3.5776564981759003,
3.5776564981759003,
3.5776564981759003,
3.61745967653339,
3.6981780389472076,
3.755720232685741,

3.9712286023902594,
3.9712286023902594,
4.519037432417192,
4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,
6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,

0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176],
[3.750846163374531e-05,
0.22285241391868899,
0.26252411232084344,
0.26252411232084344,
0.26252411232084344,
1.459545906754556,
1.4595459067545562,
1.6094835194149661,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
2.5927782099427623,
3.3552499939702187,
3.3552499939702187,
5.033284106467877,

5.033284106467877,
5.056477210452712,
6.768913439946703,
6.768913439946703,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,

-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138],
[0.003375513600300809,
0.4884127443235871,
0.6719352108785641,
0.9073170135093351,
1.4946633690060278,
1.5076957064847494,
3.872917838142945,
3.872917838142945,
3.872917838142945,
3.872917838142945,
5.047622474508209,
5.054336886650782,
5.067626550795242,
5.0733174878702085,
5.094629387163461,

5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,

-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989],
[-5.647934061474703e-18,
1.4787834906243755,
1.478918229526628,
1.478918229526628,
2.0077351764243123,
2.4302809101456218,
2.4302809101456218,
3.009709563144601,
3.009709563144601,
3.239471043198664,
3.2405283031129133,
3.2405283031129133,
3.2656598154222434,
3.5353392937639825,
6.198713539858257,

6.213604956994236,
6.73397854893658,
6.73397854893658,
6.743547791194661,
6.744609103740889,
6.744609103740889,
6.749543065628153,
6.749543065628153,
8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,

-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935],
[0.0019047787007362745,
0.7201496918445128,
0.7201496918445128,
0.7201496918445127,
0.7226708680142584,
0.8068041864232433,
1.3847174673528184,
1.6031895944534578,
2.458865333289775,
2.458865333289775,
2.721120236828938,
2.721120236828938,
3.554687304897056,
3.554687304897056,
4.537252190232355,

4.537252190232355,
5.239770593633025,
5.242363659304797,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
7.6484324534494865,
7.6484324534494865,
8.96757120343865,
9.352637118466301,
9.381932224950582,
9.966925955531599,
9.966925955531599,
0.29840004251263574,
0.3207087147269936,
-0.22478909909228914,
-1.2242611593120782,
0.061748844824383295,
1.9931455249507801,
0.02449362327304914,
-0.23402928130657288,
0.3399922671860315,
0.2355005268966802,
0.7912945290539132,
-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,

-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815],
[0.00030922277578629,
0.881375055029912,
0.881375055029912,
0.8813750550299119,
0.9803845151714838,
0.9803845151714838,
1.201629973822729,
1.2441077564052723,
1.8660677974280824,
1.8660677974280824,
2.651002314516478,
2.651002314516478,
3.233637461790533,
4.36384995486056,
4.36384995486056,

4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,
8.903406631369169,
8.903406631369169,
8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.1357940797759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,

```

0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'run_time_sec': 68892.59480500221}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
                                  'layer': 30,
                                  'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                                'r': 0},
                                  'time': {'max': 10.0, 'min': 0.0},
                                  'type': 'xy-iss'}},
'init_param': { 'value': [ 0.003375513600300809,
                             0.4884127443235871,
                             0.6719352108785641,
                             0.9073170135093351,
                             1.4946633690060278,
                             1.5076957064847494,
                             3.872917838142945,

```

3.872917838142945,
3.872917838142945,
3.872917838142945,
5.047622474508209,
5.054336886650782,
5.067626550795242,
5.0733174878702085,
5.094629387163461,
5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,

```

-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989}],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],

```


7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,


```

0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.4606782650454075],

```

```

[12, 6, 21, 8, -3.2738854608106305],
[20, 10, 35, 16, -1.6974907774474755],
[28, 14, 49, 24, -0.8939947022135477],
[36, 18, 63, 32, -0.4736436630774058]],
    'degree': 0,
    'method': 'richardson',
    'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 0.00014090538024902344,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.359960682489763,
'others': { 'beta_coefficients': [ 2.0339490779265086,
-2.250326639408051,
1.9586176305958964,
-0.9196987134972037,
0.17745864438284967],
'cost_richardson_zne': 13.914439152583324,
'sorted_expectation_vals': [ -6.4606782650454075,
-3.2738854608106305,
-1.6974907774474755,
-0.8939947022135477,
-0.4736436630774058],
'sorted_noise': [13, 47, 81, 115, 149]},
'sampled data': [ [4, 2, 7, 0, -6.4606782650454075],
[12, 6, 21, 8, -3.2738854608106305],
[20, 10, 35, 16, -1.6974907774474755],
[28, 14, 49, 24, -0.8939947022135477],
[36, 18, 63, 32, -0.4736436630774058]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.0019047787007362745,
0.7201496918445128,
0.7201496918445128,
0.7201496918445127,
0.7226708680142584,
0.8068041864232433,
1.3847174673528184,
1.6031895944534578,
2.458865333289775,
2.458865333289775,
2.721120236828938,
2.721120236828938,
3.554687304897056,
3.554687304897056,
4.537252190232355,
4.537252190232355,
5.239770593633025,
5.242363659304797,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
7.6484324534494865,
7.6484324534494865,
8.96757120343865,
9.352637118466301,
9.381932224950582,
9.966925955531599,
9.966925955531599,
0.29840004251263574,
0.3207087147269936,
-0.22478909909228914,
-1.2242611593120782,
0.061748844824383295,
1.9931455249507801,
0.02449362327304914,
-0.23402928130657288,
0.3399922671860315,

```

0.2355005268966802,
0.7912945290539132,
-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,

[illegible]

```

        'cost_richardson_zne': 13.914439152583324,
        'sorted_expectation_vals': [ -6.536182288327376,
                                       -3.4500813502488756,
                                       -1.8650932993173603,
                                       -1.0247351729167147,
                                       -0.5670015302617211],
        'sorted_noise': [13, 47, 81, 115, 149]],
    'sampled data': [ [4, 2, 7, 0, -6.536182288327376],
                      [12, 6, 21, 8, -3.4500813502488756],
                      [20, 10, 35, 16, -1.8650932993173603],
                      [28, 14, 49, 24, -1.0247351729167147],
                      [36, 18, 63, 32, -0.5670015302617211]],
    'sampling': 'default'}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
  'init_param': { 'value': [ -6.331146189926128e-17,
                              0.7069927066774304,
                              0.7069927066774304,
                              0.7069927066774305,
                              0.7069927066774304,
                              1.0044153364322925,
                              1.0092418692996383,
                              2.470333794214341,
                              2.478143749866421,
                              3.5776564981759003,
                              3.5776564981759003,
                              3.5776564981759003,
                              3.61745967653339,
                              3.6981780389472076,
                              3.755720232685741,
                              3.9712286023902594,
                              3.9712286023902594,
                              4.519037432417192,
                              4.519037432417193,
                              4.519037432417193,
                              5.280953883299406,
                              5.280953883299406,
                              6.620356980277175,
                              6.620356980277175,
                              6.620356980277175,
                              6.620356980277176,
                              8.214004051881032,
                              8.214004051881032,
                              9.105951544537387,
                              10.829366164164977,
                              1.4490195266558639,
                              1.145329650589643,
                              -0.9374272684957279,
                              -1.2910248872233119,
                              0.9701069453007352,
                              2.0238893871034924,
                              -0.6731600678051081,
                              -0.7305037462514778,
                              0.6743483023845589,
                              1.611766624162738,
                              -0.162446688913207,
                              -0.2464021583836281,
                              1.3073653685111934,
                              1.560629053845934,
                              -0.09768282308379246,
                              -0.31393750682085847,
                              0.1625705045557987,
                              -0.1344649185634244,
                              -0.03516682581352061,
                              0.15934568227883428,
                              1.4484389719925783,
                              0.31346340320241645,
                              -1.5406853809043388,
                              0.3831701903776482,
                              0.07255447956326048,

```

-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,

```

1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.311563719128683],
[12, 6, 21, 8, -3.2712898811023576],
[20, 10, 35, 16, -1.7533390824025292],
[28, 14, 49, 24, -0.966227702270059],
[36, 18, 63, 32, -0.5436512754494954]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.38690185546875e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.117886524967124,
'others': { 'beta_coeffieicients': [ 2.0339490779265086,
-2.250326639408051,
1.9586176305958964,
-0.9196987134972037,
0.17745864438284967],
'cost_richardson_zne': 13.914439152583324,
'sorted_expectation_vals': [ -6.311563719128683,
-3.2712898811023576,
-1.7533390824025292,
-0.966227702270059,
-0.5436512754494954],
'sorted_noise': [13, 47, 81, 115, 149]},
'sampled data': [ [4, 2, 7, 0, -6.311563719128683],
[12, 6, 21, 8, -3.2712898811023576],
[20, 10, 35, 16, -1.7533390824025292],
[28, 14, 49, 24, -0.966227702270059],
[36, 18, 63, 32, -0.5436512754494954]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],

```

```
        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 1.5680106786222674e-17,
    1.0238650999177359,
    1.0238650999177359,
    1.0238650999177359,
    1.7041550191762513,
    2.070931805530806,
    2.586234954306046,
    2.6265616878606783,
    2.7566136785306155,
    2.7689271757054255,
    4.271831894498143,
    4.271831894498143,
    4.271831894498143,
    5.004110362344359,
    5.004110362344359,
    5.529689350278954,
    5.529689350278954,
    5.529689350278954,
    5.5493166460873375,
    7.855838962179162,
    7.855838962179162,
    8.508023697610218,
    8.540796404638465,
    8.940056981382925,
    8.940056981382925,
    8.940056981382925,
    8.940056981382925,
    9.524293982069336,
    9.524293982069336,
    9.815045396471412,
    0.717685470636315,
    0.29125971197830336,
    0.013619559122678915,
    -0.46174394022911175,
    -0.7802589318548062,
    1.8785332138943256,
    -0.07718385116900997,
    -0.4181315830102034,
    0.3859499067718677,
    0.7218523694763775,
    0.16350787496293626,
    -0.06274337543165032,
    1.048639096314506,
    0.9454500907927006,
    -0.18328208251477485,
    0.5336296017995852,
    0.7047960543548563,
    -0.4717130073686535,
    -0.14327049451726778,
    0.5646728024946054,
    0.1377201649736573,
    1.0845662084088512,
    0.7117671017527167,
    1.8562916393289055,
    -1.0958742208007046,
    0.314756410152813,
    1.0055290931424188,
    -1.2653176839886926,
    -0.6435463058390287,
    -0.13689785210195535,
    0.5185042269644529,
    -0.07994120596020922,
    -1.3394312085309903,
    -0.01354724643952521,
    -0.8508707890759795,
    -0.7258020036409927,
    -0.1513648764845159,
    -0.5702686618598916,
    -0.20804467461705725,
    -0.059043388294177074,
    0.9724468840137966,
```


1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,

4.274742575727675,
5.343872214974395,
5.354523589318104,
5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,

```

1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],

```

```

[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0]],

'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.44845457619454],
                          [12, 6, 21, 8, -3.2446894681699203],
                          [20, 10, 35, 16, -1.6866076162118238],
                          [28, 14, 49, 24, -0.8984323471828195],
                          [36, 18, 63, 32, -0.4857082686875758]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.696846008300781e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -8.377542562540667,
                            'others': { 'beta_coefficients': [ 2.0339490779265086,
                                                              -2.250326639408051,
                                                              1.9586176305958964,
                                                              -0.9196987134972037,
                                                              0.17745864438284967],
                                          'cost_richardson_zne': 13.914439152583324,
                                          'sorted_expectation_vals': [ -6.44845457619454,
                                                                      -3.2446894681699203,
                                                                      -1.6866076162118238,
                                                                      -0.8984323471828195,
                                                                      -0.4857082686875758],
                                          'sorted_noise': [13, 47, 81, 115, 149]},
                            'sampled data': [ [4, 2, 7, 0, -6.44845457619454],
                                              [12, 6, 21, 8, -3.2446894681699203],
                                              [20, 10, 35, 16, -1.6866076162118238],
                                              [28, 14, 49, 24, -0.8984323471828195],
                                              [36, 18, 63, 32, -0.4857082686875758]],
                            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
            'init_param': { 'value': [ -5.647934061474703e-18,
                                         1.4787834906243755,
                                         1.478918229526628,
                                         1.478918229526628,
                                         2.0077351764243123,
                                         2.4302809101456218,
                                         2.4302809101456218,
                                         3.009709563144601,
                                         3.009709563144601,
                                         3.239471043198664,
                                         3.2405283031129133,
                                         3.2405283031129133,
                                         3.2656598154222434,
                                         3.5353392937639825,
                                         6.198713539858257,
                                         6.213604956994236,
                                         6.73397854893658,
                                         6.73397854893658,
                                         6.743547791194661,
                                         6.744609103740889,
                                         6.744609103740889,
                                         6.749543065628153,
                                         6.749543065628153,
                                         8.615256229309818,
                                         8.635090640841518,
                                         9.012323512360366,
                                         9.012323512360366,
                                         9.930483613907942,
                                         9.930483613907942,
                                         10.930483613907942,
                                         10.930483613907942,
                                         11.930483613907942,
                                         11.930483613907942,
                                         12.930483613907942,
                                         12.930483613907942,
                                         13.930483613907942,
                                         13.930483613907942,
                                         14.930483613907942,
                                         14.930483613907942,
                                         15.930483613907942,
                                         15.930483613907942,
                                         16.930483613907942,
                                         16.930483613907942,
                                         17.930483613907942,
                                         17.930483613907942,
                                         18.930483613907942,
                                         18.930483613907942,
                                         19.930483613907942,
                                         19.930483613907942,
                                         20.930483613907942,
                                         20.930483613907942,
                                         21.930483613907942,
                                         21.930483613907942,
                                         22.930483613907942,
                                         22.930483613907942,
                                         23.930483613907942,
                                         23.930483613907942,
                                         24.930483613907942,
                                         24.930483613907942,
                                         25.930483613907942,
                                         25.930483613907942,
                                         26.930483613907942,
                                         26.930483613907942,
                                         27.930483613907942,
                                         27.930483613907942,
                                         28.930483613907942,
                                         28.930483613907942,
                                         29.930483613907942,
                                         29.930483613907942,
                                         30.930483613907942,
                                         30.930483613907942,
                                         31.930483613907942,
                                         31.930483613907942,
                                         32.930483613907942,
                                         32.930483613907942,
                                         33.930483613907942,
                                         33.930483613907942,
                                         34.930483613907942,
                                         34.930483613907942,
                                         35.930483613907942,
                                         35.930483613907942,
                                         36.930483613907942,
                                         36.930483613907942,
                                         37.930483613907942,
                                         37.930483613907942,
                                         38.930483613907942,
                                         38.930483613907942,
                                         39.930483613907942,
                                         39.930483613907942,
                                         40.930483613907942,
                                         40.930483613907942,
                                         41.930483613907942,
                                         41.930483613907942,
                                         42.930483613907942,
                                         42.930483613907942,
                                         43.930483613907942,
                                         43.930483613907942,
                                         44.930483613907942,
                                         44.930483613907942,
                                         45.930483613907942,
                                         45.930483613907942,
                                         46.930483613907942,
                                         46.930483613907942,
                                         47.930483613907942,
                                         47.930483613907942,
                                         48.930483613907942,
                                         48.930483613907942,
                                         49.930483613907942,
                                         49.930483613907942,
                                         50.930483613907942,
                                         50.930483613907942,
                                         51.930483613907942,
                                         51.930483613907942,
                                         52.930483613907942,
                                         52.930483613907942,
                                         53.930483613907942,
                                         53.930483613907942,
                                         54.930483613907942,
                                         54.930483613907942,
                                         55.930483613907942,
                                         55.930483613907942,
                                         56.930483613907942,
                                         56.930483613907942,
                                         57.930483613907942,
                                         57.930483613907942,
                                         58.930483613907942,
                                         58.930483613907942,
                                         59.930483613907942,
                                         59.930483613907942,
                                         60.930483613907942,
                                         60.930483613907942,
                                         61.930483613907942,
                                         61.930483613907942,
                                         62.930483613907942,
                                         62.930483613907942,
                                         63.930483613907942,
                                         63.930483613907942,
                                         64.930483613907942,
                                         64.9304836
```

9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,

```

-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 0],
                                      [2, 2, 2, 0],
                                      [3, 3, 3, 0],
                                      [4, 4, 4, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
                          [12, 6, 21, 8, -3.1990719612158447],
                          [20, 10, 35, 16, -1.635137691104089],
                          [28, 14, 49, 24, -0.8526977600673817],
                          [36, 18, 63, 32, -0.44953303322849647]],
        'degree': 0,

```

```
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.315376281738281e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.377357545114267,
{others': { 'beta_coeficients': [ 2.0339490779265086,
-2.250326639408051,
1.9586176305958964,
-0.9196987134972037,
0.17745864438284967],
'cost_richardson_zne': 13.914439152583324,
'sorted_expectation_vals': [ -6.429933051303131,
-3.1990719612158447,
-1.635137691104089,
-0.8526977600673817,
-0.44953303322849647]},
'sorted_noise': [13, 47, 81, 115, 149]},
'sampled_data': [ [4, 2, 7, 0, -6.429933051303131],
[12, 6, 21, 8, -3.1990719612158447],
[20, 10, 35, 16, -1.635137691104089],
[28, 14, 49, 24, -0.8526977600673817],
[36, 18, 63, 32, -0.44953303322849647]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.33528943002425843,
0.8608925283648261,
2.8309935233222867,
2.8382233121900358,
2.847558790018716,
2.8477263794127468,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
5.127462455084763,
5.128383696390232,
5.132051241075024,
5.132231848473024,
5.132297186784173,
5.132297186784173,
5.132297186784173,
5.132320898992942,
5.134903164275234,
5.734798022200061,
5.734798022200061,
5.734798022200061,
6.71748919093495,
6.71748919093495,
9.773430860122208,
9.791656954875775,
9.791980525916317,
9.804782964874805,
9.947289619134542,
0.6818618664375632,
0.42253658736184846,
0.19067339209915313,
-2.314563573529872,
-1.7784854373672185,
-0.10945476580518654,
0.08980428543050965,
0.6265702304745648,
-0.12016219313034968,
0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
```


-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,
-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,

```

1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.333005581610797],
                          [12, 6, 21, 8, -3.1651017858597426],
                          [20, 10, 35, 16, -1.6439583689682524],
                          [28, 14, 49, 24, -0.8813072530521874],
                          [36, 18, 63, 32, -0.48318628235120475]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.172325134277344e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.253592279312013,
                           'others': { 'beta_coeffieicients': [ 2.0339490779265086,
                                                                -2.250326639408051,
                                                                1.9586176305958964,
                                                                -0.9196987134972037,
                                                                0.17745864438284967],
                                      'cost_richardson_zne': 13.914439152583324,
                                      'sorted_expectation_vals': [ -6.333005581610797,
                                                                -3.1651017858597426,
                                                                -1.6439583689682524,
                                                                -0.8813072530521874,
                                                                -0.48318628235120475,
                                                                0.17745864438284967,
                                                                1.9586176305958964,
                                                                2.250326639408051,
                                                                3.1651017858597426,
                                                                6.333005581610797]
                                    }
                           }
            }

```

```

-0.48318628235120475],
'sorted_noise': [13, 47, 81, 115, 149]],
'sampled data': [ [4, 2, 7, 0, -6.333005581610797],
                  [12, 6, 21, 8, -3.1651017858597426],
                  [20, 10, 35, 16, -1.6439583689682524],
                  [28, 14, 49, 24, -0.8813072530521874],
                  [36, 18, 63, 32, -0.48318628235120475]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.00030922277578629,
                              0.881375055029912,
                              0.881375055029912,
                              0.8813750550299119,
                              0.9803845151714838,
                              0.9803845151714838,
                              1.201629973822729,
                              1.2441077564052723,
                              1.8660677974280824,
                              1.8660677974280824,
                              2.651002314516478,
                              2.651002314516478,
                              3.233637461790533,
                              4.36384995486056,
                              4.36384995486056,
                              4.975714528653046,
                              4.975714528653046,
                              6.591840750700408,
                              6.591840750700408,
                              8.234009055219385,
                              8.238223602720305,
                              8.238223602720305,
                              8.238223602720305,
                              8.903406631369169,
                              8.903406631369169,
                              8.903406631369169,
                              9.126449234760914,
                              9.126449234760914,
                              9.441918167657187,
                              9.441918167657187,
                              0.4384156956352651,
                              -0.017407483354635628,
                              -0.13579407977759564,
                              -0.08181457210659056,
                              -0.5635031413839604,
                              0.570197125887921,
                              -0.07389454022208618,
                              0.18950484918514415,
                              0.23442314823895152,
                              0.5187197914643009,
                              0.03760777637611914,
                              1.2080522066616035,
                              -0.006796933498365149,
                              0.7597395812711474,
                              0.3165595784185247,
                              0.09234381717322694,
                              -0.047116958459538144,
                              0.02161370846364295,
                              -0.3330735946096464,
                              -0.310519881157395,
                              -0.1321730433969341,
                              -0.32004747583182785,
                              0.38880240494533963,
                              0.5587916266845782,
                              -0.04068226055752174,
                              -0.04929253976644778,
                              0.7350044083650977,
                              -0.3347542084226466,
                              0.07994172439872818,
                              0.177027499128215,

```

-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,

```

-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric5_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 0],
                                      [2, 2, 2, 0],
                                      [3, 3, 3, 0],
                                      [4, 4, 4, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.5111758480272135],
                          [12, 6, 21, 8, -3.513108400075424],
                          [20, 10, 35, 16, -1.9256894730075684],
                          [28, 14, 49, 24, -1.0648100061563792],
                          [36, 18, 63, 32, -0.5886022016151555]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.7220458984375e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.23459620137834,
                           'others': { 'beta_coefficients': [ 2.0339490779265086,
                                                             -2.250326639408051,
                                                             1.9586176305958964,
                                                             -0.9196987134972037,
                                                             0.17745864438284967],
                                       'cost_richardson_zne': 13.914439152583324,
                                       'sorted_expectation_vals': [ -6.5111758480272135,
                                                                    -3.513108400075424,
                                                                    -1.9256894730075684,
                                                                    -1.0648100061563792,
                                                                    -0.5886022016151555],
                                       'sorted_noise': [13, 47, 81, 115, 149]},
                           'sampled data': [ [4, 2, 7, 0, -6.5111758480272135],
                                              [12, 6, 21, 8, -3.513108400075424],
                                              [20, 10, 35, 16, -1.9256894730075684],
                                              [28, 14, 49, 24, -1.0648100061563792],
                                              [36, 18, 63, 32, -0.5886022016151555]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
            'init_param': { 'value': [ 0.08591881204796584,

```

1.5026023651616802,
1.9179097588555833,
1.9179097588555831,
1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,

-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406]],

'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},

[illegible]


```

'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': True}},
'zne': { 'data_points': None,
        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1 '
        'Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
        '6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
        '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
        '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
        '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
        '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
        '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
        'initial_cost_history': [ 4.118038614266066,
        2.900590937440303,
        2.874944433071395,
        3.9296977116518517,
        3.1841729388967344,
        3.695928991667305,
        3.180227821654077,
        3.0475045550195774,
        2.4669562192398296,
        3.4124916083581294],
        'noise_details': { 'gates_num': [4, 1, 1, 0],
        'identity_factors': [0, 0, 0, 0],
        'noise_level': [0, 0, 0, 0],
        'noise_profile': { 'noise_on_init_param': { 'status':
        'noise_prob': [ 0.001,
        0.001,
        0.001,
        0.001],
        'status': False,
        'type': 'depolarizing'},
        'odd_wires': 4},
        'optimized_minimum_cost': [ -8.433468022982382,
        -8.464942606128835,
        -8.43123094137328,
        -8.435401809869298,
        -7.9812014353834515,
        -8.371080050195895,
        -8.46198057547588,
        -8.19415563172473,
        -8.438078664602543,
        -8.481886373274177],
        'optimized_parameters': [ [ 0.4631008843346199,
        0.8090636887146863,
        1.1101507421405439,
        1.3400184709676968,
        1.7452332568602784,
        1.9308700883292553,
        2.1752679291102544,
        2.1784212211010603,
        3.800207092295721,
        3.800207092295721,
        3.800207092295721,
        3.800207092295721,
        3.800207092295721,
        3.800207092295721,
        3.800207092295721,
        4.798448841518561,
        4.812026604031267,
        6.02543601067661,
        6.02543601067661,
        6.5265343452987,
        6.5265343452987,
        6.526534345298701,

```

False,

6.528381158760148,
6.528381158760148,
7.113023968622042,
7.113023968622042,
7.593654062389684,
7.5949106716383605,
7.780058793305066,
9.000370275113886,
9.005799799983558,
-0.22105281285875572,
0.07196296170715184,
1.0527781347416576,
-0.4594806493129553,
1.0526039239081737,
-0.01709270509934428,
0.3199675106146978,
-0.0972220164715271,
0.9145097677998977,
0.09808103992344623,
0.5373974901276602,
-0.5786059880820694,
-0.02594763303680847,
-0.22132394348491485,
-0.7274785930689547,
-0.41090279602128,
-0.520884263329148,
-1.0260835302846771,
0.1692597388493397,
1.3250313995601488,
1.510357845739394,
-0.9233604808876308,
-0.2049157632594096,
-0.9894281051832,
0.25772120620526445,
-0.04822142329545429,
-0.3912638297585932,
-0.48486712444934904,
1.4836649713082644,
1.566996182191827,
0.08093959953023218,
-2.3562926055425963,
-0.21876618997457167,
0.10911407834571141,
-1.3776955808248483,
-0.31816531211057325,
0.7083857256813267,
0.5593482503842677,
0.7483381318097894,
0.010042025170353972,
0.05290425882351868,
0.20616876613540233,
-0.8673924682871486,
-0.08017548043713153,
-0.3056222778255953,
2.3981163154468383,
0.37079365521682633,
0.3870558721039623,
0.19341028278759992,
-0.3154248344945196,
-0.12073260037561208,
-1.5858414285227351,
0.59830347262823,
0.30388710178795286,
-1.4558796844932331,
-0.3225618511944297,
0.09243390755822743,
-0.0006426173071943948,
0.7561662109631148,
-1.3902324585417218,
0.1677701056843372,
2.7311884232201735,
-1.1122880815211051,
0.0467986017971284,
-0.020089420529231013,
0.005481851366742631,

1.3205596999819047,
-0.1290403664286252,
0.3870444394307043,
-0.6549197738372419,
-0.17656835190090397,
-0.0008013451895215781,
-0.8769883624496301,
1.6929520910231297,
0.7288671099890552,
-0.6169673343691255,
1.667217439782634,
0.2554635335546482,
-0.029515171705053127,
-0.8290153205719569,
0.042733616172741226,
-0.5148227135977747,
1.2301836750084596,
-0.9175908300980525,
0.5229779878549122,
-1.5901934710811256,
-0.07316585886439231,
1.1195018365185199,
-0.31111306362203306,
0.7254709796877604,
1.1995417479543053,
0.6908225815252583,
-0.2919630814941808,
0.15955998483086176,
1.2871422030548714,
-1.2683991057896014,
-0.3768598572966417,
1.4863954796333156,
-0.4300298743216642,
0.19151460300148976,
0.9665838873771405,
-0.07244306377732504,
-0.04169546816073936,
-0.389029635704794,
1.5218961812832925,
-0.1772066326032777,
-0.70949107362336,
-0.015222466074935298,
0.3587128629768407,
1.5809494716454662,
0.12428164288740734,
1.5083554487395696,
-0.044379618902227067,
-0.03505579308192811,
-0.6786538117837789,
-0.007428157845337949,
1.5281494732580478,
-0.14124856089608578,
2.726718429443767,
-0.003106328084445967],
[0.5270475437101116,
0.7815425625485376,
1.6469610445191378,
1.6712876404917778,
2.122904188877425,
2.131106816606521,
3.7367798442214175,
3.7367798442219424,
4.750276145421292,
4.853973384996016,
4.865098072058574,
4.873980651141647,
4.987599328262889,
5.00246239884106,
5.520902021852277,
5.793284849001337,
5.8087832398624775,
7.432729128414403,
7.432953787975026,
8.144896858715603,
8.172580744083415,

8.788071761144659,
8.792288306719156,
8.828260688191317,
9.011586808740137,
9.011995976067807,
9.538729699920484,
9.576884709106812,
10.486750161390887,
10.48675016212107,
-0.15531374899360717,
-0.0849039080739262,
-0.8434555658556069,
0.9447826609611266,
-0.19482082661427938,
2.4999420394249707,
-2.7164831941210483,
-1.5408169364026967,
-0.8621482593978547,
0.31547458537730877,
1.4220435826823787,
1.348860553316339,
0.40827474166943256,
1.063960383917828,
-0.21906798993221652,
-0.33263312803097533,
0.4889869181158972,
-0.25524084746075687,
-1.3622146811064544,
0.996105149433398,
1.4161210728452394,
-0.6491827941870941,
0.7399329880002848,
-0.11994422281276146,
3.179573825543474,
-0.034943024466848674,
0.6075464711124173,
-0.3019825794696199,
0.029392952712390567,
0.14840732209332994,
0.08592470568620861,
0.016725634393736574,
-1.3697962205589145,
-0.06155375586805023,
0.16508080354919497,
-0.5975014914051775,
0.09635736997937351,
1.254101402191796,
-0.6193953337438176,
0.47602269510847217,
-1.7251540483535392,
0.5378274350019089,
-1.0159384367751108,
1.0968223266795405,
-0.79878134010976,
0.10468891520322528,
-0.2910173593847854,
-0.7078981306941186,
-0.3510837282628223,
0.5371002854277007,
-0.4911274293534503,
1.7273270693440552,
1.3500555714793954,
0.9052059235505011,
0.4771526028272018,
-0.1893747959421651,
0.13989050072973697,
-0.5930955187300607,
0.36780210947560876,
-0.5043370473487999,
-0.4851893292987325,
0.1987440888222681,
-0.5806445532873666,
2.3171646404124093,
0.06032126128170586,
-0.6389920437730593,

0.9985359186149056,
0.04870155999532171,
0.16447399991313041,
0.05315090520136875,
0.15727255736970833,
-1.3722292706433088,
-0.33233649668322635,
0.7758243461945761,
-0.20476613404890107,
0.06972046040604139,
0.3143879960336948,
-0.01452323353492752,
-1.2901366074359686,
-0.31033000747276407,
-0.16264485166390844,
-1.3914094473719103,
-0.5273128620655095,
0.02465459151953875,
0.10942766294132272,
1.0292094835171124,
-1.882689768994833,
-1.553216293090225,
1.3287846024814622,
0.20963913600213377,
0.7195732642990699,
0.0969787732183028,
-0.9744344415633356,
1.5822487402774768,
0.7077471503862973,
-0.766345302327988,
0.3831437091022152,
-0.12028853613988252,
-0.8209919023748664,
-0.21164805512089607,
0.5290684330656528,
-0.1040878870845114,
1.1956945887281514,
0.25703931501419025,
0.6344733295912399,
-0.02497584874729833,
-0.20937697561532842,
-1.4409411147319051,
0.9707320454927133,
1.4039378085761043,
0.9760826987262126,
-0.17314228405542512,
0.16734556762922406,
0.07881248116464283,
-0.2924467918405051,
1.4946430081327182,
0.7516806171134963,
0.799700486923224,
-0.8440600324500036,
-0.01069194371181784],
[0.3388610072211759,
0.3930923343746358,
0.4059472295013711,
0.4597995545870322,
0.4597995545870322,
0.4667083780713653,
0.9924683364283498,
1.054118787032341,
1.3839145947816414,
1.4469616998432806,
3.044048229936016,
3.044048229936016,
4.682975233106521,
4.682975233106521,
5.287925895396742,
5.287925895396742,
5.625677830869467,
5.625677830869467,
6.519005703020383,
6.519005703020383,
6.519005703020383,

6.854665591577039,
6.854665591577039,
8.092962128458302,
8.097830997780322,
8.102159968176109,
8.111954675742565,
8.727038821401255,
8.727038821401255,
8.784424594255395,
-0.8636093730812797,
-0.04742973062196874,
-0.37989856483295564,
0.3750798322441465,
-0.509087961968837,
0.5366875438436245,
0.1232704940065221,
0.5249134835872401,
0.13594138148710633,
0.10802292582448766,
-0.7156772546121367,
0.34756877087651405,
-0.5704812075854907,
0.289395233495819,
0.2674862930849357,
0.8724143866961164,
0.5770602988816299,
0.658777432128999,
-0.47288232304292416,
-0.029663566814993374,
-1.323180236232167,
-0.2890901656039231,
-0.015141732930961212,
1.1846850771343012,
-0.4506688471965015,
0.22241980016839272,
-0.015748484493019736,
1.1522499025192263,
0.2996907655672081,
-1.7370631162182948,
-0.35306232532663057,
0.17322471105923057,
-1.1855180134814352,
-0.31537295018088835,
-1.8120366955290261,
0.6663892425075478,
0.21874911670355193,
1.2729693967005706,
-0.14230256773273223,
0.2012427441067739,
-2.3269317425793643,
-0.03299848401271634,
-0.4074853136565566,
-0.30371905431664253,
-0.12037074206123735,
0.8456676988075262,
-0.21838328071319946,
0.019494764922420988,
0.16383099728733078,
0.056056403895700845,
-3.5632063709291613,
0.6579904510057321,
0.5207861046034454,
-0.017180990359968364,
0.2532807007765998,
-0.03315495616728509,
0.7564905992030696,
-0.11091568858030387,
-0.9281890577202553,
-0.6082181315906088,
-0.4480511658411004,
2.1953283079942025,
-0.30446713497464617,
-0.03359602431311142,
0.28972943746372876,
0.17817064395522006,

0.7965754083210965,
-0.9520780563921613,
0.10150445450145726,
1.3510884636720888,
-0.37849615173878054,
0.17729737647006832,
0.1884683806301832,
-0.03234547530918763,
-1.338763456177623,
-0.6711058962200585,
-0.012976043431998471,
0.027578392175021227,
0.10161286285089129,
-0.00939135247962935,
-1.302164623174096,
1.3715576399721376,
0.737767307868418,
1.5939269674119705,
1.4291758244230544,
0.5601859764306084,
0.3488764286390935,
-1.093762224545935,
-1.1976831420922067,
1.4557381665683378,
0.8924472305311927,
0.5300221300338881,
1.0785280788144638,
-0.6029509417428172,
-0.6790632925762424,
0.4138865114731201,
0.2917474006869537,
1.8050154062536312,
0.05177383423831921,
0.25467777497912664,
1.4671356779039024,
0.4422247342165525,
-1.5431607378377095,
1.8101814873938125,
1.6870467346899987,
0.7517920825884995,
1.3622912362338087,
0.13653050459472865,
1.5210183141257738,
-0.004247168330480351,
0.26137218980413907,
0.5451511562235306,
-0.6705596749039198,
1.2136085126593865,
0.5231529087420831,
-0.03279466120532818,
0.2548954759886831,
1.5797820110558691,
1.075840789720537,
1.5541818084195007],
[3.821020346740443e-17,
1.386496864345847,
1.386496864345847,
1.7951472089989222,
1.8559400139285378,
2.7980544315911504,
2.7980544315911504,
2.82954781812891,
2.82954781812891,
2.82954781812891,
2.956183634338348,
3.840561875918348,
3.840561875918348,
4.79501776182173,
4.79501776182173,
5.670460018404704,
5.670460018404704,
5.670460018404704,
6.865392250052202,
6.865392250052202,
7.506826147384778,

7.506826147384778,
8.737685173635683,
8.737685173635683,
8.737685173635683,
8.737685173635683,
9.601023092540274,
9.601023092540274,
9.853738132880634,
9.853738132880634,
0.5611658219809542,
1.427331325859137,
0.8881601644240952,
-0.8937497791603879,
0.5712044928850715,
1.3195951619089188,
0.5467792160605555,
-0.6816630744322637,
0.06707477836950569,
0.6798692668930644,
0.19576168860111362,
0.5094724850631845,
1.1424414477256812,
0.30471145260454696,
-0.08158911485577636,
0.9585661478313002,
1.0836238583568754,
-0.3718197048926503,
0.7841948506971846,
0.3023515457437066,
0.7662240328883452,
0.9970048707689876,
-0.8946064024093777,
0.27649390200762114,
-0.15342626519190924,
-0.4015720487025046,
1.460926694270634,
-0.055421168101734586,
0.6077050880702092,
-0.7932081335349191,
1.3188332541813446,
0.01929555634963922,
-1.1301845164589392,
-0.1718174564221745,
2.26154163970611,
-0.9882028269623031,
0.8038643847286138,
0.6849304102208338,
0.874798983022982,
-0.016486064274954207,
0.0264221098096002,
-0.3625000384278827,
1.4376331477886695,
0.8499992740594865,
0.10042171561200945,
0.14542648933357244,
1.3513090878714353,
-0.22248922869259902,
-0.4483175327546578,
1.2345758678942387,
-0.18983522552694923,
0.019591210301167533,
-1.3607248526627624,
0.018375533581887468,
-0.3730738776375715,
1.0123537416913542,
0.7412601688930508,
-1.6525919628732422,
-0.8341682190885938,
0.04519729634722039,
-0.06718028327556408,
-1.5649719285752928,
-0.37531851249345394,
-0.11822841604899964,
0.008154106549025474,
-1.4601449692077721,

-0.0030993446167428395,
1.462573870839384,
-0.11887757571245174,
-0.09361960188417112,
-1.2543408854038138,
-1.525280802664792,
-1.241413372948363,
0.008758808605552393,
-1.6676332942965022,
0.13826170532334445,
0.7537363962746654,
0.0833397476047736,
-0.12216949704941273,
0.02084807234445201,
-0.3421949354726438,
0.0017882611822728064,
1.5186851123669147,
-0.4603163345929323,
-0.09956972276089109,
0.7247597830417082,
0.5304935691839561,
0.003193152207841171,
0.5581346990750625,
0.010542051444639388,
0.7553167297004638,
0.5956891015800863,
-0.05566254332595716,
0.5362461335652635,
0.009895282042162012,
-0.3029901133286735,
0.09399206534914867,
0.8818812838713691,
-0.06795832284115338,
-1.580933930101608,
0.3014975711017802,
1.2092022074751856,
-0.019692531397995948,
0.9250981578351692,
0.4673383297638096,
0.03344339272872125,
0.13058472572385768,
0.036262684184263205,
0.458095375934523,
0.6492846129896516,
1.3039221898344189,
0.024764812872192767,
0.7236926243101844,
0.01444840086266692,
-1.0388939706148563,
-0.08359693344439546,
-0.4698004238001188,
-0.48055768545836347,
0.4662000815300196,
-0.041204969991243484],
[0.01798320141016013,
0.3393419127512466,
0.7047697703765823,
0.7064789163091142,
0.7191656785855808,
0.7198672068604162,
0.719867206860416,
1.0840512961729545,
1.0900003904895101,
1.1926153174924548,
1.867054342490295,
1.867054342490295,
2.368094916985588,
2.3683751276938714,
3.6383003552329964,
3.642527762446636,
4.505845683894636,
4.602446412902637,
4.972427659100928,
4.972427659100928,
5.1148164150678035,

5.8066136392845,
5.820681023327439,
6.727088267818048,
8.352653016300794,
8.773334122756978,
9.065902540064004,
9.065902540064004,
9.74880092287043,
11.30145464466351,
-0.13904770480884485,
0.23493722133539818,
0.4497803172772882,
-0.640584995353188,
0.9779304528121034,
0.0011745349607135795,
-0.7175493520168287,
0.373620380836484,
0.08111255661159392,
0.6653458367314794,
0.0037967443269442437,
0.6286122183467432,
-0.03825426016359485,
0.6896824961465278,
-0.23916946602734285,
0.4624850989736848,
0.1419493525208794,
0.7145176662978019,
-0.316909964913161,
0.6605458698076423,
-0.39309909050950365,
0.7623048158369822,
0.7469736169282462,
1.2400892032597353,
0.12266455668869777,
0.19578648318487166,
-0.784043232212839,
-0.10334183265831844,
0.32774480702026265,
-0.011491906972883237,
0.322988869457402,
-0.2027870428092001,
0.1281571522083961,
-0.1947961113336468,
1.7114513053475526,
0.005425684515457419,
0.8136825826883561,
-0.25903007981373005,
-0.5015428579892063,
-0.2600978115589891,
-0.15405473476720474,
-0.4207498663011112,
1.04622682040645,
-0.8642339256273757,
0.6248691295250803,
-0.3539240130838131,
0.011689879721248103,
-0.3068283940685192,
-0.6048660879777619,
-0.11704784670887185,
-1.2745062338511486,
-1.7890038533223112,
-0.7818671277256136,
-0.8877455887286905,
0.3164818198590212,
-0.27477307924208777,
1.8083251930535746,
-0.5031479272825263,
-0.9648993597903888,
-0.41682962131770707,
0.27040705073896654,
1.0892900815418987,
0.17779960873468054,
0.020337033950376556,
1.0551529464315492,
-0.10705558118147128,

-0.8851977322505518,
0.2232945485845247,
-0.5850403569085746,
-0.7690595025978197,
-0.4391673030332381,
-0.9309566421919284,
0.2973030450025941,
0.6780545449896137,
0.14045373030826291,
-0.5109313900859794,
-0.44070010469428217,
1.6940975579604587,
0.17062304524159472,
-0.402681393200546,
-0.7091187255059627,
1.3132357546586793,
0.2729391261599054,
1.0115940175088693,
1.0046673925246778,
0.4982257464819494,
0.04454679168285006,
0.4863023994043421,
-1.1433261311581624,
-1.820308437409531,
0.6035405700431521,
0.04003907881711185,
2.99381748600971,
-1.0437523456620463,
-0.22517480393070408,
0.10746937069246698,
-0.8788513722547997,
1.286531186523234,
-1.3157160458746164,
-0.6979748907629274,
-0.16573593981190854,
0.2529194783090185,
-0.2562517800846549,
-0.7966921479109774,
-0.4446940958312971,
-1.052310031662883,
-0.1506245930250076,
0.1615975776996014,
2.9220364155549214,
-1.7197843955783065,
-0.6592103621593355,
-0.2007498223803255,
-1.6188873615119244,
-0.15193115939751467,
-0.20866967970071015,
1.2910791163230286,
-0.4917152660712325,
-0.42348313202027765,
0.3518303936962707,
-0.10034486495238693],
[0.009039772647041902,
1.1400526086001903,
1.466905028683042,
1.6524670085313564,
1.987797816914966,
1.9900950346150876,
2.1413202332332344,
2.390935514217652,
2.402429399816812,
2.81913686416337,
2.81913686416337,
3.5656810200683298,
3.5656810200683298,
3.5656810200683298,
3.76603447051701,
3.76603447051701,
5.499236323276389,
5.4992363232763894,
5.4992363232763894,
5.532789537471719,
5.532789537471719,

7.371733777832876,
7.371733777832876,
7.680783486275433,
7.706623781446164,
7.706623781446164,
8.457742149729329,
8.457742149729329,
8.95222736351662,
8.95222736351662,
0.18928393791812126,
1.0354889043700823,
0.3703113207409527,
0.30988589523494625,
-0.34669340410550803,
0.09615284519636855,
0.1011898884094612,
0.6160115432573767,
0.6148540393435122,
0.07934943483664206,
0.8557624329627399,
-0.7672633290674925,
-0.14188111026081618,
0.1995320641320393,
0.4594888802421805,
-0.03593187081620776,
1.2926591415659605,
0.31433950962447343,
0.8053272546604554,
0.6634815559763872,
0.311660786273531,
0.0294653444200854,
0.3044394736040779,
0.7599259037372076,
0.22962735656034647,
0.22562971232722423,
0.29511485598499815,
0.02260075636691185,
-0.9497096253675098,
-0.5331371571917215,
0.2192512323868535,
1.104832430155296,
0.8636617903941131,
-0.6756684630428217,
0.4169844309263729,
1.1297527651723116,
-1.160118439748996,
-0.7971836651224,
1.9169656561991728,
-0.007260796290839314,
0.3614132074474742,
0.6243022778229537,
1.8335025630018775,
0.35139722497115833,
-0.7247870129359816,
-0.6744811297738377,
0.49997538785346773,
1.4139521212248116,
-3.0799818401546553,
-0.14768564076593044,
0.06251500852230953,
-0.8547134606957393,
-0.30107856776262504,
-0.7654972812734481,
0.28603020918627825,
0.2129303248959857,
-1.2077141264570097,
-0.8488549318812054,
0.6492574804166855,
0.7751290676155554,
0.40909519807061084,
-1.2750630737105486,
0.4187798718836698,
0.33498424523924863,
-0.19017028209895162,
-0.03232931911977397,

1.0621483440417678,
0.452017889610544,
1.0192767515800205,
0.7290563184502946,
-0.5286462788982386,
-0.05512372609304388,
1.3189151919853837,
0.5865235617306253,
-0.579843833868396,
0.6980158293666034,
0.027004860579276476,
1.2876675233834314,
-0.17185617277774212,
-0.3827017785428823,
-0.7286404996479297,
1.1682112525473984,
0.22765379123755217,
1.4870969768845252,
1.1642898801914527,
-0.03976972562460009,
-0.8163296820868416,
-0.15318057329999143,
0.0482918949035788,
-0.06014345828968787,
1.1478583262090496,
-0.005141116181289305,
-0.5820859555621626,
1.086519910329415,
-1.08558016317555,
1.531511042803357,
0.042734048269580674,
0.7452599788242152,
-1.3094923903018465,
-0.9012205175381317,
-0.0755166851159581,
1.5493661520767013,
-0.2278023712186688,
0.8627212805492942,
1.8523276191392382,
-0.10961709979291216,
0.6746606852631131,
0.22566845572735392,
0.5920154791154506,
1.2709342533151342,
0.26903045177842116,
0.11724485476711638,
0.621942277674204,
0.15491924869980966,
1.472217084758381,
0.8975149794611956,
0.3601026339731485,
1.3234513983254743,
0.0182465793142515,
-0.11914786829986494],
[-5.765348397008027e-18,
0.3428984748224433,
0.3655450682584724,
0.704784031551808,
1.5409081472811441,
1.549300366088437,
2.3787666619954635,
2.3787666619954635,
2.8808650819472406,
2.884987740396357,
3.1308808834009203,
3.1363836212549696,
4.869321728501938,
4.869321728501938,
6.1150174264690085,
6.2582278136815175,
6.271636236200362,
6.551203581243674,
6.59741701640448,
8.230564833747417,
8.294003346338926,

8.31431753208387,
8.369586036193128,
8.415319617883744,
8.70447046242739,
8.709120596281233,
8.918659758853744,
8.99410216914391,
9.002256766523654,
10.218690251738174,
-0.6798627919679444,
-1.1436724543472427,
-0.1278269606092795,
-0.9286552360068459,
0.6524046490818174,
0.06134572949387961,
-0.49761179900021657,
-1.3918471119443894,
0.5858287319319103,
0.18669127045720346,
-0.4586789737568629,
0.2400608953322058,
0.022222124883641912,
0.9282004764805514,
0.6952358394741303,
0.805684979006905,
0.6146729452742722,
0.6777524139082742,
-0.8117842884117104,
0.3426849285038029,
0.7100299235979358,
0.7658186509139059,
-0.05418633990950139,
0.1761917490887145,
0.5744825473612316,
0.31705937472296297,
-0.9188523464356001,
-0.7337446596965294,
0.2061368312243915,
-0.6645694031850036,
-0.009871104765786294,
-0.14388187083546433,
-1.2485707106311008,
2.849540809212292,
-0.5119215846255833,
-0.19381894062193808,
-0.17310717674563775,
0.8465397629359774,
-0.46757240698237135,
-0.05607605828560263,
-0.40986137011816176,
0.07649505002491204,
-2.1522907800348112,
-0.6653346582594515,
0.4084801543759574,
0.4028897820911683,
0.16908117843954765,
-0.07217026591146015,
-0.72804755296126,
-0.13548738477012795,
-2.590679566047418,
-0.40743076961744734,
1.1030856800887578,
0.08388380440455438,
0.25224805905642333,
0.10644300798076774,
1.0474125729404944,
-1.1459016177859003,
-1.6916287310468103,
-0.9886067106540645,
0.24182274960650893,
-0.25703819491337826,
-1.1942818044595898,
-0.7763678643520989,
-0.916718557142268,
0.20086626876005667,

-0.3711023890817823,
-0.006962349053602254,
2.687031223814704,
0.3970791704566904,
-0.6890591654410289,
1.0504536719091986,
-1.2458238157294268,
-1.197330311624414,
-0.12155955689525438,
-0.2969718406476498,
0.15406921803787998,
0.9984735505197111,
-0.2734456365103056,
0.0100494123830882,
0.16172873483083283,
-0.6780409734277772,
0.4539997995464368,
0.7558576390626953,
0.9556605206819043,
-0.10572075101403118,
1.5564628686733721,
0.05397552837650759,
0.7415987024402694,
-0.1648500682327342,
1.052699139325594,
0.6776572023171561,
0.15219630541397192,
0.8843933196559655,
1.3948099389813178,
-0.16710315713046545,
1.2199200746863228,
-0.3825678575328573,
-0.26766985045935626,
2.5777182352113908,
0.23156136798610805,
-0.3377548185037014,
1.0689111835439342,
-0.17216022970038714,
-0.29381807805976945,
-0.5689533077540762,
0.30596827309967156,
0.5260993417881752,
-0.6768480303494689,
-0.1092430795457648,
1.5442082488768039,
-0.10826504240549732,
-0.9555598065181787,
0.5334835330029429,
0.5569227134357198,
1.2665483894271157,
-1.9677332823608626,
1.5559931182978979,
0.7709453943956464,
1.5746506000831744],
[1.7625089228179507e-05,
1.298540182320117,
1.3141870420194457,
1.3305471045820527,
2.0686735179487985,
2.23810597518972,
2.267373737425337,
2.295441377067579,
2.3325388050051976,
2.352484360014207,
2.370891483796666,
3.3117124844820194,
3.363605896627787,
3.858981460194034,
3.8589814603490344,
3.859537867870235,
3.8595381713444072,
4.26458571773699,
4.264585717738593,
6.599549149586233,
6.655963438031886,

6.939612414177192,
6.939612414268938,
7.004302986969205,
7.010544991093791,
7.97314828104894,
7.999280802101656,
8.022628296732579,
8.027896471579398,
10.077254342713585,
-1.0733166032270363,
0.4264387306682,
-0.29490479134596703,
-0.45181113351566254,
-0.9496712390342118,
-0.24412788786515252,
-0.17005038539710493,
2.131747001055919,
-0.8801477965160032,
0.1985900676052623,
-0.17829180427317373,
0.1513896817808816,
-0.2996418627583153,
0.09152392337611555,
1.0155053498240294,
0.6713280773462803,
-0.5915672184524843,
-0.21780814909773427,
-0.23676595566314207,
-0.1539223229682568,
0.9733737893230769,
-0.8325690111887051,
-0.6684349693104433,
0.944923850675747,
-0.8270803600282979,
0.4869668407293288,
0.07342964610078141,
-0.8928903181234079,
-0.04262248204617692,
-0.7602498290394286,
-0.48858825463899247,
1.7340528561515622,
0.7340502315717261,
0.5163401187759543,
-0.5015858040228415,
0.2890335055344555,
-0.46033506221820203,
-0.8185907107929625,
0.4557079379705311,
1.6601520871164626,
0.07125306723268252,
-1.4035119795986863,
-0.6741575848895364,
1.84360490583597,
1.6836338044245691,
-0.33329370458317326,
0.19854317161902654,
2.03917897859608,
-0.3342251938073038,
-1.6015864477635742,
0.35422488479060155,
0.1487121545506934,
-0.2821114888507261,
0.21869934797564045,
0.7523961573071125,
1.3021785537923125,
0.26184859002187644,
1.2441238228144538,
0.039670539596101485,
-0.15687777352236798,
1.1386642271664005,
-0.13559116971858365,
0.39167499360215746,
2.4453261619465274,
-1.0890320328395309,
-0.4289422005081893,

1.425249318116838,
-0.0016733698174337221,
0.023121567661497415,
-0.28325087531692295,
0.940069260590291,
1.163437407349777,
-0.4256527399896147,
-1.4972731615803316,
-0.12624975701746496,
-0.10144651691714318,
1.3307378561466523,
0.041778278155808674,
-1.2746374184036902,
-1.2216848741140482,
1.4048769664929757,
0.8957112974734468,
1.0058068981677006,
-3.0965653057688316,
-0.3344448337959029,
-1.0948102803209163,
-0.038478236121346335,
0.12618687504486675,
-0.0850775611793361,
1.33550885346982,
-0.04334772196309596,
0.2406908654957122,
0.7327249608913058,
0.9966290850109706,
2.2852565843192423,
-0.030570948934034583,
-0.5566407461361556,
0.813246653395058,
0.5946355273642772,
0.07519254658407079,
0.27120989268694123,
-0.15415722381241237,
-1.1239288818585278,
1.893751281002709,
0.7530205722182245,
1.4454588921082565,
0.15037300343080462,
0.1793867334099716,
0.3587259643057371,
-0.006799349268608,
1.0907160355013774,
-0.30186717991666434,
0.8507589785411056,
0.009155709268365956,
-0.8165113788656561,
0.011816515180360782,
1.116430418883973,
1.5343789763879045,
-0.1682835166498154,
-1.5032827361947747],
[0.10956969985098665,
0.4360115975982133,
0.5932904704915389,
0.6050451872391112,
1.3688142364348175,
1.3737597367805747,
1.5226773273508583,
1.5589271816733294,
3.150320547140947,
3.1613053536766147,
4.7635234433865605,
4.823148382609575,
4.823188243925802,
7.171157860626202,
7.171157878485542,
7.171158529865571,
7.175073431930883,
7.183070822592498,
8.102400589897519,
8.10392263292654,
8.103940780725882,

9.021308924987686,
9.02133947231205,
9.082831297490506,
9.094868565409332,
9.236804949782064,
9.237029573386183,
9.26430291432718,
10.00744744746223,
10.032040374677464,
-0.5042382729215367,
1.0180015436638659,
0.10423420212375085,
-0.3355687578372037,
-0.3902319222280216,
-0.09867602213718178,
0.3860296418386605,
0.8522828003851446,
-0.7730596965193267,
0.08803782521712378,
1.175277695996097,
-0.3769225558693855,
-0.4553519820435133,
0.4758962487375975,
-1.0517577693418567,
0.3712029741888282,
0.27334544561774066,
0.03397030216872572,
0.3891396532807883,
0.12152393226873999,
1.5192516885791116,
-0.5124931636551364,
-0.5667003913709334,
-0.08425393582345661,
-0.8152956002657638,
-0.008087141654348068,
-0.20833380477584987,
-0.19186344764752156,
0.309463849209865,
-0.16555810903618162,
-0.28534485935787846,
0.025710068868521854,
-0.9172960495760906,
0.027100436879331523,
-0.9806057417125867,
0.46877133554227174,
0.22597640762440013,
-1.8700902599315918,
-0.6222133609404971,
0.023922176634194332,
-0.6777524378245746,
-0.3883818313368169,
-0.19921738892417568,
-1.3418496115619054,
-0.05365669225714106,
0.38835985529462186,
0.46741217514915123,
-0.1388983196762152,
1.0002905156718758,
1.6633296566254185,
-1.0097136763815113,
1.203957772036831,
0.34033273848251194,
1.1456357751380877,
-0.4637199651146574,
-0.13624703361320328,
0.6785440200843139,
0.21691345067474185,
-0.4166036808107477,
1.6003722740710606,
-0.294881370724797,
0.04198878790454511,
-0.689532047804753,
-1.3911901894570817,
-1.1667039833298627,
0.824824800006074,

0.8047687705752488,
0.5727196814509768,
0.2385364014798047,
-0.512164832149354,
0.17925141295137917,
-0.19704824673887428,
-1.6372757558452697,
0.5441675736491897,
-0.13936070677778437,
1.3935144064781402,
0.10763856884501717,
0.6679788244416961,
-0.33118654939932857,
-0.35796695002153883,
0.2632208027041665,
1.5831503864671255,
-0.25140146740224,
1.135774876905931,
0.5897962040478675,
-0.7664904031063712,
1.092649076983624,
-0.9862462517151122,
0.010216869101717933,
0.5705833807099128,
0.09839127014019662,
-1.034770130526523,
0.20582133073730807,
-0.44428721001967564,
0.5272444819968604,
-0.2627660867388947,
-0.9857421779520247,
-1.3646538516413942,
0.14779052759756187,
1.1207932022043874,
0.30427416833245535,
-1.055109005337564,
0.30548258474769546,
0.8153937738463674,
0.5082537250510629,
-0.2973639305797175,
0.0034742209991892575,
1.5168338311712208,
0.0971904670135119,
0.5305672799342332,
-0.08170159551874663,
0.4909203029282298,
-1.1771354207712392,
0.038725619801271274,
-0.6655680746562402,
1.312626970471223,
-1.0115568894576918,
0.16290676730453105,
0.3074286838975462,
0.013117345918778788],
[0.4866654913190249,
0.4866654913190249,
1.4314116389487666,
1.4314116389487666,
1.548608655381947,
1.5552664901547235,
1.662920497545568,
1.676155881591226,
1.6952532774325564,
1.6952532774325564,
2.46741851740979,
2.50102877426778,
3.636145431120801,
3.6361454311208004,
3.6361454311208004,
3.6361454311208004,
4.0161273218482085,
4.0161273218482085,
4.5411636896776235,
4.5411636896776235,
4.922545565757144,

5.624338470516907,
5.642847901355697,
5.904037934138086,
5.915382265322609,
6.860930229950661,
6.860930229950661,
9.422517992567101,
9.467623480195014,
9.467623480195014,
0.40091998571272736,
-0.3581846438287994,
-0.32033246941904975,
0.24409866553117068,
0.176560178011825,
0.4452285262890565,
-0.37150326481454765,
-1.0185408309777293,
1.6197684873006286,
2.0095194673452026,
-0.20342804182482532,
-0.4471079471088122,
0.37127155242298737,
0.4199607045713712,
0.11145558426636207,
-0.289968410859463,
0.9752004498173846,
-0.11972987215087512,
-0.8215210043660294,
-0.4144105549344152,
0.04966095047933562,
0.6091224412599766,
-0.2481186610805342,
-0.9036735819055022,
0.22922493224752044,
-0.04567884568114995,
-0.40529506738584414,
-0.44609185937563545,
1.1000542207972743,
0.8995834380092431,
-0.4391184290327733,
-1.623814742575614,
0.11121717167805684,
-0.15344289124808663,
0.7072346226342329,
0.013911394934765222,
1.1008239636292285,
1.1342299046285917,
-0.1605718532112884,
-0.8521154510350495,
-0.9726520318623231,
-0.11078467990736285,
1.2893067940177063,
-0.8656847782514395,
-0.4554157535410911,
0.46752563535079134,
1.0115326820592876,
-0.017465678177200683,
0.7555243091795097,
0.2225241128439446,
-0.6129237770264853,
0.36249038141661666,
-0.1470816778834455,
-0.23781939745964606,
0.7410640700988803,
-0.10219672891678527,
-0.9043863030697507,
-0.054446492384892985,
-0.23618927113386157,
-0.3971482428910102,
0.42980979959963467,
-0.4511512430809565,
-0.3322123782853245,
0.15031459204805087,
-1.4287283715133035,
-0.026859055017604255,

```

0.20826999854501216,
-0.016397767746330064,
-0.026711091547853946,
0.15899307013846248,
0.18349959987180348,
-0.019181103141429404,
-1.3777401647916119,
0.05578672583947999,
-0.17191397249005735,
0.08492618339903452,
0.2781511329827625,
1.6853098582374997,
-0.7964285169174421,
0.13628687772581302,
-0.26524580358677924,
1.3615146942088054,
-0.18163849035743365,
-1.5476669195398023,
0.2740670870741208,
0.19740142577560438,
0.18567840551061798,
0.43288784764736754,
-0.8069341399605817,
1.5139336080928714,
-0.7594440385529729,
-0.20443587392639553,
0.0916888700165231,
-0.03922249489745327,
0.4135801157921025,
-0.06731598775715468,
0.5038720973518589,
-0.7705715736697758,
-1.4775324565516457,
0.061950004909204714,
-0.9555167624183505,
3.1277455162888304,
-1.137887484425651,
-0.09852947784569874,
0.042597844143281625,
0.06134264537966588,
0.3384848933516047,
0.012002169875906457,
1.5732417764972468,
-0.3550969359952667,
0.37821230776206083,
1.4914211446267915,
-1.2027836619303884,
0.22421848814312553,
-0.2072711571643093,
0.05122655754886873,
-1.5578500765322636,
1.5918641150141506,
1.2635397829742223,
-1.2452354025631063]],
'run_time_sec': 43720.81102514267}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.3773346757087749,
0.5734233884194113,
0.5734233884194113,
1.2094818154974225,
1.2528672337957094,
1.3073966674132076,
4.193348178434801,
4.228670385547864,

```

4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.274742575727675,
5.343872214974395,
5.354523589318104,
5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,

```

0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},

```



```

False,
0},
    'odd_wires': 4},
    { 'gates_num': [28, 7, 7, 24],
      'identity_factors': [3, 3, 3, 0],
      'noise_level': [28, 14, 49, 24],
      'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                        'odd_wires': 4},
      { 'gates_num': [36, 9, 9, 32],
        'identity_factors': [4, 4, 4, 0],
        'noise_level': [36, 18, 63, 32],
        'noise_profile': { 'noise_on_init_param': { 'status':
                                                      'value':
                                                      'noise_prob': [ 0.001,
                                                                      0.001,
                                                                      0.001,
                                                                      0.001],
                                                                      'status': True,
                                                                      'type': 'depolarizing'},
                            'odd_wires': 4},
        { 'gates_num': [44, 11, 11, 40],
          'identity_factors': [5, 5, 5, 0],
          'noise_level': [44, 22, 77, 40],
          'noise_profile': { 'noise_on_init_param': { 'status':
                                                        'value':
                                                        'noise_prob': [ 0.001,
                                                                      0.001,
                                                                      0.001,
                                                                      0.001],
                                                                      'status': True,
                                                                      'type': 'depolarizing'},
                              'odd_wires': 4}}],
      'run_time_sec': 68.87752628326416}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
  'init_param': { 'value': [ -5.647934061474703e-18,
                             1.4787834906243755,
                             1.478918229526628,
                             1.478918229526628,
                             2.0077351764243123,
                             2.4302809101456218,
                             2.4302809101456218,
                             3.009709563144601,
                             3.009709563144601,
                             3.239471043198664,
                             3.2405283031129133,
                             3.2405283031129133,
                             3.2656598154222434,
                             3.5353392937639825,
                             6.198713539858257,
                             6.213604956994236,
                             6.73397854893658,
                             6.73397854893658,
                             6.743547791194661,

```

6.744609103740889,
6.744609103740889,
6.749543065628153,
6.749543065628153,
8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,

```

-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0]]},
'run': 'redundant',
'state': 'dmatrix',

```

```

'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                    '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                    '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                    '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                    '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                    '0.5 X 5 X 6 + 0.5 Z 5 Z 6']}},
'output': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
                             [12, 6, 21, 8, -3.1990719612158447],
                             [20, 10, 35, 16, -1.6351376911040885],
                             [28, 14, 49, 24, -0.8526977600673817],
                             [36, 18, 63, 32, -0.44953303322849647],
                             [44, 22, 77, 40, -0.23666355366027367]],
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 7, 0],
                              'noise_profile': { 'noise_on_init_param': { 'status':

```

False,

0},

```

'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},

```

```

'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
  'identity_factors': [1, 1, 1, 0],
  'noise_level': [12, 6, 21, 8],
  'noise_profile': { 'noise_on_init_param': { 'status':

```

False,

0},

```

'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},

```

```

'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
  'identity_factors': [2, 2, 2, 0],
  'noise_level': [20, 10, 35, 16],
  'noise_profile': { 'noise_on_init_param': { 'status':

```

False,

0},

```

'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},

```

```

'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
  'identity_factors': [3, 3, 3, 0],
  'noise_level': [28, 14, 49, 24],
  'noise_profile': { 'noise_on_init_param': { 'status':

```

False,

0},

```

'noise_prob': [ 0.001,
                0.001,
                0.001,

```

```

False,
0},
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4},
    { 'gates_num': [36, 9, 9, 32],
      'identity_factors': [4, 4, 4, 0],
      'noise_level': [36, 18, 63, 32],
      'noise_profile': { 'noise_on_init_param': { 'status':
                                                              'value':
                                                              'noise_prob': [ 0.001,
                                                             0.001,
                                                             0.001,
                                                             0.001],
                                                             'status': True,
                                                             'type': 'depolarizing'},
                        'odd_wires': 4},
      { 'gates_num': [44, 11, 11, 40],
        'identity_factors': [5, 5, 5, 0],
        'noise_level': [44, 22, 77, 40],
        'noise_profile': { 'noise_on_init_param': { 'status':
                                                              'value':
                                                              'noise_prob': [ 0.001,
                                                             0.001,
                                                             0.001,
                                                             0.001],
                                                             'status': True,
                                                             'type': 'depolarizing'},
                        'odd_wires': 4}},
      'run_time_sec': 82.4823865890503}},
    { 'config': { 'ansatz': { 'gateset': 1,
                              'layer': 30,
                              'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                             'cn': [ 0.5,
                                                             0.5,
                                                             0.5,
                                                             0.5,
                                                             0.5,
                                                             0.5],
                                                             'r': 0},
                              'time': {'max': 10.0, 'min': 0.0},
                              'type': 'xy-iss'}},
      'init_param': { 'value': [ 0.003375513600300809,
                                0.4884127443235871,
                                0.6719352108785641,
                                0.9073170135093351,
                                1.4946633690060278,
                                1.5076957064847494,
                                3.872917838142945,
                                3.872917838142945,
                                3.872917838142945,
                                3.872917838142945,
                                3.872917838142945,
                                5.047622474508209,
                                5.054336886650782,
                                5.067626550795242,
                                5.0733174878702085,
                                5.094629387163461,
                                5.749571576552799,
                                5.7731554023304374,
                                7.144946717661993,
                                7.144946717661993,
                                7.144946717661993,
                                7.144946717661993,
                                7.602314289267704,
                                7.602314289267704,
                                8.018775147517832,
                                8.018775147517832,
                                8.018775147517832,
                                8.018775147517832,
                                8.018775147517832,
                                8.018775147517832,
                                9.661071929582006,
                                0.001],
                      'value':
                      'noise_prob': [ 0.001,
                                     0.001,
                                     0.001,
                                     0.001],
                      'status': True,
                      'type': 'depolarizing'},
                      'odd_wires': 4}},
                      'run_time_sec': 82.4823865890503}},
                      { 'config': { 'ansatz': { 'gateset': 1,
                                                  'layer': 30,
                                                  'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                                                     'cn': [ 0.5,
                                                                                     0.5,
                                                                                     0.5,
                                                                                     0.5,
                                                                                     0.5,
                                                                                     0.5],
                                                                                     'r': 0},
                                                  'time': {'max': 10.0, 'min': 0.0},
                                                  'type': 'xy-iss'}},
                      'init_param': { 'value': [ 0.003375513600300809,
                                                  0.4884127443235871,
                                                  0.6719352108785641,
                                                  0.9073170135093351,
                                                  1.4946633690060278,
                                                  1.5076957064847494,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  5.047622474508209,
                                                  5.054336886650782,
                                                  5.067626550795242,
                                                  5.0733174878702085,
                                                  5.094629387163461,
                                                  5.749571576552799,
                                                  5.7731554023304374,
                                                  7.144946717661993,
                                                  7.144946717661993,
                                                  7.144946717661993,
                                                  7.144946717661993,
                                                  7.602314289267704,
                                                  7.602314289267704,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  9.661071929582006,
                                                  0.001],
                      'value':
                      'noise_prob': [ 0.001,
                                     0.001,
                                     0.001,
                                     0.001],
                      'status': True,
                      'type': 'depolarizing'},
                      'odd_wires': 4}},
                      'run_time_sec': 82.4823865890503}},
                      { 'config': { 'ansatz': { 'gateset': 1,
                                                  'layer': 30,
                                                  'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                                                     'cn': [ 0.5,
                                                                                     0.5,
                                                                                     0.5,
                                                                                     0.5,
                                                                                     0.5,
                                                                                     0.5],
                                                                                     'r': 0},
                                                  'time': {'max': 10.0, 'min': 0.0},
                                                  'type': 'xy-iss'}},
                      'init_param': { 'value': [ 0.003375513600300809,
                                                  0.4884127443235871,
                                                  0.6719352108785641,
                                                  0.9073170135093351,
                                                  1.4946633690060278,
                                                  1.5076957064847494,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  5.047622474508209,
                                                  5.054336886650782,
                                                  5.067626550795242,
                                                  5.0733174878702085,
                                                  5.094629387163461,
                                                  5.749571576552799,
                                                  5.7731554023304374,
                                                  7.144946717661993,
                                                  7.144946717661993,
                                                  7.144946717661993,
                                                  7.144946717661993,
                                                  7.602314289267704,
                                                  7.602314289267704,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  9.661071929582006,
                                                  0.001],
                      'value':
                      'noise_prob': [ 0.001,
                                     0.001,
                                     0.001,
                                     0.001],
                      'status': True,
                      'type': 'depolarizing'},
                      'odd_wires': 4}},
                      'run_time_sec': 82.4823865890503}},
                      { 'config': { 'ansatz': { 'gateset': 1,
                                                  'layer': 30,
                                                  'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                                                     'cn': [ 0.5,
                                                                                     0.5,
                                                                                     0.5,
                                                                                     0.5,
                                                                                     0.5,
                                                                                     0.5],
                                                                                     'r': 0},
                                                  'time': {'max': 10.0, 'min': 0.0},
                                                  'type': 'xy-iss'}},
                      'init_param': { 'value': [ 0.003375513600300809,
                                                  0.4884127443235871,
                                                  0.6719352108785641,
                                                  0.9073170135093351,
                                                  1.4946633690060278,
                                                  1.5076957064847494,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  3.872917838142945,
                                                  5.047622474508209,
                                                  5.054336886650782,
                                                  5.067626550795242,
                                                  5.0733174878702085,
                                                  5.094629387163461,
                                                  5.749571576552799,
                                                  5.7731554023304374,
                                                  7.144946717661993,
                                                  7.144946717661993,
                                                  7.144946717661993,
                                                  7.144946717661993,
                                                  7.602314289267704,
                                                  7.602314289267704,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  8.018775147517832,
                                                  9.661071929582006,
                                                  0.001],
                      'value':
                      'noise_prob': [ 0.001,
                                     0.001,
                                     0.001,
                                     0.001],
                      'status': True,
                      'type': 'depolarizing'},
                      'odd_wires': 4}},
                      'run_time_sec
```

0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,

$\emptyset\},$

```

        'noise_prob': [ 0.001,
                        0.001,
                        0.001,
                        0.001],
        'status': True,
        'type': 'depolarizing'},
    'odd_wires': 4},
    { 'gates_num': [44, 11, 11, 40],
      'identity_factors': [5, 5, 5, 0],
      'noise_level': [44, 22, 77, 40],
      'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':

        'noise_prob': [ 0.001,
                        0.001,
                        0.001,
                        0.001],
        'status': True,
        'type': 'depolarizing'},
    'odd_wires': 4}},
    'run_time_sec': 77.34903883934021}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}},
      'init_param': { 'value': [ 0.0019047787007362745,
                                0.7201496918445128,
                                0.7201496918445128,
                                0.7201496918445127,
                                0.7226708680142584,
                                0.8068041864232433,
                                1.3847174673528184,
                                1.6031895944534578,
                                2.458865333289775,
                                2.458865333289775,
                                2.721120236828938,
                                2.721120236828938,
                                3.554687304897056,
                                3.554687304897056,
                                4.537252190232355,
                                4.537252190232355,
                                5.239770593633025,
                                5.242363659304797,
                                5.566731623636481,
                                5.566731623636481,
                                5.566731623636481,
                                5.566731623636481,
                                5.566731623636481,
                                7.6484324534494865,
                                7.6484324534494865,
                                8.96757120343865,
                                9.352637118466301,
                                9.381932224950582,
                                9.966925955531599,
                                9.966925955531599,
                                0.29840004251263574,
                                0.3207087147269936,
                                -0.22478909909228914,
                                -1.2242611593120782,
                                0.061748844824383295,
                                1.9931455249507801,
                                0.02449362327304914,
                                -0.23402928130657288,
                                0.3399922671860315,
                                0.2355005268966802,
                                0.7912945290539132,

```

-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,

```

0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.753559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.536182288327377],
[12, 6, 21, 8, -3.450081350248876],
[20, 10, 35, 16, -1.8650932993173603],
[28, 14, 49, 24, -1.0247351729167147],
[36, 18, 63, 32, -0.5670015302617211],

```

```
[44, 22, 77, 40, -0.31237897705332807]],  
    'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
                        'identity_factors': [0, 0, 0, 0],  
                        'noise_level': [4, 2, 7, 0],  
                        'noise_profile': { 'noise_on_init_param': { 'status':  
False,  
                                'value':  
                                'noise_prob': [ 0.001,  
                                                0.001,  
                                                0.001,  
                                                0.001],  
                                'status': True,  
                                'type': 'depolarizing'},  
                        'odd_wires': 4},  
                    { 'gates_num': [12, 3, 3, 8],  
                      'identity_factors': [1, 1, 1, 0],  
                      'noise_level': [12, 6, 21, 8],  
                      'noise_profile': { 'noise_on_init_param': { 'status':  
False,  
                                'value':  
                                'noise_prob': [ 0.001,  
                                                0.001,  
                                                0.001,  
                                                0.001],  
                                'status': True,  
                                'type': 'depolarizing'},  
                      'odd_wires': 4},  
                    { 'gates_num': [20, 5, 5, 16],  
                      'identity_factors': [2, 2, 2, 0],  
                      'noise_level': [20, 10, 35, 16],  
                      'noise_profile': { 'noise_on_init_param': { 'status':  
False,  
                                'value':  
                                'noise_prob': [ 0.001,  
                                                0.001,  
                                                0.001,  
                                                0.001],  
                                'status': True,  
                                'type': 'depolarizing'},  
                      'odd_wires': 4},  
                    { 'gates_num': [28, 7, 7, 24],  
                      'identity_factors': [3, 3, 3, 0],  
                      'noise_level': [28, 14, 49, 24],  
                      'noise_profile': { 'noise_on_init_param': { 'status':  
False,  
                                'value':  
                                'noise_prob': [ 0.001,  
                                                0.001,  
                                                0.001,  
                                                0.001],  
                                'status': True,  
                                'type': 'depolarizing'},  
                      'odd_wires': 4},  
                    { 'gates_num': [36, 9, 9, 32],  
                      'identity_factors': [4, 4, 4, 0],  
                      'noise_level': [36, 18, 63, 32],  
                      'noise_profile': { 'noise_on_init_param': { 'status':  
False,  
                                'value':  
                                'noise_prob': [ 0.001,  
                                                0.001,  
                                                0.001,  
                                                0.001],  
                                'status': True,  
                                'type': 'depolarizing'},  
                      'odd_wires': 4},  
                    { 'gates_num': [44, 11, 11, 40],  
                      'identity_factors': [5, 5, 5, 0],  
                      'noise_level': [44, 22, 77, 40],  
                      'noise_profile': { 'noise_on_init_param': { 'status':
```

```

False,
0},
'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4]],
'run_time_sec': 79.09394240379333}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}}},
  'init_param': { 'value': [ -6.331146189926128e-17,
                              0.7069927066774304,
                              0.7069927066774304,
                              0.7069927066774305,
                              0.7069927066774304,
                              1.0044153364322925,
                              1.0092418692996383,
                              2.470333794214341,
                              2.478143749866421,
                              3.5776564981759003,
                              3.5776564981759003,
                              3.5776564981759003,
                              3.5776564981759003,
                              3.61745967653339,
                              3.6981780389472076,
                              3.755720232685741,
                              3.9712286023902594,
                              3.9712286023902594,
                              4.519037432417192,
                              4.519037432417193,
                              4.519037432417193,
                              5.280953883299406,
                              5.280953883299406,
                              6.620356980277175,
                              6.620356980277175,
                              6.620356980277175,
                              6.620356980277176,
                              8.214004051881032,
                              8.214004051881032,
                              9.105951544537387,
                              10.829366164164977,
                              1.4490195266558639,
                              1.145329650589643,
                              -0.9374272684957279,
                              -1.2910248872233119,
                              0.9701069453007352,
                              2.0238893871034924,
                              -0.6731600678051081,
                              -0.7305037462514778,
                              0.6743483023845589,
                              1.611766624162738,
                              -0.162446688913207,
                              -0.2464021583836281,
                              1.3073653685111934,
                              1.560629053845934,
                              -0.09768282308379246,
                              -0.31393750682085847,
                              0.1625705045557987,
                              -0.1344649185634244,
                              -0.03516682581352061,
                              0.15934568227883428,
                              1.4484389719925783,
                              0.31346340320241645,

```

-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,

```

-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Z 5 Z 6']]},
'output': { 'data_points': [ [4, 2, 7, 0, -6.311563719128683],
                              [12, 6, 21, 8, -3.2712898811023576],
                              [20, 10, 35, 16, -1.7533390824025292],
                              [28, 14, 49, 24, -0.966227702270059],
                              [36, 18, 63, 32, -0.5436512754494955],
                              [44, 22, 77, 40, -0.3098340729956386]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 7, 0],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001,

```

```

False,
0},
    {
      'status': True,
      'type': 'depolarizing'},
      'odd_wires': 4},
    {
      'gates_num': [12, 3, 3, 8],
      'identity_factors': [1, 1, 1, 0],
      'noise_level': [12, 6, 21, 8],
      'noise_profile': { 'noise_on_init_param': { 'status':
        'value':
          'noise_prob': [ 0.001,
                        0.001,
                        0.001,
                        0.001],
          'status': True,
          'type': 'depolarizing'},
        'odd_wires': 4},
      {
        'gates_num': [20, 5, 5, 16],
        'identity_factors': [2, 2, 2, 0],
        'noise_level': [20, 10, 35, 16],
        'noise_profile': { 'noise_on_init_param': { 'status':
          'value':
            'noise_prob': [ 0.001,
                          0.001,
                          0.001,
                          0.001],
            'status': True,
            'type': 'depolarizing'},
          'odd_wires': 4},
        {
          'gates_num': [28, 7, 7, 24],
          'identity_factors': [3, 3, 3, 0],
          'noise_level': [28, 14, 49, 24],
          'noise_profile': { 'noise_on_init_param': { 'status':
            'value':
              'noise_prob': [ 0.001,
                            0.001,
                            0.001,
                            0.001],
              'status': True,
              'type': 'depolarizing'},
            'odd_wires': 4},
          {
            'gates_num': [36, 9, 9, 32],
            'identity_factors': [4, 4, 4, 0],
            'noise_level': [36, 18, 63, 32],
            'noise_profile': { 'noise_on_init_param': { 'status':
              'value':
                'noise_prob': [ 0.001,
                              0.001,
                              0.001,
                              0.001],
                'status': True,
                'type': 'depolarizing'},
              'odd_wires': 4},
            {
              'gates_num': [44, 11, 11, 40],
              'identity_factors': [5, 5, 5, 0],
              'noise_level': [44, 22, 77, 40],
              'noise_profile': { 'noise_on_init_param': { 'status':
                'value':
                  'noise_prob': [ 0.001,
                                0.001,
                                0.001,
                                0.001],
                  'status': True,
                  'type': 'depolarizing'},
                'odd_wires': 4}],
              'run_time_sec': 68.65868830680847}},

```



```

{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.33528943002425843,
                              0.8608925283648261,
                              2.8309935233222867,
                              2.8382233121900358,
                              2.847558790018716,
                              2.8477263794127468,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              5.127462455084763,
                              5.128383696390232,
                              5.132051241075024,
                              5.132231848473024,
                              5.132297186784173,
                              5.132297186784173,
                              5.132297186784173,
                              5.132320898992942,
                              5.134903164275234,
                              5.734798022200061,
                              5.734798022200061,
                              5.734798022200061,
                              5.734798022200061,
                              6.71748919093495,
                              6.71748919093495,
                              9.773430860122208,
                              9.791656954875775,
                              9.791980525916317,
                              9.804782964874805,
                              9.947289619134542,
                              0.6818618664375632,
                              0.42253658736184846,
                              0.19067339209915313,
                              -2.314563573529872,
                              -1.7784854373672185,
                              -0.10945476580518654,
                              0.08980428543050965,
                              0.6265702304745648,
                              -0.12016219313034968,
                              0.005450784512528824,
                              0.9621316606784164,
                              -0.01940541263687908,
                              -1.2890091107885369,
                              -0.09216272929775945,
                              -0.8615487717558121,
                              0.021600758375482327,
                              1.1068545366678577,
                              -0.06436428812394061,
                              -1.8186253546793636,
                              -0.2455724158610742,
                              0.7444478114283775,
                              1.0647343146888184,
                              -1.6388802958603008,
                              -0.05032235091912311,
                              0.7589104637111997,
                              0.0934759886788552,
                              0.7280377617711271,
                              -0.10012682423470079,
                              -2.7917715562202794,
                              0.5843828201862561,
                              1.2003794612588048,
                              -0.2782652692858759,
                              1.253725179404932,

```

-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,

```

1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.333005581610797],
                             [12, 6, 21, 8, -3.1651017858597426],
                             [20, 10, 35, 16, -1.6439583689682524],
                             [28, 14, 49, 24, -0.8813072530521873],
                             [36, 18, 63, 32, -0.48318628235120475],
                             [44, 22, 77, 40, -0.26793423014165196]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 7, 0],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                'value':
0},
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [12, 3, 3, 8],
                                'identity_factors': [1, 1, 1, 0],
                                'noise_level': [12, 6, 21, 8],
                                'noise_profile': { 'noise_on_init_param': { 'status':
False,
                                'value':
0},

```

[illegible]

```
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.08591881204796584,
1.5026023651616802,
1.9179097588555833,
1.9179097588555831,
1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
```

-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,


```

False,
0},
    'noise_prob': [ 0.001,
                    0.001,
                    0.001,
                    0.001],
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
  'identity_factors': [3, 3, 3, 0],
  'noise_level': [28, 14, 49, 24],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
    'noise_prob': [ 0.001,
                    0.001,
                    0.001,
                    0.001],
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
  'identity_factors': [4, 4, 4, 0],
  'noise_level': [36, 18, 63, 32],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
    'noise_prob': [ 0.001,
                    0.001,
                    0.001,
                    0.001],
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [44, 11, 11, 40],
  'identity_factors': [5, 5, 5, 0],
  'noise_level': [44, 22, 77, 40],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
    'noise_prob': [ 0.001,
                    0.001,
                    0.001,
                    0.001],
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4}}],
    'run_time_sec': 68.4976704120636}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
    'init_param': { 'value': [ 3.750846163374531e-05,
                                0.22285241391868899,
                                0.26252411232084344,
                                0.26252411232084344,
                                0.26252411232084344,
                                1.459545906754556,
                                1.4595459067545562,
                                1.6094835194149661,
                                2.5927782099427628,
                                2.5927782099427628,

```


2.5927782099427628,
2.5927782099427623,
3.3552499939702187,
3.3552499939702187,
5.033284106467877,
5.033284106467877,
5.056477210452712,
6.768913439946703,
6.768913439946703,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,

```

-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},

```

```

        'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0]]},

'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},

'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
         'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                    '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                    '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                    '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                    '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                    '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},

'output': { 'data_points': [ [4, 2, 7, 0, -6.460678265045408],
                             [12, 6, 21, 8, -3.2738854608106305],
                             [20, 10, 35, 16, -1.6974907774474757],
                             [28, 14, 49, 24, -0.8939947022135477],
                             [36, 18, 63, 32, -0.4736436630774058],
                             [44, 22, 77, 40, -0.24910822433441174]]},

'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 7, 0],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                'value':

                                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                    { 'gates_num': [12, 3, 3, 8],
                    'identity_factors': [1, 1, 1, 0],
                    'noise_level': [12, 6, 21, 8],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                'value':

                                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                    { 'gates_num': [20, 5, 5, 16],
                    'identity_factors': [2, 2, 2, 0],
                    'noise_level': [20, 10, 35, 16],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                'value':

                                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                    { 'gates_num': [28, 7, 7, 24],

```

```

False,
0},
'identity_factors': [3, 3, 3, 0],
'noise_level': [28, 14, 49, 24],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
'identity_factors': [4, 4, 4, 0],
'noise_level': [36, 18, 63, 32],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [44, 11, 11, 40],
'identity_factors': [5, 5, 5, 0],
'noise_level': [44, 22, 77, 40],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 67.9608006477356}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 1.5680106786222674e-17,
1.0238650999177359,
1.0238650999177359,
1.0238650999177359,
1.7041550191762513,
2.070931805530806,
2.586234954306046,
2.6265616878606783,
2.7566136785306155,
2.7689271757054255,
4.271831894498143,
4.271831894498143,
4.271831894498143,
5.004110362344359,
5.004110362344359,
5.529689350278954,
5.529689350278954,
5.529689350278954,
5.5493166460873375,
7.855838962179162,
7.855838962179162,

```

8.508023697610218,
8.540796404638465,
8.940056981382925,
8.940056981382925,
8.940056981382925,
8.940056981382925,
8.940056981382925,
9.524293982069336,
9.524293982069336,
9.815045396471412,
0.717685470636315,
0.29125971197830336,
0.013619559122678915,
-0.46174394022911175,
-0.7802589318548062,
1.8785332138943256,
-0.07718385116900997,
-0.4181315830102034,
0.3859499067718677,
0.7218523694763775,
0.16350787496293626,
-0.06274337543165032,
1.048639096314506,
0.9454500907927006,
-0.18328208251477485,
0.5336296017995852,
0.7047960543548563,
-0.4717130073686535,
-0.14327049451726778,
0.5646728024946054,
0.1377201649736573,
1.0845662084088512,
0.7117671017527167,
1.8562916393289055,
-1.0958742208007046,
0.314756410152813,
1.0055290931424188,
-1.2653176839886926,
-0.6435463058390287,
-0.13689785210195535,
0.5185042269644529,
-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,

```

0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455}},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                 'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 0],
                                      [2, 2, 2, 0],
                                      [3, 3, 3, 0],
                                      [4, 4, 4, 0],
                                      [5, 5, 5, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',

```

```

'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.515742960091852],
[12, 6, 21, 8, -3.3649211100370207],
[20, 10, 35, 16, -1.7545658776574033],
[28, 14, 49, 24, -0.9139478342631157],
[36, 18, 63, 32, -0.4680317780831103],
[44, 22, 77, 40, -0.22931309151635063],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
'identity_factors': [2, 2, 2, 0],
'noise_level': [20, 10, 35, 16],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
'identity_factors': [3, 3, 3, 0],
'noise_level': [28, 14, 49, 24],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,

```

False,

0},

False,

0},

False,

0},

False,

0},

```

                                'type': 'depolarizing'},
                                'odd_wires': 4},
                                { 'gates_num': [36, 9, 9, 32],
                                  'identity_factors': [4, 4, 4, 0],
                                  'noise_level': [36, 18, 63, 32],
                                  'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                      'value':
                                                                                      'noise_prob': [ 0.001,
                                                                                      0.001,
                                                                                      0.001,
                                                                                      0.001],
                                                                                      'status': True,
                                                                                      'type': 'depolarizing'},
                                  'odd_wires': 4},
                                  { 'gates_num': [44, 11, 11, 40],
                                    'identity_factors': [5, 5, 5, 0],
                                    'noise_level': [44, 22, 77, 40],
                                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                      'value':
                                                                                      'noise_prob': [ 0.001,
                                                                                      0.001,
                                                                                      0.001,
                                                                                      0.001],
                                                                                      'status': True,
                                                                                      'type': 'depolarizing'},
                                    'odd_wires': 4}],
                                'run_time_sec': 70.21013498306274}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                         0.5,
                                                         0.5,
                                                         0.5,
                                                         0.5,
                                                         0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.00030922277578629,
                             0.881375055029912,
                             0.881375055029912,
                             0.8813750550299119,
                             0.9803845151714838,
                             0.9803845151714838,
                             1.201629973822729,
                             1.2441077564052723,
                             1.8660677974280824,
                             1.8660677974280824,
                             2.651002314516478,
                             2.651002314516478,
                             3.233637461790533,
                             4.36384995486056,
                             4.36384995486056,
                             4.975714528653046,
                             4.975714528653046,
                             6.591840750700408,
                             6.591840750700408,
                             8.234009055219385,
                             8.238223602720305,
                             8.238223602720305,
                             8.238223602720305,
                             8.903406631369169,
                             8.903406631369169,
                             8.903406631369169,
                             9.126449234760914,
                             9.126449234760914,
                             9.441918167657187,
                             9.441918167657187,
                             0.4384156956352651,
                             -0.017407483354635628,

```


-0.13579407977759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,

```

0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisy_time_evo_ric6_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0]],
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 X 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '

```

[illegible]

```

0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [44, 11, 11, 40],
'identity_factors': [5, 5, 5, 0],
'noise_level': [44, 22, 77, 40],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'value':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 79.56445574760437}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
'file_name_prefix': 'xy_noisey_time_evo'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1 Z 2 '
+ 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z '
5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + 0.5 X 1 '
'X 2 + 0.5 Z 1 Z 2 + 0.5 X 2 X 3 + '
'0.5 Z 2 Z 3 + 0.5 X 3 X 4 + 0.5 Z 3 '
'Z 4 + 0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
'initial_cost_history': [ 3.5476707600867368,
2.3562053879970173,
1.7833099820235232,
2.4654982864165467,
3.6613143494662093,
3.6462371631182062,
3.2935768662401737,
3.7145398712422657,
2.5227446974102397,
2.4756634278387035],
'noise_details': { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},

```

```
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -6.515742960091897,
                             -6.3330055816107915,
                             -6.410492799212151,
                             -6.4484545761945915,
                             -6.311563719128753,
                             -6.46067826504545,
                             -6.397817460329919,
                             -6.429933051303193,
                             -6.536182288327426,
                             -6.511175848027232],
'optimized_parameters': [ [ 1.5680106786222674e-17,
                             1.0238650999177359,
                             1.0238650999177359,
                             1.0238650999177359,
                             1.7041550191762513,
                             2.070931805530806,
                             2.586234954306046,
                             2.6265616878606783,
                             2.7566136785306155,
                             2.7689271757054255,
                             4.271831894498143,
                             4.271831894498143,
                             4.271831894498143,
                             5.004110362344359,
                             5.004110362344359,
                             5.529689350278954,
                             5.529689350278954,
                             5.529689350278954,
                             5.529689350278954,
                             5.5493166460873375,
                             7.855838962179162,
                             7.855838962179162,
                             8.508023697610218,
                             8.540796404638465,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             9.524293982069336,
                             9.524293982069336,
                             9.815045396471412,
                             0.717685470636315,
                             0.29125971197830336,
                             0.013619559122678915,
                             -0.46174394022911175,
                             -0.7802589318548062,
                             1.8785332138943256,
                             -0.07718385116900997,
                             -0.4181315830102034,
                             0.3859499067718677,
                             0.7218523694763775,
                             0.16350787496293626,
                             -0.06274337543165032,
                             1.048639096314506,
                             0.9454500907927006,
                             -0.18328208251477485,
                             0.5336296017995852,
                             0.7047960543548563,
                             -0.4717130073686535,
                             -0.14327049451726778,
                             0.5646728024946054,
                             0.1377201649736573,
                             1.0845662084088512,
                             0.7117671017527167,
                             1.8562916393289055,
                             -1.0958742208007046,
                             0.314756410152813,
                             1.0055290931424188,
                             -1.2653176839886926,
                             -0.6435463058390287,
                             -0.13689785210195535,
                             0.5185042269644529,
```

-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,

0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455],
[0.33528943002425843,
0.8608925283648261,
2.8309935233222867,
2.8382233121900358,
2.847558790018716,
2.8477263794127468,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
5.127462455084763,
5.128383696390232,
5.132051241075024,
5.132231848473024,
5.132297186784173,
5.132297186784173,
5.132297186784173,
5.132320898992942,
5.134903164275234,
5.734798022200061,
5.734798022200061,
5.734798022200061,
6.71748919093495,
6.71748919093495,
9.773430860122208,
9.791656954875775,
9.791980525916317,
9.804782964874805,
9.947289619134542,
0.6818618664375632,
0.42253658736184846,
0.19067339209915313,
-2.314563573529872,
-1.7784854373672185,
-0.10945476580518654,
0.08980428543050965,
0.6265702304745648,
-0.12016219313034968,
0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,
-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,

-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,

1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358],
[0.08591881204796584,
1.5026023651616802,
1.9179097588555833,
1.9179097588555831,
1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,

1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,

-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406],
[0.3773346757087749,
0.5734233884194113,
0.5734233884194113,
1.2094818154974225,
1.2528672337957094,
1.3073966674132076,
4.193348178434801,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.274742575727675,
5.343872214974395,
5.354523589318104,
5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,

1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,

0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244],
[-6.331146189926128e-17,
0.7069927066774304,
0.7069927066774304,
0.7069927066774305,
0.7069927066774304,
1.0044153364322925,
1.0092418692996383,
2.470333794214341,
2.478143749866421,
3.5776564981759003,
3.5776564981759003,
3.5776564981759003,
3.61745967653339,
3.6981780389472076,
3.755720232685741,
3.9712286023902594,
3.9712286023902594,
4.519037432417192,
4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,
6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,

0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,

-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176],
[3.750846163374531e-05,
0.22285241391868899,
0.26252411232084344,
0.26252411232084344,
0.26252411232084344,
1.459545906754556,
1.4595459067545562,
1.6094835194149661,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
2.5927782099427623,
3.3552499939702187,
3.3552499939702187,
5.033284106467877,
5.033284106467877,
5.056477210452712,
6.768913439946703,
6.768913439946703,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,

0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,

0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138],
[0.003375513600300809,
0.4884127443235871,
0.6719352108785641,
0.9073170135093351,
1.4946633690060278,
1.5076957064847494,
3.872917838142945,
3.872917838142945,
3.872917838142945,
3.872917838142945,
3.872917838142945,
5.047622474508209,
5.054336886650782,
5.067626550795242,
5.0733174878702085,
5.094629387163461,
5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,

1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,

0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989],
[-5.647934061474703e-18,
1.4787834906243755,
1.478918229526628,
1.478918229526628,
2.0077351764243123,
2.4302809101456218,
2.4302809101456218,
3.009709563144601,
3.009709563144601,
3.239471043198664,
3.2405283031129133,
3.2405283031129133,
3.2656598154222434,
3.5353392937639825,
6.198713539858257,
6.213604956994236,
6.73397854893658,
6.73397854893658,
6.743547791194661,
6.744609103740889,
6.744609103740889,
6.749543065628153,
6.749543065628153,
8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,

-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,

-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935],
[0.0019047787007362745,
0.7201496918445128,
0.7201496918445128,
0.7201496918445127,
0.7226708680142584,
0.8068041864232433,
1.3847174673528184,
1.6031895944534578,
2.458865333289775,
2.458865333289775,
2.721120236828938,
2.721120236828938,
3.554687304897056,
3.554687304897056,
4.537252190232355,
4.537252190232355,
5.239770593633025,
5.242363659304797,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
7.6484324534494865,
7.6484324534494865,
8.96757120343865,
9.352637118466301,
9.381932224950582,
9.966925955531599,
9.966925955531599,
0.29840004251263574,
0.3207087147269936,
-0.22478909909228914,
-1.2242611593120782,
0.061748844824383295,
1.9931455249507801,
0.02449362327304914,
-0.23402928130657288,
0.3399922671860315,
0.2355005268966802,
0.7912945290539132,
-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,

0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,

-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815],
[0.00030922277578629,
0.881375055029912,
0.881375055029912,
0.8813750550299119,
0.9803845151714838,
0.9803845151714838,
1.201629973822729,
1.2441077564052723,
1.8660677974280824,
1.8660677974280824,
2.651002314516478,
2.651002314516478,
3.233637461790533,
4.36384995486056,
4.36384995486056,
4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,
8.903406631369169,
8.903406631369169,
8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.13579407977759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,

0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,


```

0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'run_time_sec': 68892.59480500221]]},
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
                                  'layer': 30,
                                  'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                                'r': 0},
                                  'time': {'max': 10.0, 'min': 0.0},
                                  'type': 'xy-iss'}},
'init_param': { 'value': [ 3.750846163374531e-05,
0.22285241391868899,
0.26252411232084344,
0.26252411232084344,
0.26252411232084344,
1.459545906754556,
1.4595459067545562,
1.6094835194149661,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
2.5927782099427623,
3.3552499939702187,
3.3552499939702187,
5.033284106467877,
5.033284106467877,
5.056477210452712,
6.768913439946703,
6.768913439946703,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,

```

1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,

```

-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.460678265045408],
                          [12, 6, 21, 8, -3.2738854608106305],
                          [20, 10, 35, 16, -1.6974907774474757],
                          [28, 14, 49, 24, -0.8939947022135477],
                          [36, 18, 63, 32, -0.4736436630774058],
                          [44, 22, 77, 40, -0.24910822433441174]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 6.437301635742188e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.398117369264428,
                           'others': { 'beta_coeffieicients': [ 2.1894863603561845,
                                                                -3.028013051556425,
                                                                3.513990454892641,
                                                                -2.4750715377939474,
                                                                0.9551450565312212,
                                                                -0.15553728242967432],
                                      'cost_richardson_zne': 33.37331552209775,
                                      'sorted_expectation_vals': [ -6.460678265045408,
                                                                -3.2738854608106305,
                                                                -1.6974907774474757,
                                                                -0.8939947022135477,
                                                                -0.4736436630774058,
                                                                -0.24910822433441174],
                                      'sorted_noise': [13, 47, 81, 115, 149, 183]},
                           'sampled data': [ [4, 2, 7, 0, -6.460678265045408],
                                              [12, 6, 21, 8, -3.2738854608106305],
                                              [20, 10, 35, 16, -1.6974907774474757],

```

```

[28, 14, 49, 24, -0.8939947022135477],
[36, 18, 63, 32, -0.4736436630774058],
[44, 22, 77, 40, -0.24910822433441174]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 1.5680106786222674e-17,
1.0238650999177359,
1.0238650999177359,
1.0238650999177359,
1.7041550191762513,
2.070931805530806,
2.586234954306046,
2.6265616878606783,
2.7566136785306155,
2.7689271757054255,
4.271831894498143,
4.271831894498143,
4.271831894498143,
5.004110362344359,
5.004110362344359,
5.529689350278954,
5.529689350278954,
5.529689350278954,
5.529689350278954,
5.5493166460873375,
7.855838962179162,
7.855838962179162,
8.508023697610218,
8.540796404638465,
8.940056981382925,
8.940056981382925,
8.940056981382925,
8.940056981382925,
8.940056981382925,
9.524293982069336,
9.524293982069336,
9.815045396471412,
0.717685470636315,
0.29125971197830336,
0.013619559122678915,
-0.46174394022911175,
-0.7802589318548062,
1.8785332138943256,
-0.07718385116900997,
-0.4181315830102034,
0.3859499067718677,
0.7218523694763775,
0.16350787496293626,
-0.06274337543165032,
1.048639096314506,
0.9454500907927006,
-0.18328208251477485,
0.5336296017995852,
0.7047960543548563,
-0.4717130073686535,
-0.14327049451726778,
0.5646728024946054,
0.1377201649736573,
1.0845662084088512,
0.7117671017527167,
1.8562916393289055,
-1.0958742208007046,
0.314756410152813,
1.0055290931424188,
-1.2653176839886926,
-0.6435463058390287,
-0.13689785210195535,
0.5185042269644529,
-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,

```

-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.2290356822623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,


```
'init_param': { 'value': [ 0.00030922277578629,  
0.881375055029912,  
0.881375055029912,  
0.8813750550299119,  
0.9803845151714838,  
0.9803845151714838,  
1.201629973822729,  
1.2441077564052723,  
1.8660677974280824,  
1.8660677974280824,  
2.651002314516478,  
2.651002314516478,  
3.233637461790533,  
4.36384995486056,  
4.36384995486056,  
4.975714528653046,  
4.975714528653046,  
6.591840750700408,  
6.591840750700408,  
8.234009055219385,  
8.238223602720305,  
8.238223602720305,  
8.238223602720305,  
8.903406631369169,  
8.903406631369169,  
8.903406631369169,  
9.126449234760914,  
9.126449234760914,  
9.441918167657187,  
9.441918167657187,  
0.4384156956352651,  
-0.017407483354635628,  
-0.13579407977759564,  
-0.08181457210659056,  
-0.5635031413839604,  
0.570197125887921,  
-0.07389454022208618,  
0.18950484918514415,  
0.23442314823895152,  
0.5187197914643009,  
0.03760777637611914,  
1.2080522066616035,  
-0.006796933498365149,  
0.7597395812711474,  
0.3165595784185247,  
0.09234381717322694,  
-0.047116958459538144,  
0.02161370846364295,  
-0.3330735946096464,  
-0.310519881157395,  
-0.1321730433969341,  
-0.32004747583182785,  
0.38880240494533963,  
0.5587916266845782,  
-0.04068226055752174,  
-0.04929253976644778,  
0.7350044083650977,  
-0.3347542084226466,  
0.07994172439872818,  
0.177027499128215,  
-0.03555523462670161,  
0.33584282142298794,  
-0.06474358479167515,  
-0.23271669416276414,  
0.27707249457309313,  
-1.8841797053563085,  
-0.39250057722196113,  
0.0604718224769807,  
-0.15658247224402194,  
1.097732135880844,  
-0.47958374693582667,  
-0.13884788062191106,  
1.2517060190766023,  
0.08912037934357776,  
1.8722824819152626,
```

0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]]},


```

'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0.001, 0.001],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.5111758480272135],
                          [12, 6, 21, 8, -3.513108400075424],
                          [20, 10, 35, 16, -1.9256894730075684],
                          [28, 14, 49, 24, -1.0648100061563792],
                          [36, 18, 63, 32, -0.5886022016151555],
                          [44, 22, 77, 40, -0.3212779442729232]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.7206878662109375e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -8.261995894794591,
                          'others': { 'beta_coefficients': [ 2.1894863603561845,
                                                            -3.028013051556425,
                                                            3.513990454892641,
                                                            -2.4750715377939474,
                                                            0.9551450565312212,
                                                            -0.15553728242967432],
                                      'cost_richardson_zne': 33.37331552209775,
                                      'sorted_expectation_vals': [ -6.5111758480272135,
                                                                    -3.513108400075424,
                                                                    -1.9256894730075684,
                                                                    -1.0648100061563792,
                                                                    -0.5886022016151555,
                                                                    -0.3212779442729232],
                                      'sorted_noise': [13, 47, 81, 115, 149, 183]},
                          'sampled data': [ [4, 2, 7, 0, -6.5111758480272135],
                                              [12, 6, 21, 8, -3.513108400075424],
                                              [20, 10, 35, 16, -1.9256894730075684],
                                              [28, 14, 49, 24, -1.0648100061563792],
                                              [36, 18, 63, 32, -0.5886022016151555],
                                              [44, 22, 77, 40, -0.3212779442729232]],
                          'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                       'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                       'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}}},
  'init_param': { 'value': [ -6.331146189926128e-17,
                              0.7069927066774304,
                              0.7069927066774304,
                              0.7069927066774305,
                              0.7069927066774304,
                              1.0044153364322925,
                              1.0092418692996383,
                              2.470333794214341,
                              2.478143749866421,
                              3.5776564981759003,
                              3.5776564981759003,

```

3.5776564981759003,
3.61745967653339,
3.6981780389472076,
3.755720232685741,
3.9712286023902594,
3.9712286023902594,
4.519037432417192,
4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,
6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,

```

0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#5'},

```


5.566731623636481,
7.6484324534494865,
7.6484324534494865,
8.96757120343865,
9.352637118466301,
9.381932224950582,
9.966925955531599,
9.966925955531599,
0.29840004251263574,
0.3207087147269936,
-0.22478909909228914,
-1.2242611593120782,
0.061748844824383295,
1.9931455249507801,
0.02449362327304914,
-0.23402928130657288,
0.3399922671860315,
0.2355005268966802,
0.7912945290539132,
-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,
0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,

```

-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 0],
                                      [2, 2, 2, 0],
                                      [3, 3, 3, 0],
                                      [4, 4, 4, 0],
                                      [5, 5, 5, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,

```

```

        'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.536182288327377],
                          [12, 6, 21, 8, -3.450081350248876],
                          [20, 10, 35, 16, -1.8650932993173603],
                          [28, 14, 49, 24, -1.0247351729167147],
                          [36, 18, 63, 32, -0.5670015302617211],
                          [44, 22, 77, 40, -0.31237897705332807]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.744529724121094e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.374599934128453,
                           'others': { 'beta_coefficients': [ 2.1894863603561845,
                                                                -3.028013051556425,
                                                                3.513990454892641,
                                                                -2.4750715377939474,
                                                                0.9551450565312212,
                                                                -0.15553728242967432],
                                       'cost_richardson_zne': 33.37331552209775,
                                       'sorted_expectation_vals': [ -6.536182288327377,
                                                                    -3.450081350248876,
                                                                    -1.8650932993173603,
                                                                    -1.0247351729167147,
                                                                    -0.5670015302617211,
                                                                    -0.31237897705332807],
                                       'sorted_noise': [13, 47, 81, 115, 149, 183]},
                           'sampled data': [ [4, 2, 7, 0, -6.536182288327377],
                                              [12, 6, 21, 8, -3.450081350248876],
                                              [20, 10, 35, 16, -1.8650932993173603],
                                              [28, 14, 49, 24, -1.0247351729167147],
                                              [36, 18, 63, 32, -0.5670015302617211],
                                              [44, 22, 77, 40, -0.31237897705332807]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.33528943002425843,
                              0.8608925283648261,
                              2.8309935233222867,
                              2.8382233121900358,
                              2.847558790018716,
                              2.8477263794127468,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              5.127462455084763,
                              5.128383696390232,
                              5.132051241075024,
                              5.132231848473024,
                              5.132297186784173,
                              5.132297186784173,
                              5.132297186784173,
                              5.132320898992942,
                              5.134903164275234,
                              5.734798022200061,
                              5.734798022200061,
                              5.734798022200061,
                              6.71748919093495,
                              6.71748919093495,
                              9.773430860122208,
                              9.791656954875775,
                              9.791980525916317,
                              9.804782964874805,
                              9.947289619134542,
                              0.6818618664375632,
                              0.42253658736184846,
                              0.19067339209915313,

```

-2.314563573529872,
-1.7784854373672185,
-0.10945476580518654,
0.08980428543050965,
0.6265702304745648,
-0.12016219313034968,
0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,
-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,


```

-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.333005581610797],
                          [12, 6, 21, 8, -3.1651017858597426],
                          [20, 10, 35, 16, -1.6439583689682524],
                          [28, 14, 49, 24, -0.8813072530521873],
                          [36, 18, 63, 32, -0.48318628235120475],
                          [44, 22, 77, 40, -0.26793423014165196]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,

```

```
'run_time_sec': 8.177757263183594e-05,
'zne_values': { 'degree': 0,
                 'extrapolated_value': -8.297454569568384,
                 'others': { 'beta_coeffieicients': [ 2.1894863603561845,
                                                       -3.028013051556425,
                                                       3.513990454892641,
                                                       -2.4750715377939474,
                                                       0.9551450565312212,
                                                       -0.15553728242967432],
                             'cost_richardson_zne': 33.37331552209775,
                             'sorted_expectation_vals': [ -6.333005581610797,
                                                           -3.1651017858597426,
                                                           -1.6439583689682524,
                                                           -0.8813072530521873,
                                                           -0.48318628235120475,
                                                           -0.26793423014165196],
                             'sorted_noise': [13, 47, 81, 115, 149, 183]},
                 'sampled data': [ [4, 2, 7, 0, -6.333005581610797],
                                   [12, 6, 21, 8, -3.1651017858597426],
                                   [20, 10, 35, 16, -1.6439583689682524],
                                   [28, 14, 49, 24, -0.8813072530521873],
                                   [36, 18, 63, 32, -0.48318628235120475],
                                   [44, 22, 77, 40, -0.26793423014165196]],
                 'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.3773346757087749,
                              0.5734233884194113,
                              0.5734233884194113,
                              1.2094818154974225,
                              1.2528672337957094,
                              1.3073966674132076,
                              4.193348178434801,
                              4.228670385547864,
                              4.228670385547864,
                              4.228670385547864,
                              4.228670385547864,
                              4.228670385547864,
                              4.274742575727675,
                              5.343872214974395,
                              5.354523589318104,
                              5.449133113124401,
                              5.449133113124401,
                              5.449133113124401,
                              5.449133113124401,
                              6.208341300319731,
                              6.208341300319731,
                              6.738723848778546,
                              6.7473872692498285,
                              8.515456463543387,
                              8.552181104315652,
                              8.552181104315652,
                              8.552181104315652,
                              9.363042150968939,
                              9.363042150968939,
                              11.478464454984158,
                              -0.2683676450988651,
                              -0.9370321768196137,
                              0.03790413469005285,
                              -0.3520989510779511,
                              0.5131804204688105,
                              0.767748989456867,
                              -1.1529933056041268,
                              1.0195402939041247,
                              -0.5189860890713169,
                              1.700377968800765,
                              0.48912124484584857,
                              1.506730848787325,
                              0.4485612726223177,
                              -1.6248073049289613,
```

0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,

```

-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.44845457619454],
                          [12, 6, 21, 8, -3.2446894681699203],
                          [20, 10, 35, 16, -1.686607616211824],
                          [28, 14, 49, 24, -0.8984323471828196],
                          [36, 18, 63, 32, -0.4857082686875758],
                          [44, 22, 77, 40, -0.2634879429635577]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 0.0003571510314941406,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.419819668611884,
                           'others': { 'beta_coefficients': [ 2.1894863603561845,
                                                             -3.028013051556425,
                                                             3.513990454892641,
                                                             -2.4750715377939474,
                                                             0.9551450565312212,
                                                             -0.15553728242967432],
                                       'cost_richardson_zne': 33.37331552209775,
                                       'sorted_expectation_vals': [ -6.44845457619454,

```

```

-3.2446894681699203,
-1.686607616211824,
-0.8984323471828196,
-0.4857082686875758,
-0.2634879429635577],
'sorted_noise': [13, 47, 81, 115, 149, 183]],
'sampled data': [ [4, 2, 7, 0, -6.44845457619454],
[12, 6, 21, 8, -3.2446894681699203],
[20, 10, 35, 16, -1.686607616211824],
[28, 14, 49, 24, -0.8984323471828196],
[36, 18, 63, 32, -0.4857082686875758],
[44, 22, 77, 40, -0.2634879429635577]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.08591881204796584,
1.5026023651616802,
1.9179097588555833,
1.9179097588555831,
1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,

```

0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,
1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,

```

0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.4104927992121],
[12, 6, 21, 8, -3.2577133670016694],
[20, 10, 35, 16, -1.674813432318219],
[28, 14, 49, 24, -0.8591076530612254],
[36, 18, 63, 32, -0.43058689568319575],
[44, 22, 77, 40, -0.20313281829021995]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.601478576660156e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.309891686714744,
'others': { 'beta_coefficients': [ 2.1894863603561845,
-3.028013051556425,
3.513990454892641,
-2.4750715377939474,
0.9551450565312212,
-0.15553728242967432],
'cost_richardson_zne': 33.37331552209775,
'sorted_expectation_vals': [ -6.4104927992121,
-3.2577133670016694,
-1.674813432318219,
-0.8591076530612254,
-0.43058689568319575,
-0.20313281829021995],
'sorted_noise': [13, 47, 81, 115, 149, 183]}},
'sampled data': [ [4, 2, 7, 0, -6.4104927992121],
[12, 6, 21, 8, -3.2577133670016694],
[20, 10, 35, 16, -1.674813432318219],
[28, 14, 49, 24, -0.8591076530612254],
[36, 18, 63, 32, -0.43058689568319575],

```

```
[44, 22, 77, 40, -0.20313281829021995]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.003375513600300809,
                              0.4884127443235871,
                              0.6719352108785641,
                              0.9073170135093351,
                              1.4946633690060278,
                              1.5076957064847494,
                              3.872917838142945,
                              3.872917838142945,
                              3.872917838142945,
                              3.872917838142945,
                              5.047622474508209,
                              5.054336886650782,
                              5.067626550795242,
                              5.0733174878702085,
                              5.094629387163461,
                              5.749571576552799,
                              5.7731554023304374,
                              7.144946717661993,
                              7.144946717661993,
                              7.144946717661993,
                              7.144946717661993,
                              7.602314289267704,
                              7.602314289267704,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              8.018775147517832,
                              9.661071929582006,
                              0.2735947252101626,
                              2.471262065567802,
                              0.18954222370300397,
                              0.2633058348660961,
                              0.07494334922808346,
                              1.4980157261935538,
                              0.3538857424122425,
                              0.26341770410626436,
                              0.17174804107002956,
                              -0.22912718472194316,
                              0.11060714764109056,
                              -0.30804859927082595,
                              2.8406850614439736,
                              1.0823654569610508,
                              0.9668529413216363,
                              1.8727663516543385,
                              -0.0901659930896485,
                              0.05868560690604535,
                              0.24212775097742179,
                              1.0092515209898247,
                              -0.7556969386491817,
                              0.3279350524218167,
                              1.078117272505244,
                              0.10391815458190773,
                              -0.7434930272741371,
                              -1.6965388530236032,
                              -1.360748554100242,
                              0.34939560927302815,
                              0.03188634842980519,
                              -1.2447472675125983,
                              0.06092342743494014,
                              1.017082923902293,
                              -1.007905562333318,
                              -0.8576057837691866,
                              0.5321038189070756,
                              1.300652060018237,
```


-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,

```

0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.397817460329862],
[12, 6, 21, 8, -3.211571673080875],
[20, 10, 35, 16, -1.641958817033255],
[28, 14, 49, 24, -0.8464644349040009],
[36, 18, 63, 32, -0.4337371704190119],
[44, 22, 77, 40, -0.216007189653039]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.696846008300781e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.33870544603468,
'others': { 'beta_coeffieicients': [ 2.1894863603561845,
-3.028013051556425,
3.513990454892641,
-2.4750715377939474,
0.9551450565312212,
-0.15553728242967432],
'cost_richardson_zne': 33.37331552209775,
'sorted_expectation_vals': [ -6.397817460329862,
-3.211571673080875,
-1.641958817033255,
-0.8464644349040009,
-0.4337371704190119,
-0.216007189653039],
'sorted_noise': [13, 47, 81, 115, 149, 183]},
'sampled data': [ [4, 2, 7, 0, -6.397817460329862],
[12, 6, 21, 8, -3.211571673080875],
[20, 10, 35, 16, -1.641958817033255],
[28, 14, 49, 24, -0.8464644349040009],
[36, 18, 63, 32, -0.4337371704190119],
[44, 22, 77, 40, -0.216007189653039]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ -5.647934061474703e-18,
1.4787834906243755,

```

1.478918229526628,
1.478918229526628,
2.0077351764243123,
2.4302809101456218,
2.4302809101456218,
3.009709563144601,
3.009709563144601,
3.239471043198664,
3.2405283031129133,
3.2405283031129133,
3.2656598154222434,
3.5353392937639825,
6.198713539858257,
6.213604956994236,
6.73397854893658,
6.73397854893658,
6.743547791194661,
6.744609103740889,
6.744609103740889,
6.749543065628153,
6.749543065628153,
8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,

```
-1.249293985363804,  
-2.5283772945229797,  
-1.0735775181393186,  
-0.628541665091616,  
-1.1357659634071433,  
-0.5734897531715744,  
-1.6096579963890296,  
-0.8342055948126037,  
-1.5447846874108708,  
-1.1270394146554403,  
0.06392732561319403,  
0.7711512775812011,  
-1.8472478314151175,  
-0.46160141905881696,  
1.8963279721150403,  
-2.313663424814036,  
0.060952702837845166,  
-1.4971949982077863,  
1.3040610818975555,  
0.10605679464403613,  
-1.138800572315487,  
-0.03947636943653289,  
0.07839455865010819,  
-0.6470277815644087,  
0.946101637008616,  
-0.32144060478582315,  
-0.6371693991779968,  
0.30580409265523,  
-0.2617129381706641,  
0.4094287336545931,  
0.49837362733711055,  
-0.9058714721495094,  
0.6137893604415533,  
0.8546844695081749,  
-0.8205470722456557,  
-0.598852289145785,  
0.46823811794119646,  
1.2205247300232143,  
0.031629319060699435,  
1.1369587010491427,  
0.12298062775535874,  
-0.016960252964031076,  
1.1127338626212067,  
0.37972118532018134,  
-0.1475616440844684,  
0.32749925367563326,  
0.030050642250761914,  
0.17169542743574548,  
-0.5825729547476518,  
0.7052714548722974,  
0.9884296874822006,  
-0.500384308215816,  
0.01722431287494931,  
0.40290712722731054,  
-0.029200452046427447,  
1.2704708243944813,  
2.3709704014259345,  
-2.0267551524865275,  
0.7347873374577798,  
-0.34767223966060473,  
0.009778943736386289,  
0.221000841842164,  
-0.010995780695540691,  
-0.481889271583797,  
0.7008500159073542,  
2.3944065221124076,  
-0.08320445199579081,  
2.02470278988152,  
0.02142760118000664,  
1.6794429086669174,  
1.6046551807968945,  
0.2831685070910348,  
1.5805135598594935]],  
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},  
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
```

```

        'status': True,
        'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric6_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0]]},

'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
                          [12, 6, 21, 8, -3.1990719612158447],
                          [20, 10, 35, 16, -1.6351376911040885],
                          [28, 14, 49, 24, -0.8526977600673817],
                          [36, 18, 63, 32, -0.44953303322849647],
                          [44, 22, 77, 40, -0.23666355366027367]],

        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 6.842613220214844e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -8.419348593542587,
                          'others': { 'beta_coefficients': [ 2.1894863603561845,
                                                            -3.028013051556425,
                                                            3.513990454892641,
                                                            -2.4750715377939474,
                                                            0.9551450565312212,
                                                            -0.15553728242967432],
                                      'cost_richardson_zne': 33.37331552209775,
                                      'sorted_expectation_vals': [ -6.429933051303131,
                                                                  -3.1990719612158447,
                                                                  -1.6351376911040885,
                                                                  -0.8526977600673817,
                                                                  -0.44953303322849647,
                                                                  -0.23666355366027367],
                                      'sorted_noise': [13, 47, 81, 115, 149, 183]},
                          'sampled data': [ [4, 2, 7, 0, -6.429933051303131],
                                             [12, 6, 21, 8, -3.1990719612158447],
                                             [20, 10, 35, 16, -1.6351376911040885],
                                             [28, 14, 49, 24, -0.8526977600673817],
                                             [36, 18, 63, 32, -0.44953303322849647],
                                             [44, 22, 77, 40, -0.23666355366027367]],
                          'sampling': 'default'}}}],
'xy-ric7': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
                                                    'layer': 30,
                                                    'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                                                   'cn': [ 0.5,
                                                                                   0.5,
                                                                                   0.5,
                                                                                   0.5,
                                                                                   0.5,
                                                                                   0.5],
                                                                                   'r': 0},
                                                    'time': {'max': 10.0, 'min': 0.0},
                                                    'type': 'xy-iss'}},
                              'init_param': {'value': 'random'},
                              'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                                                'noise_prob': [0.001, 0.001, 0.001, 0.001],
                                                'status': False,
                                                'type': 'depolarizing'},
                              'nqubits': 7,
                              'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],

```

```

        'r': 1},
        'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
            'file_name_prefix': 'xy_noise_off'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': True}},
'zne': { 'data_points': None,
        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1 '
                                'Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                    '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                    '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                    '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                    '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                    '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 4.118038614266066,
                                     2.900590937440303,
                                     2.874944433071395,
                                     3.9296977116518517,
                                     3.1841729388967344,
                                     3.695928991667305,
                                     3.180227821654077,
                                     3.0475045550195774,
                                     2.4669562192398296,
                                     3.4124916083581294],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [0, 0, 0, 0],
                              'noise_profile': { 'noise_on_init_param': { 'status':
                                                                              'value': 0},
                                                'noise_prob': [ 0.001,
                                                                0.001,
                                                                0.001,
                                                                0.001],
                                                'status': False,
                                                'type': 'depolarizing'},
                              'odd_wires': 4},
            'optimized_minimum_cost': [ -8.433468022982382,
                                       -8.464942606128835,
                                       -8.43123094137328,
                                       -8.435401809869298,
                                       -7.9812014353834515,
                                       -8.371080050195895,
                                       -8.46198057547588,
                                       -8.19415563172473,
                                       -8.438078664602543,
                                       -8.481886373274177],
            'optimized_parameters': [ [ 0.4631008843346199,
                                       0.8090636887146863,
                                       1.1101507421405439,
                                       1.3400184709676968,
                                       1.7452332568602784,
                                       1.9308700883292553,
                                       2.1752679291102544,
                                       2.1784212211010603,
                                       3.800207092295721,
                                       3.800207092295721,
                                       3.800207092295721,
                                       3.800207092295721,
                                       3.800207092295721,
                                       3.800207092295721,
                                       3.800207092295721,
                                       4.798448841518561,
                                       4.812026604031267,
                                       6.02543601067661,

```

False,

6.02543601067661,
6.5265343452987,
6.5265343452987,
6.526534345298701,
6.528381158760148,
6.528381158760148,
7.113023968622042,
7.113023968622042,
7.593654062389684,
7.5949106716383605,
7.780058793305066,
9.000370275113886,
9.005799799983558,
-0.22105281285875572,
0.07196296170715184,
1.0527781347416576,
-0.4594806493129553,
1.0526039239081737,
-0.01709270509934428,
0.3199675106146978,
-0.0972220164715271,
0.9145097677998977,
0.09808103992344623,
0.5373974901276602,
-0.5786059880820694,
-0.02594763303680847,
-0.22132394348491485,
-0.7274785930689547,
-0.41090279602128,
-0.520884263329148,
-1.0260835302846771,
0.1692597388493397,
1.3250313995601488,
1.510357845739394,
-0.9233604808876308,
-0.2049157632594096,
-0.9894281051832,
0.25772120620526445,
-0.04822142329545429,
-0.3912638297585932,
-0.48486712444934904,
1.4836649713082644,
1.566996182191827,
0.08093959953023218,
-2.3562926055425963,
-0.21876618997457167,
0.10911407834571141,
-1.3776955808248483,
-0.31816531211057325,
0.7083857256813267,
0.5593482503842677,
0.7483381318097894,
0.010042025170353972,
0.05290425882351868,
0.20616876613540233,
-0.8673924682871486,
-0.08017548043713153,
-0.3056222778255953,
2.3981163154468383,
0.37079365521682633,
0.3870558721039623,
0.19341028278759992,
-0.3154248344945196,
-0.12073260037561208,
-1.5858414285227351,
0.59830347262823,
0.30388710178795286,
-1.4558796844932331,
-0.3225618511944297,
0.09243390755822743,
-0.0006426173071943948,
0.7561662109631148,
-1.3902324585417218,
0.1677701056843372,
2.7311884232201735,

-1.1122880815211051,
0.0467986017971284,
-0.020089420529231013,
0.005481851366742631,
1.3205596999819047,
-0.1290403664286252,
0.3870444394307043,
-0.6549197738372419,
-0.17656835190090397,
-0.0008013451895215781,
-0.8769883624496301,
1.6929520910231297,
0.7288671099890552,
-0.6169673343691255,
1.667217439782634,
0.2554635335546482,
-0.029515171705053127,
-0.8290153205719569,
0.042733616172741226,
-0.5148227135977747,
1.2301836750084596,
-0.9175908300980525,
0.5229779878549122,
-1.5901934710811256,
-0.07316585886439231,
1.1195018365185199,
-0.31111306362203306,
0.7254709796877604,
1.1995417479543053,
0.6908225815252583,
-0.2919630814941808,
0.15955998483086176,
1.2871422030548714,
-1.2683991057896014,
-0.3768598572966417,
1.4863954796333156,
-0.4300298743216642,
0.19151460300148976,
0.9665838873771405,
-0.07244306377732504,
-0.04169546816073936,
-0.389029635704794,
1.5218961812832925,
-0.1772066326032777,
-0.70949107362336,
-0.015222466074935298,
0.3587128629768407,
1.5809494716454662,
0.12428164288740734,
1.5083554487395696,
-0.044379618902227067,
-0.03505579308192811,
-0.6786538117837789,
-0.007428157845337949,
1.5281494732580478,
-0.14124856089608578,
2.726718429443767,
-0.003106328084445967],
[0.5270475437101116,
0.7815425625485376,
1.6469610445191378,
1.6712876404917778,
2.122904188877425,
2.131106816606521,
3.7367798442214175,
3.7367798442219424,
4.750276145421292,
4.853973384996016,
4.865098072058574,
4.873980651141647,
4.987599328262889,
5.00246239884106,
5.520902021852277,
5.793284849001337,
5.8087832398624775,

7.432729128414403,
7.432953787975026,
8.144896858715603,
8.172580744083415,
8.788071761144659,
8.792288306719156,
8.828260688191317,
9.011586808740137,
9.011995976067807,
9.538729699920484,
9.576884709106812,
10.486750161390887,
10.48675016212107,
-0.15531374899360717,
-0.0849039080739262,
-0.8434555658556069,
0.9447826609611266,
-0.19482082661427938,
2.4999420394249707,
-2.7164831941210483,
-1.5408169364026967,
-0.8621482593978547,
0.31547458537730877,
1.4220435826823787,
1.348860553316339,
0.40827474166943256,
1.063960383917828,
-0.21906798993221652,
-0.33263312803097533,
0.4889869181158972,
-0.25524084746075687,
-1.3622146811064544,
0.996105149433398,
1.4161210728452394,
-0.6491827941870941,
0.7399329880002848,
-0.11994422281276146,
3.179573825543474,
-0.034943024466848674,
0.6075464711124173,
-0.3019825794696199,
0.029392952712390567,
0.14840732209332994,
0.08592470568620861,
0.016725634393736574,
-1.3697962205589145,
-0.06155375586805023,
0.16508080354919497,
-0.5975014914051775,
0.09635736997937351,
1.254101402191796,
-0.6193953337438176,
0.47602269510847217,
-1.7251540483535392,
0.5378274350019089,
-1.0159384367751108,
1.0968223266795405,
-0.79878134010976,
0.10468891520322528,
-0.2910173593847854,
-0.7078981306941186,
-0.3510837282628223,
0.5371002854277007,
-0.4911274293534503,
1.7273270693440552,
1.3500555714793954,
0.9052059235505011,
0.4771526028272018,
-0.1893747959421651,
0.13989050072973697,
-0.5930955187300607,
0.36780210947560876,
-0.5043370473487999,
-0.4851893292987325,
0.1987440888222681,

-0.5806445532873666,
2.3171646404124093,
0.06032126128170586,
-0.6389920437730593,
0.9985359186149056,
0.04870155999532171,
0.16447399991313041,
0.05315090520136875,
0.15727255736970833,
-1.3722292706433088,
-0.33233649668322635,
0.7758243461945761,
-0.20476613404890107,
0.06972046040604139,
0.3143879960336948,
-0.01452323353492752,
-1.2901366074359686,
-0.31033000747276407,
-0.16264485166390844,
-1.3914094473719103,
-0.5273128620655095,
0.02465459151953875,
0.10942766294132272,
1.0292094835171124,
-1.882689768994833,
-1.553216293090225,
1.3287846024814622,
0.20963913600213377,
0.7195732642990699,
0.0969787732183028,
-0.9744344415633356,
1.5822487402774768,
0.7077471503862973,
-0.766345302327988,
0.3831437091022152,
-0.12028853613988252,
-0.8209919023748664,
-0.21164805512089607,
0.5290684330656528,
-0.1040878870845114,
1.1956945887281514,
0.25703931501419025,
0.6344733295912399,
-0.02497584874729833,
-0.20937697561532842,
-1.4409411147319051,
0.9707320454927133,
1.4039378085761043,
0.9760826987262126,
-0.17314228405542512,
0.16734556762922406,
0.07881248116464283,
-0.2924467918405051,
1.4946430081327182,
0.7516806171134963,
0.799700486923224,
-0.8440600324500036,
-0.01069194371181784],
[0.3388610072211759,
0.3930923343746358,
0.4059472295013711,
0.4597995545870322,
0.4597995545870322,
0.4667083780713653,
0.9924683364283498,
1.054118787032341,
1.3839145947816414,
1.4469616998432806,
3.044048229936016,
3.044048229936016,
4.682975233106521,
4.682975233106521,
5.287925895396742,
5.287925895396742,
5.625677830869467,

5.625677830869467,
6.519005703020383,
6.519005703020383,
6.519005703020383,
6.854665591577039,
6.854665591577039,
8.092962128458302,
8.097830997780322,
8.102159968176109,
8.111954675742565,
8.727038821401255,
8.727038821401255,
8.784424594255395,
-0.8636093730812797,
-0.04742973062196874,
-0.37989856483295564,
0.3750798322441465,
-0.509087961968837,
0.5366875438436245,
0.1232704940065221,
0.5249134835872401,
0.13594138148710633,
0.10802292582448766,
-0.7156772546121367,
0.34756877087651405,
-0.5704812075854907,
0.289395233495819,
0.2674862930849357,
0.8724143866961164,
0.5770602988816299,
0.658777432128999,
-0.47288232304292416,
-0.029663566814993374,
-1.323180236232167,
-0.2890901656039231,
-0.015141732930961212,
1.1846850771343012,
-0.4506688471965015,
0.22241980016839272,
-0.015748484493019736,
1.1522499025192263,
0.2996907655672081,
-1.7370631162182948,
-0.35306232532663057,
0.17322471105923057,
-1.1855180134814352,
-0.31537295018088835,
-1.8120366955290261,
0.6663892425075478,
0.21874911670355193,
1.2729693967005706,
-0.14230256773273223,
0.2012427441067739,
-2.3269317425793643,
-0.03299848401271634,
-0.4074853136565566,
-0.30371905431664253,
-0.12037074206123735,
0.8456676988075262,
-0.21838328071319946,
0.019494764922420988,
0.16383099728733078,
0.056056403895700845,
-3.5632063709291613,
0.6579904510057321,
0.5207861046034454,
-0.017180990359968364,
0.2532807007765998,
-0.03315495616728509,
0.7564905992030696,
-0.11091568858030387,
-0.9281890577202553,
-0.6082181315906088,
-0.4480511658411004,
2.1953283079942025,

-0.30446713497464617,
-0.03359602431311142,
0.28972943746372876,
0.17817064395522006,
0.7965754083210965,
-0.9520780563921613,
0.10150445450145726,
1.3510884636720888,
-0.37849615173878054,
0.17729737647006832,
0.1884683806301832,
-0.03234547530918763,
-1.338763456177623,
-0.6711058962200585,
-0.012976043431998471,
0.027578392175021227,
0.10161286285089129,
-0.00939135247962935,
-1.302164623174096,
1.3715576399721376,
0.737767307868418,
1.5939269674119705,
1.4291758244230544,
0.5601859764306084,
0.3488764286390935,
-1.093762224545935,
-1.1976831420922067,
1.4557381665683378,
0.8924472305311927,
0.5300221300338881,
1.0785280788144638,
-0.6029509417428172,
-0.6790632925762424,
0.4138865114731201,
0.2917474006869537,
1.8050154062536312,
0.05177383423831921,
0.25467777497912664,
1.4671356779039024,
0.4422247342165525,
-1.5431607378377095,
1.8101814873938125,
1.6870467346899987,
0.7517920825884995,
1.3622912362338087,
0.13653050459472865,
1.5210183141257738,
-0.004247168330480351,
0.26137218980413907,
0.5451511562235306,
-0.6705596749039198,
1.2136085126593865,
0.5231529087420831,
-0.03279466120532818,
0.2548954759886831,
1.5797820110558691,
1.075840789720537,
1.5541818084195007],
[3.821020346740443e-17,
1.386496864345847,
1.386496864345847,
1.7951472089989222,
1.8559400139285378,
2.7980544315911504,
2.7980544315911504,
2.82954781812891,
2.82954781812891,
2.82954781812891,
2.956183634338348,
3.840561875918348,
3.840561875918348,
4.79501776182173,
4.79501776182173,
5.670460018404704,
5.670460018404704,

5.670460018404704,
6.865392250052202,
6.865392250052202,
7.506826147384778,
7.506826147384778,
8.737685173635683,
8.737685173635683,
8.737685173635683,
8.737685173635683,
9.601023092540274,
9.601023092540274,
9.853738132880634,
9.853738132880634,
0.5611658219809542,
1.427331325859137,
0.8881601644240952,
-0.8937497791603879,
0.5712044928850715,
1.3195951619089188,
0.5467792160605555,
-0.6816630744322637,
0.06707477836950569,
0.6798692668930644,
0.19576168860111362,
0.5094724850631845,
1.1424414477256812,
0.30471145260454696,
-0.08158911485577636,
0.9585661478313002,
1.0836238583568754,
-0.3718197048926503,
0.7841948506971846,
0.3023515457437066,
0.7662240328883452,
0.9970048707689876,
-0.8946064024093777,
0.27649390200762114,
-0.15342626519190924,
-0.4015720487025046,
1.460926694270634,
-0.055421168101734586,
0.6077050880702092,
-0.7932081335349191,
1.3188332541813446,
0.01929555634963922,
-1.1301845164589392,
-0.1718174564221745,
2.26154163970611,
-0.9882028269623031,
0.8038643847286138,
0.6849304102208338,
0.874798983022982,
-0.016486064274954207,
0.0264221098096002,
-0.3625000384278827,
1.4376331477886695,
0.8499992740594865,
0.10042171561200945,
0.14542648933357244,
1.3513090878714353,
-0.22248922869259902,
-0.4483175327546578,
1.2345758678942387,
-0.18983522552694923,
0.019591210301167533,
-1.3607248526627624,
0.018375533581887468,
-0.3730738776375715,
1.0123537416913542,
0.7412601688930508,
-1.6525919628732422,
-0.8341682190885938,
0.04519729634722039,
-0.06718028327556408,
-1.5649719285752928,

-0.37531851249345394,
-0.11822841604899964,
0.008154106549025474,
-1.4601449692077721,
-0.0030993446167428395,
1.462573870839384,
-0.11887757571245174,
-0.09361960188417112,
-1.2543408854038138,
-1.525280802664792,
-1.241413372948363,
0.008758808605552393,
-1.6676332942965022,
0.13826170532334445,
0.7537363962746654,
0.0833397476047736,
-0.12216949704941273,
0.02084807234445201,
-0.3421949354726438,
0.0017882611822728064,
1.5186851123669147,
-0.4603163345929323,
-0.09956972276089109,
0.7247597830417082,
0.5304935691839561,
0.003193152207841171,
0.5581346990750625,
0.010542051444639388,
0.7553167297004638,
0.5956891015800863,
-0.05566254332595716,
0.5362461335652635,
0.009895282042162012,
-0.3029901133286735,
0.09399206534914867,
0.8818812838713691,
-0.06795832284115338,
-1.580933930101608,
0.3014975711017802,
1.2092022074751856,
-0.019692531397995948,
0.9250981578351692,
0.4673383297638096,
0.03344339272872125,
0.13058472572385768,
0.036262684184263205,
0.458095375934523,
0.6492846129896516,
1.3039221898344189,
0.024764812872192767,
0.7236926243101844,
0.01444840086266692,
-1.0388939706148563,
-0.08359693344439546,
-0.4698004238001188,
-0.48055768545836347,
0.4662000815300196,
-0.041204969991243484],
[0.01798320141016013,
0.3393419127512466,
0.7047697703765823,
0.7064789163091142,
0.7191656785855808,
0.7198672068604162,
0.719867206860416,
1.0840512961729545,
1.0900003904895101,
1.1926153174924548,
1.867054342490295,
1.867054342490295,
2.368094916985588,
2.3683751276938714,
3.6383003552329964,
3.642527762446636,
4.505845683894636,

4.602446412902637,
4.972427659100928,
4.972427659100928,
5.1148164150678035,
5.8066136392845,
5.820681023327439,
6.727088267818048,
8.352653016300794,
8.773334122756978,
9.065902540064004,
9.065902540064004,
9.74880092287043,
11.30145464466351,
-0.13904770480884485,
0.23493722133539818,
0.4497803172772882,
-0.640584995353188,
0.9779304528121034,
0.0011745349607135795,
-0.7175493520168287,
0.373620380836484,
0.08111255661159392,
0.6653458367314794,
0.0037967443269442437,
0.6286122183467432,
-0.03825426016359485,
0.6896824961465278,
-0.23916946602734285,
0.4624850989736848,
0.1419493525208794,
0.7145176662978019,
-0.316909964913161,
0.6605458698076423,
-0.39309909050950365,
0.7623048158369822,
0.7469736169282462,
1.2400892032597353,
0.12266455668869777,
0.19578648318487166,
-0.784043232212839,
-0.10334183265831844,
0.32774480702026265,
-0.011491906972883237,
0.322988869457402,
-0.2027870428092001,
0.1281571522083961,
-0.1947961113336468,
1.7114513053475526,
0.005425684515457419,
0.8136825826883561,
-0.25903007981373005,
-0.5015428579892063,
-0.2600978115589891,
-0.15405473476720474,
-0.4207498663011112,
1.04622682040645,
-0.8642339256273757,
0.6248691295250803,
-0.3539240130838131,
0.011689879721248103,
-0.3068283940685192,
-0.6048660879777619,
-0.11704784670887185,
-1.2745062338511486,
-1.7890038533223112,
-0.7818671277256136,
-0.8877455887286905,
0.3164818198590212,
-0.27477307924208777,
1.8083251930535746,
-0.5031479272825263,
-0.9648993597903888,
-0.41682962131770707,
0.27040705073896654,
1.0892900815418987,

0.17779960873468054,
0.020337033950376556,
1.0551529464315492,
-0.10705558118147128,
-0.8851977322505518,
0.2232945485845247,
-0.5850403569085746,
-0.7690595025978197,
-0.4391673030332381,
-0.9309566421919284,
0.2973030450025941,
0.6780545449896137,
0.14045373030826291,
-0.5109313900859794,
-0.44070010469428217,
1.6940975579604587,
0.17062304524159472,
-0.402681393200546,
-0.7091187255059627,
1.3132357546586793,
0.2729391261599054,
1.0115940175088693,
1.0046673925246778,
0.4982257464819494,
0.04454679168285006,
0.4863023994043421,
-1.1433261311581624,
-1.820308437409531,
0.6035405700431521,
0.04003907881711185,
2.99381748600971,
-1.0437523456620463,
-0.22517480393070408,
0.10746937069246698,
-0.8788513722547997,
1.286531186523234,
-1.3157160458746164,
-0.6979748907629274,
-0.16573593981190854,
0.2529194783090185,
-0.2562517800846549,
-0.7966921479109774,
-0.4446940958312971,
-1.052310031662883,
-0.1506245930250076,
0.1615975776996014,
2.9220364155549214,
-1.7197843955783065,
-0.6592103621593355,
-0.2007498223803255,
-1.6188873615119244,
-0.15193115939751467,
-0.20866967970071015,
1.2910791163230286,
-0.4917152660712325,
-0.42348313202027765,
0.3518303936962707,
-0.10034486495238693],
[0.009039772647041902,
1.1400526086001903,
1.466905028683042,
1.6524670085313564,
1.987797816914966,
1.9900950346150876,
2.1413202332332344,
2.390935514217652,
2.402429399816812,
2.81913686416337,
2.81913686416337,
3.5656810200683298,
3.5656810200683298,
3.5656810200683298,
3.76603447051701,
3.76603447051701,
5.499236323276389,

5.4992363232763894,
5.4992363232763894,
5.532789537471719,
5.532789537471719,
7.371733777832876,
7.371733777832876,
7.680783486275433,
7.706623781446164,
7.706623781446164,
8.457742149729329,
8.457742149729329,
8.95222736351662,
8.95222736351662,
0.18928393791812126,
1.0354889043700823,
0.3703113207409527,
0.30988589523494625,
-0.34669340410550803,
0.09615284519636855,
0.1011898884094612,
0.6160115432573767,
0.6148540393435122,
0.07934943483664206,
0.8557624329627399,
-0.7672633290674925,
-0.14188111026081618,
0.1995320641320393,
0.4594888802421805,
-0.03593187081620776,
1.2926591415659605,
0.31433950962447343,
0.8053272546604554,
0.6634815559763872,
0.311660786273531,
0.0294653444200854,
0.3044394736040779,
0.7599259037372076,
0.22962735656034647,
0.22562971232722423,
0.29511485598499815,
0.02260075636691185,
-0.9497096253675098,
-0.5331371571917215,
0.2192512323868535,
1.104832430155296,
0.8636617903941131,
-0.6756684630428217,
0.4169844309263729,
1.1297527651723116,
-1.160118439748996,
-0.7971836651224,
1.9169656561991728,
-0.007260796290839314,
0.3614132074474742,
0.6243022778229537,
1.8335025630018775,
0.35139722497115833,
-0.7247870129359816,
-0.6744811297738377,
0.49997538785346773,
1.4139521212248116,
-3.0799818401546553,
-0.14768564076593044,
0.06251500852230953,
-0.8547134606957393,
-0.30107856776262504,
-0.7654972812734481,
0.28603020918627825,
0.2129303248959857,
-1.2077141264570097,
-0.8488549318812054,
0.6492574804166855,
0.7751290676155554,
0.40909519807061084,
-1.2750630737105486,

0.4187798718836698,
0.33498424523924863,
-0.19017028209895162,
-0.03232931911977397,
1.0621483440417678,
0.452017889610544,
1.0192767515800205,
0.7290563184502946,
-0.5286462788982386,
-0.05512372609304388,
1.3189151919853837,
0.5865235617306253,
-0.579843833868396,
0.6980158293666034,
0.027004860579276476,
1.2876675233834314,
-0.17185617277774212,
-0.3827017785428823,
-0.7286404996479297,
1.1682112525473984,
0.22765379123755217,
1.4870969768845252,
1.1642898801914527,
-0.03976972562460009,
-0.8163296820868416,
-0.15318057329999143,
0.0482918949035788,
-0.06014345828968787,
1.1478583262090496,
-0.005141116181289305,
-0.5820859555621626,
1.086519910329415,
-1.08558016317555,
1.531511042803357,
0.042734048269580674,
0.7452599788242152,
-1.3094923903018465,
-0.9012205175381317,
-0.0755166851159581,
1.5493661520767013,
-0.2278023712186688,
0.8627212805492942,
1.8523276191392382,
-0.10961709979291216,
0.6746606852631131,
0.22566845572735392,
0.5920154791154506,
1.2709342533151342,
0.26903045177842116,
0.11724485476711638,
0.621942277674204,
0.15491924869980966,
1.472217084758381,
0.8975149794611956,
0.3601026339731485,
1.3234513983254743,
0.0182465793142515,
-0.11914786829986494],
[-5.765348397008027e-18,
0.3428984748224433,
0.3655450682584724,
0.704784031551808,
1.5409081472811441,
1.549300366088437,
2.3787666619954635,
2.3787666619954635,
2.8808650819472406,
2.884987740396357,
3.1308808834009203,
3.1363836212549696,
4.869321728501938,
4.869321728501938,
6.1150174264690085,
6.2582278136815175,
6.271636236200362,

6.551203581243674,
6.59741701640448,
8.230564833747417,
8.294003346338926,
8.31431753208387,
8.369586036193128,
8.415319617883744,
8.70447046242739,
8.709120596281233,
8.918659758853744,
8.99410216914391,
9.002256766523654,
10.218690251738174,
-0.6798627919679444,
-1.1436724543472427,
-0.1278269606092795,
-0.9286552360068459,
0.6524046490818174,
0.06134572949387961,
-0.49761179900021657,
-1.3918471119443894,
0.5858287319319103,
0.18669127045720346,
-0.4586789737568629,
0.2400608953322058,
0.022222124883641912,
0.9282004764805514,
0.6952358394741303,
0.805684979006905,
0.6146729452742722,
0.6777524139082742,
-0.8117842884117104,
0.3426849285038029,
0.7100299235979358,
0.7658186509139059,
-0.05418633990950139,
0.1761917490887145,
0.5744825473612316,
0.31705937472296297,
-0.9188523464356001,
-0.7337446596965294,
0.2061368312243915,
-0.6645694031850036,
-0.009871104765786294,
-0.14388187083546433,
-1.2485707106311008,
2.849540809212292,
-0.5119215846255833,
-0.19381894062193808,
-0.17310717674563775,
0.8465397629359774,
-0.46757240698237135,
-0.05607605828560263,
-0.40986137011816176,
0.07649505002491204,
-2.1522907800348112,
-0.6653346582594515,
0.4084801543759574,
0.4028897820911683,
0.16908117843954765,
-0.07217026591146015,
-0.72804755296126,
-0.13548738477012795,
-2.590679566047418,
-0.40743076961744734,
1.1030856800887578,
0.08388380440455438,
0.25224805905642333,
0.10644300798076774,
1.0474125729404944,
-1.1459016177859003,
-1.6916287310468103,
-0.9886067106540645,
0.24182274960650893,
-0.25703819491337826,

-1.1942818044595898,
-0.7763678643520989,
-0.916718557142268,
0.20086626876005667,
-0.3711023890817823,
-0.006962349053602254,
2.687031223814704,
0.3970791704566904,
-0.6890591654410289,
1.0504536719091986,
-1.2458238157294268,
-1.197330311624414,
-0.12155955689525438,
-0.2969718406476498,
0.15406921803787998,
0.9984735505197111,
-0.2734456365103056,
0.0100494123830882,
0.16172873483083283,
-0.6780409734277772,
0.4539997995464368,
0.7558576390626953,
0.9556605206819043,
-0.10572075101403118,
1.5564628686733721,
0.05397552837650759,
0.7415987024402694,
-0.1648500682327342,
1.052699139325594,
0.6776572023171561,
0.15219630541397192,
0.8843933196559655,
1.3948099389813178,
-0.16710315713046545,
1.2199200746863228,
-0.3825678575328573,
-0.26766985045935626,
2.5777182352113908,
0.23156136798610805,
-0.3377548185037014,
1.0689111835439342,
-0.17216022970038714,
-0.29381807805976945,
-0.5689533077540762,
0.30596827309967156,
0.5260993417881752,
-0.6768480303494689,
-0.1092430795457648,
1.5442082488768039,
-0.10826504240549732,
-0.9555598065181787,
0.5334835330029429,
0.5569227134357198,
1.2665483894271157,
-1.9677332823608626,
1.5559931182978979,
0.7709453943956464,
1.5746506000831744],
[1.7625089228179507e-05,
1.298540182320117,
1.3141870420194457,
1.3305471045820527,
2.0686735179487985,
2.23810597518972,
2.267373737425337,
2.295441377067579,
2.3325388050051976,
2.352484360014207,
2.370891483796666,
3.3117124844820194,
3.363605896627787,
3.858981460194034,
3.8589814603490344,
3.859537867870235,
3.8595381713444072,

4.26458571773699,
4.264585717738593,
6.599549149586233,
6.655963438031886,
6.939612414177192,
6.939612414268938,
7.004302986969205,
7.010544991093791,
7.97314828104894,
7.999280802101656,
8.022628296732579,
8.027896471579398,
10.077254342713585,
-1.0733166032270363,
0.4264387306682,
-0.29490479134596703,
-0.45181113351566254,
-0.9496712390342118,
-0.24412788786515252,
-0.17005038539710493,
2.131747001055919,
-0.8801477965160032,
0.1985900676052623,
-0.17829180427317373,
0.1513896817808816,
-0.2996418627583153,
0.09152392337611555,
1.0155053498240294,
0.6713280773462803,
-0.5915672184524843,
-0.21780814909773427,
-0.23676595566314207,
-0.1539223229682568,
0.9733737893230769,
-0.8325690111887051,
-0.6684349693104433,
0.944923850675747,
-0.8270803600282979,
0.4869668407293288,
0.07342964610078141,
-0.8928903181234079,
-0.04262248204617692,
-0.7602498290394286,
-0.48858825463899247,
1.7340528561515622,
0.7340502315717261,
0.5163401187759543,
-0.5015858040228415,
0.2890335055344555,
-0.46033506221820203,
-0.8185907107929625,
0.4557079379705311,
1.6601520871164626,
0.07125306723268252,
-1.4035119795986863,
-0.6741575848895364,
1.84360490583597,
1.6836338044245691,
-0.33329370458317326,
0.19854317161902654,
2.03917897859608,
-0.3342251938073038,
-1.6015864477635742,
0.35422488479060155,
0.1487121545506934,
-0.2821114888507261,
0.21869934797564045,
0.7523961573071125,
1.3021785537923125,
0.26184859002187644,
1.2441238228144538,
0.039670539596101485,
-0.15687777352236798,
1.1386642271664005,
-0.13559116971858365,

0.39167499360215746,
2.4453261619465274,
-1.0890320328395309,
-0.4289422005081893,
1.425249318116838,
-0.0016733698174337221,
0.023121567661497415,
-0.28325087531692295,
0.940069260590291,
1.163437407349777,
-0.4256527399896147,
-1.4972731615803316,
-0.12624975701746496,
-0.10144651691714318,
1.3307378561466523,
0.041778278155808674,
-1.2746374184036902,
-1.2216848741140482,
1.4048769664929757,
0.8957112974734468,
1.0058068981677006,
-3.0965653057688316,
-0.3344448337959029,
-1.0948102803209163,
-0.038478236121346335,
0.12618687504486675,
-0.0850775611793361,
1.33550885346982,
-0.04334772196309596,
0.2406908654957122,
0.7327249608913058,
0.9966290850109706,
2.2852565843192423,
-0.030570948934034583,
-0.5566407461361556,
0.813246653395058,
0.5946355273642772,
0.07519254658407079,
0.27120989268694123,
-0.15415722381241237,
-1.1239288818585278,
1.893751281002709,
0.7530205722182245,
1.4454588921082565,
0.15037300343080462,
0.1793867334099716,
0.3587259643057371,
-0.006799349268608,
1.0907160355013774,
-0.30186717991666434,
0.8507589785411056,
0.009155709268365956,
-0.8165113788656561,
0.011816515180360782,
1.116430418883973,
1.5343789763879045,
-0.1682835166498154,
-1.5032827361947747],
[0.10956969985098665,
0.4360115975982133,
0.5932904704915389,
0.6050451872391112,
1.3688142364348175,
1.3737597367805747,
1.5226773273508583,
1.5589271816733294,
3.150320547140947,
3.1613053536766147,
4.7635234433865605,
4.823148382609575,
4.823188243925802,
7.171157860626202,
7.171157878485542,
7.171158529865571,
7.175073431930883,

7.183070822592498,
8.102400589897519,
8.10392263292654,
8.103940780725882,
9.021308924987686,
9.02133947231205,
9.082831297490506,
9.094868565409332,
9.236804949782064,
9.237029573386183,
9.26430291432718,
10.00744744746223,
10.032040374677464,
-0.5042382729215367,
1.0180015436638659,
0.10423420212375085,
-0.3355687578372037,
-0.3902319222280216,
-0.09867602213718178,
0.3860296418386605,
0.8522828003851446,
-0.7730596965193267,
0.08803782521712378,
1.175277695996097,
-0.3769225558693855,
-0.4553519820435133,
0.4758962487375975,
-1.0517577693418567,
0.3712029741888282,
0.27334544561774066,
0.03397030216872572,
0.3891396532807883,
0.12152393226873999,
1.5192516885791116,
-0.5124931636551364,
-0.5667003913709334,
-0.08425393582345661,
-0.8152956002657638,
-0.008087141654348068,
-0.20833380477584987,
-0.19186344764752156,
0.309463849209865,
-0.16555810903618162,
-0.28534485935787846,
0.025710068868521854,
-0.9172960495760906,
0.027100436879331523,
-0.9806057417125867,
0.46877133554227174,
0.22597640762440013,
-1.8700902599315918,
-0.6222133609404971,
0.023922176634194332,
-0.6777524378245746,
-0.3883818313368169,
-0.19921738892417568,
-1.3418496115619054,
-0.05365669225714106,
0.38835985529462186,
0.46741217514915123,
-0.1388983196762152,
1.0002905156718758,
1.6633296566254185,
-1.0097136763815113,
1.203957772036831,
0.34033273848251194,
1.1456357751380877,
-0.4637199651146574,
-0.13624703361320328,
0.6785440200843139,
0.21691345067474185,
-0.4166036808107477,
1.6003722740710606,
-0.294881370724797,
0.04198878790454511,

-0.689532047804753,
-1.3911901894570817,
-1.1667039833298627,
0.824824800006074,
0.8047687705752488,
0.5727196814509768,
0.2385364014798047,
-0.512164832149354,
0.17925141295137917,
-0.19704824673887428,
-1.6372757558452697,
0.5441675736491897,
-0.13936070677778437,
1.3935144064781402,
0.10763856884501717,
0.6679788244416961,
-0.33118654939932857,
-0.35796695002153883,
0.2632208027041665,
1.5831503864671255,
-0.25140146740224,
1.135774876905931,
0.5897962040478675,
-0.7664904031063712,
1.092649076983624,
-0.9862462517151122,
0.010216869101717933,
0.5705833807099128,
0.09839127014019662,
-1.034770130526523,
0.20582133073730807,
-0.44428721001967564,
0.5272444819968604,
-0.2627660867388947,
-0.9857421779520247,
-1.3646538516413942,
0.14779052759756187,
1.1207932022043874,
0.30427416833245535,
-1.055109005337564,
0.30548258474769546,
0.8153937738463674,
0.5082537250510629,
-0.2973639305797175,
0.0034742209991892575,
1.5168338311712208,
0.0971904670135119,
0.5305672799342332,
-0.08170159551874663,
0.4909203029282298,
-1.1771354207712392,
0.038725619801271274,
-0.6655680746562402,
1.312626970471223,
-1.0115568894576918,
0.16290676730453105,
0.3074286838975462,
0.013117345918778788],
[0.4866654913190249,
0.4866654913190249,
1.4314116389487666,
1.4314116389487666,
1.548608655381947,
1.5552664901547235,
1.662920497545568,
1.676155881591226,
1.6952532774325564,
1.6952532774325564,
2.46741851740979,
2.50102877426778,
3.636145431120801,
3.6361454311208004,
3.6361454311208004,
3.6361454311208004,
4.0161273218482085,

4.0161273218482085,
4.5411636896776235,
4.5411636896776235,
4.922545565757144,
5.624338470516907,
5.642847901355697,
5.904037934138086,
5.915382265322609,
6.860930229950661,
6.860930229950661,
9.422517992567101,
9.467623480195014,
9.467623480195014,
0.40091998571272736,
-0.3581846438287994,
-0.32033246941904975,
0.24409866553117068,
0.176560178011825,
0.4452285262890565,
-0.37150326481454765,
-1.0185408309777293,
1.6197684873006286,
2.0095194673452026,
-0.20342804182482532,
-0.4471079471088122,
0.37127155242298737,
0.4199607045713712,
0.11145558426636207,
-0.289968410859463,
0.9752004498173846,
-0.11972987215087512,
-0.8215210043660294,
-0.4144105549344152,
0.04966095047933562,
0.6091224412599766,
-0.2481186610805342,
-0.9036735819055022,
0.22922493224752044,
-0.04567884568114995,
-0.40529506738584414,
-0.44609185937563545,
1.1000542207972743,
0.8995834380092431,
-0.4391184290327733,
-1.623814742575614,
0.11121717167805684,
-0.15344289124808663,
0.7072346226342329,
0.013911394934765222,
1.1008239636292285,
1.1342299046285917,
-0.1605718532112884,
-0.8521154510350495,
-0.9726520318623231,
-0.11078467990736285,
1.2893067940177063,
-0.8656847782514395,
-0.4554157535410911,
0.46752563535079134,
1.0115326820592876,
-0.017465678177200683,
0.7555243091795097,
0.2225241128439446,
-0.6129237770264853,
0.36249038141661666,
-0.1470816778834455,
-0.23781939745964606,
0.7410640700988803,
-0.10219672891678527,
-0.9043863030697507,
-0.054446492384892985,
-0.23618927113386157,
-0.3971482428910102,
0.42980979959963467,
-0.4511512430809565,

```

-0.3322123782853245,
0.15031459204805087,
-1.4287283715133035,
-0.026859055017604255,
0.20826999854501216,
-0.016397767746330064,
-0.026711091547853946,
0.15899307013846248,
0.18349959987180348,
-0.019181103141429404,
-1.3777401647916119,
0.05578672583947999,
-0.17191397249005735,
0.08492618339903452,
0.2781511329827625,
1.6853098582374997,
-0.7964285169174421,
0.13628687772581302,
-0.26524580358677924,
1.3615146942088054,
-0.18163849035743365,
-1.5476669195398023,
0.2740670870741208,
0.19740142577560438,
0.18567840551061798,
0.43288784764736754,
-0.8069341399605817,
1.5139336080928714,
-0.7594440385529729,
-0.20443587392639553,
0.0916888700165231,
-0.03922249489745327,
0.4135801157921025,
-0.06731598775715468,
0.5038720973518589,
-0.7705715736697758,
-1.4775324565516457,
0.061950004909204714,
-0.9555167624183505,
3.1277455162888304,
-1.137887484425651,
-0.09852947784569874,
0.042597844143281625,
0.06134264537966588,
0.3384848933516047,
0.012002169875906457,
1.5732417764972468,
-0.3550969359952667,
0.37821230776206083,
1.4914211446267915,
-1.2027836619303884,
0.22421848814312553,
-0.2072711571643093,
0.05122655754886873,
-1.5578500765322636,
1.5918641150141506,
1.2635397829742223,
-1.2452354025631063]],
'run_time_sec': 43720.81102514267}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.08591881204796584,
1.5026023651616802,
1.9179097588555833,
1.9179097588555831,

```

1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,

```
-0.007720607194202524,  
0.7816530161823243,  
-0.8246572495389679,  
-0.28480932751570914,  
1.8751826609713358,  
-0.20740645402102328,  
-0.04894031202761418,  
-1.0170156037248201,  
-0.035476380392734465,  
0.9940259955458774,  
-0.657126951929564,  
-0.4786936406484008,  
1.4564140457987391,  
-1.062151832750044,  
-0.01496017231630948,  
0.16259290081999456,  
0.601493806263198,  
-0.20485253219607644,  
-1.2421776762345744,  
1.152288606322054,  
1.1883019471730756,  
0.9817659575699377,  
0.9653429636905305,  
0.5382122852444811,  
0.722384759736696,  
-0.13079560936986118,  
-0.9638314109375934,  
-0.13358671856975504,  
-0.3949417844115237,  
1.01780227278121,  
0.6746960094686067,  
0.14295933107831804,  
-0.37974724013569766,  
0.3641245559377578,  
0.12972678280008856,  
0.5949873255000262,  
0.8499370211382984,  
-0.0792654431528575,  
1.9402087793428133,  
-0.5583113020357903,  
0.6535986429024293,  
1.0008805619835928,  
0.16556690201288055,  
1.1071973338235597,  
0.07194244573338314,  
-0.9146859400815048,  
-0.005186820541518637,  
0.4795762978809806,  
0.6475066336984044,  
0.11860845746693519,  
0.08044881503693782,  
0.5402817334108434,  
1.5532870492056725,  
0.36643388228039625,  
1.6150985923432208,  
0.22271977417212363,  
-0.04956133994172771,  
-0.6258533610916538,  
-0.6517224443292691,  
0.2594825887888975,  
-0.1286341552665023,  
0.48749759828242367,  
-0.009813928289633511,  
0.5249350004928881,  
-0.01015655683116892,  
-0.48671658518654837,  
-0.588311268126242,  
-0.25978871509233853,  
0.9039610340757382,  
1.5549616218831699,  
-0.05330158954092406]],  
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},  
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],  
                    'status': True,  
                    'type': 'depolarizing'},
```


[illegible]

```
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.3773346757087749,
0.5734233884194113,
0.5734233884194113,
1.2094818154974225,
1.2528672337957094,
1.3073966674132076,
4.193348178434801,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.274742575727675,
5.343872214974395,
5.354523589318104,
5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
```

0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,


```
'noise_level': [20, 10, 35, 16],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
'identity_factors': [3, 3, 3, 0],
'noise_level': [28, 14, 49, 24],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
'identity_factors': [4, 4, 4, 0],
'noise_level': [36, 18, 63, 32],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [44, 11, 11, 40],
'identity_factors': [5, 5, 5, 0],
'noise_level': [44, 22, 77, 40],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [52, 13, 13, 48],
'identity_factors': [6, 6, 6, 0],
'noise_level': [52, 26, 91, 48],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}],
'run_time_sec': 115.31898212432861}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
```

```
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.00030922277578629,
0.881375055029912,
0.881375055029912,
0.8813750550299119,
0.9803845151714838,
0.9803845151714838,
1.201629973822729,
1.2441077564052723,
1.8660677974280824,
1.8660677974280824,
2.651002314516478,
2.651002314516478,
3.233637461790533,
4.36384995486056,
4.36384995486056,
4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,
8.903406631369169,
8.903406631369169,
8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.13579407977759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
```

1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,

```

0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_xy_noisy_time_evo_ric7_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0],
[6, 6, 6, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.5111758480272135],
[12, 6, 21, 8, -3.513108400075424],
[20, 10, 35, 16, -1.9256894730075684],
[28, 14, 49, 24, -1.064810006156379],
[36, 18, 63, 32, -0.5886022016151555],
[44, 22, 77, 40, -0.3212779442729232],
[52, 26, 91, 48, -0.16991504362792267]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001,

```

```

False,
0},
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
  'identity_factors': [2, 2, 2, 0],
  'noise_level': [20, 10, 35, 16],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                        'status': True,
                        'type': 'depolarizing'},
                    'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
  'identity_factors': [3, 3, 3, 0],
  'noise_level': [28, 14, 49, 24],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                        'status': True,
                        'type': 'depolarizing'},
                    'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
  'identity_factors': [4, 4, 4, 0],
  'noise_level': [36, 18, 63, 32],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                        'status': True,
                        'type': 'depolarizing'},
                    'odd_wires': 4},
{ 'gates_num': [44, 11, 11, 40],
  'identity_factors': [5, 5, 5, 0],
  'noise_level': [44, 22, 77, 40],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                        'status': True,
                        'type': 'depolarizing'},
                    'odd_wires': 4},
{ 'gates_num': [52, 13, 13, 48],
  'identity_factors': [6, 6, 6, 0],
  'noise_level': [52, 26, 91, 48],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                        'status': True,
                        'type': 'depolarizing'},
                    'odd_wires': 4}],
'run_time_sec': 119.58362412452698}},

```

```

{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
  'init_param': { 'value': [ 3.750846163374531e-05,
                              0.22285241391868899,
                              0.26252411232084344,
                              0.26252411232084344,
                              0.26252411232084344,
                              1.459545906754556,
                              1.4595459067545562,
                              1.6094835194149661,
                              2.5927782099427628,
                              2.5927782099427628,
                              2.5927782099427628,
                              2.5927782099427623,
                              3.3552499939702187,
                              3.3552499939702187,
                              5.033284106467877,
                              5.033284106467877,
                              5.056477210452712,
                              6.768913439946703,
                              6.768913439946703,
                              7.2782492670747585,
                              7.2782492670747585,
                              7.2782492670747585,
                              7.2782492670747585,
                              7.2782492670747585,
                              8.934198246659292,
                              8.937369801451712,
                              9.544406408519471,
                              9.544406408519471,
                              10.23170722265548,
                              -0.4501443853573418,
                              0.06519513561037263,
                              -0.13386490845883003,
                              -0.01542333988020039,
                              0.3988203537526196,
                              0.9163371790808124,
                              0.08159154039220383,
                              -0.11665429084087826,
                              -0.21130662780886203,
                              1.515236372470638,
                              0.4436142129357345,
                              -0.28780404896369716,
                              0.3681425383234808,
                              0.648176841676201,
                              0.3046875502986337,
                              0.05897820457762899,
                              0.09316951897603123,
                              1.5277657654344678,
                              -0.655141628591867,
                              0.073084468489131,
                              -0.11423944001370272,
                              0.3986898081963471,
                              1.0974832156349317,
                              1.8462557861113,
                              0.5998463368708868,
                              -0.0014214294560794885,
                              0.4026665434776887,
                              -0.03860309342262787,
                              0.3728818533964108,
                              1.7322816292529104,
                              0.42156018976657544,
                              0.9872712539907578,
                              0.5749190583224287,

```

0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,


```

0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0],
[6, 6, 6, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.4606782650454075],
[12, 6, 21, 8, -3.273885460810631],
[20, 10, 35, 16, -1.6974907774474757],
[28, 14, 49, 24, -0.8939947022135478],
[36, 18, 63, 32, -0.47364366307740574],
[44, 22, 77, 40, -0.2491082243344117],
[52, 26, 91, 48, -0.12744297687087136]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
False,

```

```

0},
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
  'identity_factors': [2, 2, 2, 0],
  'noise_level': [20, 10, 35, 16],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
  'identity_factors': [3, 3, 3, 0],
  'noise_level': [28, 14, 49, 24],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
  'identity_factors': [4, 4, 4, 0],
  'noise_level': [36, 18, 63, 32],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [44, 11, 11, 40],
  'identity_factors': [5, 5, 5, 0],
  'noise_level': [44, 22, 77, 40],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [52, 13, 13, 48],
  'identity_factors': [6, 6, 6, 0],
  'noise_level': [52, 26, 91, 48],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [60, 15, 15, 56],
  'identity_factors': [7, 7, 7, 0],
  'noise_level': [60, 30, 105, 56],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [68, 17, 17, 64],
  'identity_factors': [8, 8, 8, 0],
  'noise_level': [68, 34, 119, 64],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [76, 19, 19, 72],
  'identity_factors': [9, 9, 9, 0],
  'noise_level': [76, 38, 133, 72],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [84, 21, 21, 80],
  'identity_factors': [10, 10, 10, 0],
  'noise_level': [84, 42, 147, 80],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [92, 23, 23, 88],
  'identity_factors': [11, 11, 11, 0],
  'noise_level': [92, 46, 161, 88],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [100, 25, 25, 96],
  'identity_factors': [12, 12, 12, 0],
  'noise_level': [100, 50, 175, 96],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [108, 27, 27, 104],
  'identity_factors': [13, 13, 13, 0],
  'noise_level': [108, 54, 189, 104],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [116, 29, 29, 112],
  'identity_factors': [14, 14, 14, 0],
  'noise_level': [116, 58, 203, 112],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [124, 31, 31, 120],
  'identity_factors': [15, 15, 15, 0],
  'noise_level': [124, 62, 217, 120],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [132, 33, 33, 128],
  'identity_factors': [16, 16, 16, 0],
  'noise_level': [132, 66, 231, 128],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [140, 35, 35, 136],
  'identity_factors': [17, 17, 17, 0],
  'noise_level': [140, 70, 245, 136],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [148, 37, 37, 144],
  'identity_factors': [18, 18, 18, 0],
  'noise_level': [148, 74, 259, 144],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [156, 39, 39, 152],
  'identity_factors': [19, 19, 19, 0],
  'noise_level': [156, 78, 273, 152],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [164, 41, 41, 160],
  'identity_factors': [20, 20, 20, 0],
  'noise_level': [164, 82, 287, 160],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [172, 43, 43, 168],
  'identity_factors': [21, 21, 21, 0],
  'noise_level': [172, 86, 301, 168],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [180, 45, 45, 176],
  'identity_factors': [22, 22, 22, 0],
  'noise_level': [180, 90, 315, 176],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [188, 47, 47, 184],
  'identity_factors': [23, 23, 23, 0],
  'noise_level': [188, 94, 329, 184],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [196, 49, 49, 192],
  'identity_factors': [24, 24, 24, 0],
  'noise_level': [196, 98, 343, 192],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [204, 51, 51, 200],
  'identity_factors': [25, 25, 25, 0],
  'noise_level': [204, 102, 357, 200],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [212, 53, 53, 208],
  'identity_factors': [26, 26, 26, 0],
  'noise_level': [212, 106, 371, 208],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value':
                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.001,
                                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
    'odd_wires': 4},
{ 'gates_num': [220, 55, 55, 216],
  'identity_factors': [27, 27, 27, 0],
  'noise_level': [220, 110, 385, 216],

```

```

                                0.001],
                                'status': True,
                                'type': 'depolarizing'},
                                'odd_wires': 4]],
                                'run_time_sec': 116.51207256317139}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                         'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.0019047787007362745,
                             0.7201496918445128,
                             0.7201496918445128,
                             0.7201496918445127,
                             0.7226708680142584,
                             0.8068041864232433,
                             1.3847174673528184,
                             1.6031895944534578,
                             2.458865333289775,
                             2.458865333289775,
                             2.721120236828938,
                             2.721120236828938,
                             3.554687304897056,
                             3.554687304897056,
                             4.537252190232355,
                             4.537252190232355,
                             5.239770593633025,
                             5.242363659304797,
                             5.566731623636481,
                             5.566731623636481,
                             5.566731623636481,
                             5.566731623636481,
                             5.566731623636481,
                             7.6484324534494865,
                             7.6484324534494865,
                             8.96757120343865,
                             9.352637118466301,
                             9.381932224950582,
                             9.966925955531599,
                             9.966925955531599,
                             0.29840004251263574,
                             0.3207087147269936,
                             -0.22478909909228914,
                             -1.2242611593120782,
                             0.061748844824383295,
                             1.9931455249507801,
                             0.02449362327304914,
                             -0.23402928130657288,
                             0.3399922671860315,
                             0.2355005268966802,
                             0.7912945290539132,
                             -0.9147564287830934,
                             0.12280724330167514,
                             0.42016875545313104,
                             0.6183328009701774,
                             0.9098148543333162,
                             0.2439503202728901,
                             0.4941740976390827,
                             0.0721485641264495,
                             0.20051214721509877,
                             0.14273239312798908,
                             -0.03846070968482946,
                             0.11474641891065286,
                             0.1552743298359438,
                             -0.050455466832234855,
                             0.24385918138398863,
                             0.0646818550964626,
                             1.0379575003266166,

```

0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,

```

-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0.001, 0.001],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                  'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                  'r': 1},
               'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0],
                                     [6, 6, 6, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                '5 X 6 + 1 Z 5 + 1 Z 6',
          'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                    '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                    '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                    '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                    '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                    '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.536182288327376],
                             [12, 6, 21, 8, -3.450081350248876],
                             [20, 10, 35, 16, -1.8650932993173606],
                             [28, 14, 49, 24, -1.0247351729167145],
                             [36, 18, 63, 32, -0.5670015302617211],
                             [44, 22, 77, 40, -0.31237897705332807],
                             [52, 26, 91, 48, -0.16869939858614091]],
          'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 7, 0],
                              'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},

```

```

False,
0},

{ 'gates_num': [12, 3, 3, 8],
  'identity_factors': [1, 1, 1, 0],
  'noise_level': [12, 6, 21, 8],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':

                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                    'status': True,
                                                    'type': 'depolarizing'},
  'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
  'identity_factors': [2, 2, 2, 0],
  'noise_level': [20, 10, 35, 16],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':

                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                    'status': True,
                                                    'type': 'depolarizing'},
  'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
  'identity_factors': [3, 3, 3, 0],
  'noise_level': [28, 14, 49, 24],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':

                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                    'status': True,
                                                    'type': 'depolarizing'},
  'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
  'identity_factors': [4, 4, 4, 0],
  'noise_level': [36, 18, 63, 32],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':

                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                    'status': True,
                                                    'type': 'depolarizing'},
  'odd_wires': 4},
{ 'gates_num': [44, 11, 11, 40],
  'identity_factors': [5, 5, 5, 0],
  'noise_level': [44, 22, 77, 40],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':

                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                    'status': True,
                                                    'type': 'depolarizing'},
  'odd_wires': 4},
{ 'gates_num': [52, 13, 13, 48],
  'identity_factors': [6, 6, 6, 0],
  'noise_level': [52, 26, 91, 48],
  'noise_profile': { 'noise_on_init_param': { 'status':

```

```

0},
'noise_prob': [ 0.001,
                0.001,
                0.001,
                0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4]],
'run_time_sec': 112.39416193962097}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
'init_param': { 'value': [ -5.647934061474703e-18,
                            1.4787834906243755,
                            1.478918229526628,
                            1.478918229526628,
                            2.0077351764243123,
                            2.4302809101456218,
                            2.4302809101456218,
                            3.009709563144601,
                            3.009709563144601,
                            3.239471043198664,
                            3.2405283031129133,
                            3.2405283031129133,
                            3.2656598154222434,
                            3.5353392937639825,
                            6.198713539858257,
                            6.213604956994236,
                            6.73397854893658,
                            6.73397854893658,
                            6.743547791194661,
                            6.744609103740889,
                            6.744609103740889,
                            6.749543065628153,
                            6.749543065628153,
                            8.615256229309818,
                            8.635090640841518,
                            9.012323512360366,
                            9.012323512360366,
                            9.930483613907942,
                            9.930483613907942,
                            11.782061805135282,
                            0.6870051227789642,
                            0.5234095533264759,
                            0.11216801162302814,
                            0.044591843687496906,
                            -0.747159787599118,
                            1.5727279855217766,
                            -0.16958009141159439,
                            -0.7234855106318131,
                            0.45999589631996707,
                            1.3613822619347415,
                            0.5234464983029046,
                            0.18337466193510712,
                            1.0660408011194182,
                            1.2813123812864906,
                            0.34899885807903364,
                            1.2204971120849089,
                            0.37250493843195553,
                            0.3861568591798728,
                            -0.4881036169368887,
                            0.23720347697856276,
                            0.8136369801063428,
                            -0.8595161765585049,
                            0.6710512785075716,

```

0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,


```

-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0],
[6, 6, 6, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
[12, 6, 21, 8, -3.1990719612158447],
[20, 10, 35, 16, -1.635137691104089],
[28, 14, 49, 24, -0.8526977600673816],
[36, 18, 63, 32, -0.4495330332284964],
[44, 22, 77, 40, -0.23666355366027367],
[52, 26, 91, 48, -0.12224019497382062]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value':
0},
'noise_prob': [ 0.001,
0.001,

```

```

0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
'identity_factors': [2, 2, 2, 0],
'noise_level': [20, 10, 35, 16],
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
'identity_factors': [3, 3, 3, 0],
'noise_level': [28, 14, 49, 24],
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
'identity_factors': [4, 4, 4, 0],
'noise_level': [36, 18, 63, 32],
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [44, 11, 11, 40],
'identity_factors': [5, 5, 5, 0],
'noise_level': [44, 22, 77, 40],
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},

```

False,
0},

```
{ 'gates_num': [52, 13, 13, 48],  
  'identity_factors': [6, 6, 6, 0],  
  'noise_level': [52, 26, 91, 48],  
  'noise_profile': { 'noise_on_init_param': { 'status':  
                                                    'value':  
                                                    'noise_prob': [ 0.001,  
                                                            0.001,  
                                                            0.001,  
                                                            0.001],  
                                                    'status': True,  
                                                    'type': 'depolarizing'},  
                      'odd_wires': 4}},  
  'run_time_sec': 113.48871517181396}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],  
                                                         'cn': [ 0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5],  
                                                         'r': 0},  
                                      'time': {'max': 10.0, 'min': 0.0},  
                                      'type': 'xy-iss'}},  
    'init_param': { 'value': [ 0.33528943002425843,  
                                0.8608925283648261,  
                                2.8309935233222867,  
                                2.8382233121900358,  
                                2.847558790018716,  
                                2.8477263794127468,  
                                3.9337993695450666,  
                                3.9337993695450666,  
                                3.9337993695450666,  
                                3.9337993695450666,  
                                3.9337993695450666,  
                                5.127462455084763,  
                                5.128383696390232,  
                                5.132051241075024,  
                                5.132231848473024,  
                                5.132297186784173,  
                                5.132297186784173,  
                                5.132297186784173,  
                                5.132320898992942,  
                                5.134903164275234,  
                                5.734798022200061,  
                                5.734798022200061,  
                                5.734798022200061,  
                                6.71748919093495,  
                                6.71748919093495,  
                                9.773430860122208,  
                                9.791656954875775,  
                                9.791980525916317,  
                                9.804782964874805,  
                                9.947289619134542,  
                                0.6818618664375632,  
                                0.42253658736184846,  
                                0.19067339209915313,  
                                -2.314563573529872,  
                                -1.7784854373672185,  
                                -0.10945476580518654,  
                                0.08980428543050965,  
                                0.6265702304745648,  
                                -0.12016219313034968,  
                                0.005450784512528824,  
                                0.9621316606784164,  
                                -0.01940541263687908,  
                                -1.2890091107885369,  
                                -0.09216272929775945,  
                                -0.8615487717558121,  
                                0.021600758375482327,  
                                1.1068545366678577,  
                                -0.06436428812394061,
```

-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,

```

-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0],
[6, 6, 6, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.333005581610796],
[12, 6, 21, 8, -3.1651017858597426],
[20, 10, 35, 16, -1.6439583689682524],
[28, 14, 49, 24, -0.8813072530521873],
[36, 18, 63, 32, -0.4831862823512047],
[44, 22, 77, 40, -0.26793423014165196],
[52, 26, 91, 48, -0.14824931816458392]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':

```

```
False,
    0},
    'noise_prob': [ 0.001,
                    0.001,
                    0.001,
                    0.001],
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4},
    { 'gates_num': [12, 3, 3, 8],
      'identity_factors': [1, 1, 1, 0],
      'noise_level': [12, 6, 21, 8],
      'noise_profile': { 'noise_on_init_param': { 'status':
False,
    0},
    'noise_prob': [ 0.001,
                    0.001,
                    0.001,
                    0.001],
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4},
    { 'gates_num': [20, 5, 5, 16],
      'identity_factors': [2, 2, 2, 0],
      'noise_level': [20, 10, 35, 16],
      'noise_profile': { 'noise_on_init_param': { 'status':
False,
    0},
    'noise_prob': [ 0.001,
                    0.001,
                    0.001,
                    0.001],
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4},
    { 'gates_num': [28, 7, 7, 24],
      'identity_factors': [3, 3, 3, 0],
      'noise_level': [28, 14, 49, 24],
      'noise_profile': { 'noise_on_init_param': { 'status':
False,
    0},
    'noise_prob': [ 0.001,
                    0.001,
                    0.001,
                    0.001],
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4},
    { 'gates_num': [36, 9, 9, 32],
      'identity_factors': [4, 4, 4, 0],
      'noise_level': [36, 18, 63, 32],
      'noise_profile': { 'noise_on_init_param': { 'status':
False,
    0},
    'noise_prob': [ 0.001,
                    0.001,
                    0.001,
                    0.001],
    'status': True,
    'type': 'depolarizing'},
    'odd_wires': 4},
    { 'gates_num': [44, 11, 11, 40],
      'identity_factors': [5, 5, 5, 0],
      'noise_level': [44, 22, 77, 40],
      'noise_profile': { 'noise_on_init_param': { 'status':
False,
    0},
    'noise_prob': [ 0.001,
                    0.001,
```

```

0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [52, 13, 13, 48],
'identity_factors': [6, 6, 6, 0],
'noise_level': [52, 26, 91, 48],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'value':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 88.97754573822021}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.003375513600300809,
0.4884127443235871,
0.6719352108785641,
0.9073170135093351,
1.4946633690060278,
1.5076957064847494,
3.872917838142945,
3.872917838142945,
3.872917838142945,
3.872917838142945,
3.872917838142945,
5.047622474508209,
5.054336886650782,
5.067626550795242,
5.0733174878702085,
5.094629387163461,
5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,

```

1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,


```

-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0],
                                     [6, 6, 6, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                  '1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
                                  '5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 7, 0, -6.397817460329862],
                              [12, 6, 21, 8, -3.2115716730808748],
                              [20, 10, 35, 16, -1.6419588170332553],
                              [28, 14, 49, 24, -0.8464644349040009],
                              [36, 18, 63, 32, -0.4337371704190119],
                              [44, 22, 77, 40, -0.216007189653039],

```

```
[52, 26, 91, 48, -0.10025204405746856]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 7, 0],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                  { 'gates_num': [12, 3, 3, 8],
                    'identity_factors': [1, 1, 1, 0],
                    'noise_level': [12, 6, 21, 8],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                  { 'gates_num': [20, 5, 5, 16],
                    'identity_factors': [2, 2, 2, 0],
                    'noise_level': [20, 10, 35, 16],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                  { 'gates_num': [28, 7, 7, 24],
                    'identity_factors': [3, 3, 3, 0],
                    'noise_level': [28, 14, 49, 24],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                  { 'gates_num': [36, 9, 9, 32],
                    'identity_factors': [4, 4, 4, 0],
                    'noise_level': [36, 18, 63, 32],
                    'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.001,
                                                            0.001],
                                                                    'status': True,
                                                                    'type': 'depolarizing'},
                    'odd_wires': 4},
                  { 'gates_num': [44, 11, 11, 40],
                    'identity_factors': [5, 5, 5, 0],
                    'noise_level': [44, 22, 77, 40],
                    'noise_profile': { 'noise_on_init_param': { 'status':
```

```

False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [52, 13, 13, 48],
'identity_factors': [6, 6, 6, 0],
'noise_level': [52, 26, 91, 48],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 118.73719573020935}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ -6.331146189926128e-17,
0.7069927066774304,
0.7069927066774304,
0.7069927066774305,
0.7069927066774304,
1.0044153364322925,
1.0092418692996383,
2.470333794214341,
2.478143749866421,
3.5776564981759003,
3.5776564981759003,
3.5776564981759003,
3.61745967653339,
3.6981780389472076,
3.755720232685741,
3.9712286023902594,
3.9712286023902594,
4.519037432417192,
4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,
6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,

```

0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,

```

-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0],
[6, 6, 6, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.311563719128683],

```

```

[12, 6, 21, 8, -3.2712898811023576],
[20, 10, 35, 16, -1.7533390824025292],
[28, 14, 49, 24, -0.966227702270059],
[36, 18, 63, 32, -0.5436512754494955],
[44, 22, 77, 40, -0.3098340729956386],
[52, 26, 91, 48, -0.17724371427497804]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 3, 8],
'identity_factors': [1, 1, 1, 0],
'noise_level': [12, 6, 21, 8],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 5, 16],
'identity_factors': [2, 2, 2, 0],
'noise_level': [20, 10, 35, 16],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 7, 24],
'identity_factors': [3, 3, 3, 0],
'noise_level': [28, 14, 49, 24],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [36, 9, 9, 32],
'identity_factors': [4, 4, 4, 0],
'noise_level': [36, 18, 63, 32],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'type': 'depolarizing'},

```

```

False,
0},

'odd_wires': 4},
{ 'gates_num': [44, 11, 11, 40],
  'identity_factors': [5, 5, 5, 0],
  'noise_level': [44, 22, 77, 40],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                    'status': True,
                                                    'type': 'depolarizing'}},
  'odd_wires': 4},
{ 'gates_num': [52, 13, 13, 48],
  'identity_factors': [6, 6, 6, 0],
  'noise_level': [52, 26, 91, 48],
  'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value':
                                                    'noise_prob': [ 0.001,
                                                                    0.001,
                                                                    0.001,
                                                                    0.001],
                                                    'status': True,
                                                    'type': 'depolarizing'}},
  'odd_wires': 4}},
  'run_time_sec': 118.2682409286499}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
  'init_param': { 'value': [ 1.5680106786222674e-17,
                              1.0238650999177359,
                              1.0238650999177359,
                              1.0238650999177359,
                              1.7041550191762513,
                              2.070931805530806,
                              2.586234954306046,
                              2.6265616878606783,
                              2.7566136785306155,
                              2.7689271757054255,
                              4.271831894498143,
                              4.271831894498143,
                              4.271831894498143,
                              5.004110362344359,
                              5.004110362344359,
                              5.529689350278954,
                              5.529689350278954,
                              5.529689350278954,
                              5.5493166460873375,
                              7.855838962179162,
                              7.855838962179162,
                              8.508023697610218,
                              8.540796404638465,
                              8.940056981382925,
                              8.940056981382925,
                              8.940056981382925,
                              8.940056981382925,
                              9.524293982069336,
                              9.524293982069336,
                              9.815045396471412,
                              0.717685470636315,
                              0.29125971197830336,
                              0.013619559122678915,

```

-0.46174394022911175,
-0.7802589318548062,
1.8785332138943256,
-0.07718385116900997,
-0.4181315830102034,
0.3859499067718677,
0.7218523694763775,
0.16350787496293626,
-0.06274337543165032,
1.048639096314506,
0.9454500907927006,
-0.18328208251477485,
0.5336296017995852,
0.7047960543548563,
-0.4717130073686535,
-0.14327049451726778,
0.5646728024946054,
0.1377201649736573,
1.0845662084088512,
0.7117671017527167,
1.8562916393289055,
-1.0958742208007046,
0.314756410152813,
1.0055290931424188,
-1.2653176839886926,
-0.6435463058390287,
-0.13689785210195535,
0.5185042269644529,
-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,


```
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0],
[6, 6, 6, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 +
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 +
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 +

```

```

'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 7, 0, -6.515742960091852],
                              [12, 6, 21, 8, -3.36492111003702],
                              [20, 10, 35, 16, -1.7545658776574036],
                              [28, 14, 49, 24, -0.9139478342631155],
                              [36, 18, 63, 32, -0.46803177808311025],
                              [44, 22, 77, 40, -0.22931309151635063],
                              [52, 26, 91, 48, -0.10153016436427796]],
          'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 7, 0],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                               'value':
                                                                                               'noise_prob': [ 0.001,
                                                                                               0.001,
                                                                                               0.001,
                                                                                               0.001],
                                                                                               'status': True,
                                                                                               'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [12, 3, 3, 8],
                                'identity_factors': [1, 1, 1, 0],
                                'noise_level': [12, 6, 21, 8],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                               'value':
                                                                                               'noise_prob': [ 0.001,
                                                                                               0.001,
                                                                                               0.001,
                                                                                               0.001],
                                                                                               'status': True,
                                                                                               'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [20, 5, 5, 16],
                                'identity_factors': [2, 2, 2, 0],
                                'noise_level': [20, 10, 35, 16],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                               'value':
                                                                                               'noise_prob': [ 0.001,
                                                                                               0.001,
                                                                                               0.001,
                                                                                               0.001],
                                                                                               'status': True,
                                                                                               'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [28, 7, 7, 24],
                                'identity_factors': [3, 3, 3, 0],
                                'noise_level': [28, 14, 49, 24],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                               'value':
                                                                                               'noise_prob': [ 0.001,
                                                                                               0.001,
                                                                                               0.001,
                                                                                               0.001],
                                                                                               'status': True,
                                                                                               'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [36, 9, 9, 32],
                                'identity_factors': [4, 4, 4, 0],
                                'noise_level': [36, 18, 63, 32],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                               'value':
                                                                                               'noise_prob': [ 0.001,
                                                                                               0.001,
                                                                                               0.001,
                                                                                               0.001],
                                                                                               'status': True,
                                                                                               'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [44, 11, 11, 40],
                                'identity_factors': [5, 5, 5, 0],
                                'noise_level': [44, 22, 77, 40],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                               'value':
                                                                                               'noise_prob': [ 0.001,
                                                                                               0.001,
                                                                                               0.001,
                                                                                               0.001],
                                                                                               'status': True,
                                                                                               'type': 'depolarizing'},
                                'odd_wires': 4},
                              { 'gates_num': [52, 13, 13, 48],
                                'identity_factors': [6, 6, 6, 0],
                                'noise_level': [52, 26, 91, 48],
                                'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                               'value':
                                                                                               'noise_prob': [ 0.001,
                                                                                               0.001,
                                                                                               0.001,
                                                                                               0.001],
                                                                                               'status': True,
                                                                                               'type': 'depolarizing'},
                                'odd_wires': 4}
                              ]
        }
    }
}

```

```

0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [44, 11, 11, 40],
'identity_factors': [5, 5, 5, 0],
'noise_level': [44, 22, 77, 40],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
{ 'gates_num': [52, 13, 13, 48],
'identity_factors': [6, 6, 6, 0],
'noise_level': [52, 26, 91, 48],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
0},
'noise_prob': [ 0.001,
0.001,
0.001,
0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4}},
'run_time_sec': 92.21969175338745}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
'file_name_prefix': 'xy_noisey_time_evo'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1 Z 2 '
'+ 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z '
'5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + 0.5 X 1 '
'X 2 + 0.5 Z 1 Z 2 + 0.5 X 2 X 3 + '
'0.5 Z 2 Z 3 + 0.5 X 3 X 4 + 0.5 Z 3 '
'Z 4 + 0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
'initial_cost_history': [ 3.5476707600867368,

```

```

2.3562053879970173,
1.7833099820235232,
2.4654982864165467,
3.6613143494662093,
3.6462371631182062,
3.2935768662401737,
3.7145398712422657,
2.5227446974102397,
2.4756634278387035],
'noise_details': { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 7, 0],
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -6.515742960091897,
-6.3330055816107915,
-6.410492799212151,
-6.4484545761945915,
-6.311563719128753,
-6.46067826504545,
-6.397817460329919,
-6.429933051303193,
-6.536182288327426,
-6.511175848027232],
'optimized_parameters': [ [ 1.5680106786222674e-17,
1.0238650999177359,
1.0238650999177359,
1.0238650999177359,
1.7041550191762513,
2.070931805530806,
2.586234954306046,
2.6265616878606783,
2.7566136785306155,
2.7689271757054255,
4.271831894498143,
4.271831894498143,
4.271831894498143,
5.004110362344359,
5.004110362344359,
5.529689350278954,
5.529689350278954,
5.529689350278954,
5.5493166460873375,
7.855838962179162,
7.855838962179162,
8.508023697610218,
8.540796404638465,
8.940056981382925,
8.940056981382925,
8.940056981382925,
8.940056981382925,
9.524293982069336,
9.524293982069336,
9.815045396471412,
0.717685470636315,
0.29125971197830336,
0.013619559122678915,
-0.46174394022911175,
-0.7802589318548062,
1.8785332138943256,
-0.07718385116900997,
-0.4181315830102034,
0.3859499067718677,
0.7218523694763775,
0.16350787496293626,
-0.06274337543165032,
1.048639096314506,
0.9454500907927006,
-0.18328208251477485,
0.5336296017995852,
0.7047960543548563,

```

-0.4717130073686535,
-0.14327049451726778,
0.5646728024946054,
0.1377201649736573,
1.0845662084088512,
0.7117671017527167,
1.8562916393289055,
-1.0958742208007046,
0.314756410152813,
1.0055290931424188,
-1.2653176839886926,
-0.6435463058390287,
-0.13689785210195535,
0.5185042269644529,
-0.07994120596020922,
-1.3394312085309903,
-0.01354724643952521,
-0.8508707890759795,
-0.7258020036409927,
-0.1513648764845159,
-0.5702686618598916,
-0.20804467461705725,
-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,

0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,
0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455],
[0.33528943002425843,
0.8608925283648261,
2.8309935233222867,
2.8382233121900358,
2.847558790018716,
2.8477263794127468,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
3.9337993695450666,
5.127462455084763,
5.128383696390232,
5.132051241075024,
5.132231848473024,
5.132297186784173,
5.132297186784173,
5.132297186784173,
5.132320898992942,
5.134903164275234,
5.734798022200061,
5.734798022200061,
5.734798022200061,
6.71748919093495,
6.71748919093495,
9.773430860122208,
9.791656954875775,
9.791980525916317,
9.804782964874805,
9.947289619134542,
0.6818618664375632,
0.42253658736184846,
0.19067339209915313,
-2.314563573529872,
-1.7784854373672185,
-0.10945476580518654,
0.08980428543050965,
0.6265702304745648,
-0.12016219313034968,
0.005450784512528824,
0.9621316606784164,
-0.01940541263687908,
-1.2890091107885369,
-0.09216272929775945,
-0.8615487717558121,
0.021600758375482327,
1.1068545366678577,

-0.06436428812394061,
-1.8186253546793636,
-0.2455724158610742,
0.7444478114283775,
1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,

-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,
-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358],
[0.08591881204796584,
1.5026023651616802,
1.9179097588555833,
1.9179097588555831,
1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,
2.258056275075482,
-0.5160508087356981,

0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,
1.0008805619835928,
0.16556690201288055,

1.1071973338235597,
0.07194244573338314,
-0.9146859400815048,
-0.005186820541518637,
0.4795762978809806,
0.6475066336984044,
0.11860845746693519,
0.08044881503693782,
0.5402817334108434,
1.5532870492056725,
0.36643388228039625,
1.6150985923432208,
0.22271977417212363,
-0.04956133994172771,
-0.6258533610916538,
-0.6517224443292691,
0.2594825887888975,
-0.1286341552665023,
0.48749759828242367,
-0.009813928289633511,
0.5249350004928881,
-0.01015655683116892,
-0.48671658518654837,
-0.588311268126242,
-0.25978871509233853,
0.9039610340757382,
1.5549616218831699,
-0.05330158954092406],
[0.3773346757087749,
0.5734233884194113,
0.5734233884194113,
1.2094818154974225,
1.2528672337957094,
1.3073966674132076,
4.193348178434801,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.274742575727675,
5.343872214974395,
5.354523589318104,
5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,

0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,
1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,

1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,
0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244],
[-6.331146189926128e-17,
0.7069927066774304,
0.7069927066774304,
0.7069927066774305,
0.7069927066774304,
1.0044153364322925,
1.0092418692996383,
2.470333794214341,
2.478143749866421,
3.5776564981759003,
3.5776564981759003,
3.5776564981759003,
3.61745967653339,
3.6981780389472076,
3.755720232685741,
3.9712286023902594,
3.9712286023902594,
4.519037432417192,
4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,
6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,

-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,

0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176],
[3.750846163374531e-05,
0.22285241391868899,
0.26252411232084344,
0.26252411232084344,
0.26252411232084344,
1.459545906754556,
1.4595459067545562,
1.6094835194149661,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
2.5927782099427623,
3.3552499939702187,
3.3552499939702187,
5.033284106467877,
5.033284106467877,
5.056477210452712,
6.768913439946703,
6.768913439946703,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,

1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,
-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,

0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138],
[0.003375513600300809,
0.4884127443235871,
0.6719352108785641,
0.9073170135093351,
1.4946633690060278,
1.5076957064847494,
3.872917838142945,
3.872917838142945,
3.872917838142945,
3.872917838142945,
5.047622474508209,
5.054336886650782,
5.067626550795242,
5.0733174878702085,
5.094629387163461,
5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,

0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,
1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,

-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989],
[-5.647934061474703e-18,
1.4787834906243755,
1.478918229526628,
1.478918229526628,
2.0077351764243123,
2.4302809101456218,
2.4302809101456218,
3.009709563144601,
3.009709563144601,
3.239471043198664,
3.2405283031129133,
3.2405283031129133,
3.2656598154222434,
3.5353392937639825,
6.198713539858257,
6.213604956994236,
6.73397854893658,
6.73397854893658,
6.743547791194661,
6.744609103740889,
6.744609103740889,
6.749543065628153,
6.749543065628153,
8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,

0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,
-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,

0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,
1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935],
[0.0019047787007362745,
0.7201496918445128,
0.7201496918445128,
0.7201496918445127,
0.7226708680142584,
0.8068041864232433,
1.3847174673528184,
1.6031895944534578,
2.458865333289775,
2.458865333289775,
2.721120236828938,
2.721120236828938,
3.554687304897056,
3.554687304897056,
4.537252190232355,
4.537252190232355,
5.239770593633025,
5.242363659304797,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
5.566731623636481,
7.6484324534494865,
7.6484324534494865,
8.96757120343865,
9.352637118466301,
9.381932224950582,
9.966925955531599,
9.966925955531599,
0.29840004251263574,
0.3207087147269936,
-0.22478909909228914,
-1.2242611593120782,
0.061748844824383295,
1.9931455249507801,
0.02449362327304914,
-0.23402928130657288,
0.3399922671860315,
0.2355005268966802,
0.7912945290539132,
-0.9147564287830934,
0.12280724330167514,
0.42016875545313104,
0.6183328009701774,
0.9098148543333162,
0.2439503202728901,

0.4941740976390827,
0.0721485641264495,
0.20051214721509877,
0.14273239312798908,
-0.03846070968482946,
0.11474641891065286,
0.1552743298359438,
-0.050455466832234855,
0.24385918138398863,
0.0646818550964626,
1.0379575003266166,
0.4438839418702803,
0.6652939620943935,
0.6574583263339632,
0.6386444070388955,
0.30923240052603657,
-0.34636230410321134,
0.23860747048870812,
-0.8837469955189079,
0.25272330748816335,
-0.05865013289270936,
0.324900593333006,
0.14494577696571714,
-0.2063591527406891,
0.03616506891850059,
0.47262902670638013,
-1.0341740644891497,
0.05484745701468325,
0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,

0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815],
[0.00030922277578629,
0.881375055029912,
0.881375055029912,
0.8813750550299119,
0.9803845151714838,
0.9803845151714838,
1.201629973822729,
1.2441077564052723,
1.8660677974280824,
1.8660677974280824,
2.651002314516478,
2.651002314516478,
3.233637461790533,
4.36384995486056,
4.36384995486056,
4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,
8.903406631369169,
8.903406631369169,
8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.1357940797759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,
0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,

0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,
1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,

```

0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'run_time_sec': 68892.59480500221}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
                                'layer': 30,
                                'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                            'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                            'r': 0},
                                'time': { 'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.00030922277578629,
0.881375055029912,
0.881375055029912,
0.8813750550299119,
0.9803845151714838,
0.9803845151714838,
1.201629973822729,
1.2441077564052723,
1.8660677974280824,
1.8660677974280824,
2.651002314516478,
2.651002314516478,
3.233637461790533,
4.36384995486056,
4.36384995486056,
4.975714528653046,
4.975714528653046,
6.591840750700408,
6.591840750700408,
8.234009055219385,
8.238223602720305,
8.238223602720305,
8.238223602720305,
8.903406631369169,
8.903406631369169,
8.903406631369169,
9.126449234760914,
9.126449234760914,
9.441918167657187,
9.441918167657187,
0.4384156956352651,
-0.017407483354635628,
-0.13579407977759564,
-0.08181457210659056,
-0.5635031413839604,
0.570197125887921,
-0.07389454022208618,
0.18950484918514415,
0.23442314823895152,

```


0.5187197914643009,
0.03760777637611914,
1.2080522066616035,
-0.006796933498365149,
0.7597395812711474,
0.3165595784185247,
0.09234381717322694,
-0.047116958459538144,
0.02161370846364295,
-0.3330735946096464,
-0.310519881157395,
-0.1321730433969341,
-0.32004747583182785,
0.38880240494533963,
0.5587916266845782,
-0.04068226055752174,
-0.04929253976644778,
0.7350044083650977,
-0.3347542084226466,
0.07994172439872818,
0.177027499128215,
-0.03555523462670161,
0.33584282142298794,
-0.06474358479167515,
-0.23271669416276414,
0.27707249457309313,
-1.8841797053563085,
-0.39250057722196113,
0.0604718224769807,
-0.15658247224402194,
1.097732135880844,
-0.47958374693582667,
-0.13884788062191106,
1.2517060190766023,
0.08912037934357776,
1.8722824819152626,
0.5767878477607333,
0.4375958673531614,
0.017187174100509652,
-1.172587468420202,
0.736957937590481,
-1.8818047463381875,
-0.29954118659750983,
-2.8240783717504883,
-0.0009188056875444823,
-0.014480253731157836,
0.8344869723837732,
-0.235921407746533,
-0.2726490308679638,
-0.4874528967327087,
0.05731267695545898,
0.32920686175801595,
0.0470036094807545,
0.6942117349751494,
-0.24738597040458754,
-0.3530344251488351,
-0.7207866325012796,
0.6865004571010461,
-0.022934954570734595,
-0.4703523591474931,
0.00010065187353926284,
-0.25737840979321513,
-0.37927040299996573,
0.3323512575736423,
-0.7162822410895011,
-0.8849129065134153,
-0.02984860970928172,
0.9857401879405846,
0.13869536095324345,
-0.4901093246549395,
-0.15185390643388375,
-1.1107951730061545,
1.1255217267520845,
-0.36113008327046914,
0.12498769155032201,

```

1.105653484876161,
-0.0826241142760939,
-0.6079720837354008,
-0.4798331114275297,
-1.1230771992973987,
0.13634359721928585,
-1.0117198549615587,
0.0420371772323395,
0.4704689604818435,
-0.1136051809968673,
-0.4158299646895606,
0.7682233881623428,
-0.8567828943049521,
-1.0844985071567337,
0.8118563616911731,
-1.0011580890685694,
0.06773503500153778,
-0.9972068257582826,
0.22064841214728595,
1.015101892093672,
-0.023903188302909387,
-1.5448535965900865,
0.40727537812840897,
0.6273868528756265,
-0.06403282214251436,
0.1402221940396834,
0.011807494022614938,
0.19970219531306482,
-0.3159297567283321,
0.5243897804672414,
0.6302439365729751,
0.4184253509022523,
0.38422115408735924,
0.8610939738936156,
0.3457693810952046,
0.19105880264763628]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0],
                                     [6, 6, 6, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.5111758480272135],
                          [12, 6, 21, 8, -3.513108400075424],
                          [20, 10, 35, 16, -1.9256894730075684],
                          [28, 14, 49, 24, -1.064810006156379],
                          [36, 18, 63, 32, -0.5886022016151555],
                          [44, 22, 77, 40, -0.3212779442729232],
                          [52, 26, 91, 48, -0.16991504362792267]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.745887756347656e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.274964541089247,
                           'others': { 'beta_coefficients': [ 2.3290124519475084,

```

```

-3.8651696011043746,
5.6068818287625195,
-5.265593369620453,
3.048036430401101,
-0.9926938319776267,
0.13952609159132542],
'cost_richardson_zne': 89.82286707785958,
'sorted_expectation_vals': [ -6.5111758480272135,
-3.513108400075424,
-1.9256894730075684,
-1.064810006156379,
-0.5886022016151555,
-0.3212779442729232,
-0.16991504362792267],
'sorted_noise': [13, 47, 81, 115, 149, 183, 217]],
'sampled data': [ [4, 2, 7, 0, -6.5111758480272135],
[12, 6, 21, 8, -3.513108400075424],
[20, 10, 35, 16, -1.9256894730075684],
[28, 14, 49, 24, -1.064810006156379],
[36, 18, 63, 32, -0.5886022016151555],
[44, 22, 77, 40, -0.3212779442729232],
[52, 26, 91, 48, -0.16991504362792267]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.08591881204796584,
1.5026023651616802,
1.9179097588555833,
1.9179097588555831,
1.9179097588555831,
1.9179097588555831,
2.8800636946459592,
2.9523818900173797,
2.9523818900173797,
2.954896407422686,
3.0487820562708485,
3.0487820562708485,
3.9462134206000328,
3.9462134206000328,
5.170248653928018,
5.170248653928018,
6.5066905594049205,
6.5066905594049205,
6.5066905594049205,
6.914119522580601,
6.914119522580601,
6.9842551957162815,
7.300516913785898,
8.782483894556266,
8.782483894556266,
8.999633503017922,
8.999633503017922,
8.999633503017922,
9.414122362394847,
9.414122362394847,
0.09854676674938925,
-0.05173289612974198,
0.30094803169630163,
-1.1984142138894447,
0.08610670053567002,
0.2851928692216035,
0.26177875477440216,
0.11764473243628538,
0.3495067876706816,
-0.9678438601279628,
-0.08762050677252062,
0.014826088936787857,
-0.30347729159143555,
-1.1730530788097624,
-0.12550631929596553,

```

2.258056275075482,
-0.5160508087356981,
0.004541093619226378,
-0.9345420797218197,
0.47154522212338745,
-0.9129446894819914,
0.2768683391341781,
0.29975752794333804,
-0.16177915588279423,
0.6021148977823381,
0.35405737535162146,
0.8149645603473178,
-0.7074306860917636,
-0.8281028342440522,
-0.20840122350984505,
-0.19593362207831977,
1.2208184674045053,
-0.10408257731449277,
0.9878855326441833,
0.6508295623732647,
-0.31520021014255994,
-1.502007043998756,
-0.44913621684685756,
-0.2831231513937543,
-1.048170079126895,
-0.10057118739311635,
-0.4440190475287521,
0.5925479388494825,
0.1500100176310508,
-0.4655542365338734,
0.09847924242741647,
-0.25695620197439545,
-0.5449201592551612,
0.4987130755500597,
-0.007720607194202524,
0.7816530161823243,
-0.8246572495389679,
-0.28480932751570914,
1.8751826609713358,
-0.20740645402102328,
-0.04894031202761418,
-1.0170156037248201,
-0.035476380392734465,
0.9940259955458774,
-0.657126951929564,
-0.4786936406484008,
1.4564140457987391,
-1.062151832750044,
-0.01496017231630948,
0.16259290081999456,
0.601493806263198,
-0.20485253219607644,
-1.2421776762345744,
1.152288606322054,
1.1883019471730756,
0.9817659575699377,
0.9653429636905305,
0.5382122852444811,
0.722384759736696,
-0.13079560936986118,
-0.9638314109375934,
-0.13358671856975504,
-0.3949417844115237,
1.01780227278121,
0.6746960094686067,
0.14295933107831804,
-0.37974724013569766,
0.3641245559377578,
0.12972678280008856,
0.5949873255000262,
0.8499370211382984,
-0.0792654431528575,
1.9402087793428133,
-0.5583113020357903,
0.6535986429024293,

[illegible]

```

        'cost_richardson_zne': 89.82286707785958,
        'sorted_expectation_vals': [ -6.4104927992121,
                                      -3.25771336700167,
                                      -1.674813432318219,
                                      -0.8591076530612253,
                                      -0.43058689568319575,
                                      -0.20313281829021995,
                                      -0.08257115165007212],
        'sorted_noise': [13, 47, 81, 115, 149, 183, 217]],
    'sampled data': [ [4, 2, 7, 0, -6.4104927992121],
                      [12, 6, 21, 8, -3.25771336700167],
                      [20, 10, 35, 16, -1.674813432318219],
                      [28, 14, 49, 24, -0.8591076530612253],
                      [36, 18, 63, 32, -0.43058689568319575],
                      [44, 22, 77, 40, -0.20313281829021995],
                      [52, 26, 91, 48, -0.08257115165007212]],
    'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.33528943002425843,
                              0.8608925283648261,
                              2.8309935233222867,
                              2.8382233121900358,
                              2.847558790018716,
                              2.8477263794127468,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              3.9337993695450666,
                              5.127462455084763,
                              5.128383696390232,
                              5.132051241075024,
                              5.132231848473024,
                              5.132297186784173,
                              5.132297186784173,
                              5.132297186784173,
                              5.132320898992942,
                              5.134903164275234,
                              5.734798022200061,
                              5.734798022200061,
                              5.734798022200061,
                              6.71748919093495,
                              6.71748919093495,
                              9.773430860122208,
                              9.791656954875775,
                              9.791980525916317,
                              9.804782964874805,
                              9.947289619134542,
                              0.6818618664375632,
                              0.42253658736184846,
                              0.19067339209915313,
                              -2.314563573529872,
                              -1.7784854373672185,
                              -0.10945476580518654,
                              0.08980428543050965,
                              0.6265702304745648,
                              -0.12016219313034968,
                              0.005450784512528824,
                              0.9621316606784164,
                              -0.01940541263687908,
                              -1.2890091107885369,
                              -0.09216272929775945,
                              -0.8615487717558121,
                              0.021600758375482327,
                              1.1068545366678577,
                              -0.06436428812394061,
                              -1.8186253546793636,
                              -0.2455724158610742,
                              0.7444478114283775,

```

1.0647343146888184,
-1.6388802958603008,
-0.05032235091912311,
0.7589104637111997,
0.0934759886788552,
0.7280377617711271,
-0.10012682423470079,
-2.7917715562202794,
0.5843828201862561,
1.2003794612588048,
-0.2782652692858759,
1.253725179404932,
-0.831140674623039,
-1.3898002860152843,
1.7689493816619246,
-0.768539740515663,
-0.5265628815801298,
-0.8222194498348508,
-0.4525275176973082,
0.8944486767447871,
1.1352373194126446,
0.06348552937029919,
-0.7478835175458556,
-0.44914670649658045,
0.5247278033135997,
0.5948349944294522,
-0.5743460857513,
0.873550995352649,
0.8797938271812764,
-0.37270941226617715,
0.20553479070949218,
0.6188766136751805,
-0.4012501825920459,
-0.4763388197192611,
-0.34431485887187113,
-1.1697954599661295,
0.5556509722807279,
-0.5397649164737537,
-0.41722730509388967,
0.9275888618509817,
0.4371362266023078,
-1.4110895986752616,
0.6716489600966955,
0.08900519900900147,
0.7708643633651343,
0.0944251249629154,
-0.053131619179539805,
-0.132883145931867,
0.44414428660563765,
-0.3164026614336652,
0.9835391768228284,
-0.49696354121947833,
0.3510514174127117,
-0.24991518010085176,
-0.7780713458689107,
1.3387260017849976,
0.712191280671364,
-0.4436070432657242,
1.090195713122469,
0.3876483875477464,
1.113683201871723,
0.09724940056823322,
-1.3265075307540413,
-1.8530581309545013,
0.335564434910613,
0.16061631243062188,
0.6994545842241634,
-0.0055568765936962455,
1.8148369905661879,
-0.9149620921846291,
-0.5341673651010566,
-0.6583203681162046,
-0.05148037848607484,
-0.1302834828824637,
-0.2083684341417298,

```

-0.7142637766159184,
0.4980789266978527,
-2.030263033934291,
-0.002330467371510849,
-0.7835259213136883,
-0.18712783689065152,
-0.271594041353006,
0.606962231033315,
0.19917323339650786,
0.581852597021436,
1.0547240409345577,
0.07111303544026976,
1.4877210254684077,
0.08669172360020491,
0.8024105605127613,
0.8315802366658298,
1.117305276662395,
-1.0742437197416552,
0.30921812342286065,
-0.06994991433225481,
1.0419800264617582,
1.4507701003592177,
0.2408413974211676,
1.6573761327725358]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 0],
                                      [2, 2, 2, 0],
                                      [3, 3, 3, 0],
                                      [4, 4, 4, 0],
                                      [5, 5, 5, 0],
                                      [6, 6, 6, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.333005581610796],
                          [12, 6, 21, 8, -3.1651017858597426],
                          [20, 10, 35, 16, -1.6439583689682524],
                          [28, 14, 49, 24, -0.8813072530521873],
                          [36, 18, 63, 32, -0.4831862823512047],
                          [44, 22, 77, 40, -0.26793423014165196],
                          [52, 26, 91, 48, -0.14824931816458392]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.7220458984375e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.320345710242359,
                           'others': { 'beta_coeffieicients': [ 2.3290124519475084,
                                                                -3.8651696011043746,
                                                                5.6068818287625195,
                                                                -5.265593369620453,
                                                                3.048036430401101,
                                                                -0.9926938319776267,
                                                                0.13952609159132542],
                                  'cost_richardson_zne': 89.82286707785958,
                                  'sorted_expectation_vals': [ -6.333005581610796,
                                                                -3.1651017858597426,
                                                                -1.6439583689682524,
                                                                -0.8813072530521873,
                                                                -0.4831862823512047,
                                                                0.26793423014165196,
                                                                0.8813072530521873,
                                                                1.6439583689682524,
                                                                3.048036430401101,
                                                                5.265593369620453,
                                                                -5.6068818287625195,
                                                                -3.8651696011043746,
                                                                -2.3290124519475084]
                                }
                           }
            }

```



```

-0.26793423014165196,
-0.14824931816458392]],
'sorted_noise': [13, 47, 81, 115, 149, 183, 217]],
'sampled data': [ [4, 2, 7, 0, -6.333005581610796],
[12, 6, 21, 8, -3.1651017858597426],
[20, 10, 35, 16, -1.6439583689682524],
[28, 14, 49, 24, -0.8813072530521873],
[36, 18, 63, 32, -0.4831862823512047],
[44, 22, 77, 40, -0.26793423014165196],
[52, 26, 91, 48, -0.14824931816458392]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ -5.647934061474703e-18,
1.4787834906243755,
1.478918229526628,
1.478918229526628,
2.0077351764243123,
2.4302809101456218,
2.4302809101456218,
3.009709563144601,
3.009709563144601,
3.239471043198664,
3.2405283031129133,
3.2405283031129133,
3.2656598154222434,
3.5353392937639825,
6.198713539858257,
6.213604956994236,
6.73397854893658,
6.73397854893658,
6.743547791194661,
6.744609103740889,
6.744609103740889,
6.749543065628153,
6.749543065628153,
8.615256229309818,
8.635090640841518,
9.012323512360366,
9.012323512360366,
9.930483613907942,
9.930483613907942,
11.782061805135282,
0.6870051227789642,
0.5234095533264759,
0.11216801162302814,
0.044591843687496906,
-0.747159787599118,
1.5727279855217766,
-0.16958009141159439,
-0.7234855106318131,
0.45999589631996707,
1.3613822619347415,
0.5234464983029046,
0.18337466193510712,
1.0660408011194182,
1.2813123812864906,
0.34899885807903364,
1.2204971120849089,
0.37250493843195553,
0.3861568591798728,
-0.4881036169368887,
0.23720347697856276,
0.8136369801063428,
-0.8595161765585049,
0.6710512785075716,
0.5103741134651811,
-0.48491784736765847,
-1.939727949979543,
0.5221979500550953,

```

-0.39618913325979604,
-0.5258671175956018,
0.08753210163723862,
1.3619602585721242,
-1.2518915867901346,
0.1520244579503878,
-1.126921262549515,
-1.1285674400831274,
-0.7045257843816373,
-0.4468310259761979,
0.6091022847926961,
0.4198375873878555,
0.5765253059135943,
-2.8026328359302406,
-1.7254430077855836,
0.37020611055835473,
0.4465655907565091,
-1.4308928069625704,
1.6254306194316424,
-0.16298810299835312,
-1.249293985363804,
-2.5283772945229797,
-1.0735775181393186,
-0.628541665091616,
-1.1357659634071433,
-0.5734897531715744,
-1.6096579963890296,
-0.8342055948126037,
-1.5447846874108708,
-1.1270394146554403,
0.06392732561319403,
0.7711512775812011,
-1.8472478314151175,
-0.46160141905881696,
1.8963279721150403,
-2.313663424814036,
0.060952702837845166,
-1.4971949982077863,
1.3040610818975555,
0.10605679464403613,
-1.138800572315487,
-0.03947636943653289,
0.07839455865010819,
-0.6470277815644087,
0.946101637008616,
-0.32144060478582315,
-0.6371693991779968,
0.30580409265523,
-0.2617129381706641,
0.4094287336545931,
0.49837362733711055,
-0.9058714721495094,
0.6137893604415533,
0.8546844695081749,
-0.8205470722456557,
-0.598852289145785,
0.46823811794119646,
1.2205247300232143,
0.031629319060699435,
1.1369587010491427,
0.12298062775535874,
-0.016960252964031076,
1.1127338626212067,
0.37972118532018134,
-0.1475616440844684,
0.32749925367563326,
0.030050642250761914,
0.17169542743574548,
-0.5825729547476518,
0.7052714548722974,
0.9884296874822006,
-0.500384308215816,
0.01722431287494931,
0.40290712722731054,
-0.029200452046427447,

```

1.2704708243944813,
2.3709704014259345,
-2.0267551524865275,
0.7347873374577798,
-0.34767223966060473,
0.009778943736386289,
0.221000841842164,
-0.010995780695540691,
-0.481889271583797,
0.7008500159073542,
2.3944065221124076,
-0.08320445199579081,
2.02470278988152,
0.02142760118000664,
1.6794429086669174,
1.6046551807968945,
0.2831685070910348,
1.5805135598594935]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0],
[6, 6, 6, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.429933051303131],
[12, 6, 21, 8, -3.1990719612158447],
[20, 10, 35, 16, -1.635137691104089],
[28, 14, 49, 24, -0.8526977600673816],
[36, 18, 63, 32, -0.4495330332284964],
[44, 22, 77, 40, -0.23666355366027367],
[52, 26, 91, 48, -0.12224019497382062]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 7.414817810058594e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.440816890128959,
'others': { 'beta_coefficients': [ 2.3290124519475084,
-3.8651696011043746,
5.6068818287625195,
-5.265593369620453,
3.048036430401101,
-0.9926938319776267,
0.13952609159132542],
'cost_richardson_zne': 89.82286707785958,
'sorted_expectation_vals': [ -6.429933051303131,
-3.1990719612158447,
-1.635137691104089,
-0.8526977600673816,
-0.4495330332284964,
-0.23666355366027367,
-0.12224019497382062],
'sorted_noise': [13, 47, 81, 115, 149, 183, 217]},
'sampled data': [ [4, 2, 7, 0, -6.429933051303131],
[12, 6, 21, 8, -3.1990719612158447],
[20, 10, 35, 16, -1.635137691104089],

```

```

[28, 14, 49, 24, -0.8526977600673816],
[36, 18, 63, 32, -0.4495330332284964],
[44, 22, 77, 40, -0.23666355366027367],
[52, 26, 91, 48, -0.12224019497382062]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.3773346757087749,
0.5734233884194113,
0.5734233884194113,
1.2094818154974225,
1.2528672337957094,
1.3073966674132076,
4.193348178434801,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.228670385547864,
4.274742575727675,
5.343872214974395,
5.354523589318104,
5.449133113124401,
5.449133113124401,
5.449133113124401,
5.449133113124401,
6.208341300319731,
6.208341300319731,
6.738723848778546,
6.7473872692498285,
8.515456463543387,
8.552181104315652,
8.552181104315652,
8.552181104315652,
9.363042150968939,
9.363042150968939,
11.478464454984158,
-0.2683676450988651,
-0.9370321768196137,
0.03790413469005285,
-0.3520989510779511,
0.5131804204688105,
0.767748989456867,
-1.1529933056041268,
1.0195402939041247,
-0.5189860890713169,
1.700377968800765,
0.48912124484584857,
1.506730848787325,
0.4485612726223177,
-1.6248073049289613,
0.10079320215400162,
1.7454590745233427,
1.474284546585253,
0.08837986403943651,
1.215839965531947,
-0.42666568442503866,
0.04298065245990749,
-1.495484471060033,
0.39055472334393304,
-0.06187227108331559,
1.0010296249342066,
-0.33745943355443286,
-0.19046242319241546,
-0.4884459619425607,
0.8930564792122229,
0.09947441181287657,
-1.4049039336860292,
1.0413688199593338,
0.027348986225077598,

```

1.5457890290583323,
0.8022774572900748,
-1.0377116853398,
0.5652694406258009,
0.9990396087940023,
-0.3409174955760714,
-0.8590373420111477,
0.3662952964918326,
0.6492392290096071,
-1.0735392409833318,
-0.620775369578654,
0.5064494688850749,
0.6065178361322348,
0.31795568319369283,
-0.4388081557471054,
0.07128954327413695,
-0.07460186306053893,
-1.5916050363108065,
0.40348292639904854,
0.7429600006983637,
0.13897118193917846,
0.13192395570874205,
-0.560408177185694,
-0.7293430452459922,
1.2393938765994381,
0.16520016983754293,
-0.04218705839108043,
1.9844775109402493,
-0.40811731341033075,
0.40494641878472026,
0.44317505546879765,
-0.14541224425415586,
-1.3628454492957067,
0.35572573997739254,
-0.17363781014762678,
-0.21765464091811726,
0.25867396534698067,
-0.19299239511743832,
0.842583631219517,
0.01538577194684443,
-1.3780712180305699,
0.30168991607554185,
0.03816118508668757,
-0.6330896266111891,
-0.14462412237866193,
0.22751230177420415,
2.198362599080823,
0.5538971259685714,
1.6735676153080048,
0.01161307703421222,
0.15241612900647253,
-0.4114895408906573,
0.020024246948836234,
-0.361021382333633,
1.2366726872792735,
-0.8652479233612058,
-0.38157323774943186,
-1.2043495043519374,
0.03294448415473864,
1.864475600371399,
-0.048446953541905424,
1.483038428004856,
0.02119809714572118,
-0.20756566215919184,
1.2566509021622863,
0.4040298326697209,
0.0492062183426151,
0.3396501220758691,
0.09097515161528406,
-0.4545275805495163,
1.5327906096069877,
-0.47568898039994184,
0.7447654713950665,
0.7179833985093875,
-0.03692927531761856,

```

0.11509825242982577,
0.020898935398901888,
0.2387893529411216,
1.5195379458286031,
-0.5337205856413728,
-0.5741707032781822,
1.298358172715442,
-0.010829860462227588,
-1.9037204771139875,
1.5774751765228583,
-0.28550943128124906,
1.5531884118568244]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0],
[6, 6, 6, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.448454576194539],
[12, 6, 21, 8, -3.24468946816992],
[20, 10, 35, 16, -1.6866076162118238],
[28, 14, 49, 24, -0.8984323471828195],
[36, 18, 63, 32, -0.4857082686875758],
[44, 22, 77, 40, -0.26348794296355765],
[52, 26, 91, 48, -0.14141822148386676]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 6.246566772460938e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.44171126504305,
'others': { 'beta_coefficients': [ 2.3290124519475084,
-3.8651696011043746,
5.6068818287625195,
-5.265593369620453,
3.048036430401101,
-0.9926938319776267,
0.13952609159132542],
'cost_richardson_zne': 89.82286707785958,
'sorted_expectation_vals': [ -6.448454576194539,
-3.24468946816992,
-1.6866076162118238,
-0.8984323471828195,
-0.4857082686875758,
-0.26348794296355765,
-0.14141822148386676],
'sorted_noise': [13, 47, 81, 115, 149, 183, 217]},
'sampled data': [ [4, 2, 7, 0, -6.448454576194539],
[12, 6, 21, 8, -3.24468946816992],
[20, 10, 35, 16, -1.6866076162118238],
[28, 14, 49, 24, -0.8984323471828195],
[36, 18, 63, 32, -0.4857082686875758],
[44, 22, 77, 40, -0.26348794296355765],
[52, 26, 91, 48, -0.14141822148386676]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,

```

```
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                              'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                              'r': 0},
          'time': {'max': 10.0, 'min': 0.0},
          'type': 'xy-iss'}},
'init_param': { 'value': [ 1.5680106786222674e-17,
                             1.0238650999177359,
                             1.0238650999177359,
                             1.0238650999177359,
                             1.7041550191762513,
                             2.070931805530806,
                             2.586234954306046,
                             2.6265616878606783,
                             2.7566136785306155,
                             2.7689271757054255,
                             4.271831894498143,
                             4.271831894498143,
                             4.271831894498143,
                             5.004110362344359,
                             5.004110362344359,
                             5.529689350278954,
                             5.529689350278954,
                             5.529689350278954,
                             5.5493166460873375,
                             7.855838962179162,
                             7.855838962179162,
                             8.508023697610218,
                             8.540796404638465,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             8.940056981382925,
                             9.524293982069336,
                             9.524293982069336,
                             9.815045396471412,
                             0.717685470636315,
                             0.29125971197830336,
                             0.013619559122678915,
                             -0.46174394022911175,
                             -0.7802589318548062,
                             1.8785332138943256,
                             -0.07718385116900997,
                             -0.4181315830102034,
                             0.3859499067718677,
                             0.7218523694763775,
                             0.16350787496293626,
                             -0.06274337543165032,
                             1.048639096314506,
                             0.9454500907927006,
                             -0.18328208251477485,
                             0.5336296017995852,
                             0.7047960543548563,
                             -0.4717130073686535,
                             -0.14327049451726778,
                             0.5646728024946054,
                             0.1377201649736573,
                             1.0845662084088512,
                             0.7117671017527167,
                             1.8562916393289055,
                             -1.0958742208007046,
                             0.314756410152813,
                             1.0055290931424188,
                             -1.2653176839886926,
                             -0.6435463058390287,
                             -0.13689785210195535,
                             0.5185042269644529,
                             -0.07994120596020922,
                             -1.3394312085309903,
                             -0.01354724643952521,
                             -0.8508707890759795,
                             -0.7258020036409927,
                             -0.1513648764845159,
                             -0.5702686618598916,
                             -0.20804467461705725,
```

-0.059043388294177074,
0.9724468840137966,
1.6889388278999584,
0.19460354666786345,
1.52688995899303,
0.40690710189505636,
0.054839646577847824,
-1.9384169999074634,
-0.8955556931339368,
0.22344394192624842,
-2.4039574731682265,
-0.22903568222623788,
-0.008629533947631484,
1.3318794505555018,
-0.04476169800775707,
-0.5314892190926975,
-0.05384939938166128,
0.5067524952638964,
0.7555528952477042,
0.6437292330207226,
0.02590432636550948,
0.44592066957482307,
-0.0809779454755679,
-1.1390483539597829,
0.4898473400025203,
0.11381426982162568,
0.6089836387514223,
0.17594570893388972,
0.012948010393531988,
-0.547404644839424,
0.028969189369927453,
1.1006601522977721,
-0.6406183594181203,
-0.12562720825840668,
2.0392738219317867,
0.9906966768182601,
0.03179376057566713,
1.5460159269414353,
-0.01674613470972367,
-1.1841027420803907,
0.24496917335434354,
-0.07332300856328518,
0.25897293698241475,
0.35918978324443196,
-0.007942172382781155,
0.4618323110818234,
-0.08547471514820712,
-0.5115221242285567,
0.4563488682048878,
-1.2086759202201718,
0.14282176979415442,
0.48263581465206523,
0.09855500655515351,
0.5327824526475502,
-0.19110982084320108,
-0.775795154813047,
-0.31907838177549175,
0.03477100089693282,
0.7436201132387485,
-0.022209896476399357,
-0.14741027177548285,
-0.10110024079398225,
0.20359400710495562,
-0.15623022021775929,
0.1964090589605299,
0.18042072654891128,
0.8910666012551952,
0.03518277211157266,
0.11858050209575345,
0.6912090997145024,
0.05142750966379351,
1.2523961498884204,
-1.1477698487953947,
-0.16063831357643202,
-1.0074319345942784,


```

0.29535289246662044,
-0.002280412475169233,
0.45599081133710123,
-1.6715207166067523,
0.3320227490012926,
1.5421306314740455]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0],
[6, 6, 6, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.515742960091852],
[12, 6, 21, 8, -3.36492111003702],
[20, 10, 35, 16, -1.7545658776574036],
[28, 14, 49, 24, -0.9139478342631155],
[36, 18, 63, 32, -0.46803177808311025],
[44, 22, 77, 40, -0.22931309151635063],
[52, 26, 91, 48, -0.10153016436427796]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 5.14984130859375e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.407527909298077,
'others': { 'beta_coefficients': [ 2.3290124519475084,
-3.8651696011043746,
5.6068818287625195,
-5.265593369620453,
3.048036430401101,
-0.9926938319776267,
0.13952609159132542],
'cost_richardson_zne': 89.82286707785958,
'sorted_expectation_vals': [ -6.515742960091852,
-3.36492111003702,
-1.7545658776574036,
-0.9139478342631155,
-0.46803177808311025,
-0.22931309151635063,
-0.10153016436427796],
'sorted_noise': [13, 47, 81, 115, 149, 183, 217]},
'sampled data': [ [4, 2, 7, 0, -6.515742960091852],
[12, 6, 21, 8, -3.36492111003702],
[20, 10, 35, 16, -1.7545658776574036],
[28, 14, 49, 24, -0.9139478342631155],
[36, 18, 63, 32, -0.46803177808311025],
[44, 22, 77, 40, -0.22931309151635063],
[52, 26, 91, 48, -0.10153016436427796]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},

```

```
'init_param': { 'value': [ 0.0019047787007362745,  
0.7201496918445128,  
0.7201496918445128,  
0.7201496918445127,  
0.7226708680142584,  
0.8068041864232433,  
1.3847174673528184,  
1.6031895944534578,  
2.458865333289775,  
2.458865333289775,  
2.721120236828938,  
2.721120236828938,  
3.554687304897056,  
3.554687304897056,  
4.537252190232355,  
4.537252190232355,  
5.239770593633025,  
5.242363659304797,  
5.566731623636481,  
5.566731623636481,  
5.566731623636481,  
5.566731623636481,  
5.566731623636481,  
7.6484324534494865,  
7.6484324534494865,  
8.96757120343865,  
9.352637118466301,  
9.381932224950582,  
9.966925955531599,  
9.966925955531599,  
0.29840004251263574,  
0.3207087147269936,  
-0.22478909909228914,  
-1.2242611593120782,  
0.061748844824383295,  
1.9931455249507801,  
0.02449362327304914,  
-0.23402928130657288,  
0.3399922671860315,  
0.2355005268966802,  
0.7912945290539132,  
-0.9147564287830934,  
0.12280724330167514,  
0.42016875545313104,  
0.6183328009701774,  
0.9098148543333162,  
0.2439503202728901,  
0.4941740976390827,  
0.0721485641264495,  
0.20051214721509877,  
0.14273239312798908,  
-0.03846070968482946,  
0.11474641891065286,  
0.1552743298359438,  
-0.050455466832234855,  
0.24385918138398863,  
0.0646818550964626,  
1.0379575003266166,  
0.4438839418702803,  
0.6652939620943935,  
0.6574583263339632,  
0.6386444070388955,  
0.30923240052603657,  
-0.34636230410321134,  
0.23860747048870812,  
-0.8837469955189079,  
0.25272330748816335,  
-0.05865013289270936,  
0.324900593333006,  
0.14494577696571714,  
-0.2063591527406891,  
0.03616506891850059,  
0.47262902670638013,  
-1.0341740644891497,  
0.05484745701468325,
```

0.6503839058002072,
0.9852671145762593,
-0.15013660688285613,
-1.1623207819211936,
-0.09407804105158574,
-0.7141485635922941,
-0.20002766061477661,
0.256318263131488,
1.811390397828268,
-0.4480297381108074,
-0.0031154151062657686,
0.685008095432428,
-0.007804627058269528,
-2.9100406031766375,
-0.17918863042787767,
-0.29620353111227854,
-0.2953351498676352,
0.4434234593316091,
0.0018987515325079429,
0.5290648024022053,
0.011057994967698829,
0.03103232328752182,
-0.21693206073792237,
-0.07883582131375938,
-0.2554965378040217,
1.978711034134111,
-0.02822992907519939,
1.110830733101554,
0.04155227001102173,
0.17370289874025888,
-0.08435695708516765,
0.38686640173882947,
1.0727525781247964,
-1.4871438236674412,
0.1437881416009117,
-0.15769165641037303,
0.46096388290194723,
0.3881288491041242,
0.3113157866411551,
-0.21237825892214532,
1.3408904937177344,
0.033287678968192874,
-0.22654084728278404,
0.09162287503278096,
1.1250326534568733,
0.670327239820091,
-1.4911819217025961,
0.8417198858779599,
0.018685303562101165,
-0.5955098452611822,
1.37350461224772,
-0.5503685346151788,
0.07415629279393592,
0.22396953453102358,
-0.02219350017922496,
2.5136787648879593,
1.5560141766956836,
-0.037039417521638966,
-1.5578667717297374,
-0.15174825264689498,
-0.0075908588231686416,
-0.1347910987366526,
0.7535559035800633,
2.9519825568174283,
-0.8640992588734872,
-0.486197602889753,
-0.029079605859607876,
0.1393557947212105,
-0.03201203543340934,
-0.17255788414719309,
-0.6220837667998668,
0.10441375911567917,
0.7019930428559267,
1.822960884928266,
0.0010256621400676815]]},

```

'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0.001, 0.001],
                  'status': True,
                  'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                  'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                  'r': 1},
               'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0],
                                     [6, 6, 6, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.536182288327376],
                          [12, 6, 21, 8, -3.450081350248876],
                          [20, 10, 35, 16, -1.8650932993173606],
                          [28, 14, 49, 24, -1.0247351729167145],
                          [36, 18, 63, 32, -0.5670015302617211],
                          [44, 22, 77, 40, -0.31237897705332807],
                          [52, 26, 91, 48, -0.16869939858614091]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 6.270408630371094e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -8.390901982465364,
                          'others': { 'beta_coefficients': [ 2.3290124519475084,
                                                            -3.8651696011043746,
                                                            5.6068818287625195,
                                                            -5.265593369620453,
                                                            3.048036430401101,
                                                            -0.9926938319776267,
                                                            0.13952609159132542],
                          'cost_richardson_zne': 89.82286707785958,
                          'sorted_expectation_vals': [ -6.536182288327376,
                                                        -3.450081350248876,
                                                        -1.8650932993173606,
                                                        -1.0247351729167145,
                                                        -0.5670015302617211,
                                                        -0.31237897705332807,
                                                        -0.16869939858614091],
                          'sorted_noise': [13, 47, 81, 115, 149, 183, 217]},
                          'sampled data': [ [4, 2, 7, 0, -6.536182288327376],
                                              [12, 6, 21, 8, -3.450081350248876],
                                              [20, 10, 35, 16, -1.8650932993173606],
                                              [28, 14, 49, 24, -1.0247351729167145],
                                              [36, 18, 63, 32, -0.5670015302617211],
                                              [44, 22, 77, 40, -0.31237897705332807],
                                              [52, 26, 91, 48, -0.16869939858614091]],
                          'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                    'r': 0},
                        'time': { 'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}}},
  'init_param': { 'value': [ 3.750846163374531e-05,
                             0.22285241391868899,
                             0.26252411232084344,
                             0.26252411232084344,
                             0.26252411232084344,
                             1.459545906754556,

```

1.4595459067545562,
1.6094835194149661,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
2.5927782099427628,
2.5927782099427623,
3.3552499939702187,
3.3552499939702187,
5.033284106467877,
5.033284106467877,
5.056477210452712,
6.768913439946703,
6.768913439946703,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
7.2782492670747585,
8.934198246659292,
8.937369801451712,
9.544406408519471,
9.544406408519471,
10.23170722265548,
-0.4501443853573418,
0.06519513561037263,
-0.13386490845883003,
-0.01542333988020039,
0.3988203537526196,
0.9163371790808124,
0.08159154039220383,
-0.11665429084087826,
-0.21130662780886203,
1.515236372470638,
0.4436142129357345,
-0.28780404896369716,
0.3681425383234808,
0.648176841676201,
0.3046875502986337,
0.05897820457762899,
0.09316951897603123,
1.5277657654344678,
-0.655141628591867,
0.073084468489131,
-0.11423944001370272,
0.3986898081963471,
1.0974832156349317,
1.8462557861113,
0.5998463368708868,
-0.0014214294560794885,
0.4026665434776887,
-0.03860309342262787,
0.3728818533964108,
1.7322816292529104,
0.42156018976657544,
0.9872712539907578,
0.5749190583224287,
0.1654949387796794,
1.0577776390475342,
-0.5892377479966974,
0.4324416138472369,
0.7909276553155474,
1.0132171075838867,
0.008626434202972095,
-1.4776636473003866,
-0.0743310361925915,
1.1363109745761366,
0.32003332101567866,
0.08535303128967886,
1.1165485958252095,
0.476282782575683,
0.11414859665640108,
0.8927652783900955,
0.027791184984770272,
0.8970588198442948,

```

-0.04579755391586939,
0.10596016247070955,
0.4721342517962854,
0.3142323996684511,
-0.0207983447823891,
1.3304831617683006,
-0.20639835153288702,
0.2928189752635581,
-0.4120973709154422,
-0.16251962266938982,
-0.928532755338732,
0.13709877636127538,
0.09386847134985479,
-0.9415336108192559,
-1.3861285848764584,
0.5116476525314603,
-1.5694556777668651,
0.31948518026769196,
-0.0136576069296156,
0.8752899744519326,
-0.448175539102687,
-0.8567162669878051,
0.737642795798484,
-0.5233142967805873,
-0.020239952918795164,
0.9696407297918527,
1.064479355466651,
-0.43102633401249074,
0.3234367312145231,
0.02606795235445944,
0.2525608175326581,
0.09544790103129135,
0.30399471295861424,
0.6638935631613432,
2.095756590649309,
-0.5383535092385583,
-0.1523671776118538,
-0.49733805370513506,
1.0814777965289457,
-0.5541810322110113,
0.4702060218904802,
0.1941018040115748,
0.027389458133440587,
-0.6339976476553459,
0.824073822166477,
-0.8904328829691108,
0.5852778403101284,
-0.37705413851354036,
-0.20234246488211116,
0.19390975159216547,
-0.03854894368679599,
-0.5885610444718222,
0.2892174263372239,
-0.7158388932188376,
-0.03853087013753846,
0.36228239756074476,
0.06448164730191597,
0.8103021451607575,
0.04195704491201313,
2.1946244272306323,
1.1078290402756337,
-0.0920096926231485,
0.7957980849005134,
0.10683086015182242,
0.04139972880547422,
1.068477454710041,
1.5812878062267348,
1.0292383703670558,
1.5822631942162138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],

```

```

        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 1},
        'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 0],
                                     [2, 2, 2, 0],
                                     [3, 3, 3, 0],
                                     [4, 4, 4, 0],
                                     [5, 5, 5, 0],
                                     [6, 6, 6, 0]]},

'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.4606782650454075],
                          [12, 6, 21, 8, -3.273885460810631],
                          [20, 10, 35, 16, -1.6974907774474757],
                          [28, 14, 49, 24, -0.8939947022135478],
                          [36, 18, 63, 32, -0.47364366307740574],
                          [44, 22, 77, 40, -0.2491082243344117],
                          [52, 26, 91, 48, -0.12744297687087136]],

        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 6.556510925292969e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -8.41727174756573,
                          'others': { 'beta_coefficients': [ 2.3290124519475084,
                                                            -3.8651696011043746,
                                                            5.6068818287625195,
                                                            -5.265593369620453,
                                                            3.048036430401101,
                                                            -0.9926938319776267,
                                                            0.13952609159132542],
                          'cost_richardson_zne': 89.82286707785958,
                          'sorted_expectation_vals': [ -6.4606782650454075,
                                                        -3.273885460810631,
                                                        -1.6974907774474757,
                                                        -0.8939947022135478,
                                                        -0.47364366307740574,
                                                        -0.2491082243344117,
                                                        -0.12744297687087136],
                          'sorted_noise': [13, 47, 81, 115, 149, 183, 217]},
                          'sampled data': [ [4, 2, 7, 0, -6.4606782650454075],
                                              [12, 6, 21, 8, -3.273885460810631],
                                              [20, 10, 35, 16, -1.6974907774474757],
                                              [28, 14, 49, 24, -0.8939947022135478],
                                              [36, 18, 63, 32, -0.47364366307740574],
                                              [44, 22, 77, 40, -0.2491082243344117],
                                              [52, 26, 91, 48, -0.12744297687087136]],
                          'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                    'r': 0},
                        'time': { 'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.003375513600300809,
                             0.4884127443235871,
                             0.6719352108785641,
                             0.9073170135093351,
                             1.4946633690060278,
                             1.5076957064847494,
                             3.872917838142945,
                             3.872917838142945,
                             3.872917838142945,
                             3.872917838142945,
                             5.047622474508209,
                             5.054336886650782,

```

5.067626550795242,
5.0733174878702085,
5.094629387163461,
5.749571576552799,
5.7731554023304374,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.144946717661993,
7.602314289267704,
7.602314289267704,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
8.018775147517832,
9.661071929582006,
0.2735947252101626,
2.471262065567802,
0.18954222370300397,
0.2633058348660961,
0.07494334922808346,
1.4980157261935538,
0.3538857424122425,
0.26341770410626436,
0.17174804107002956,
-0.22912718472194316,
0.11060714764109056,
-0.30804859927082595,
2.8406850614439736,
1.0823654569610508,
0.9668529413216363,
1.8727663516543385,
-0.0901659930896485,
0.05868560690604535,
0.24212775097742179,
1.0092515209898247,
-0.7556969386491817,
0.3279350524218167,
1.078117272505244,
0.10391815458190773,
-0.7434930272741371,
-1.6965388530236032,
-1.360748554100242,
0.34939560927302815,
0.03188634842980519,
-1.2447472675125983,
0.06092342743494014,
1.017082923902293,
-1.007905562333318,
-0.8576057837691866,
0.5321038189070756,
1.300652060018237,
-1.8122591145188616,
-1.247002191580596,
1.2529274198051785,
0.3750417535596218,
0.2898066740735119,
1.589675559301449,
2.3222772390172888,
-1.568740409290771,
0.5925151941094322,
-0.20655148780097043,
-0.5059482271844034,
-0.9561046882137155,
-0.432249774596058,
1.8189144734584217,
0.6791516075394335,
-0.10885247153326365,
-2.2650870540994883,
0.2658818907332787,
-0.9372958671062981,
-1.0664167087973948,
-0.7139434901419379,


```

1.3816698348804732,
2.008892916411081,
0.15285666374274742,
-1.782425531898792,
0.07791440554206944,
-1.28397958204954,
-0.7247696039786395,
0.5612678373457747,
-0.004762916486662008,
0.4386089298525658,
0.010283368626678616,
0.8990610129088383,
0.03840864905637026,
-1.2418081984826976,
0.33190310279028407,
0.3061008622829172,
-0.06950158836023225,
0.93859456058615,
-0.02489786142317165,
0.9927840878492759,
-0.0709710381209458,
-1.4546153026559692,
-0.6707274766895803,
1.0492831078613678,
1.59883886587945,
-0.7948795957139008,
0.011914265512354442,
0.29611851635537545,
-0.026452865362460363,
0.17911271593442915,
-0.268102313362264,
-0.31698956355176716,
0.5150163737561883,
-0.2617344493849528,
-0.08207987775663798,
-0.7031192006515304,
0.10296103394319861,
-1.3322481995457,
-0.8096620548125795,
0.22894179493051453,
-0.6404807766018014,
0.5981001234262467,
0.6725836248662437,
-0.24769451099401643,
-1.5367942204876157,
0.32496629998060117,
0.8231756699235411,
0.27229759145944715,
-0.4283887626472357,
0.6783210911797265,
1.178917544999848,
-0.48124012627536317,
-0.23642700938756703,
-0.22983768395119897,
0.7841497681166887,
0.5963513861028734,
-0.8505831212819382,
0.6516631671421783,
0.6698405071431395,
0.7826636181190869,
-1.6023915303224998,
0.6733061844063541,
1.5755992624240989]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.001, 0.001],
                    'status': True,
                    'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],

```

```

[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0],
[6, 6, 6, 0]],

'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 7, 0, -6.397817460329862],
                           [12, 6, 21, 8, -3.2115716730808748],
                           [20, 10, 35, 16, -1.6419588170332553],
                           [28, 14, 49, 24, -0.8464644349040009],
                           [36, 18, 63, 32, -0.4337371704190119],
                           [44, 22, 77, 40, -0.216007189653039],
                           [52, 26, 91, 48, -0.10025204405746856]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.4345855712890625e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -8.358064334291676,
                            'others': { 'beta_coefficients': [ 2.3290124519475084,
                                                                -3.8651696011043746,
                                                                5.6068818287625195,
                                                                -5.265593369620453,
                                                                3.048036430401101,
                                                                -0.9926938319776267,
                                                                0.13952609159132542],
                            'cost_richardson_zne': 89.82286707785958,
                            'sorted_expectation_vals': [ -6.397817460329862,
                                                          -3.2115716730808748,
                                                          -1.6419588170332553,
                                                          -0.8464644349040009,
                                                          -0.4337371704190119,
                                                          -0.216007189653039,
                                                          -0.10025204405746856],
                            'sorted_noise': [13, 47, 81, 115, 149, 183, 217]},
                            'sampled data': [ [4, 2, 7, 0, -6.397817460329862],
                                                [12, 6, 21, 8, -3.2115716730808748],
                                                [20, 10, 35, 16, -1.6419588170332553],
                                                [28, 14, 49, 24, -0.8464644349040009],
                                                [36, 18, 63, 32, -0.4337371704190119],
                                                [44, 22, 77, 40, -0.216007189653039],
                                                [52, 26, 91, 48, -0.10025204405746856]],
                            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                          'time': { 'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}}},
  'init_param': { 'value': [ -6.331146189926128e-17,
                              0.7069927066774304,
                              0.7069927066774304,
                              0.7069927066774305,
                              0.7069927066774304,
                              1.0044153364322925,
                              1.0092418692996383,
                              2.470333794214341,
                              2.478143749866421,
                              3.5776564981759003,
                              3.5776564981759003,
                              3.5776564981759003,
                              3.61745967653339,
                              3.6981780389472076,
                              3.755720232685741,
                              3.9712286023902594,
                              3.9712286023902594,
                              4.519037432417192,

```

4.519037432417193,
4.519037432417193,
5.280953883299406,
5.280953883299406,
6.620356980277175,
6.620356980277175,
6.620356980277175,
6.620356980277176,
8.214004051881032,
8.214004051881032,
9.105951544537387,
10.829366164164977,
1.4490195266558639,
1.145329650589643,
-0.9374272684957279,
-1.2910248872233119,
0.9701069453007352,
2.0238893871034924,
-0.6731600678051081,
-0.7305037462514778,
0.6743483023845589,
1.611766624162738,
-0.162446688913207,
-0.2464021583836281,
1.3073653685111934,
1.560629053845934,
-0.09768282308379246,
-0.31393750682085847,
0.1625705045557987,
-0.1344649185634244,
-0.03516682581352061,
0.15934568227883428,
1.4484389719925783,
0.31346340320241645,
-1.5406853809043388,
0.3831701903776482,
0.07255447956326048,
-0.4619144712511035,
0.6220101497739178,
-0.08645510294766677,
-0.036774035023737585,
-0.02433291891195538,
1.2745777688613118,
0.806643386980706,
-0.1734465019831428,
-1.474099767063678,
-1.2534035864969235,
0.024086803725753666,
-0.5946515656594247,
3.047855036387052,
0.561263952510902,
-0.27353577574812815,
0.476385213099692,
-0.4183083264972014,
-0.07584344603997495,
-0.04004176630390785,
0.6379867024754904,
3.306912023625799,
0.6758488609505069,
-0.49379826249780634,
0.04089533617940984,
0.4062429454438471,
-0.2647421675477994,
0.1974729438666808,
0.043470303912556706,
0.2301819741695647,
1.9247102260310271,
0.19401136489510468,
0.17784496580796996,
-0.12037519976757159,
0.10596613899479704,
0.02480260398862852,
0.33616247382037256,
-0.2960626605489573,
1.2038327046358004,

```
-0.4266046665751028,
-0.14331073423696536,
0.33240571018120446,
0.13743208352564132,
0.21308235615415505,
-0.8816113951868988,
-0.03958129872975694,
1.7385124490728345,
1.3953860705003918,
0.7193393529381484,
-0.5890148767931709,
1.6845229617798814,
-0.04059142172715042,
-0.6588344885662398,
-1.5492107393137406,
0.9886630320432893,
1.3955090248591298,
-1.7521400718196618,
-0.02012078102450614,
1.5598528019020075,
-0.041272015671692544,
0.6659176073691222,
-0.10884895615623923,
-0.8860283635501729,
-0.042494102069769425,
-0.5241060846215897,
-0.049429521011899694,
-1.6158837397685775,
-0.29703279066650246,
0.6940812648197017,
-1.0869617874060826,
-1.1891726572683667,
0.03935191545498333,
-1.7727595774952096,
-0.0005664350349098603,
-1.2288438066739866,
2.182693040857262,
1.2648198022740724,
0.636776362006322,
-1.1418938134635428,
0.04892386539982865,
-0.6595482134655696,
-0.016038133219537475,
-1.53365152672916,
-0.5818287115590742,
-0.8537662908823758,
1.3418170063691075,
-0.32888813707646736,
-0.018460135902194714,
-0.8323242421777916,
1.5026664500941838,
1.809610363246633,
-1.5283405786971713,
0.5470095670143573,
-1.4753657998350065,
0.8122840148585063,
1.5317878779716176]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.001, 0.001],
'status': True,
'type': 'depolarizing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy_noisy_time_evo_ric7_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0],
[1, 1, 1, 0],
[2, 2, 2, 0],
[3, 3, 3, 0],
[4, 4, 4, 0],
[5, 5, 5, 0],
[6, 6, 6, 0]]},
```

